

Owner comparison matrix - beam-v03 (EC3) vs MasterSeries

63 cases, 118 runs, generated 2026-09-20. Basis: S275, 6.0 m simply supported fork ends, EC3 + UK NA, ULS 1.35G + 1.5Q, SLS 1.0Q; base loading G UDL 10 kN/m + Q UDL 8 kN/m + Q 30 kN at 2.0 m. One brief per run follows in this order. Verdicts: PASS 21, FAIL 89, NOT VERIFIED 4, REFUSED 4, ERROR 0.

Method: eigen = FE eigenvalue M_cr; standard = SN003a / P385 closed forms; full = fully restrained; refused = not run. Annex A = EN 1993-6 LTB + torsion (≥ 99 (capacity exhausted) = the amplifier is unbounded, $M_{y,Ed} \geq M_{cr}$). A channel's standard-route M_cr is the kappa-chain value (SN003a value in brackets where offered). N_Ed and M_z,Ed of L5..L7 are ULS design values entered directly. Comparison column: MasterSeries figures.

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b,Rd	Annex A	defl.	Comparison
UB-L1	UB 457 x 191 x 82	L1: no torsion	eigen	PASS	LTB M_Ed/M_b,Rd (0.581)	393.1	290.4	-	0.193	
UB-L1	UB 457 x 191 x 82	L1: no torsion	standard	PASS	LTB M_Ed/M_b,Rd (0.575)	398.4	293.4	-	0.193	
UB-L2	UB 457 x 191 x 82	L2: horizontal ecc. e = +40	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	393.1	290.4	0.897	0.193	
UB-L2	UB 457 x 191 x 82	L2: horizontal ecc. e = +40	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-L3t	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	eigen	PASS	LTB M_Ed/M_b,Rd (0.719)	283.9	234.8	-	0.193	
UB-L3t	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	standard	NOT VERIFIED	LTB M_Ed/M_b,Rd (0.575)	398.4	293.4	-	0.193	
UB-L3b	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	eigen	PASS	LTB M_Ed/M_b,Rd (0.491)	542.7	343.8	-	0.193	
UB-L3b	UB 457 x 191 x 82	L3: vertical ecc. z_g = +/-D/2	standard	PASS	LTB M_Ed/M_b,Rd (0.575)	398.4	293.4	-	0.193	
UB-L4	UB 457 x 191 x 82	L4: e + z_g	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.104)	283.9	234.8	1.104	0.193	
UB-L4	UB 457 x 191 x 82	L4: e + z_g	standard	NOT VERIFIED	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-L5	UB 457 x 191 x 82	L5: e + z_g + N_Ed	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.104)	283.9	234.8	1.104	0.193	
UB-L5	UB 457 x 191 x 82	L5: e + z_g + N_Ed	standard	NOT VERIFIED	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-L6	UB 457 x 191 x 82	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.700)	292.8	241.1	1.700	0.277	
UB-L6	UB 457 x 191 x 82	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Member buckling z-z (Eq 6.62) (1.423)	416.4	303.6	1.243	0.277	
UB-L7	UB 457 x 191 x 82	L7: L6 + couple 40 kN.m	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.959)	297.7	243.5	1.959	0.316	
UB-L7	UB 457 x 191 x 82	L7: L6 + couple 40 kN.m	standard	FAIL	Member buckling z-z (Eq 6.62) (1.502)	415.5	303.1	1.359	0.316	
UC-L1	UC 203 x 203 x 60	L1: no torsion	eigen	FAIL	Deflection (1.169)	277.9	151.8	-	1.169	
UC-L1	UC 203 x 203 x 60	L1: no torsion	standard	FAIL	Deflection (1.169)	281.9	153.0	-	1.169	
UC-L2	UC 203 x 203 x 60	L2: horizontal ecc. e = +40	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.647)	277.9	151.8	1.647	1.169	
UC-L2	UC 203 x 203 x 60	L2: horizontal ecc. e = +40	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.636)	281.9	153.0	1.636	1.169	
UC-L3t	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	LTB M_Ed/M_b,Rd (1.189)	219.6	141.0	-	1.169	
UC-L3t	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (1.169)	281.9	153.0	-	1.169	
UC-L3b	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (1.169)	350.7	160.5	-	1.169	
UC-L3b	UC 203 x 203 x 60	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (1.169)	281.9	153.0	-	1.169	
UC-L4	UC 203 x 203 x 60	L4: e + z_g	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.921)	219.6	141.0	1.921	1.169	
UC-L4	UC 203 x 203 x 60	L4: e + z_g	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.636)	281.9	153.0	1.636	1.169	
UC-L5	UC 203 x 203 x 60	L5: e + z_g + N_Ed	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.921)	219.6	141.0	1.921	1.169	
UC-L5	UC 203 x 203 x 60	L5: e + z_g + N_Ed	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.636)	281.9	153.0	1.636	1.169	
UC-L6	UC 203 x 203 x 60	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (7.159)	226.9	144.0	7.159	1.678	
UC-L6	UC 203 x 203 x 60	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (2.392)	294.8	156.4	2.392	1.678	
UC-L7	UC 203 x 203 x 60	L7: L6 + couple 40 kN.m	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	228.6	143.7	>= 99 (capacity exhausted)	1.913	
UC-L7	UC 203 x 203 x 60	L7: L6 + couple 40 kN.m	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (2.724)	294.1	156.3	2.724	1.913	
RHS-L1	RHS 200 x 100 x 8.0 (HF)	L1: no torsion	eigen	FAIL	Deflection (3.210)	899.1	77.5	-	3.210	
RHS-L1	RHS 200 x 100 x 8.0 (HF)	L1: no torsion	standard	FAIL	Deflection (3.210)	916.2	77.5	-	3.210	
RHS-L2	RHS 200 x 100 x 8.0 (HF)	L2: horizontal ecc. e = +40	eigen	FAIL	Deflection (3.210)	899.1	77.5	-	3.210	
RHS-L2	RHS 200 x 100 x 8.0 (HF)	L2: horizontal ecc. e = +40	standard	FAIL	Deflection (3.210)	916.2	77.5	-	3.210	

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b,Rd	Annex A	defl.	Comparison
RHS-L3t	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (3.210)	876.7	77.5		- 3.210	
RHS-L3t	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (3.210)	916.2	77.5		- 3.210	
RHS-L3b	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (3.210)	921.9	77.5		- 3.210	
RHS-L3b	RHS 200 x 100 x 8.0 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (3.210)	916.2	77.5		- 3.210	
RHS-L4	RHS 200 x 100 x 8.0 (HF)	L4: e + z_g	eigen	FAIL	Deflection (3.210)	876.7	77.5		- 3.210	
RHS-L4	RHS 200 x 100 x 8.0 (HF)	L4: e + z_g	standard	FAIL	Deflection (3.210)	916.2	77.5		- 3.210	
RHS-L5	RHS 200 x 100 x 8.0 (HF)	L5: e + z_g + N_Ed	eigen	FAIL	Bending+axial cross-section (6.2.9) (3.621)	876.7	77.5		- 3.210	
RHS-L5	RHS 200 x 100 x 8.0 (HF)	L5: e + z_g + N_Ed	standard	FAIL	Bending+axial cross-section (6.2.9) (3.621)	916.2	77.5		- 3.210	
RHS-L6	RHS 200 x 100 x 8.0 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (6.070)	910.8	77.5		- 4.605	
RHS-L6	RHS 200 x 100 x 8.0 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Biaxial bending + axial (6.2.9.1) (6.070)	958.8	77.5		- 4.605	
RHS-L7	RHS 200 x 100 x 8.0 (HF)	L7: L6 + couple 40 kN.m	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (7.033)	895.8	77.5		- 5.249	
RHS-L7	RHS 200 x 100 x 8.0 (HF)	L7: L6 + couple 40 kN.m	standard	FAIL	Biaxial bending + axial (6.2.9.1) (7.033)	956.0	77.5		- 5.249	
SHS-L1	SHS 150 x 150 x 6.3 (HF)	L1: no torsion	eigen	FAIL	Deflection (5.867)	1190.1	52.8		- 5.867	
SHS-L1	SHS 150 x 150 x 6.3 (HF)	L1: no torsion	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L2	SHS 150 x 150 x 6.3 (HF)	L2: horizontal ecc. e = +40	eigen	FAIL	Deflection (5.867)	1190.1	52.8		- 5.867	
SHS-L2	SHS 150 x 150 x 6.3 (HF)	L2: horizontal ecc. e = +40	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L3t	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (5.867)	1162.3	52.8		- 5.867	
SHS-L3t	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L3b	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	Deflection (5.867)	1218.4	52.8		- 5.867	
SHS-L3b	SHS 150 x 150 x 6.3 (HF)	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L4	SHS 150 x 150 x 6.3 (HF)	L4: e + z_g	eigen	FAIL	Deflection (5.867)	1162.3	52.8		- 5.867	
SHS-L4	SHS 150 x 150 x 6.3 (HF)	L4: e + z_g	standard	FAIL	Deflection (5.867)	1212.8	52.8		- 5.867	
SHS-L5	SHS 150 x 150 x 6.3 (HF)	L5: e + z_g + N_Ed	eigen	FAIL	Bending+axial cross-section (6.2.9) (7.031)	1162.3	52.8		- 5.867	
SHS-L5	SHS 150 x 150 x 6.3 (HF)	L5: e + z_g + N_Ed	standard	FAIL	Bending+axial cross-section (6.2.9) (7.031)	1212.8	52.8		- 5.867	
SHS-L6	SHS 150 x 150 x 6.3 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (11.579)	1207.9	52.8		- 8.417	
SHS-L6	SHS 150 x 150 x 6.3 (HF)	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Biaxial bending + axial (6.2.9.1) (11.579)	1269.4	52.8		- 8.417	
SHS-L7	SHS 150 x 150 x 6.3 (HF)	L7: L6 + couple 40 kN.m	eigen	FAIL	Biaxial bending + axial (6.2.9.1) (13.483)	1187.5	52.8		- 9.595	
SHS-L7	SHS 150 x 150 x 6.3 (HF)	L7: L6 + couple 40 kN.m	standard	FAIL	Biaxial bending + axial (6.2.9.1) (13.483)	1265.5	52.8		- 9.595	
PFC-L1	PFC 200 x 75 x 23	L1: no torsion	eigen	FAIL	LTB M_Ed/M_b,Rd (6.475)	35.8	25.6		- 3.652	
PFC-L1	PFC 200 x 75 x 23	L1: no torsion	standard	FAIL	LTB M_Ed/M_b,Rd (18.794)	9.1 (35.7)	8.8		- 3.652	
PFC-L2	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-L2	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b.Rd	Annex A	defl.	Comparison
PFC-L2w	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-L2w	PFC 200 x 75 x 23	L2: horizontal ecc. e = +40	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L3t	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	30.9	23.0	>= 99 (capacity exhausted)	3.652	
PFC-L3t	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L3b	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	41.4	28.2	>= 99 (capacity exhausted)	3.652	
PFC-L3b	PFC 200 x 75 x 23	L3: vertical ecc. z_g = +/-D/2	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L4	PFC 200 x 75 x 23	L4: e + z_g	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	30.9	23.0	>= 99 (capacity exhausted)	3.652	
PFC-L4	PFC 200 x 75 x 23	L4: e + z_g	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L5	PFC 200 x 75 x 23	L5: e + z_g + N_Ed	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	30.9	23.0	>= 99 (capacity exhausted)	3.652	
PFC-L5	PFC 200 x 75 x 23	L5: e + z_g + N_Ed	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-L6	PFC 200 x 75 x 23	L6: L5 + inclined load (approx., M_z,Ed)	eigen	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	32.0	23.7	>= 99 (capacity exhausted)	5.239	
PFC-L6	PFC 200 x 75 x 23	L6: L5 + inclined load (approx., M_z,Ed)	standard	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	5.239	
PFC-L7	PFC 200 x 75 x 23	L7: L6 + couple 40 kN.m	eigen	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	31.9	23.6	>= 99 (capacity exhausted)	5.972	
PFC-L7	PFC 200 x 75 x 23	L7: L6 + couple 40 kN.m	standard	FAIL	Pure shear failure at M-V check point (6.2.6) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	5.972	
UB-R1	UB 457 x 191 x 82	R1: fully restrained	full	PASS	Bending+torsion cross-section (P385 3.1.2) (0.408)	-	503.3	-	0.193	
UB-R2	UB 457 x 191 x 82	R2: unrestrained (reference)	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	393.1	290.4	0.897	0.193	
UB-R2	UB 457 x 191 x 82	R2: unrestrained (reference)	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-R3	UB 457 x 191 x 82	R3: restrained at 2, 4 m	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.564)	2355.7	485.7	0.564	0.193	
UB-R3	UB 457 x 191 x 82	R3: restrained at 2, 4 m	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-R4	UB 457 x 191 x 82	R4: warping restrained	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.587)	669.5	372.0	0.587	0.193	
UB-R4	UB 457 x 191 x 82	R4: warping restrained	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.739)	398.4	293.4	0.739	0.193	
UB-R5	UB 457 x 191 x 82	R5: warping free	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	393.1	290.4	0.897	0.193	
UB-R5	UB 457 x 191 x 82	R5: warping free	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-R6	UB 457 x 191 x 82	R6: destabilising switch, z_g = 0	eigen	NOT VERIFIED	LTB+torsion (EN 1993-6 Annex A) (0.897)	393.1	290.4	0.897	0.193	
UB-R6	UB 457 x 191 x 82	R6: destabilising switch, z_g = 0	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (1.043)	306.3	248.1	1.043	0.193	
UB-R7	UB 457 x 191 x 82	R7: stabilising z_g = -D/2	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.776)	542.7	343.8	0.776	0.193	
UB-R7	UB 457 x 191 x 82	R7: stabilising z_g = -D/2	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.890)	398.4	293.4	0.890	0.193	
UB-R8	UB 457 x 191 x 82	R8: L_E override 0.85 L	eigen	PASS	LTB+torsion (EN 1993-6 Annex A) (0.897)	393.1	290.4	0.897	0.193	
UB-R8	UB 457 x 191 x 82	R8: L_E override 0.85 L	standard	PASS	LTB+torsion (EN 1993-6 Annex A) (0.794)	510.6	335.1	0.794	0.193	
UB-R9	UB 457 x 191 x 82	R9: top flange restrained (= full)	full	PASS	Bending+torsion cross-section (P385 3.1.2) (0.408)	-	503.3	-	0.193	
UB-R10	UB 457 x 191 x 82	R10: bottom flange restrained	refused	REFUSED	-	-	-	-	-	
UB-R11	UB 457 x 191 x 82	R11: partially restrained	refused	REFUSED	-	-	-	-	-	
PFC-R1	PFC 200 x 75 x 23	R1: fully restrained	full	FAIL	Bending+torsion cross-section (P385 3.1.2) (14.062)	-	62.4	-	3.652	
PFC-R2	PFC 200 x 75 x 23	R2: unrestrained (reference)	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-R2	PFC 200 x 75 x 23	R2: unrestrained (reference)	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R3	PFC 200 x 75 x 23	R3: restrained at 2, 4 m	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	119.0	46.3	>= 99 (capacity exhausted)	3.652	
PFC-R3	PFC 200 x 75 x 23	R3: restrained at 2, 4 m	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R4	PFC 200 x 75 x 23	R4: warping restrained	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	41.6	28.2	>= 99 (capacity exhausted)	3.652	
PFC-R4	PFC 200 x 75 x 23	R4: warping restrained	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R5	PFC 200 x 75 x 23	R5: warping free	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-R5	PFC 200 x 75 x 23	R5: warping free	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R6	PFC 200 x 75 x 23	R6: destabilising switch, z_g = 0	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-R6	PFC 200 x 75 x 23	R6: destabilising switch, z_g = 0	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	6.3	6.3	>= 99 (capacity exhausted)	3.652	

id	section	owner case	method	verdict	governing (util.)	M_cr	M_b.Rd	Annex A	defl.	Comparison
PFC-R7	PFC 200 x 75 x 23	R7: stabilising z_g = -D/2	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	41.4	28.2	>= 99 (capacity exhausted)	3.652	
PFC-R7	PFC 200 x 75 x 23	R7: stabilising z_g = -D/2	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	9.1	8.8	>= 99 (capacity exhausted)	3.652	
PFC-R8	PFC 200 x 75 x 23	R8: L_E override 0.85 L	eigen	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	35.8	25.6	>= 99 (capacity exhausted)	3.652	
PFC-R8	PFC 200 x 75 x 23	R8: L_E override 0.85 L	standard	FAIL	LTB+torsion (EN 1993-6 Annex A) (>= 99 (capacity exhausted))	12.6	11.5	>= 99 (capacity exhausted)	3.652	
PFC-R9	PFC 200 x 75 x 23	R9: top flange restrained (= full)	full	FAIL	Bending+torsion cross-section (P385 3.1.2) (14.062)	-	62.4	-	3.652	
PFC-R10	PFC 200 x 75 x 23	R10: bottom flange restrained	refused	REFUSED	-	-	-	-	-	
PFC-R11	PFC 200 x 75 x 23	R11: partially restrained	refused	REFUSED	-	-	-	-	-	

UB 457 x 191 x 82, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0$, $z_g = 0$)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

The diagram illustrates a simply supported beam of length $L = 6.00 \text{ m}$. The beam is subjected to a point load of 30 kN acting downwards at the center, and two uniformly distributed loads: 8 kN/m (labeled Q) and 10 kN/m (labeled G), both acting downwards along the entire length of the beam. Below the beam, the deflection curve is shown, indicating a maximum deflection of 168.87 kN.m at the center of the beam.

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_{cr} > 0.5$, else 0	6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

F_{cr} = 0.9k_F.E.t_w³/h_w

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

F_{Ed}/F_{Rd}

η₂ + 0.8η₁ ≤ 1.4

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

0.9 x 3.402 x 210000 x 9.90³/428.00

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

√(192.79 x 9.90 x 275/1457.610e3)

0.5/0.600 = 0.833 ; 0.833 x 192.79

275 x 160.64 x 9.90/1.0

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

3.402

1457.610 kN

100.00 mm

192.79 mm

0.600

160.64 mm

437.340 kN

0.251

0.179

Fig 6.1(c)

6.4(1)

6.5(4)

6.5(4)

6.4(1)

6.4(1)

6.2(1)

OK

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.147

1.162

k_c only

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

λ_{LT} = √W.f_y/M_{cr}

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

M_{b.Rd} = χW_{pl.y}.f_y ≤ M_{c.y.Rd}

M_{y.Ed}/M_{b.Rd}

End 1 v, φ = 0; End 2 v, φ = 0

64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached

√ 1830.0 x 275 / 393.107

1.131, 1.159, 0.49

0.562, 1.131, 0.934, 0.974

0.577 x 1830.0 x 275 ≤ 503.250 =

168.868 / 290.421

6 m

393.107 kN.m

1.131

0.562

0.577

290.421 kN.m

0.581

FE

converged

Curve c

6.3.2.3

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 3.22 mm OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.581

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.581$

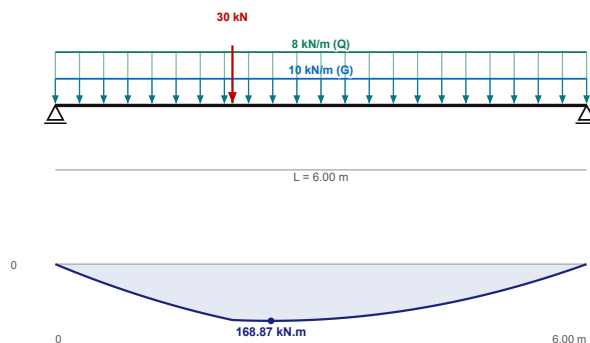
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0$, $z_g = 0$)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

the moment and shear envelopes keep the entered γ_G , so add a



Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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Auto Design Load Cases 1; SLS 1

$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 1830.0/1	503.250 kN.m	
$M_{v,Ed} / M_{c,v,Rd}$	168.868 / 503.250 =	0.336	OK

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$$

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	3.22 mm	OK
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.575

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.575$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L2 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.897
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at e = +40 mm (torque), z_g = 0

Beam & Beam-Portion (Member)

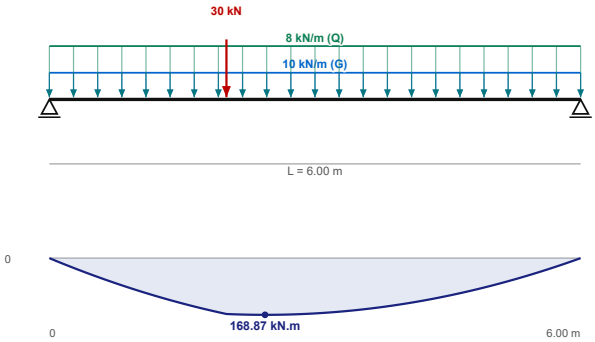
Member UB-L2 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = F_n(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}R_d

V_{y,Ed}/V_{pl,y}R_d 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y}R_d 109.758 / 756.260 = 0.145 OK
V_{pl,y}R_d = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y}R_d = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y}R_d 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x (428.00/16.00)² = 14.311 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

3.402

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 3.402 x 210000 x 9.90³/428.00

1457.610 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

192.79 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(192.79 x 9.90 x 275/1457.610e3)

0.600

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.600 = 0.833 ; 0.833 x 192.79

160.64 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 160.64 x 9.90/1.0

437.340 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251

OK

η₂ + 0.8η₁ ≤ 1.4

0.251 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.179

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

1.147

k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.162

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

End 1 v, φ = 0; End 2 v, φ = 0

6 m

FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached

393.107 kN.m

converged

λ_{LT} = √W.f_y/M_{cr}

√ 1830.0 x 275 / 393.107

1.131

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

1.131, 1.159, 0.49

0.562

Curve c

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

0.562, 1.131, 0.934, 0.974

0.577

6.3.2.3

M_{b.Rd} = χW_{pl.y}.f_y ≤ M_{c.y.Rd}

0.577 x 1830.0 x 275 ≤ 503.250 =

290.421 kN.m

M_{y.Ed}/M_{b.Rd}

168.868 / 290.421

0.581

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34 \text{ m}$: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $\tau_t = 27.22 \text{ N/mm}^2$; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2.365 \text{ m}$: 168.802/282.919 + 1.000x9.873/83.600 + 0.665x0.882x1.753x7.403/41.800	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660 \text{ kN.m}$	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.788 \text{ m}$, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67 \text{ mm}$ @ $x = 2.85 \text{ m}$	3.22 mm	OK
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.408	0.156	0.897	0.897

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L2 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.890
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at e = +40 mm (torque), z_g = 0

Beam & Beam-Portion (Member)

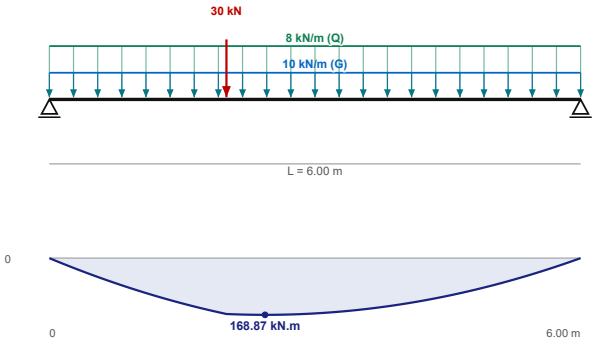
Member UB-L2 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = F_n(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}R_d

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 756.260 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x (428.00/16.00)² = 14.311 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ" (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.365 m: 168.802/285.039 + 0.890 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.408	0.156	0.890	0.890

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L3t · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **PASS** · governing LTB M_{Ed}/M_{b,Rd} = 0.719
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +230 mm (D/2, destabilising), e = 0

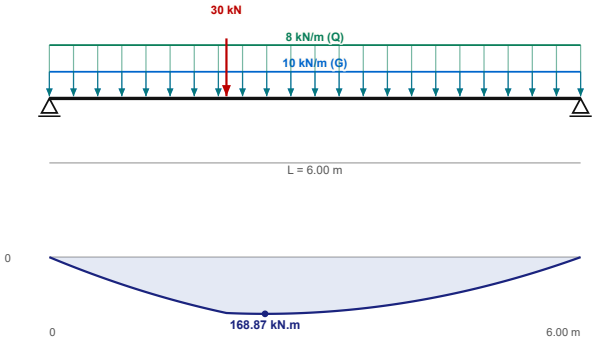
Beam & Beam-Portion (Member)
Member UB-L3t UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 0 mm z_g = 230 mm (kN/m)
Q UDL 8.000 0–6 m e = 0 mm z_g = 230 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = 230 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}R_d

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

F_{cr} = 0.9k_F.E.t_w³/h_w

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

F_{Ed}/F_{Rd}

η₂ + 0.8η₁ ≤ 1.4

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

0.9 x 3.402 x 210000 x 9.90³/428.00

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

√(192.79 x 9.90 x 275/1457.610e3)

0.5/0.600 = 0.833 ; 0.833 x 192.79

275 x 160.64 x 9.90/1.0

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

3.402

1457.610 kN

100.00 mm

192.79 mm

0.600

160.64 mm

437.340 kN

0.251

0.179

Fig 6.1(c)

6.4(1)

6.5(4)

6.5(4)

6.4(1)

6.4(1)

6.2(1)

OK

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

C₁ = M_{cr}/M_{cr,uniform}

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.147

1.162

k_c only

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

λ_{LT} = √W.f_y/M_{cr}

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd}

M_{y,Ed}/M_{b,Rd}

End 1 v, φ = 0; End 2 v, φ = 0

64 elements, 0.000 %; z_g = 230, 0 mm; 1 combination(s), 1 solve(s) + 2 cached

√ 1830.0 x 275 / 283.858

1.332, 1.393, 0.49

0.460, 1.332, 0.934, 0.986

0.467 x 1830.0 x 275 ≤ 503.250 =

168.868 / 234.796

6 m

283.858 kN.m

1.332

0.460

0.467

234.796 kN.m

0.719

FE

converged

Curve c

6.3.2.3

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 3.22 mm OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.336	0.719	0.193	0.145	0.251	0.257	0.719

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.719$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

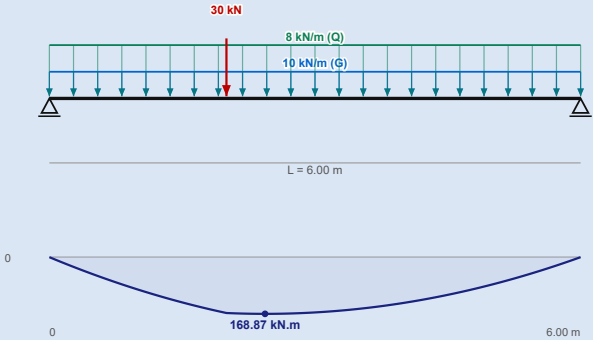
UB-L3t · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **NOT VERIFIED** · governing LTB M_{Ed}/M_{b,Rd} = 0.575
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +230 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (NOT VERIFIED)
Member UB-L3t UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 0 mm z_g = 230 mm (kN/m)
Q UDL 8.000 0–6 m e = 0 mm z_g = 230 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = 230 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	0.00	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)
Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/(√3Y_{M0}) 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 756.260 kN 6.2.6
mm²
M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m
-------------	----------------	-----

$M_{cr} = F_n(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = +230$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	398.376 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m 3.22 mm OK

MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.575

NOT VERIFIED — governing LTB $M_{Ed}/M_{b,Rd} = 0.575$

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L3b · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **PASS** · governing LTB M_{Ed}/M_{b,Rd} = 0.491
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -230 mm (-D/2, stabilising), e = 0

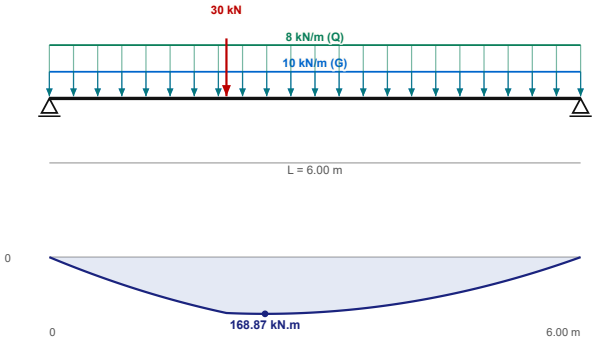
Beam & Beam-Portion (Member)
Member UB-L3b UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 0 mm z_g = -230 mm (kN/m)
Q UDL 8.000 0-6 m e = 0 mm z_g = -230 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = -230 mm (kN)
G SW 0.8044 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	0.00	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

F_{cr} = 0.9k_F.E.t_w³/h_w

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

F_{Ed}/F_{Rd}

η₂ + 0.8η₁ ≤ 1.4

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

0.9 x 3.402 x 210000 x 9.90³/428.00

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

√(192.79 x 9.90 x 275/1457.610e3)

0.5/0.600 = 0.833 ; 0.833 x 192.79

275 x 160.64 x 9.90/1.0

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

3.402

1457.610 kN

100.00 mm

192.79 mm

0.600

160.64 mm

437.340 kN

0.251

0.179

Fig 6.1(c)

6.4(1)

6.5(4)

6.5(4)

6.4(1)

6.4(1)

6.2(1)

OK

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.147

1.162

k_c only

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

λ_{LT} = √W.f_y/M_{cr}

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd}

M_{y,Ed}/M_{b,Rd}

End 1 v, φ = 0; End 2 v, φ = 0

64 elements, 0.000 %; z_g = -230, 0 mm; 1 combination(s), 1 solve(s) + 2 cached

√ 1830.0 x 275 / 542.681

0.963, 0.986, 0.49

0.662, 0.963, 0.934, 0.969

0.683 x 1830.0 x 275 ≤ 503.250 =

168.868 / 343.812

6 m

542.681 kN.m

0.963

0.662

0.683

343.812 kN.m

0.491

FE

converged

Curve c

6.3.2.3

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $3.22 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 3.22 mm OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.336	0.491	0.193	0.145	0.251	0.257	0.491

PASS — governing LTB $M_{Ed}/M_{b,Rd} = 0.491$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L3b · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **PASS** · governing LTB M_{Ed}/M_{b,Rd} = 0.575
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -230 mm (-D/2, stabilising), e = 0

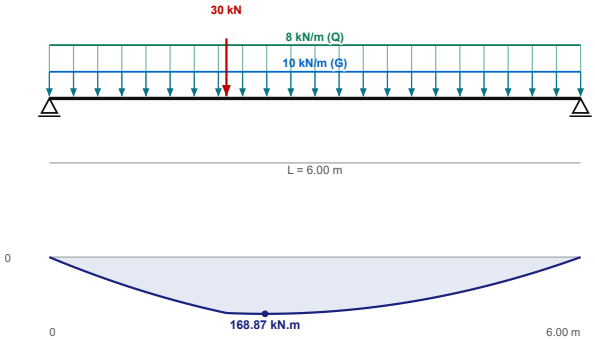
Beam & Beam-Portion (Member)
Member UB-L3b UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 0 mm z_g = -230 mm (kN/m)
Q UDL 8.000 0-6 m e = 0 mm z_g = -230 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = -230 mm (kN)
G SW 0.8044 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	0.00	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/(√3 y_{M0}) 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ η h_w t_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y,Rd} = f_y · W_{pl,y}/y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf} · b_f/(f_{yw} · t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = -230$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _{-(y,Ed)} /M _{-(b,Rd)}	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.575

PASS — governing LTB M_{Ed}/M_{b,Rd} = 0.575

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L4 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.104
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at e = +40 mm and z_g = +230 mm (top flange)

Beam & Beam-Portion (Member) (FAIL)

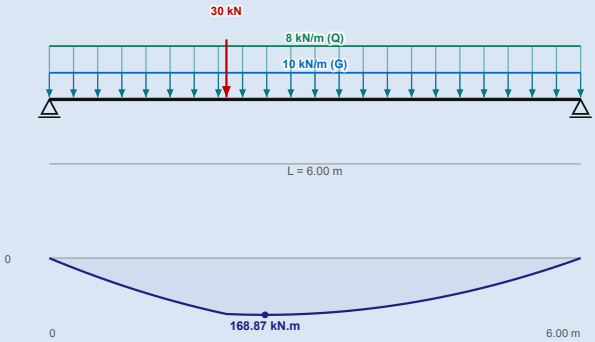
Member UB-L4 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 230 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 230 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 230 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = F_n(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V _{y,Ed} /V _{pl,y} .R _d	0.006 / 756.260 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y} .R _d	109.758 / 756.260 =	0.145	OK
V _{pl,y} .R _d = A _v f _y /√3Y _{M0}	4763.2 mm ² x 275/(√3 x 1) ; A _v ≥ ηh _w t _w = 4237.2 mm ²	756.260 kN	6.2.6
M _{c,y} .R _d = f _y .W _{pl,y} /Y _{M0}	275 x 1830.0/1	503.250 kN.m	
M _{y,Ed} /M _{c,y} .R _d	168.868 / 503.250 =	0.336	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}.b_f/(f_{yw}.t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x (428.00/16.00)² = 14.311 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00 3.402 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 3.402 x 210000 x 9.90³/428.00 1457.610 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f√(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f√(m₁ + m₂)] m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79] 192.79 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(192.79 x 9.90 x 275/1457.610e3) 0.600 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.600 = 0.833 ; 0.833 x 192.79 160.64 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 160.64 x 9.90/1.0 437.340 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 = 0.251 OK

η₂ + 0.8η₁ ≤ 1.4 0.251 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.179 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 393.107 / 342.775 kN.m (shape only: z_g = z_j = 0) 1.147 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN·m 1.162 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; z_g = 230, 0 mm; 1 combination(s), 0 solve(s) + 3 cached 283.858 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 1830.0 x 275 / 283.858 1.332

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 1.332, 1.393, 0.49 0.460 Curve c

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.460, 1.332, 0.934, 0.986 0.467 6.3.2.3

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd} 0.467 x 1830.0 x 275 ≤ 503.250 = 234.796 kN.m

M_{y,Ed}/M_{b,Rd} 168.868 / 234.796 0.719 OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
φ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2.29$ m: 168.587/231.414 + 1.000x9.749/83.600 + 0.664x0.883x2.469x7.472/41.800	1.104	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.788$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm	OK
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.719	0.193	0.145	0.251	0.257	0.408	0.156	1.104	1.104

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.104

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

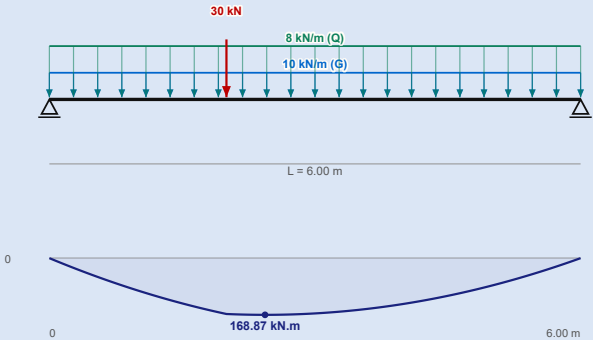
UB-L4 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **NOT VERIFIED** · governing LTB+torsion (EN 1993-6 Annex A) = 0.890
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center
UB 457 x 191 x 82, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at e = +40 mm and z_g = +230 mm (top flange)

Beam & Beam-Portion (Member) (NOT VERIFIED)
Member UB-L4 UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 230 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 230 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 230 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/(\sqrt{3}Y_{M0}) 4763.2 mm² x 275/(\sqrt{3} x 1) ; A_v ≥ ηh_wt_w = 4237.2 756.260 kN 6.2.6
mm²
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m
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$M_{cr} = F_n(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = +230$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	398.376 kN.m	BLOCKED
$\bar{\lambda}_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = F_n(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = F_n(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
$\bar{\lambda}_{LT}$ (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\bar{\lambda}_z\sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0} , S_{w1}	212 cm ³ , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
φ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi'''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.365$ m: 168.802/285.039 + 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800	0.890	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.788$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm	OK
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.408	0.156	0.890	0.890

NOT VERIFIED — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
UB 457 x 191 x 82, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z g = +230 mm

756.260 kN 6.2.6

$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 1830.0/1	503.250 kN.m	
$M_{y.Ed} / M_{c.y.Rd}$	168.868 / 503.250 =	0.336	OK
$V_{z.Ed} / V_{pl.z.Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 304.0/1	83.600 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	104.00 x 275/1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c.Rd} = A_{eff} \cdot f_y / \gamma_{M0}$	100.66 x 275/1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c.Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	1830.0 x 275/1	503.250 kN.m	6.2.9.1(4)
$(M_{y.Ed} / M_{N.y.Rd})^\alpha + (M_{z.Ed} / M_{N.z.Rd})^\beta$	$(168.868/503.250)^2 + (0)^1 =$	0.113	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 0.000/503.250 + N_{Ed} / N_{pl.Rd} = 150.000/2860.000 = 0.052) = 0.293 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.209	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.209	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.287	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.185	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
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A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff} f_y$; $\lambda = \sqrt{(A_{eff} f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff} f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	$100.66 \times 0.963 \times 275 / 10 / 1 =$	2664.536 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff} f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	$100.66 \times 0.306 \times 275 / 10 / 1 =$	845.783 kN	Curve b

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 164.64$, $\psi = 0.859$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 164.64$, $\psi = 0.859$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 230$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	283.858 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 283.858}$	1.332	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.332, 1.393, 0.49	0.460	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.460, 1.332, 0.934, 0.986	0.467	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.467 \times 1830.0 \times 275 \leq 503.250 =$	234.796 kN.m	
$M_{y,Ed} / M_{b,Rd}$	$168.868 / 234.796$	0.719	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff} f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (X_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 2664.536$ ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (X_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 845.783$	0.177	OK
$U_{M,y} = M_{y,Ed} / (X_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$168.868 / 206.569$ ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	0.817	OK
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$0.000 / 53.900$	0.000	OK
$k_{yy} = C_{my} \{1 + 0.6 \lambda_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 (1 + 0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 (1 + 0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 0.817 + 1.106 \times 0.000$	0.884	OK

$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	0.177+0.988x0.817+1.106x0.000	0.985	OK
Torsion Design			
J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A
Torsion Bending Design @ 2.34 m			
T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 3.23	P385 App C
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	3.435 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	7.736 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	10.129 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 41.800, M _{pl,z} = 83.600	0.408	OK P385 3.1.2
V _{pl,T,Rd} = √(1 - T _t /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _t = 27.22 N/mm ² ; V _{pl,Rd} = 756.260	702.492 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	109.758 / 702.492	0.156	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.29 m: 168.587/231.414 + 1.000x9.749/83.600 + 0.664x0.883x2.469x7.472/41.800	1.104	Warning EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.260 / 3.660 kN.m	1.902 / 1.724 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK
NOT VERIFIED		NOT VERIFIED Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	3.22 mm	OK
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N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M _{6.61}	Ax+M _{6.62}	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.054	0.113	0.177	0.719	0.884	0.985	0.193	0.145	0.336	0.719	0.251	0.287	0.408	0.156	1.104 1.104

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.104

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
UB 457 x 191 x 82, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +230 mm

6.2.6

$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 1830.0/1	503.250 kN.m	
$M_{y.Ed} / M_{c.y.Rd}$	168.868 / 503.250 =	0.336	OK
$V_{z.Ed} / V_{pl.z.Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 304.0/1	83.600 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	104.00 x 275/1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c.Rd} = A_{eff} \cdot f_y / \gamma_{M0}$	100.66 x 275/1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c.Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	1830.0 x 275/1	503.250 kN.m	6.2.9.1(4)
$(M_{y.Ed} / M_{N.y.Rd})^\alpha + (M_{z.Ed} / M_{N.z.Rd})^\beta$	$(168.868/503.250)^2 + (0)^1 =$	0.113	OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]			
s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 0.000/503.250 + N_{Ed} / N_{pl.Rd} = 150.000/2860.000 = 0.052) = 0.293 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.209	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.209	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.287	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.185	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A _{eff} (Class-4 web in uniform compression)	N _{Rk} = A _{eff} f _y ; λ = √(A _{eff} f _y /N _{cr}) (6.3.1.3(1)); Table 6.7 Class-4 column with W _{eff,y} = W _{el,y} (flanges Class ≤ 3; web Class 4 only in uniform compression, e _N = 0) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
λ _y = √A _{eff} .f _y /N _{cr}	√100.66x275/21359.47 (N _{cr,y} = π ² EI _y /L _{ey} ²)	0.362	
N _{b,y,Rd} = A _{eff} .X.f _y /γ _{M1}	100.66x0.963x275/10/1 =	2664.536 kN	Curve a
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A _{eff} .f _y /N _{crz}	√100.66x275/1076.61 (N _{cr,z} = π ² EI _z /L _{ez} ²)	1.607	
N _{b,z,Rd} = A _{eff} .X.f _y /γ _{M1}	100.66x0.306x275/10/1 =	845.783 kN	Curve b

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M _{max} =168.9 kN·m		SN003a
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 164.64, ψ = 0.859, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 0.000, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 164.64, ψ = 0.859, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.162, 6.000, 1870, 69.2, 0.922, 210000; z _g = +230 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	398.376 kN.m	BLOCKED
λ _{LT} = √W.f _y /M _{cr}	√ 1830.0 x 275 / 398.376	1.124	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.124, 1.151, 0.49	0.566	Curve c
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.583 x 1830.0 x 275 ≤ 503.250 =	293.432 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.364; χ _{LT,mod} = 0.451; M _{y,Ed} /M _{b,Rd,simp} = 0.744	226.988 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff,y} \cdot f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} \cdot f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 2664.536 ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 845.783	0.177	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	168.868 / 258.156 ($M_{y,Rk} = 442.750$; $W_{el,y}/W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	0.654	OK
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	0.000 / 53.900	0.000	OK
$k_{yy} = C_{my} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 \{1 + 0.6 \times 0.056\}$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 \{1 + 0.6 \times 0.177\}$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 0.654 + 1.106 \times 0.000$	0.718	OK
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.177 + 0.988 \times 0.654 + 1.106 \times 0.000$	0.824	OK

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ³ , 1620 cm ³		P385 App A

Torsion Bending Design @ 2.34 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
φ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
B _{Ed} = EI _w φ" (max)	@ $x = 2$ m	3.435 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	7.736 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ $x = 2.365$ m: 168.802/285.039 + 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800	0.890	OK EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at $x = 0$ / $x = L$; total T = GI _T φ' - EI _w φ''' = 4.260 / 3.660 kN.m	1.902 / 1.724 kN.m	
θ _{ser} ≤ θ _{limit}	@ $x = 2.788$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	3.22 ≤ 6000 / 360 = 16.67 mm @ $x = 2.85$ m	3.22 mm	OK
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N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.054	0.113	0.177	0.575	0.718	0.824	0.193	0.145	0.336	0.575	0.251	0.287	0.408	0.156	0.890

NOT VERIFIED — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

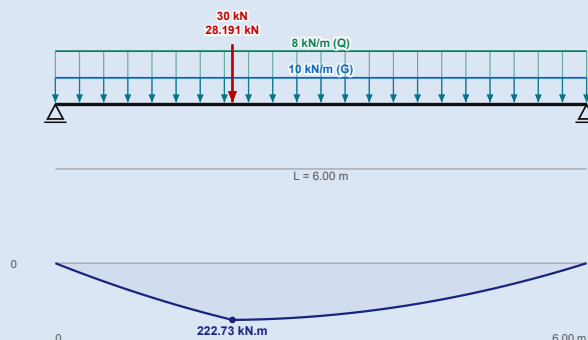
ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +230, factored by the combination) plus an entered CONSTANT M_z,Ed = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_mz = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.230 = 2.36 kN.m characteristic) is not modelled

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

- uplift / hold-down forces and the web-bearing reactions;



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	2.46	137.95	0.00	0.00	20.52	222.73	20.52	-4.61	137.95	—
	End 2, x = 6	150.00C	-2.11	-108.85	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	108.85	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z/I_{z,z}$: outstand compressed by M_z : root 169.3, tip 257.5 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq 9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 136.2, tip 48.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
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$A_{eff} = A - (1 - \rho) b t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (b/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$	2.510 / 756.260 =	0.003	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	137.949 / 756.260 =	0.182	OK
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} Y_{M0})$	4763.2 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$M_{c,y,Rd} = f_y W_{pl,y} / Y_{M0}$	275 x 1830.0 / 1	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	222.726 / 503.250 =	0.443	OK
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / Y_{M0}$	275 x 304.0 / 1	83.600 kN.m	
$N_{pl,Rd} = A_g f_y / Y_{M0}$	104.00 x 275 / 1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} f_y / Y_{M0}$	100.66 x 275 / 1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c,Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / Y_{M0}$	1830.0 x 275 / 1	503.250 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / Y_{M0}$	304.0 x 275 / 1	83.600 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(222.726/503.250)^2 + (20.521/83.600)^1 =$	0.441	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^3 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / Y_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 137.949 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.315	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.315 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 + N_{Ed} / N_{pl,Rd} = 150.000 / 2860.000 = 0.052) = 0.357 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.255	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	137.949	ULS: 1.35G + 1.5Q	0.315	0.255	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.416	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	108.853	ULS: 1.35G + 1.5Q	0.249	0.208	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A _{eff} (Class-4 web in uniform compression)	N _{Rk} = A _{eff} .f _y ; λ = √(A _{eff} .f _y /N _{cr}) (6.3.1.3(1)); Table 6.7 Class-4 column with W _{eff,y} = W _{el,y} (flanges Class ≤ 3; web Class 4 only in uniform compression, e _N = 0) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
λ _y = √A _{eff} .f _y /N _{cr}	√100.66x275/21359.47 (N _{cr,y} = π ² EI _y /L _{ey} ²)	0.362	
N _{b,y,Rd} = A _{eff} .X.f _y /γ _{M1}	100.66x0.963x275/10/1 =	2664.536 kN	Curve a
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A _{eff} .f _y /N _{crz}	√100.66x275/1076.61 (N _{cr,z} = π ² EI _z /L _{ez} ²)	1.607	
N _{b,z,Rd} = A _{eff} .X.f _y /γ _{M1}	100.66x0.306x275/10/1 =	845.783 kN	Curve b

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = M _{cr} /M _{cr,uniform}	408.966 / 342.775 kN.m (shape only: z _g = z _j = 0)	1.193	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 206.9, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=177.0, M(L/2)=206.9, M(3L/4)=133.4, M _{max} =222.7 kN.m	1.215	Serna
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 206.92, ψ = 0.785, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 206.92, ψ = 0.785, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 230, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	292.804 kN.m	converged
λ _{LT} = √W.f _y /M _{cr}	√ 1830.0 x 275 / 292.804	1.311	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.311, 1.368, 0.49	0.469	Curve c
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.469, 1.311, 0.916, 0.980	0.479	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.479 x 1830.0 x 275 ≤ 503.250 =	241.095 kN.m	
M _{y,Ed} /M _{b,Rd}	222.726 / 241.095	0.924	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff,y} \cdot f_y = 2768.024 \text{ kN}$; $M_{y,Rk} = W_{eff,y} \cdot f_y$ with $W_{eff,y} = W_{el,y} = 1610.0 \text{ cm}^3$ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 2664.536 ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 845.783	0.177	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	222.726 / 212.111 ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	1.050	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 53.900	0.381	OK
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 (1 + 0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 (1 + 0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zz}$		1.106	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.056 + 1.012 \times 1.050 + 1.106 \times 0.381$	1.540	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.177 + 0.988 \times 1.050 + 1.106 \times 0.381$	1.636	Warning

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 3.23	P385 App C
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0771 rad = 4.42°	
B _{Ed} = E I _w φ'' (max)	@ x = 2 m	4.807 kN.m ²	
M _{w,Ed} = E I _w φ'' / (h - t _f)	max over span	10.827 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	16.609 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2 m: 222.726, 10.827, 15.608 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 41.800, M _{pl,z} = 83.600	0.642	OK P385 3.1.2
V _{pl,T,Rd} = √(1 - T _t /(1.25 f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _t = 35.20 N/mm ² ; V _{pl,Rd} = 756.260	685.927 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	137.949 / 685.927	0.201	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2 m: 222.726/236.229 + 1.000 × 15.608/83.600 + 0.648 × 0.813 × 4.178 × 10.827/41.800	1.700	Warning EN 1993-6 A
End torques T _t	St Venant part G I _T φ' at x = 0 / x = L; total T = G I _T φ' - E I _w φ''' = 5.388 / 4.224 kN.m	2.460 / 2.113 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.714 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	2.08°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	4.61 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	4.61 mm	OK
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N _{Ed} /N _{(pl,Rd)cal}	UNyz	UMyz	Ax+M _{6.61} Ax+M _{6.62}	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.054	0.441	0.177	0.924	1.540 1.636	0.277	0.182	0.443	0.924	0.315	0.416	0.642	0.201	1.700 1.700

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.700

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +230, factored by the combination) plus an entered CONSTANT M_z,Ed = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_mz = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.230 = 2.36 kN.m characteristic) is not modelled

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	5.03, 41.20, 275, 150.00, 222.73, 20.52		Class 1
	(Axial: Slender web)		
Web classified for	bending + compression: $\alpha = 0.568$, $\psi = -0.895$; limits $62.1\epsilon / 71.5\epsilon / 112.1\epsilon$ (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2
Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z y/I_z$: outstand compressed by M_z : root 169.3, tip 257.5 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\epsilon/\alpha, 10\epsilon/\alpha, 21\epsilon\sqrt{k_\sigma} \geq$ $9\epsilon/10\epsilon/14\epsilon$ bound kept); opposite outstand: root 136.2, tip 48.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2

$A_{eff} = A - (1 - \rho)\delta t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\epsilon = 38.83$; $\lambda_p = (b/t)/(28.4\epsilon\sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{eff} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm^2	EN 1993-1-5 4.4
Auto Design Load Cases	1; SLS 1		

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$	$2.510 / 756.260 =$	0.003	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	$137.949 / 756.260 =$	0.182	OK
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$4763.2 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$; $A_v \geq \eta h_w t_w = 4237.2 \text{ mm}^2$	756.260 kN	6.2.6
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 1830.0 / 1$	503.250 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$222.726 / 503.250 =$	0.443	OK
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 978.476 = (A_{v,z} = 6162.8 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 304.0 / 1$	83.600 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	$104.00 \times 275 / 1 =$ (No bearing / block tearing design)	2860.000 kN	
$N_{c,Rd} = A_{eff} \cdot f_y / \gamma_{M0}$	$100.66 \times 275 / 1$ (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 2860.000 =$ (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c,Rd}$	$150.000 / 2768.024 =$	0.054	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	$1830.0, 47.632, 0.052$	1830.0 cm^3	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	$1830.0 \times 275 / 1$	503.250 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	$304.0, 61.628, 0.052$	304.0 cm^3	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	$304.0 \times 275 / 1$	83.600 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(222.726 / 503.250)^2 + (20.521 / 83.600)^1 =$	0.441	OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$428.00, 9.90, 16.00, 191.30 \text{ mm}$ ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_w)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0		6.5(1)
$a =$ distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000 \text{ mm}$, the full member length (conservative)		6.4(1)

Governing station $x = 0 \text{ m}$: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F	$s_s = 100.00 \text{ mm}$ (entered); $c = 0.00 \text{ mm}$ ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 137.949 \text{ kN}$ (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.315	OK

$$\eta_2 + 0.8\eta_1 \leq 1.4 \quad 0.315 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 + N_{Ed}/N_{pl,Rd} = 150.000/2860.000 = 0.052) = 0.357 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.255 \quad \text{OK 7.2}$$

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	137.949	ULS: 1.35G + 1.5Q	0.315	0.255	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.416	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	108.853	ULS: 1.35G + 1.5Q	0.249	0.208	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y ·L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A _{eff} (Class-4 web in uniform compression)	N _{Rk} = A _{eff} f _y ; λ = √(A _{eff} f _y /N _{cr}) (6.3.1.3(1)); Table 6.7 Class-4 column with W _{eff,y} = W _{el,y} (flanges Class ≤ 3; web Class 4 only in uniform compression, e _N = 0) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
λ _y = √(A _{eff} ·f _y /N _{cr})	√(100.66x275/21359.47 (N _{cr,y} = π ² EI _y /L _{ey} ²))	0.362	
N _{b,y,Rd} = A _{eff} ·X·f _y /γ _{M1}	100.66x0.963x275/10/1 =	2664.536 kN	Curve a
L _{ez} = K _z ·L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √(A _{eff} ·f _y /N _{crz})	√(100.66x275/1076.61 (N _{cr,z} = π ² EI _z /L _{ez} ²))	1.607	
N _{b,z,Rd} = A _{eff} ·X·f _y /γ _{M1}	100.66x0.306x275/10/1 =	845.783 kN	Curve b

Equivalent Uniform Moment Factors C₁, C_{mLT}, C_{mz}, and C_{my}

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 206.9, 1.000, 300.000	1.215	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=177.0, M(L/2)=206.9, M(3L/4)=133.4, M _{max} =222.7 kN·m		SN003a
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 206.92, ψ = 0.785, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 206.92, ψ = 0.785, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.215, 6.000, 1870, 69.2, 0.922, 210000; z _g = +230 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	416.357 kN.m	BLOCKED
λ _{LT} = √(W _y ·f _y /M _{cr})	√(1830.0 x 275 / 416.357)	1.099	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.099, 1.125, 0.49	0.580	Curve c
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.580, 1.099, 0.907, 0.962	0.603	6.3.2.3
M _{b,Rd} = χW _{pl,y} ·f _y ≤ M _{c,y,Rd}	0.603 x 1830.0 x 275 ≤ 503.250 =	303.596 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.334; χ _{LT,mod} = 0.468; M _{y,Ed} /M _{b,Rd,simp} = 0.946	235.463 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	222.726 / 303.596	0.734	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

Table 6.7 Class-4 column

N_{Rk} = A_{eff}f_y = 2768.024 kN; M_{y,Rk} = W_{eff,y}f_y with 6.3.3

W_{eff,y} = W_{el,y} = 1610.0 cm³ (flanges Class ≤ 3, web Class 4 only in uniform compression, ΔM = 0)

[assumption]; k_{ij} from the Class 3/4 rows

U_{N,y} = N_{Ed}/(χ_y·N_{Rk}/γ_{M1}) 150.000 / 2664.536 (N_{Rk} = 2768.024) 0.056 OK

U_{N,z} = N_{Ed}/(χ_z·N_{Rk}/γ_{M1}) 150.000 / 845.783 0.177 OK

U_{M,y} = M_{y,Ed}/(χ_{LT}·M_{y,Rk}/γ_{M1}) 222.726 / 267.098 (M_{y,Rk} = 442.750; W_{el,y}/W_{pl,y} = 0.880 applied to M_{b,Rd}) 0.834 OK

U_{M,z} = M_{z,Ed}/(M_{z,Rk}/γ_{M1}) 20.521 / 53.900 0.381 OK

k_{yy} = C_{my}{1+0.6χ_yU_{N,y}} 1.000{1+0.6x0.362x0.056} ≤ 1.000(1+0.6x0.056) 1.012 Table B.1 (Class 3)

k_{zz} = C_{mz}{1+0.6χ_zU_{N,z}} 1.000{1+0.6x1.607x0.177} ≤ 1.000(1+0.6x0.177) 1.106

k_{yz} = k_{zz} 1.106

k_{zy} = 1 - 0.1χ_zU_{N,z}/(C_{mLT}-0.25) 0.988 Table B.2

U_{Ny}+k_{yy}·U_{M,y}+k_{yz}·U_{M,z} 0.056+1.012x0.834+1.106x0.381 1.322 Warning

U_{Nz}+k_{zy}·U_{M,y}+k_{zz}·U_{M,z} 0.177+0.988x0.834+1.106x0.381 1.423 Warning

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A

W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 5.388 kN.m

Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C

φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0771 rad = 4.42°

B_{Ed} = EI_wφ'' (max) @ x = 2 m 4.807 kN.m²

M_{w,Ed} = EI_wφ''/(h-t_f) max over span 10.827 kN.m

M_{z,Ed} = φ·M_{y,Ed} max coincident 16.609 kN.m

(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2 m: 222.726, 10.827, 15.608 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.642 OK P385 3.1.2

V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3))·V_{pl,Rd} @ x = 0 m: τ_t = 35.20 N/mm²; V_{pl,Rd} = 756.260 685.927 kN 6.2.7(9)

V_{Ed}/V_{pl,T,Rd} 137.949 / 685.927 0.201 OK

M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_αM_w/M_{f,Rk} @ x = 2 m: 222.726/292.053 + 1.000x15.608/83.600 + 0.648x0.813x2.150x10.827/41.800 1.243 Warning EN 1993-6 A

End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 5.388 / 4.224 kN.m 2.460 / 2.113 kN.m

θ_{ser} ≤ θ_{limit} @ x = 2.714 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 2.08° review

NOT VERIFIED Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked. **NOT VERIFIED**

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $4.61 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 4.61 mm OK

N_Ed/N_Rd	pl.Rd	UNyz	UMyz	Ax+M_6.6.1	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.054	0.441	0.177	0.734	1.322	1.423	0.277	0.182	0.443	0.734	0.315	0.416	0.642	0.201	1.243	1.423

FAIL — governing Member buckling z-z (Eq 6.62) = 1.423

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, unrestrained; $L7 = L6 + \text{"EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure } M_y \text{ in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z } q); only forces with a lever arm do$

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

$A_{\text{eff}} = A - (1 - \rho) \bar{\sigma} t_w$	web Class 4 in uniform compression: $d/t = 41.20 > 42\varepsilon = 38.83$; $\lambda_p = (\bar{\sigma} t)/(28.4\varepsilon \sqrt{k_\sigma}) = 41.20/(28.4 \times 0.924 \times \sqrt{4}) = 0.785$; $\rho = (\lambda_p - 0.055(3+\psi))/\lambda_p^2 = 0.917$ ($\psi = 1$); $b_{\text{eff}} = 373.82 \text{ mm} = b_{e1} + b_{e2} = 2 \times 186.91$; $104.00 - 3.34 \text{ cm}^2$ ($e_N = 0$, symmetric)	100.66 cm ²	EN 1993-1-5:2004, 4.4.6.2
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Local Capacity Check

$V_{y.Ed}/V_{pl.y.Rd}$	0.046 / 756.260 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	147.949 / 756.260 =	0.196	OK
$V_{pl.y.Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	4763.2 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 4237.2$ mm ²	756.260 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 1830.0 / 1	503.250 kN.m	
$M_{y.Ed} / M_{c.y.Rd}$	243.781 / 503.250 =	0.484	OK
$V_{z.Ed} / V_{pl.z.Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 304.0 / 1	83.600 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	104.00 x 275 / 1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c.Rd} = A_{eff} \cdot f_y / \gamma_{M0}$	100.66 x 275 / 1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed} / N_{c.Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	1830.0 x 275 / 1	503.250 kN.m	6.2.9.1(4)
$W_{pl.N.z} = F_n(W_{pl.z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N.z.Rd} = W_{pl.N.z} \cdot f_y / \gamma_{M0}$	304.0 x 275 / 1	83.600 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^\alpha + (M_{z.Ed} / M_{N.z.Rd})^\beta$	$(243.781 / 503.250)^2 + (20.521 / 83.600)^1 =$	0.480	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00 / 16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \text{e}3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 147.949$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 437.340 =	0.338	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.338 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 0.000 / 503.250 + N_{Ed} / N_{pl.Rd} = 150.000 / 2860.000 = 0.052) = 0.380 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.272	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	147.949	ULS: 1.35G + 1.5Q	0.338	0.272	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.438	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	98.853	ULS: 1.35G + 1.5Q	0.226	0.191	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A_{eff} (Class-4 web in uniform compression)	$N_{Rk} = A_{eff,y} \cdot f_y$; $\lambda = \sqrt{(A_{eff,y} \cdot f_y / N_{cr})}$ (6.3.1.3(1)); Table 6.7 Class-4 column with $W_{eff,y} = W_{el,y}$ (flanges Class ≤ 3 ; web Class 4 only in uniform compression, $e_N = 0$) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
$\lambda_y = \sqrt{A_{eff,y} \cdot f_y / N_{cr}}$	$\sqrt{100.66 \times 275 / 21359.47}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.362	
$N_{b,y,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	100.66 x 0.963 x 275 / 10 / 1 =	2664.536 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A_{eff,z} \cdot f_y / N_{crz}}$	$\sqrt{100.66 \times 275 / 1076.61}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.607	
$N_{b,z,Rd} = A_{eff} \cdot X \cdot f_y / \gamma_{M1}$	100.66 x 0.306 x 275 / 10 / 1 =	845.783 kN	Curve b

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	402.212 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.173	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 236.9, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=192.0$, $M(L/2)=236.9$, $M(3L/4)=118.4$, $M_{max}=243.8$ kN.m	1.212	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 236.92$, $\psi = 0.664$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 236.92$, $\psi = 0.664$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 230$, 0 mm; 1 combination(s), 3 solve(s)	297.660 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 297.660}$	1.300	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.300, 1.355, 0.49	0.475	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.475, 1.300, 0.923, 0.981	0.484	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.484 \times 1830.0 \times 275 \leq 503.250 =$	243.466 kN.m	
$M_{y,Ed} / M_{b,Rd}$	243.781 / 243.466	1.001	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff,y} \cdot f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} \cdot f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed} / (X_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 2664.536 ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed} / (X_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 845.783	0.177	OK
$U_{M,y} = M_{y,Ed} / (X_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	243.781 / 214.197 ($M_{y,Rk} = 442.750$; $W_{el,y} / W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	1.138	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 53.900	0.381	OK

$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.362 \times 0.056\} \leq 1.000(1+0.6 \times 0.056)$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N,z}\}$	$1.000\{1+0.6 \times 1.607 \times 0.177\} \leq 1.000(1+0.6 \times 0.177)$	1.106	
$k_{yz} = k_{zy}$		1.106	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.056 + 1.012 \times 1.138 + 1.106 \times 0.381$	1.630	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.177 + 0.988 \times 1.138 + 1.106 \times 0.381$	1.723	Warning

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴		P385 App A

Torsion Bending Design @ 2.091 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 3.23	P385 App C
φ _{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0771 rad = 4.42°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	4.807 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	10.827 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	18.592 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.091 m: 243.299, 10.593, 17.426 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 41.800, M _{pl,z} = 83.600	0.696	OK P385 3.1.2
V _{pl,T,Rd} = √(1 - T _f /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _f = 35.20 N/mm ² ; V _{pl,Rd} = 756.260	685.927 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	147.949 / 685.927	0.216	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2 m: 242.726/238.794 + 1.000 × 17.010/83.600 + 0.648 × 0.797 × 5.524 × 10.827/41.800	1.959	Warning EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 5.388 / 4.224 kN.m	2.460 / 2.113 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.714 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	2.08°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	5.26 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	5.26 mm	OK
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N_Ed/N_(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max	
0.054	0.480	0.177	1.001	1.630	1.723	0.316	0.196	0.484	1.001	0.338	0.438	0.696	0.216	1.959	1.959

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.959

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 2.1° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-L7 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **FAIL** · governing Member buckling z-z (Eq 6.62) = 1.502

Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER

UB 457 x 191 x 82, 6 m SS, unrestrained; L7 = L6 + "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

Axial with Moments (Member) (FAIL)
Member UB-L7 UB 457 x 191 x 82
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 230 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 230 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 230 mm (kN)

Q PY 28.191 @ 2 m e = 40 mm z_g = 230 mm (kN)

Q M 40.000 @ 4 m (kN.m)

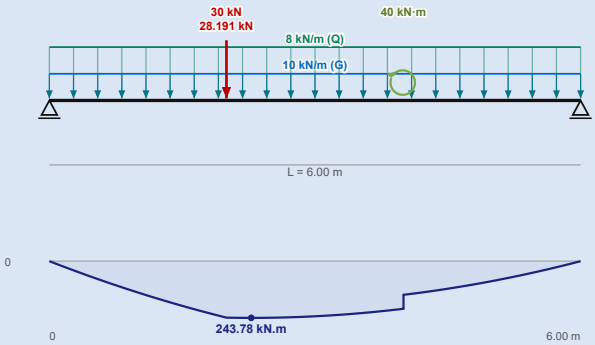
G SW 0.8044 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	2.46	147.95	0.00	0.00	20.52	243.78	20.52	-5.26	147.95	—
	End 2, x = 6	150.00C	-2.11	-98.85	0.00	0.00	20.52	@ 2.280	@ —	@ 2.850	98.85	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m)	457 x 191 x 82 UB [S275]		
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	5.03, 41.20, 275, 150.00, 243.78, 20.52 (Axial: Slender web)		Class 1
Web classified for	bending + compression: α = 0.568, ψ = -0.895; limits 62.1ε / 71.5ε / 112.1ε (M _z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2
Flange outstands under M _y + M _z (+N)	σ = N/A + M _y /W _{el,y} ± M _z /I _z : outstand compressed by M _z : root 182.4, tip 270.5 N/mm ² (both compressive, α = 1 → 9ε/α, 10ε/α, 21ε√k _σ ≥ 9ε/10ε/14ε bound kept); opposite outstand: root 149.3, tip 61.1 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
A _{eff} = A - (1 - ρ)δt _w	web Class 4 in uniform compression: d/t = 41.20 > 42ε = 38.83; λ _p = (b/t)/(28.4ε√k _σ) = 41.20/(28.4 x 0.924 x √4) = 0.785; ρ = (λ _p - 0.055(3+ψ))/λ _p ² = 0.917 (ψ = 1); b _{eff} = 373.82 mm = b _{e1} + b _{e2} = 2 x 186.91; 104.00 - 3.34 cm ² (e _N = 0, symmetric)	100.66 cm ²	EN 1993-1-5 4.4

Local Capacity Check

$V_{y.Ed}/V_{pl.y.Rd}$	0.046 / 756.260 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	147.949 / 756.260 =	0.196	OK
$V_{pl.y.Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	4763.2 mm ² x 275 / ($\sqrt{3}$ x 1) ; $A_v \geq \eta h_w t_w = 4237.2$ mm ²	756.260 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 1830.0/1	503.250 kN.m	
$M_{y.Ed}/M_{c.y.Rd}$	243.781 / 503.250 =	0.484	OK
$V_{z.Ed}/V_{pl.z.Rd}$	0 / 978.476 = ($A_{v,z} = 6162.8$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 304.0/1	83.600 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	104.00 x 275/1 = (No bearing / block tearing design)	2860.000 kN	
$N_{c.Rd} = A_{eff} \cdot f_y / \gamma_{M0}$	100.66 x 275/1 (Class-4 web in uniform compression, cl 6.2.4(2))	2768.024 kN	6.2.4
$n = N_{Ed}/N_{pl.Rd}$	150.000 / 2860.000 = (parameter of the 6.2.9 interaction)	0.052	
$N_{Ed}/N_{c.Rd}$	150.000 / 2768.024 =	0.054	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	1830.0, 47.632, 0.052	1830.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	1830.0 x 275/1	503.250 kN.m	6.2.9.1(4)
$W_{pl.N.z} = F_n(W_{pl.z}, A_{vz}, n)$	304.0, 61.628, 0.052	304.0 cm ³	
$M_{N.z.Rd} = W_{pl.N.z} \cdot f_y / \gamma_{M0}$	304.0 x 275/1	83.600 kN.m	
$(M_{y.Ed}/M_{N.y.Rd})^\alpha + (M_{z.Ed}/M_{N.z.Rd})^\beta$	(243.781/503.250) ² + (20.521/83.600) ¹ =	0.480	OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	275 x 191.30 / (275 x 9.90) = 19.323 ; 0.02 x (428.00/16.00) ² = 14.311 if $\lambda_{\bar{F}} > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	0.9 x 3.402 x 210000 x 9.90 ³ / 428.00	1457.610 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	3.402 x 210000 x 9.90 ² / (2 x 275 x 428.00) = 297.44 ≤ 100.00	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_{\bar{F}} > 0.5$); min[227.03, 192.79]	192.79 mm	6.5(4)
$\lambda_{\bar{F}} = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5 / \lambda_{\bar{F}} \leq 1$; $L_{eff} = \chi_F \cdot l_y$	0.5/0.600 = 0.833 ; 0.833 x 192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	275 x 160.64 x 9.90 / 1.0	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 147.949$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.338	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	0.338 + 0.8 x ($M_{Ed}/M_{c.y.Rd} = 0.000/503.250 + N_{Ed}/N_{pl.Rd} = 150.000/2860.000 = 0.052$) = 0.380 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.272	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	147.949	ULS: 1.35G + 1.5Q	0.338	0.272	governs
2	point load 3+4	(a)	0.00	6.010	172.67	0.427	1.000	470.084	87.286	ULS: 1.35G + 1.5Q	0.186	0.438	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	98.853	ULS: 1.35G + 1.5Q	0.226	0.191	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
A _{eff} (Class-4 web in uniform compression)	N _{Rk} = A _{eff} .f _y ; λ = √(A _{eff} .f _y /N _{cr}) (6.3.1.3(1)); Table 6.7 Class-4 column with W _{eff,y} = W _{el,y} (flanges Class ≤ 3; web Class 4 only in uniform compression, e _N = 0) [assumption printed]	100.66 cm ²	EN 1993-1-5 4.4
λ _y = √A _{eff} .f _y /N _{cr}	√100.66x275/21359.47 (N _{cr,y} = π ² EI _y /L _{ey} ²)	0.362	
N _{b,y,Rd} = A _{eff} .X.f _y /γ _{M1}	100.66x0.963x275/10/1 =	2664.536 kN	Curve a
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A _{eff} .f _y /N _{crz}	√100.66x275/1076.61 (N _{cr,z} = π ² EI _z /L _{ez} ²)	1.607	
N _{b,z,Rd} = A _{eff} .X.f _y /γ _{M1}	100.66x0.306x275/10/1 =	845.783 kN	Curve b

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 236.9, 1.000, 300.000	1.212	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=192.0, M(L/2)=236.9, M(3L/4)=118.4, M _{max} =243.8 kN.m		SN003a
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 236.92, ψ = 0.664, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 236.92, ψ = 0.664, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.212, 6.000, 1870, 69.2, 0.922, 210000; z _g = +230 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	415.519 kN.m	BLOCKED
λ _{LT} = √W.f _y /M _{cr}	√ 1830.0 x 275 / 415.519	1.101	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.101, 1.126, 0.49	0.580	Curve c
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.580, 1.101, 0.908, 0.962	0.602	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.602 x 1830.0 x 275 ≤ 503.250 =	303.127 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.336; χ _{LT,mod} = 0.467; M _{y,Ed} /M _{b,Rd,simp} = 1.037	235.066 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	243.781 / 303.127	0.804	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

Table 6.7 Class-4 column	$N_{Rk} = A_{eff,y} \cdot f_y = 2768.024$ kN; $M_{y,Rk} = W_{eff,y} \cdot f_y$ with $W_{eff,y} = W_{el,y} = 1610.0$ cm ³ (flanges Class ≤ 3 , web Class 4 only in uniform compression, $\Delta M = 0$) [assumption]; k_{ij} from the Class 3/4 rows		6.3.3
$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 2664.536 ($N_{Rk} = 2768.024$)	0.056	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 845.783	0.177	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	243.781 / 266.685 ($M_{y,Rk} = 442.750$; $W_{el,y}/W_{pl,y} = 0.880$ applied to $M_{b,Rd}$)	0.914	OK
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	20.521 / 53.900	0.381	OK
$k_{yy} = C_{my} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 0.362 \times 0.056\} \leq 1.000 \{1 + 0.6 \times 0.056\}$	1.012	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \lambda_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 1.607 \times 0.177\} \leq 1.000 \{1 + 0.6 \times 0.177\}$	1.106	
$k_{yz} = k_{zy}$		1.106	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.607$, $U_{N,z} = 0.177$, $C_{mLT} = 1.000$	0.988	Table B.2
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.056 + 1.012 \times 0.914 + 1.106 \times 0.381$	1.403	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.177 + 0.988 \times 0.914 + 1.106 \times 0.381$	1.502	Warning

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —		P385 App A
W _{n0} , S _{w1}	212 cm ³ , 1620 cm ³		P385 App A

Torsion Bending Design @ 2.091 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
φ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0771 rad = 4.42°	
B _{Ed} = E I _w φ'' (max)	@ $x = 2$ m	4.807 kN.m ²	
M _{w,Ed} = E I _w $\varphi''/(h - t_f)$	max over span	10.827 kN.m	
M _{z,Ed} = $\varphi \cdot M_{y,Ed}$	max coincident	18.592 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.091$ m: 243.299, 10.593, 17.426 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.696	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25 f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 35.20$ N/mm ² ; $V_{pl,Rd} = 756.260$	685.927 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	147.949 / 685.927	0.216	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ $x = 2$ m: 242.726/291.733 + $1.000 \times 17.010/83.600 + 0.648 \times 0.797 \times 2.419 \times 10.827/41.800$	1.359	Warning EN 1993-6 A
End torques T _t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 5.388 / 4.224$ kN.m	2.460 / 2.113 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.714$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	2.08°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$5.26 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	5.26 mm	OK
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N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M _{Ed} 6.6.1	Ax+M _{Ed} 6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.054	0.480	0.177	0.804	1.403	1.502	0.316	0.196	0.484	0.804	0.338	0.438	0.696	0.216	1.359 1.502

FAIL — governing Member buckling z-z (Eq 6.62) = 1.502

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +230$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 2.1^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L1 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 1.169

Owner case: vertical loads resulting in no torsion

UC 203 x 203 x 60, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre (e = 0, z_g = 0)

Beam & Beam-Portion (Member) (FAIL)

Member UC-L1 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

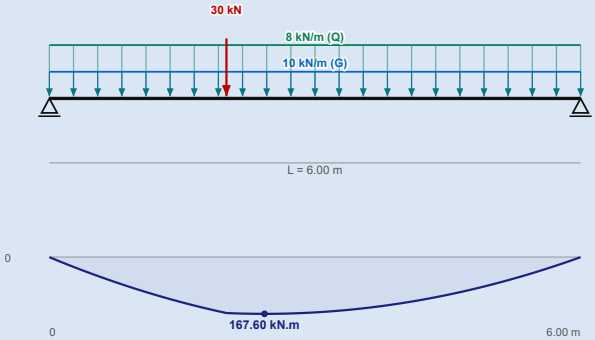
G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/(√3Y_{M0}) 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30εt_f = 403.20); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/181.20 5.311 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.311 x 210000 x 9.40³/181.20 4601.343 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.311 x 210000 x 9.40²/(2 x 275 x 181.20) = 988.90 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.310 ≤ 0.5 with m₂ = 3.257, so m₂ = 0); min[210.49, 166.44] 166.44 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(166.44 x 9.40 x 275/4601.343e3) 0.306 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.306 = 1.635 → 1.000 ; 1.000 x 166.44 166.44 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 166.44 x 9.40/1.0 430.254 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 108.884 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 = 0.253 OK

η₂ + 0.8η₁ ≤ 1.4 0.253 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/180.400 = 0.000) = 0.253 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.181 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 277.879 / 242.476 kN.m (shape only: z_g = z_j = 0) 1.146 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M_{max}=167.6 kN.m 1.163 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached 277.879 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 656.0 x 275 / 277.879 0.806

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 0.806, 0.812, 0.34 0.814 Curve b

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.814, 0.806, 0.934, 0.967 0.842 6.3.2.3

M_{b.Rd} = χW_{pl.y}.f_y ≤ M_{c.y.Rd} 0.842 x 656.0 x 275 ≤ 180.400 = 151.844 kN.m

M_{y.Ed}/M_{b.Rd} 167.604 / 151.844 1.104 **Warning**

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f.Ed} = M _{Ed} /h (kN)	2.5 % N _{f.Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 209.60 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 19.49 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.104	1.169	0.309	0.253	0.600	1.104

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$$

$$275 \times 205.80 / (275 \times 9.40) = 21.894 ; 0.02 \times (181.20/14.20)^2 = 3.257 \text{ if } \lambda_F > 0.5, \text{ else } 0$$

6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2/(2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275/4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 =$	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = 0$ (load through the shear centre)	281.930 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60 \text{ mm}$. $N_{fEd} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{fEd} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	19.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.096

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L2 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.647
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
UC 203 x 203 x 60, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at e = +40 mm (torque), z_g = 0

Beam & Beam-Portion (Member) (FAIL)

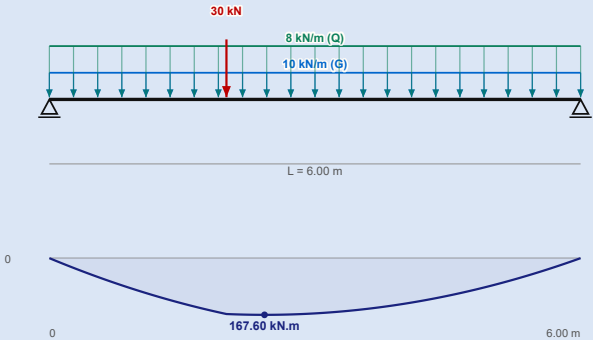
Member UC-L2 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.5886 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	2.94	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	-2.57	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V _{y,Ed} /V _{pl,y} .R _d	0.012 / 352.225 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y} .R _d	108.884 / 352.225 =	0.309	OK
V _{pl,y} .R _d = A _v f _y /√3Y _{M0}	2218.4 mm ² x 275/(√3 x 1) ; A _v ≥ ηh _w t _w = 1703.3 mm ²	352.225 kN	6.2.6
M _{c,y} .R _d = f _y .W _{pl,y} /Y _{M0}	275 x 656.0/1	180.400 kN.m	
M _{y,Ed} /M _{c,y} .R _d	167.604 / 180.400 =	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30εt_f = 403.20); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}.b_f/(f_{yw}.t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/181.20 5.311 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.311 x 210000 x 9.40³/181.20 4601.343 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.311 x 210000 x 9.40²/(2 x 275 x 181.20) = 988.90 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.310 ≤ 0.5 with m₂ = 3.257, so m₂ = 0); min[210.49, 166.44] 166.44 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(166.44 x 9.40 x 275/4601.343e3) 0.306 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.306 = 1.635 → 1.000 ; 1.000 x 166.44 166.44 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 166.44 x 9.40/1.0 430.254 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 108.884 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 = 0.253 0.253 OK

η₂ + 0.8η₁ ≤ 1.4 0.253 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/180.400 = 0.000) = 0.253 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.181 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 277.879 / 242.476 kN.m (shape only: z_g = z_j = 0) 1.146 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M_{max}=167.6 kN.m 1.163 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached 277.879 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 656.0 x 275 / 277.879 0.806

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 0.806, 0.812, 0.34 0.814 Curve b

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.814, 0.806, 0.934, 0.967 0.842 6.3.2.3

M_{b,Rd} = χW_{pl.y}.f_y ≤ M_{c.y,Rd} 0.842 x 656.0 x 275 ≤ 180.400 = 151.844 kN.m

M_{y,Ed}/M_{b,Rd} 167.604 / 151.844 1.104 **Warning**

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0}, S_{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.315 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 5.77$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	1.835 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315 \text{ m}$: 167.433, 8.337, 21.602 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.938$, $M_{pl,z} = 83.875$	1.318	Warning P385 3.1.2
$V_{pl.T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $T_t = 58.60 \text{ N/mm}^2$; $V_{pl,Rd} = 352.225$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl.T,Rd}$	108.884 / 295.686	0.368	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2.116 \text{ m}$: 166.313/146.844 + 1.000x20.748/83.875 + 0.657x0.753x2.520x8.969/41.938	1.647	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660 \text{ kN.m}$	2.943 / 2.568 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.664 \text{ m}$, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	3.19°	review

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	19.49 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.929	1.104	1.169	0.309	0.253	0.600	1.318	0.368	1.647	1.647

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.647

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L2 · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.636
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
UC 203 x 203 x 60, 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at e = +40 mm (torque), z_g = 0

Beam & Beam-Portion (Member) (FAIL)

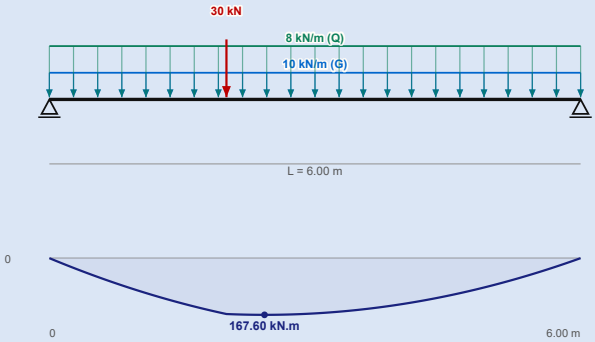
Member UC-L2 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.5886 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	2.94	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	-2.57	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V_{y,Ed}/V_{pl,y}.R_d 0.012 / 352.225 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y}.R_d 108.884 / 352.225 = 0.309 OK
V_{pl,y}.R_d = A_vf_y/√3Y_{M0} 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6
M_{c,y}.R_d = f_y.W_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m
M_{y,Ed}/M_{c,y}.R_d 167.604 / 180.400 = 0.929 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30εt_f = 403.20); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}.b_f/(f_{yw}.t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2/(2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275/4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 =$	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = 0$ (load through the shear centre)	281.930 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 209.60 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 47.2 cm⁴, 0.197 dm⁶, 1040 mm, —, — P385 App A
W_{n0}, S_{w1} 101 cm², 734 cm⁴ P385 App A

Torsion Bending Design @ 2.315 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 5.77 P385 App C
φ_{max} (ULS) L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10) 0.1331 rad = 7.62°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 1.835 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 9.391 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 22.160 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.315 m: 167.433, 8.337, 21.602 kN.m; M_{pl,y} = 180.400, M_{pl,f} = 41.938, M_{pl,z} = 83.875 1.318 **Warning** P385 3.1.2
V_{pl,T,Rd} = √(1 - t_f/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 58.60 N/mm²; V_{pl,Rd} = 352.225 295.686 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 108.884 / 295.686 0.368 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.141 m: 166.510/147.421 + 1.636 **Warning** EN 1993-6 A
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x20.876/83.875 + 0.658x0.751x2.466x8.883/41.938
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 2.943 / 2.568 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.664 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 3.19° review

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 19.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 19.49 mm **Warning**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.318	0.368	1.636	1.636

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.636

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 3.2° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L3t · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB M_{Ed}/M_{b,Rd} = 1.189

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

UC 203 x 203 x 60, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +104.8 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

Member UC-L3t UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 0 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0–6 m e = 0 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = 105 mm (kN)

G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

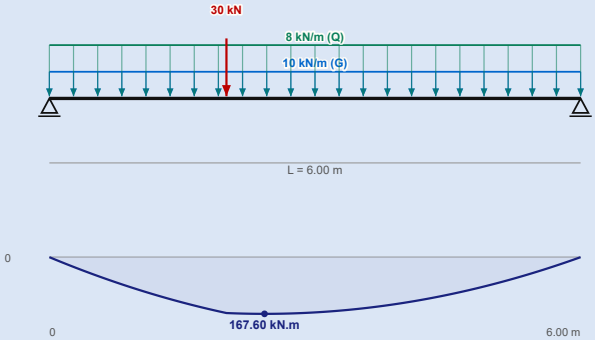
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/(√3Y_{M0}) 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30εt_f = 403.20); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

F_{cr} = 0.9k_F.E.t_w³/h_w

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

F_{Ed}/F_{Rd}

η₂ + 0.8η₁ ≤ 1.4

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/181.20

0.9 x 5.311 x 210000 x 9.40³/181.20

5.311 x 210000 x 9.40²/(2 x 275 x 181.20) = 988.90 ≤ 100.00

m₂ = 0.000 (first pass λ_F = 0.310 ≤ 0.5 with m₂ = 3.257, so m₂ = 0); min[210.49, 166.44]

√(166.44 x 9.40 x 275/4601.343e3)

0.5/0.306 = 1.635 → 1.000 ; 1.000 x 166.44

275 x 166.44 x 9.40/1.0

F_{Ed} = 108.884 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =

0.253 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/180.400 = 0.000) = 0.253 ≤ 1.4 (ULS: 1.35G + 1.5Q)

5.311

4601.343 kN

100.00 mm

166.44 mm

0.306

166.44 mm

430.254 kN

0.253

0.181

Fig 6.1(c)

6.4(1)

6.5(4)

6.5(4)

6.4(1)

6.4(1)

6.2(1)

OK

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

277.879 / 242.476 kN.m (shape only: z_g = z_j = 0)

0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M_{max}=167.6 kN.m

1.146

1.163

k_c only

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

λ_{LT} = √W.f_y/M_{cr}

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd}

M_{y,Ed}/M_{b,Rd}

End 1 v, φ = 0; End 2 v, φ = 0

64 elements, 0.000 %; z_g = 105, 0 mm; 1 combination(s), 1 solve(s) + 2 cached

√ 656.0 x 275 / 219.624

0.906, 0.894, 0.34

0.756, 0.906, 0.934, 0.968

0.781 x 656.0 x 275 ≤ 180.400 =

167.604 / 140.966

6 m

219.624 kN.m

0.906

0.756

0.781

140.966 kN.m

1.189

FE

converged

Curve b

6.3.2.3

Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 209.60 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 19.49 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.189	1.169	0.309	0.253	0.600	1.189

FAIL — governing LTB $M_{Ed}/M_{b,Rd} = 1.189$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L3t · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 1.169

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

UC 203 x 203 x 60, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +104.8 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

Member UC-L3t UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 0 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0–6 m e = 0 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = 105 mm (kN)

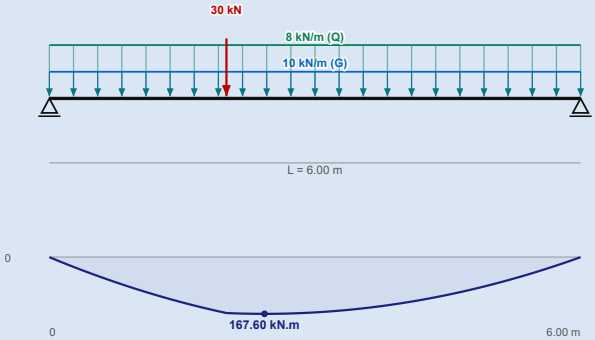
G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6.00C	0.00	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/(√3Y_{M0}) 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343)} = 0.306$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 = 0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m
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$M_{cr} = F_n(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = +105$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	281.930 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m 19.49 mm **Warning**

MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.096

FAIL — governing Deflection = 1.169

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L3b · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 1.169
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
UC 203 x 203 x 60, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -104.8 mm (-D/2, stabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

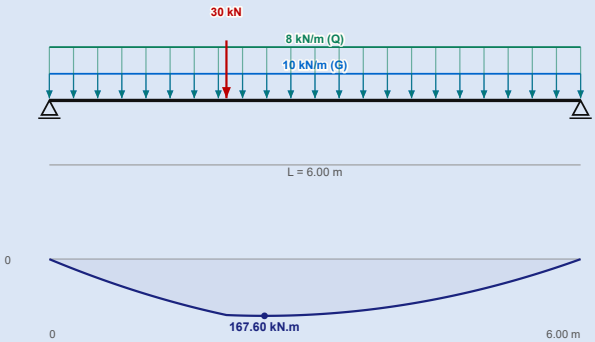
Member UC-L3b UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 0 mm z_g = -105 mm (kN/m)
Q UDL 8.000 0-6 m e = 0 mm z_g = -105 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = -105 mm (kN)
G SW 0.5886 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V_{y,Ed}/V_{pl,y}.R_d 0.012 / 352.225 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y}.R_d 108.884 / 352.225 = 0.309 OK
V_{pl,y}.R_d = A_vf_y/(√3γ_{M0}) 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6
M_{c,y}.R_d = f_y·W_{pl,y}/γ_{M0} 275 x 656.0/1 180.400 kN.m
M_{y,Ed}/M_{c,y}.R_d 167.604 / 180.400 = 0.929 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30εt_f = 403.20); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/181.20 5.311 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.311 x 210000 x 9.40³/181.20 4601.343 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.311 x 210000 x 9.40²/(2 x 275 x 181.20) = 988.90 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.310 ≤ 0.5 with m₂ = 3.257, so m₂ = 0); min[210.49, 166.44] 166.44 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(166.44 x 9.40 x 275/4601.343e3) 0.306 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.306 = 1.635 → 1.000 ; 1.000 x 166.44 166.44 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 166.44 x 9.40/1.0 430.254 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 108.884 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 = 0.253 OK

η₂ + 0.8η₁ ≤ 1.4 0.253 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/180.400 = 0.000) = 0.253 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.181 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 277.879 / 242.476 kN.m (shape only: z_g = z_j = 0) 1.146 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M_{max}=167.6 kN.m 1.163 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; z_g = -105, 0 mm; 1 combination(s), 1 solve(s) + 2 cached 350.653 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 656.0 x 275 / 350.653 0.717

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 0.717, 0.747, 0.34 0.861 Curve b

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.861, 0.717, 0.934, 0.968 0.890 6.3.2.3

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd} 0.890 x 656.0 x 275 ≤ 180.400 = 160.527 kN.m

M_{y,Ed}/M_{b,Rd} 167.604 / 160.527 1.044 **Warning**

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 209.60 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 19.49 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.044	1.169	0.309	0.253	0.600	1.044

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L3b · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 1.169

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

UC 203 x 203 x 60, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -104.8 mm (-D/2, stabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

Member UC-L3b UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 0 mm z_g = -105 mm (kN/m)

Q UDL 8.000 0–6 m e = 0 mm z_g = -105 mm (kN/m)

Q PY 30.000 @ 2 m e = 0 mm z_g = -105 mm (kN)

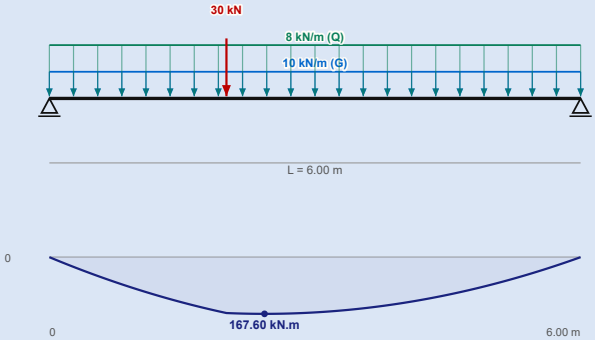
G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	0.00	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30εt_f = 403.20); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2/(2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275/4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 =$	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	$1.163, 6.000, 2060, 47.2, 0.197, 210000$; $z_g = -105$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	281.930 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	$0.800, 0.808, 0.34$	0.817	Curve b
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	$0.817, 0.800, 0.927, 0.964$	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	$167.604 / 152.975$	1.096	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60 \text{ mm}$. $N_{fEd} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{fEd} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	19.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.096

FAIL — governing Deflection = 1.169

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L4 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.921
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center
UC 203 x 203 x 60, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at e = +40 mm and z_g = +104.8 mm (top flange)

Beam & Beam-Portion (Member) (FAIL)

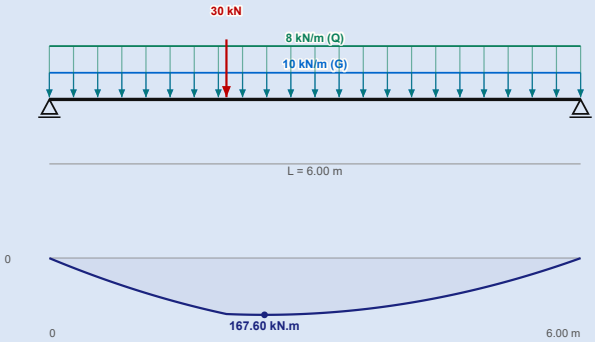
Member UC-L4 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 105 mm (kN)
G SW 0.5886 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	2.94	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	-2.57	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]
Class = F_n(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V _{y,Ed} /V _{pl,y} .R _d	0.012 / 352.225 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y} .R _d	108.884 / 352.225 =	0.309	OK
V _{pl,y} .R _d = A _v f _y /√3Y _{M0}	2218.4 mm ² x 275/(√3 x 1) ; A _v ≥ ηh _w t _w = 1703.3 mm ²	352.225 kN	6.2.6
M _{c,y} .R _d = f _y .W _{pl,y} /Y _{M0}	275 x 656.0/1	180.400 kN.m	
M _{y,Ed} /M _{c,y} .R _d	167.604 / 180.400 =	0.929	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 181.20, 9.40, 14.20, 205.80 mm (B = 205.80 ≤ t_w + 30ε_t = 403.20); f_{yw} = f_y = 275; 1 web Fig 5.1
m₁ = f_{yf}.b_f/(f_{yw}.t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 205.80/(275 x 9.40) = 21.894 ; 0.02 x (181.20/14.20)² = 3.257 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/181.20 5.311 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.311 x 210000 x 9.40³/181.20 4601.343 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.311 x 210000 x 9.40²/(2 x 275 x 181.20) = 988.90 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.310 ≤ 0.5 with m₂ = 3.257, so m₂ = 0); min[210.49, 166.44] 166.44 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(166.44 x 9.40 x 275/4601.343e3) 0.306 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.306 = 1.635 → 1.000 ; 1.000 x 166.44 166.44 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 166.44 x 9.40/1.0 430.254 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 108.884 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 = 0.253 OK

η₂ + 0.8η₁ ≤ 1.4 0.253 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/180.400 = 0.000) = 0.253 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.181 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 277.879 / 242.476 kN.m (shape only: z_g = z_j = 0) 1.146 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=133.7, M(L/2)=163.3, M(3L/4)=111.2, M_{max}=167.6 kN·m 1.163 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; z_g = 105, 0 mm; 1 combination(s), 0 solve(s) + 3 cached 219.624 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 656.0 x 275 / 219.624 0.906

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 0.906, 0.894, 0.34 0.756 Curve b

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.756, 0.906, 0.934, 0.968 0.781 6.3.2.3

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd} 0.781 x 656.0 x 275 ≤ 180.400 = 140.966 kN.m

M_{y,Ed}/M_{b,Rd} 167.604 / 140.966 1.189 **Warning**

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0}, S_{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.315 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 5.77$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	1.835 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315 \text{ m}$: 167.433, 8.337, 21.602 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.938$, $M_{pl,z} = 83.875$	1.318	Warning P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $T_t = 58.60 \text{ N/mm}^2$; $V_{pl,Rd} = 352.225$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	108.884 / 295.686	0.368	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2 \text{ m}$: 165.178/136.428 + 1.000x20.068/83.875 + 0.655x0.761x4.222x9.391/41.938	1.921	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660 \text{ kN.m}$	2.943 / 2.568 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.664 \text{ m}$, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	3.19°	review

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	19.49 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.929	1.189	1.169	0.309	0.253	0.600	1.318	0.368	1.921	1.921

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.921

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member UC-L4 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 105 mm (kN)

G SW 0.5886 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

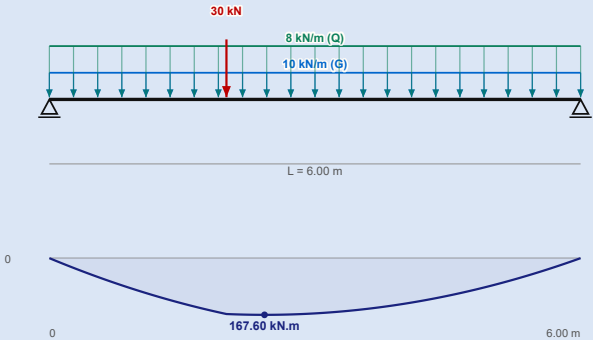
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	2.94	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	0.00C	-2.57	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 0.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}R_d

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/((√3)Y_{M0}) 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3/181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343)} = 0.306$	0.306	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40/1.0$	430.254 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $430.254 = 0.253 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/180.400 = 0.000) = 0.253 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.181	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.181	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.600	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.156	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m
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$M_{cr} = F_n(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = +105$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	281.930 kN.m	BLOCKED
$\lambda_{LT} = \sqrt{W_{fy}/M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = F_n(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} f_y \leq M_{c,y,Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z\sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y,Ed}/M_{b,Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	167.604 / 152.975	1.096	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0} , S_{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.315 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 5.77$	P385 App C
ϕ_{max} (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.835 kN.m ²	
$M_{w,Ed} = EI_w \phi'''/(h-t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315$ m: 167.433, 8.337, 21.602 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.938$, $M_{pl,z} = 83.875$	1.318	Warning P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3})).V_{pl,Rd}}$	@ $x = 0$ m: $T_t = 58.60$ N/mm ² ; $V_{pl,Rd} = 352.225$	295.686 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	108.884 / 295.686	0.368	OK
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	@ $x = 2.141$ m: 166.510/147.421 + 1.000x20.876/83.875 + 0.658x0.751x2.466x8.883/41.938	1.636	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.260 / 3.660$ kN.m	2.943 / 2.568 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.664$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	3.19°	review

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	19.49 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.929	1.096	1.169	0.309	0.253	0.600	1.318	0.368	1.636	1.636

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.636

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\text{inf}}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L5 · UC 203 x 203 x 60 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.921

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam

UC 203 x 203 x 60, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +104.8 mm

Axial with Moments (Member) (FAIL)
Member UC-L5 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 105 mm (kN)

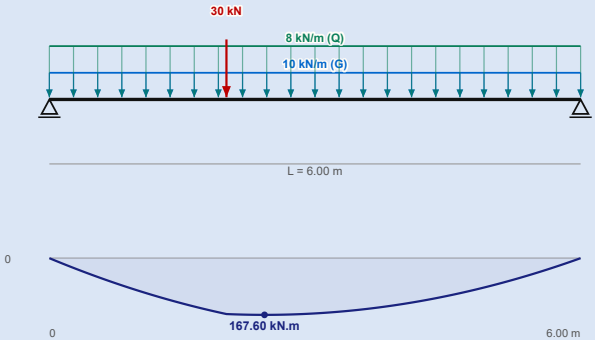
G SW 0.5886 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	2.94	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	150.00C	-2.57	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 150.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)

Web classified for bending + compression: α = 0.680, ψ = -0.857; Class 1 Table 5.2
limits 50.5ε / 58.1ε / 108.5ε

Auto Design Load Cases 1; SLS 1

Local Capacity Check

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/(√3Y_{M0}) 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

V_{z,Ed}/V_{pl,z,Rd} 0 / 942.581 = (A_{v,z} = 5936.7 mm²; no minor-axis shear in the single-plane model) 0.000 Low Shear

M_{c,z,Rd} = f_y·W_{pl,z}/Y_{M0} 275 x 305.0/1 83.875 kN.m

$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	76.40 x 275/1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	656.0 x 275/1	180.400 kN.m	6.2.9.1(4)
$(M_{y,Ed} / M_{N,y,Rd})^{\alpha} + (M_{z,Ed} / M_{N,z,Rd})^{\beta}$	$(167.604/180.400)^2 + (0)^1 =$	0.863	OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.310 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.222	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.222	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.641	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.197	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10/1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10/1 =$	791.578 kN	Curve c

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	277.879 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.146	k_c only
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$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 163.3, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{\max}=167.6 \text{ kN}\cdot\text{m}$	1.163	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 163.33$, $\psi = 0.856$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 163.33$, $\psi = 0.856$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 105$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	219.624 kN.m	converged
$\lambda_{LT} = \sqrt{W_{fy}/M_{cr}}$	$\sqrt{656.0 \times 275 / 219.624}$	0.906	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.906, 0.894, 0.34	0.756	Curve b
$\chi_{LT, \text{mod}} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.756, 0.906, 0.934, 0.968	0.781	6.3.2.3
$M_{b, \text{Rd}} = \chi W_{pl, y} \cdot f_y \leq M_{c, y, \text{Rd}}$	$0.781 \times 656.0 \times 275 \leq 180.400 =$	140.966 kN.m	
$M_{y, \text{Ed}}/M_{b, \text{Rd}}$	167.604 / 140.966	1.189	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f, \text{Ed}} = M_{\text{Ed}}/h$ (kN)	2.5 % $N_{f, \text{Ed}}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60 \text{ mm}$. $N_{f, \text{Ed}} = M_{\text{Ed}}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f, \text{Ed}}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{\text{Ed}} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N, y} = N_{\text{Ed}}/(X_y \cdot N_{\text{Rk}}/\gamma_{M1})$	150.000 / 1558.850 ($N_{\text{Rk}} = 2101.000$)	0.096	OK
$U_{N, z} = N_{\text{Ed}}/(X_z \cdot N_{\text{Rk}}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M, y} = M_{y, \text{Ed}}/(X_{LT} \cdot M_{y, \text{Rk}}/\gamma_{M1})$	167.604 / 140.966 ($M_{y, \text{Rk}} = 180.400$)	1.189	Warning
$U_{M, z} = M_{z, \text{Ed}}/(M_{z, \text{Rk}}/\gamma_{M1})$	0.000 / 83.875	0.000	OK
$k_{yy} = C_{my}\{1 + (\lambda_y - 0.2)U_{N, y}\}$	$1.000\{1 + (0.771 - 0.2) \times 0.096\} \leq 1.000(1 + 0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1 + (2\lambda_z - 0.6)U_{N, z}\}$	$1.000\{1 + (2 \times 1.329 - 0.6) \times 0.189\} \leq 1.000(1 + 1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\lambda_z U_{N, z}/(C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.329$, $U_{N, z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny} + k_{yy} \cdot U_{M, y} + k_{yz} \cdot U_{M, z}$	$0.096 + 1.055 \times 1.189 + 0.759 \times 0.000$	1.351	Warning
$U_{Nz} + k_{zy} \cdot U_{M, y} + k_{zz} \cdot U_{M, z}$	$0.189 + 0.975 \times 1.189 + 1.265 \times 0.000$	1.348	Warning

Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0} , S_{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.315 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
ϕ_{max} (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1331 rad = 7.62°	
$B_{\text{Ed}} = EI_w \phi''$ (max)	@ $x = 2 \text{ m}$	1.835 kN.m ²	
$M_{w, \text{Ed}} = EI_w \phi''/(h - t_f)$	max over span	9.391 kN.m	
$M_{z, \text{Ed}} = \phi \cdot M_{y, \text{Ed}}$	max coincident	22.160 kN.m	
$(M_y/M_{pl, y})^2 + M_w/M_{pl, f} + M_z/M_{pl, z}$	@ $x = 2.315 \text{ m}$: 167.433, 8.337, 21.602 kN.m; $M_{pl, y} = 180.400$, $M_{pl, f} = 41.938$, $M_{pl, z} = 83.875$	1.318	Warning P385 3.1.2
$V_{pl, \text{T.Rd}} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl, \text{Rd}}$	@ $x = 0 \text{ m}$: $T_t = 58.60 \text{ N/mm}^2$; $V_{pl, \text{Rd}} = 352.225$	295.686 kN	6.2.7(9)

$V_{Ed}/V_{pl.T.Rd}$	108.884 / 295.686	0.368	OK
$M_y/M_{b.Rd} + C_{mz}M_z/M_{z.Rk} + k_wk_{zw}k_{\alpha}M_w/M_{f.Rk}$	@ x = 2 m: 165.178/136.428 + 1.000x20.068/83.875 + 0.655x0.761x4.222x9.391/41.938	1.921	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T\phi'$ at x = 0 / x = L; total T = $GI_T\phi' - EI_w\phi''' = 4.260 / 3.660$ kN.m	2.943 / 2.568 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ x = 2.664 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	3.19°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$19.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	19.49 mm	Warning
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N_Ed/N_(pl.Rd)	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max	
0.071	0.863	0.189	1.189	1.351	1.348	1.169	0.309	0.929	1.189	0.253	0.641	1.318	0.368	1.921	1.921

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.921

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC-L5 · UC 203 x 203 x 60 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = 1.636

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam

UC 203 x 203 x 60, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +104.8 mm

Axial with Moments (Member) (FAIL)
Member UC-L5 UC 203 x 203 x 60

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 105 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 105 mm (kN)

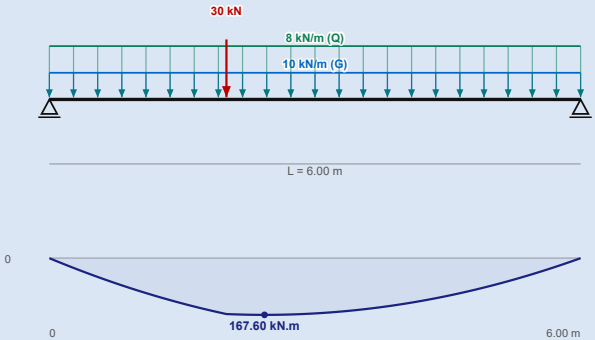
G SW 0.5886 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	2.94	108.88	0.00	0.00	0.00	167.60	0.00	-19.49	108.88	—
	End 2, x = 6	150.00C	-2.57	-93.88	0.00	0.00	0.00	@ 2.430	@ —	@ 2.850	93.88	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (60.0 kg/m) 203 x 203 x 60 UC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 6.20, 17.10, 275, 150.00, 167.60, 0.00 Class 1
(Axial: Non-Slender)

Web classified for bending + compression: α = 0.680, ψ = -0.857; Class 1 Table 5.2
limits 50.5ε / 58.1ε / 108.5ε

Auto Design Load Cases 1; SLS 1

Local Capacity Check

V_{y,Ed}/V_{pl,y,Rd} 0.012 / 352.225 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 108.884 / 352.225 = 0.309 OK

V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 2218.4 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 1703.3 mm² 352.225 kN 6.2.6

M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 656.0/1 180.400 kN.m

M_{y,Ed}/M_{c,y,Rd} 167.604 / 180.400 = 0.929 OK

V_{z,Ed}/V_{pl,z,Rd} 0 / 942.581 = (A_{v,z} = 5936.7 mm²; no minor-axis shear in the single-plane model) 0.000 Low Shear

M_{c,z,Rd} = f_yW_{pl,z}/Y_{M0} 275 x 305.0/1 83.875 kN.m

$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	76.40 x 275/1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	656.0 x 275/1	180.400 kN.m	6.2.9.1(4)
$(M_{y,Ed} / M_{N,y,Rd})^{\alpha} + (M_{z,Ed} / M_{N,z,Rd})^{\beta}$	$(167.604/180.400)^2 + (0)^1 =$	0.863	OK

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20/14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^3 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^3 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 108.884$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.253	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.253 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000/180.400 + N_{Ed} / N_{pl,Rd} = 150.000/2101.000 = 0.071) = 0.310 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.222	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	108.884	ULS: 1.35G + 1.5Q	0.253	0.222	governs
2	point load 3	(a)	0.00	6.002	161.29	0.283	1.000	416.923	45.000	ULS: 1.35G + 1.5Q	0.108	0.641	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	93.884	ULS: 1.35G + 1.5Q	0.218	0.197	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.771	

$N_{b.y.Rd} = \text{Area} \cdot \chi_{f,y} / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10 / 1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\chi_z = \sqrt{A \cdot f_y / N_{cr,z}}$	$\sqrt{76.40 \times 275 / 1186.00} (N_{cr,z} = \pi^2 E I_z / L_{ez}^2)$	1.329	
$N_{b.z.Rd} = \text{Area} \cdot \chi_{f,z} / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10 / 1 =$	791.578 kN	Curve c

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 163.3, 1.000, 300.000	1.163	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=133.7$, $M(L/2)=163.3$, $M(3L/4)=111.2$, $M_{max}=167.6$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 163.33$, $\psi = 0.856$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 163.33$, $\psi = 0.856$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.163, 6.000, 2060, 47.2, 0.197, 210000; $z_g = +105$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	281.930 kN.m	BLOCKED
$\chi_{LT} = \sqrt{W_{f,y} / M_{cr}}$	$\sqrt{656.0 \times 275 / 281.930}$	0.800	
$\chi_{LT} = \text{Fn}(\chi_{LT}, \Phi_{LT}, \alpha_{LT})$	0.800, 0.808, 0.34	0.817	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \chi_{LT}, k_c, f)$	0.817, 0.800, 0.927, 0.964	0.848	6.3.2.3
$M_{b.Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y.Rd}$	$0.848 \times 656.0 \times 275 \leq 180.400 =$	152.975 kN.m	
χ_{LT} (P362 6.55 simplified) ; $M_{b.Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\chi_z \sqrt{\beta_w} = 1.109$; $\chi_{LT,mod} = 0.652$; $M_{y.Ed} / M_{b.Rd,simp} = 1.425$	117.633 kN.m	P362 6.55 comparison
$M_{y.Ed} / M_{b.Rd}$	167.604 / 152.975	1.096	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f.Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f.Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60$ mm. $N_{f.Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f.Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y.Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	167.604 / 152.975 ($M_{y,Rk} = 180.400$)	1.096	Warning
$U_{M,z} = M_{z.Ed} / (M_{z,Rk} / \gamma_{M1})$	0.000 / 83.875	0.000	OK
$k_{yy} = C_{my} \{1 + (\chi_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.771 - 0.2) \times 0.096\} \leq 1.000 (1 + 0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz} \{1 + (2\chi_z - 0.6) U_{N,z}\}$	$1.000 \{1 + (2 \times 1.329 - 0.6) \times 0.189\} \leq 1.000 (1 + 1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6 k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.096 + 1.055 \times 1.096 + 0.759 \times 0.000$	1.252	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.189 + 0.975 \times 1.096 + 1.265 \times 0.000$	1.257	Warning

Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W_{n0} , S_{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2.315 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
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Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 5.77$	P385 App C
$\phi_{\max}$ (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	$0.1331 \text{ rad} = 7.62^\circ$	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2 \text{ m}$	1.835 kN.m^2	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	9.391 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	22.160 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.315 \text{ m}$: $167.433, 8.337, 21.602 \text{ kN.m}$; $M_{pl,y} = 180.400, M_{pl,f} = 41.938, M_{pl,z} = 83.875$	1.318	Warning P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $T_t = 58.60 \text{ N/mm}^2$; $V_{pl,Rd} = 352.225$	295.686 kN	$6.2.7(9)$
$V_{Ed}/V_{pl,T,Rd}$	$108.884 / 295.686$	0.368	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ $x = 2.141 \text{ m}$: $166.510/147.421 + 1.000 \times 20.876/83.875 + 0.658 \times 0.751 \times 2.466 \times 8.883/41.938$	1.636	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \phi'$ at $x = 0 / x = L$; total $T = GI_T \phi' - EI_w \phi''' = 4.260 / 3.660 \text{ kN.m}$	$2.943 / 2.568 \text{ kN.m}$	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.664 \text{ m}$, SLS: $1.0Q$; limit 2° (P385 guidance, advisory)	3.19°	review
NOT VERIFIED Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.			

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$				19.49 $\leq$ 6000 / 360 = 16.67 mm @ x = 2.85 m							19.49 mm		Warning		
N_Ed/N_(pl,Rd)	UNyz	UMyz	Ax+M_6.6.1	Ax+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max	
0.071	0.863	0.189	1.096	1.252	1.257	1.169	0.309	0.929	1.096	0.253	0.641	1.318	0.368	1.636	1.636

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.636

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 3.2^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC 203 x 203 x 60, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +104.8, factored by the combination) plus an entered CONSTANT M_z,Ed = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_mz = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.105 = 1.08 kN.m characteristic) is not modelled

Auto Design Load Cases 1; SLS 1

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$	$2.801 / 352.225 =$	0.008	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	$137.075 / 352.225 =$	0.389	OK
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$221.560 / 180.400 =$	1.228	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 942.581 = (A_{v,z} = 5936.7 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / \gamma_{M0}$	$275 \times 305.0 / 1$	83.875 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$76.40 \times 275 / 1 = (\text{No bearing / block tearing design})$	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 2101.000 =$	0.071	OK
$W_{pl,N,y} = F_n (W_{pl,y}, A_{vy}, n)$	$656.0, 22.184, 0.071$	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$656.0 \times 275 / 1$	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n (W_{pl,z}, A_{vz}, n)$	$305.0, 59.367, 0.071$	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} f_y / \gamma_{M0}$	$305.0 \times 275 / 1$	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(221.560 / 180.400)^2 + (20.521 / 83.875)^1 =$	1.753	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm} (B = 205.80 \leq t_w + 30t_f = 403.20); f_{yw} = f_{yf} = 275; 1 \text{ web}$	Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w) ; m_2 = 0.02(h_w / t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894 ; 0.02 \times (181.20 / 14.20)^2 = 3.257 \text{ if } \lambda_F > 0.5, \text{ else } 0$	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000 \text{ (first pass } \lambda_F = 0.310 \leq 0.5 \text{ with } m_2 = 3.257, \text{ so } m_2 = 0); \min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343e3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000 ; 1.000 \times 166.44$	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 137.075 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) } / 430.254 =$	0.319	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.319 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.376 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q)$	0.268	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	137.075	ULS: 1.35G + 1.5Q	0.319	0.268	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.892	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	107.979	ULS: 1.35G + 1.5Q	0.251	0.220	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 10 / 1 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 10 / 1 =$	791.578 kN	Curve c

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	289.130 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.192	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 205.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=176.0$, $M(L/2)=205.6$, $M(3L/4)=132.4$, $M_{max}=221.6$ kN.m	1.216	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 205.61$, $\psi = 0.781$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 205.61$, $\psi = 0.781$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 105$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	226.932 kN.m	converged
$\lambda_{LT} = \sqrt{W_y \cdot f_y / M_{cr}}$	$\sqrt{656.0 \times 275 / 226.932}$	0.892	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.892, 0.882, 0.34	0.765	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.765, 0.892, 0.916, 0.959	0.798	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.798 \times 656.0 \times 275 \leq 180.400 =$	143.956 kN.m	
$M_{y,Ed} / M_{b,Rd}$	221.560 / 143.956	1.539	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	221.560 / 143.956 ($M_{y,Rk} = 180.400$)	1.539	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 83.875	0.245	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.771 - 0.2) \times 0.096\} \leq 1.000 (1 + 0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz} \{1 + (2\lambda_z - 0.6) U_{N,z}\}$	$1.000 \{1 + (2 \times 1.329 - 0.6) \times 0.189\} \leq 1.000 (1 + 1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6 k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.096 + 1.055 \times 1.539 + 0.759 \times 0.245$	1.906	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.189 + 0.975 \times 1.539 + 1.265 \times 0.245$	1.999	Warning

Torsion Design

J, H, a, Q_f , Q_w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W_{n0} , S_{w1}	101 cm ³ , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 5.77$	P385 App C
φ_{max} (ULS)	$L/a = 5.77$; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	2.696 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	13.796 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	36.730 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2$ m: 221.560, 13.796, 35.001 kN.m; $M_{pl,y} = 180.400$, $M_{pl,f} = 41.938$, $M_{pl,z} = 83.875$	2.255	Warning P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 76.13$ N/mm ² ; $V_{pl,Rd} = 352.225$	276.531 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	137.075 / 276.531	0.496	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_d M_w/M_{f,Rk}$	@ $x = 2$ m: 221.560/137.995 + 1.000x35.001/83.875 + 0.634x0.583x42.246x13.796/41.938	7.159	Warning EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 5.388 / 4.224$ kN.m	3.823 / 3.096 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.564$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	4.62°	review

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$27.97 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	27.97 mm	Warning
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N _{Ed} /N _{Ed,Ed}	N _{Ed} /N _{Ed,Ed}	UNyz	UMyz	Ax+M _{6.61}	Ax+M _{6.62}	Deflection	V/Vpl	MA/Mc	LTB	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.071	1.753	0.189	1.539	1.906	1.999	1.678	0.389	1.228	1.539	0.319	0.892	2.255	0.496	7.159	7.159

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 7.159

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 4.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UC 203 x 203 x 60, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +104.8, factored by the combination) plus an entered CONSTANT M_z,Ed = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_mz = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.105 = 1.08 kN.m characteristic) is not modelled

Auto Design Load Cases 1; SLS 1

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$	$2.801 / 352.225 =$	0.008	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	$137.075 / 352.225 =$	0.389	OK
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	$2218.4 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1) ; A_v \geq \eta h_w t_w = 1703.3 \text{ mm}^2$	352.225 kN	6.2.6
$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$	$275 \times 656.0 / 1$	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$221.560 / 180.400 =$	1.228	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	$0 / 942.581 = (A_{v,z} = 5936.7 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
$M_{c,z,Rd} = f_y W_{pl,z} / \gamma_{M0}$	$275 \times 305.0 / 1$	83.875 kN.m	
$N_{pl,Rd} = A_g f_y / \gamma_{M0}$	$76.40 \times 275 / 1 = (\text{No bearing / block tearing design})$	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	$150.000 / 2101.000 =$	0.071	OK
$W_{pl,N,y} = F_n (W_{pl,y}, A_{vy}, n)$	$656.0, 22.184, 0.071$	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} f_y / \gamma_{M0}$	$656.0 \times 275 / 1$	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n (W_{pl,z}, A_{vz}, n)$	$305.0, 59.367, 0.071$	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} f_y / \gamma_{M0}$	$305.0 \times 275 / 1$	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(221.560 / 180.400)^2 + (20.521 / 83.875)^1 =$	1.753	Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	$181.20, 9.40, 14.20, 205.80 \text{ mm} (B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20); f_{yw} = f_{yf} = 275; 1 \text{ web}$		Fig 5.1
$m_1 = f_{yf} b_f / (f_{yw} t_w) ; m_2 = 0.02 (h_w / t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894 ; 0.02 \times (181.20 / 14.20)^2 = 3.257 \text{ if } \lambda_F > 0.5, \text{ else } 0$		6.5(1)
$a = \text{distance between transverse stiffeners}$	$\text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)}$		6.4(1)

Governing station $x = 0 \text{ m}$: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c ; k_F$	$s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F E t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F E t_w^2 / (2 f_{yw} h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000 \text{ (first pass } \lambda_F = 0.310 \leq 0.5 \text{ with } m_2 = 3.257, \text{ so } m_2 = 0); \min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y t_w f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343 \text{e}3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000 ; 1.000 \times 166.44$	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} L_{eff} t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 137.075 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) } / 430.254 =$	0.319	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.319 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.376 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q)$	0.268	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	137.075	ULS: 1.35G + 1.5Q	0.319	0.268	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.892	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	107.979	ULS: 1.35G + 1.5Q	0.251	0.220	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √A.f _y /N _{cr}	√76.40x275/3523.45 (N _{cr,y} = π²EI _y /L _{ey} ²)	0.771	
N _{b,y,Rd} = Area.χ.f _y /γ _{M1}	76.40x0.742x275/10/1 =	1558.850 kN	Curve b
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A.f _y /N _{crz}	√76.40x275/1186.00 (N _{cr,z} = π²EI _z /L _{ez} ²)	1.329	
N _{b,z,Rd} = Area.χ.f _y /γ _{M1}	76.40x0.377x275/10/1 =	791.578 kN	Curve c

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 205.6, 1.000, 300.000	1.216	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=176.0, M(L/2)=205.6, M(3L/4)=132.4, M _{max} =221.6 kN·m		SN003a
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 205.61, ψ = 0.781, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 205.61, ψ = 0.781, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.216, 6.000, 2060, 47.2, 0.197, 210000; z _g = +105 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	294.827 kN.m	BLOCKED
λ _{LT} = √W.f _y /M _{cr}	√ 656.0 x 275 / 294.827	0.782	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.782, 0.794, 0.34	0.827	Curve b
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.827, 0.782, 0.907, 0.953	0.867	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.867 x 656.0 x 275 ≤ 180.400 =	156.441 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.085; χ _{LT,mod} = 0.674; M _{y,Ed} /M _{b,Rd,simp} = 1.822	121.608 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	221.560 / 156.441	1.416	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 209.60 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	221.560 / 156.441 ($M_{y,Rk} = 180.400$)	1.416	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	20.521 / 83.875	0.245	OK
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\lambda_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq 1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.1 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.096+1.055 \times 1.416+0.759 \times 0.245$	1.776	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.189+0.975 \times 1.416+1.265 \times 0.245$	1.880	Warning

Torsion Design

J, H, a, Q _f , Q _w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W _{n0} , S _{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 5.77	P385 App C
φ_{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	2.696 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	13.796 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	36.730 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2 m: 221.560, 13.796, 35.001 kN.m; M _{pl,y} = 180.400, M _{pl,f} = 41.938, M _{pl,z} = 83.875	2.255	Warning P385 3.1.2
V _{pl,T,Rd} = √(1 - T _t /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _t = 76.13 N/mm ² ; V _{pl,Rd} = 352.225	276.531 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	137.075 / 276.531	0.496	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2 m: 221.560/149.162 + 1.000x35.001/83.875 + 0.634x0.583x4.024x13.796/41.938	2.392	Warning EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 5.388 / 4.224 kN.m	3.823 / 3.096 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.564 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	4.62°	review

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	27.97 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	27.97 mm	Warning
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N _{Ed} /N _(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max	
0.071	1.753	0.189	1.416	1.776	1.880	1.678	0.389	1.228	1.416	0.319	0.892	2.255	0.496	2.392	2.392

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 2.392

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +105 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 4.6° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

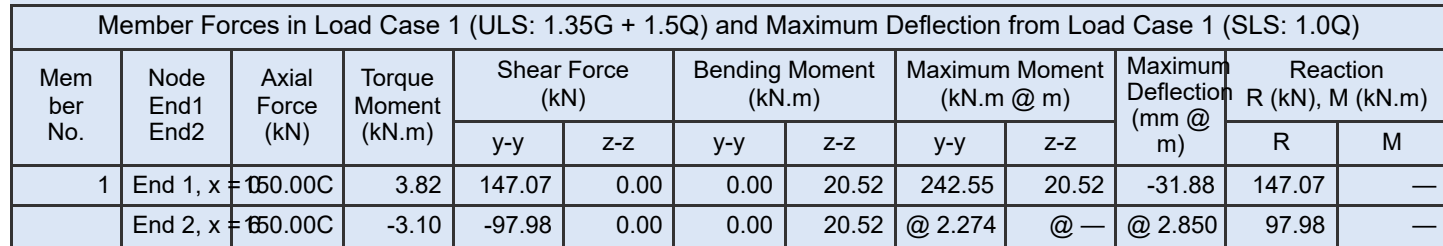
Axial with Moments (Member) (FAIL)
Member UC-L7 UC 203 x 203 x 60
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)
End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

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Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm zg = 105 mm ( kN/m )
Q UDL 8.000 0-6 m e = 40 mm zg = 105 mm ( kN/m )
Q PY 30.000 @ 2 m e = 40 mm zg = 105 mm ( kN )
Q PY 28.191 @ 2 m e = 40 mm zg = 105 mm ( kN )
Q M 40.000 @ 4 m ( kN.m )
G SW 0.5886 kN/m 0-6 m ( automatic )

YG,inf companions (reactions only): 2 - G at 1.0 (STR
set B) and 0.9 (EQU set A) of every ULS combination
with G > 1.0, variable factors as entered
    ULS C1: ULS: 1.35G + 1.5Q [YG,inf = 1.0, STR set B]
    ULS C2: ULS: 1.35G + 1.5Q [YG,inf = 0.9, EQU set A]
YG,inf companions: 2 ULS combination(s) re-solved with G at
1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at
0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only

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Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,\text{inf}}$ companions with G at 1.0 and 0.9)	$R_{\min} \geq 0$	OK
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Section (60.0 kg/m)	203 x 203 x 60 UC [S275]		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	6.20, 17.10, 275, 150.00, 242.55, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	bending + compression: $\alpha = 0.680, \psi = -0.857$; limits $50.5\varepsilon / 58.1\varepsilon / 108.5\varepsilon$ (M_z does not stress the web of an I/H section: it lies on the z-z axis)	Class 1	Table 5.2
Flange outstands under $M_y + M_z (+N)$	$\sigma = N/A + M_y/W_{el,y} \pm M_z y/I_z$: outstand compressed by M_z : root 449.7, tip 537.0 N/mm ² (both compressive, $\alpha = 1 \rightarrow 9\varepsilon/\alpha, 10\varepsilon/\alpha, 21\varepsilon/\sqrt{k_\sigma} \geq$ $9\varepsilon/10\varepsilon/14\varepsilon$ bound kept); opposite outstand: root 420.2, tip 332.9 N/mm ² (wholly in compression, laxer limits, does not govern)	Class 1	Table 5.2 sheet 2
Auto Design Load Cases	1; SLS 1		

$V_{y,Ed}/V_{pl,y,Rd}$	$0.006 / 352.225 =$	0.000	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	$147.075 / 352.225 =$	0.418	OK

$V_{pl.y.Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2218.4 mm ² x 275 / ($\sqrt{3} \times 1$) ; $A_v \geq \eta h_w t_w = 1703.3$ mm ²	352.225 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 656.0 / 1	180.400 kN.m	
$M_{y.Ed} / M_{c.y.Rd}$	242.546 / 180.400 =	1.344	Warning
$(M_{y.Ed} / M_{N.V.y.Rd})^\alpha + (M_{z.Ed} / M_{N.V.z.Rd})^\beta @ x$	@ x = 0.33 m (ULS: 1.35G + 1.5Q): V = 138.398, $\rho = 0.000$; $N_{V.Rd} = (A - \rho A_v) f_y = 2100.999$, $n_v = 0.071$; $M_{v.y.Rd} = (W_{pl.y} - \rho A_v t_w / 4) f_y / \gamma_{M0} = 180.400$; $M_{v.z.Rd} = (W_{pl.z} - \rho A_v t_w / 4) f_y / \gamma_{M0} = 83.875$; $a_v = 0.235$; 6.2.9.1(4): no reduction ($N \leq 0.25 N_{V.Rd}$, $N \leq 0.5 h_w t_w (1 - \rho) f_y$); $M_{N.V.y.Rd} = 180.400$, $M_{N.V.z.Rd} = 83.875$, $\alpha = 2$, $\beta = 1$; $M_y = 47.103$, $M_z = 20.521$, $N = 150.000$ (plastic, 6.2.9.1 with $(1 - \rho) f_y$ on A_v)	0.313	OK 6.2.10
$V_{z.Ed} / V_{pl.z.Rd}$	0 / 942.581 = ($A_{v,z} = 5936.7$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 305.0 / 1	83.875 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	76.40 x 275 / 1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	656.0 x 275 / 1	180.400 kN.m	6.2.9.1(4)
$W_{pl.N.z} = F_n(W_{pl.z}, A_{vz}, n)$	305.0, 59.367, 0.071	305.0 cm ³	
$M_{N.z.Rd} = W_{pl.N.z} \cdot f_y / \gamma_{M0}$	305.0 x 275 / 1	83.875 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^\alpha + (M_{z.Ed} / M_{N.z.Rd})^\beta$	$(242.546 / 180.400)^2 + (20.521 / 83.875)^1 =$	2.052	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	275 x 205.80 / (275 x 9.40) = 21.894 ; 0.02 x (181.20 / 14.20) ² = 3.257 if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c ; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	0.9 x 5.311 x 210000 x 9.40 ³ / 181.20	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	5.311 x 210000 x 9.40 ² / (2 x 275 x 181.20) = 988.90 ≤ 100.00	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343 \text{e}3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	0.5 / 0.306 = 1.635 → 1.000 ; 1.000 x 166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	275 x 166.44 x 9.40 / 1.0	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 147.075$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.342	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	0.342 + 0.8 x ($M_{Ed} / M_{c.y.Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl.Rd} = 150.000 / 2101.000 = 0.071$) = 0.399 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.285	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8 \eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	147.075	ULS: 1.35G + 1.5Q	0.342	0.285	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.955	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	97.979	ULS: 1.35G + 1.5Q	0.228	0.203	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2 h_w / 3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an

end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{76.40 \times 275 / 3523.45}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.771	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.742 \times 275 / 101 =$	1558.850 kN	Curve b
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{76.40 \times 275 / 1186.00}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	1.329	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$76.40 \times 0.377 \times 275 / 101 =$	791.578 kN	Curve c

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	284.105 / 242.476 kN.m (shape only: $z_g = z_j = 0$)	1.172	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 235.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=191.0$, $M(L/2)=235.6$, $M(3L/4)=117.4$, $M_{max}=242.5$ kN.m	1.213	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 235.61$, $\psi = 0.659$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 235.61$, $\psi = 0.659$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e =$ portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
$M_{cr} =$ FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 105$, 0 mm; 1 combination(s), 3 solve(s)	228.573 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{656.0 \times 275 / 228.573}$	0.888	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.888, 0.879, 0.34	0.767	Curve b
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.767, 0.888, 0.924, 0.963	0.797	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.797 \times 656.0 \times 275 \leq 180.400 =$	143.723 kN.m	
$M_{y,Ed} / M_{b,Rd}$	242.546 / 143.723	1.688	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 209.60$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	242.546 / 143.723 ($M_{y,Rk} = 180.400$)	1.688	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 83.875	0.245	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.771 - 0.2) \times 0.096\} \leq 1.000 (1 + 0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz} \{1 + (2\lambda_z - 0.6) U_{N,z}\}$	$1.000 \{1 + (2 \times 1.329 - 0.6) \times 0.189\} \leq 1.000 (1 + 1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6 k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1 \lambda_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.1 \cdot \min(\lambda_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\lambda_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.096 + 1.055 \times 1.688 + 0.759 \times 0.245$	2.062	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.189 + 0.975 \times 1.688 + 1.265 \times 0.245$	2.144	Warning

Torsion Design

J, H, a, Q _f , Q _w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —	P385 App A
W _{n0} , S _{w1}	101 cm ² , 734 cm ⁴	P385 App A

Torsion Bending Design @ 2.091 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 5.77	P385 App C
φ _{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	2.696 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	13.796 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	40.987 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.091 m: 242.108, 13.089, 39.011 kN.m; M _{pl,y} = 180.400, M _{pl,f} = 41.938, M _{pl,z} = 83.875	2.578	Warning P385 3.1.2
V _{pl,T,Rd} = √(1 - τ _t /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: τ _t = 76.13 N/mm ² ; V _{pl,Rd} = 352.225	276.531 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	147.075 / 276.531	0.532	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 228.573 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 5.388 / 4.224 kN.m	3.823 / 3.096 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.564 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	4.62°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _α is unbounded; the member is inadequate as arranged.	NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	31.88 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	31.88 mm	Warning
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N _{Ed} /N _{pl,Rd}	Ed	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.071	2.052	0.189	1.688	2.062	2.144	1.913	0.418	1.344	1.688	0.313	0.342	0.955	2.578	0.532	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 4.6° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} \gamma_{M0})$	2218.4 mm ² x 275 / ($\sqrt{3}$ x 1) ; $A_v \geq \eta h_w t_w = 1703.3$ mm ²	352.225 kN	6.2.6
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 656.0 / 1	180.400 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	242.546 / 180.400 =	1.344	Warning
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta @ x$	@ x = 0.33 m (ULS: 1.35G + 1.5Q): V = 138.398, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 2100.999$, $n_v = 0.071$; $M_{v,y,Rd} = (W_{pl,y} - \rho A_v^2 / 4 t_w) f_y / \gamma_{M0} = 180.400$; $M_{v,z,Rd} = (W_{pl,z} - \rho A_v t_w / 4) f_y / \gamma_{M0} = 83.875$; $a_v = 0.235$; 6.2.9.1(4): no reduction ($N \leq 0.25 N_{V,Rd}$, $N \leq 0.5 h_w t_w (1 - \rho) f_y$); $M_{N,y,Rd} = 180.400$, $M_{N,z,Rd} = 83.875$, $\alpha = 2$, $\beta = 1$; $M_y = 47.103$, $M_z = 20.521$, $N = 150.000$ (plastic, 6.2.9.1 with $(1 - \rho) f_y$ on A_v)	0.313	OK 6.2.10
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 942.581 = ($A_{v,z} = 5936.7$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 305.0 / 1	83.875 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	76.40 x 275 / 1 = (No bearing / block tearing design)	2101.000 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 2101.000 =	0.071	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	656.0, 22.184, 0.071	656.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	656.0 x 275 / 1	180.400 kN.m	6.2.9.1(4)
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	305.0, 59.367, 0.071	305.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	305.0 x 275 / 1	83.875 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^\alpha + (M_{z,Ed} / M_{N,z,Rd})^\beta$	$(242.546 / 180.400)^2 + (20.521 / 83.875)^1 =$	2.052	Warning
NOT VERIFIED	Standard (closed-form) M_{cr} : a destabilising load height $z_g = +105$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.		NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	181.20, 9.40, 14.20, 205.80 mm ($B = 205.80 \leq t_w + 30 \epsilon t_f = 403.20$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w / t_f)^2$	$275 \times 205.80 / (275 \times 9.40) = 21.894$; $0.02 \times (181.20 / 14.20)^2 = 3.257$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]			
s_s , c ; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c) / h_w \leq 6 = 2 + 6 \times 100.00 / 181.20$	5.311	Fig 6.1(c)
$F_{cr} = 0.9 k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.311 \times 210000 \times 9.40^3 / 181.20$	4601.343 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2 f_{yw} \cdot h_w) \leq s_s + c$	$5.311 \times 210000 \times 9.40^2 / (2 \times 275 \times 181.20) = 988.90 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1 / 2 + (l_e / t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.310 \leq 0.5$ with $m_2 = 3.257$, so $m_2 = 0$); $\min[210.49, 166.44]$	166.44 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(166.44 \times 9.40 \times 275 / 4601.343 \text{e}3)}$	0.306	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.306 = 1.635 \rightarrow 1.000$; 1.000×166.44	166.44 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 166.44 \times 9.40 / 1.0$	430.254 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 147.075$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 430.254 =	0.342	OK
$\eta_2 + 0.8 \eta_1 \leq 1.4$	$0.342 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 180.400 + N_{Ed} / N_{pl,Rd} = 150.000 / 2101.000 = 0.071) = 0.399 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.285	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	147.075	ULS: 1.35G + 1.5Q	0.342	0.285	governs
2	point load 3+4	(a)	0.00	6.002	161.29	0.283	1.000	416.923	87.286	ULS: 1.35G + 1.5Q	0.209	0.955	OK
6	End 2 reaction	(c)	100.00	5.311	166.44	0.306	1.000	430.254	97.979	ULS: 1.35G + 1.5Q	0.228	0.203	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √A.f _y /N _{cr}	√76.40x275/3523.45 (N _{cr,y} = π²EI _y /L _{ey} ²)	0.771	
N _{b,y,Rd} = Area.χ.f _y /γ _{M1}	76.40x0.742x275/10/1 =	1558.850 kN	Curve b
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A.f _y /N _{crz}	√76.40x275/1186.00 (N _{cr,z} = π²EI _z /L _{ez} ²)	1.329	
N _{b,z,Rd} = Area.χ.f _y /γ _{M1}	76.40x0.377x275/10/1 =	791.578 kN	Curve c

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 235.6, 1.000, 300.000	1.213	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=191.0, M(L/2)=235.6, M(3L/4)=117.4, M _{max} =242.5 kN·m		SN003a
C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 235.61, ψ = 0.659, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 235.61, ψ = 0.659, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.213, 6.000, 2060, 47.2, 0.197, 210000; z _g = +105 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	294.111 kN.m	BLOCKED
λ _{LT} = √W.f _y /M _{cr}	√ 656.0 x 275 / 294.111	0.783	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	0.783, 0.795, 0.34	0.826	Curve b
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.826, 0.783, 0.908, 0.954	0.866	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.866 x 656.0 x 275 ≤ 180.400 =	156.254 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.086; χ _{LT,mod} = 0.673; M _{y,Ed} /M _{b,Rd,simp} = 1.998	121.390 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	242.546 / 156.254	1.552	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 209.60 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 1558.850 ($N_{Rk} = 2101.000$)	0.096	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 791.578	0.189	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	242.546 / 156.254 ($M_{y,Rk} = 180.400$)	1.552	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	20.521 / 83.875	0.245	OK
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(0.771-0.2) \times 0.096\} \leq 1.000(1+0.8 \times 0.096)$	1.055	Table B.1
$k_{zz} = C_{mz}\{1+(2\lambda_z-0.6)U_{N,z}\}$	$1.000\{1+(2 \times 1.329-0.6) \times 0.189\} \leq 1.000(1+1.4 \times 0.189)$	1.265	
$k_{yz} = 0.6k_{zz}$	0.6×1.265	0.759	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.1 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 1.329$, $U_{N,z} = 0.189$, $C_{mLT} = 1.000$	0.975	Table B.2
$U_{Ny}+k_{yy} \cdot U_{My}+k_{yz} \cdot U_{Mz}$	$0.096+1.055 \times 1.552+0.759 \times 0.245$	1.920	Warning
$U_{Nz}+k_{zy} \cdot U_{My}+k_{zz} \cdot U_{Mz}$	$0.189+0.975 \times 1.552+1.265 \times 0.245$	2.012	Warning

Torsion Design

J, H, a, Q _f , Q _w	47.2 cm ⁴ , 0.197 dm ⁶ , 1040 mm, —, —		P385 App A
W _{n0} , S _{w1}	101 cm ² , 734 cm ⁴		P385 App A

Torsion Bending Design @ 2.091 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 5.77	P385 App C
φ_{max} (ULS)	L/a = 5.77; fork ends, warping free (P385 Cases 3/4/10)	0.1701 rad = 9.75°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	2.696 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	13.796 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	40.987 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.091 m: 242.108, 13.089, 39.011 kN.m; M _{pl,y} = 180.400, M _{pl,f} = 41.938, M _{pl,z} = 83.875	2.578	Warning P385 3.1.2
V _{pl,T,Rd} = √(1 - T _t /(1.25f _y /√3)) · V _{pl,Rd}	@ x = 0 m: T _t = 76.13 N/mm ² ; V _{pl,Rd} = 352.225	276.531 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	147.075 / 276.531	0.532	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2 m: 241.560/149.069 + 1.000x38.160/83.875 + 0.634x0.545x5.704x13.796/41.938	2.724	Warning EN 1993-6 A
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 5.388 / 4.224 kN.m	3.823 / 3.096 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.564 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	4.62°	review

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	31.88 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	31.88 mm	Warning
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N _{Ed} /N _(pl,Rd)	UNy	UMy	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.071	2.052	0.189	1.552	1.920	2.012	1.913	0.418	1.344	1.552	0.313	0.342	0.955	2.578	0.532	2.724 2.724

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 2.724

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +105 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 4.6° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

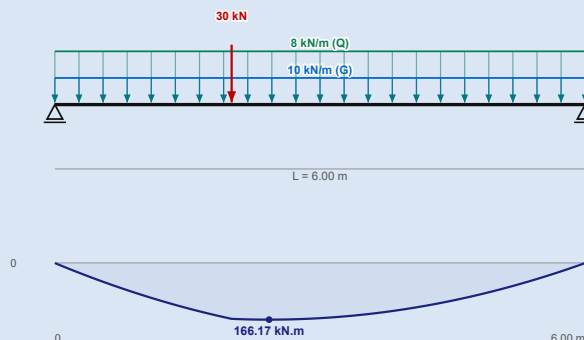
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS 200 x 100 x 8.0 (HF), 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0$, $z_g = 0$)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model: M_{-} is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	9.50, 22.00, 275, 0.00, 166.17, 0.00	Class 1
	(Axial: Non-Slender)	

Auto Design Load Cases 1: SLS 1

$V_{y,Ed}/V_{pl,y,Rd}$	$0.044 / 474.197 =$	0.000	Low Shear OK 6.2.6
$V_{y,Ed,max}/V_{pl,y,Rd}$	$107.894 / 474.197 =$	0.228	
$V_{pl,y,Rd} = A_v f_y / (\sqrt{3} Y_{M0})$	$2986.7 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	474.197 kN	
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / Y_{M0}$	$275 \times 282.0 / 1$	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$166.174 / 77.550 =$	2.143	

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15\epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig. 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017 \times 10^3)}$	0.298 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 528.000 =	0.204 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.146 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	899.055 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 899.055}$	0.294	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.294 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.294, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed} / M_{b,Rd}$	$166.174 / 77.550$	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L1 · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 3.210

Owner case: vertical loads resulting in no torsion

RHS 200 x 100 x 8.0 (HF), 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre (e = 0, z_g = 0)

Beam & Beam-Portion (Member) (FAIL)

Member RHS-L1 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

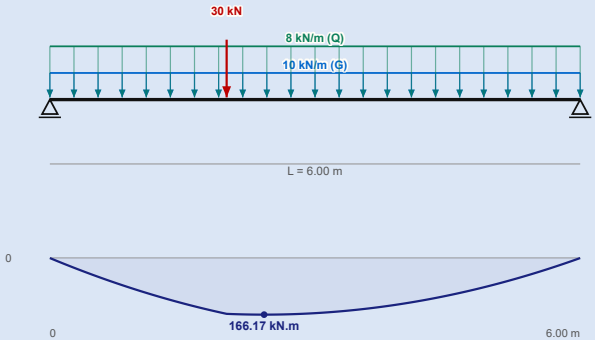
G SW 0.3443 kN/m 0–6 m (automatic)

$Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf}$ = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf}$ = 0.9, EQU set A]

$Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m) 200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender) Class 1

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$	0.044 / 474.197 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	107.894 / 474.197 =	0.228	OK
$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} Y_{MO})$	2986.7 mm ² x 275 / ($\sqrt{3}$ x 1)	474.197 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / Y_{MO}$	275 x 282.0/1	77.550 kN.m	
$M_{y.Ed}/M_{c.y.Rd}$	166.174 / 77.550 =	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017 \text{e}3)}$	0.298 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = 0$ (load through the shear centre)	916.230 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed} / h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. $N_{f,Ed} = M_{Ed} / h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

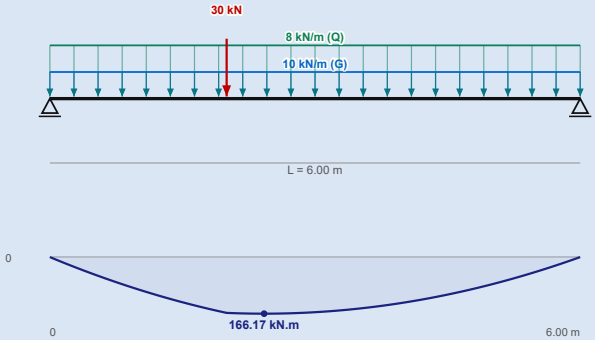
RHS-L2 · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 3.210
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L2 "horizontal eccentricity from shear center": all loads at e = +40 mm (torque), z_g = 0

Beam & Beam-Portion (Member) (FAIL)
Member RHS-L2 RHS 200 x 100 x 8.0 (HF)
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.3443 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m) 200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200
B=100 t=8
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 9.50, 22.00, 275, 0.00, 166.17, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.044 / 474.197 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.894 / 474.197 =	0.228	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	2986.7 mm ² x 275/(√3 x 1)	474.197 kN	6.2.6
M _{c,y,Rd} = f _y W _{pl,y} /V _{M0}	275 x 282.0/1	77.550 kN.m	
M _{y,Ed} /M _{c,y,Rd}	166.174 / 77.550 =	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm (B/2 = 50.00 ≤ t_w + 15t_f = 118.93); f_{yw} = f_{yf} = 275; 2 webs (flat depth from the section table, corner geometry) Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 50.00/(275 x 8.00) = 6.250 ; 0.02 x (176.00/8.00)² = 9.680 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
s _s , c ; k _F	s _s = 0.00 mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; k _F = 6 + 2(h _w /a) ² = 6 + 2(176.00/6000) ²	6.002 Fig 6.1(a)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 6.002 x 210000 x 8.00 ³ /176.00	3299.855 kN 6.4(1)
I _y = s _s + 2t _f (1 + √(m ₁ + m ₂)) ≤ a	m ₂ = 0.000 (first pass λ _F = 0.231 ≤ 0.5 with m ₂ = 9.680, so m ₂ = 0); 0.00 + 2 x 8.00 x (1 + √(6.250 + 0.000)) = 56.00	56.00 mm 6.5(3)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(56.00 x 8.00 x 275/3299.855e3)	0.193 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.193 = 2.588 → 1.000 ; 1.000 x 56.00	56.00 mm 6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 56.00 x 8.00/1.0 per web; load share to this web = 0.935 (lever rule, e = 40 mm); F _{Rd} for the load = 131.795 kN	123.200 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 45.000 kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.341 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 163.859/77.550 = 2.113) = 2.032 ≤ 1.4 (ULS: 1.35G + 1.5Q)	1.451 Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	899.055 / 787.606 kN.m (shape only: z _g = z _j = 0)	1.142	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.6, M(L/2)=161.8, M(3L/4)=110.1, M _{max} =166.2 kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	899.055 kN.m	converged
λ _{LT} = √W.f _y /M _{cr}	√ 282.0 x 275 / 899.055	0.294	
λ _{LT} ≤ λ _{LT,0}	0.294 ≤ 0.4 : LTB may be ignored	χ _{LT} = 1.000	6.3.2.2(4)
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	1.000, 0.294, 0.936, 1.000	1.000	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	1.000 x 282.0 x 275 ≤ 77.550 =	77.550 kN.m	
M _{y,Ed} /M _{b,Rd}	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 251/(√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 39.852	0.107	OK
V _{pl.T.Rd} = [1 - T _t /(f _y √3)].V _{pl.Rd}	@ x = 0 m: T _t = 16.97 N/mm ²	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl.T.Rd}	107.894 / 423.507	0.255	OK
θ (SLS)	T _{Ed,SLS} = 1.760 kN.m; @ x = 2 m	0.11°	advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	M _{-(y.Ed)/M_{-(b.Rd)}}	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Torsion	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	0.255	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member RHS-L2 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.3443 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

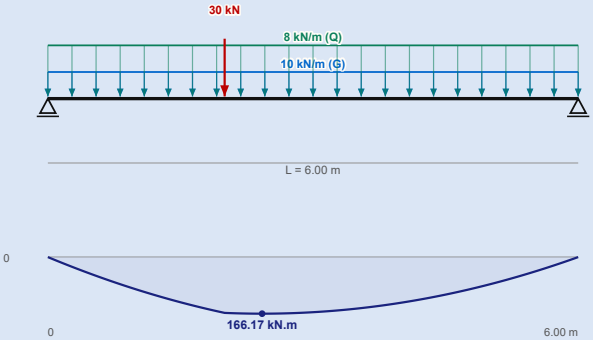
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m) 200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200
B=100 t=8

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 9.50, 22.00, 275, 0.00, 166.17, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.044 / 474.197 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.894 / 474.197 =	0.228	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	2986.7 mm ² x 275/(√3 x 1)	474.197 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 282.0/1	77.550 kN.m	
M _{y,Ed} /M _{c,y,Rd}	166.174 / 77.550 =	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm (B/2 = 50.00 ≤ t_w + 15ε_t = 118.93); f_{yw} = f_{yf} = 275; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 50.00/(275 x 8.00) = 6.250 ; 0.02 x (176.00/8.00)² = 9.680 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
s _s , c ; k _F	s _s = 0.00 mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; k _F = 6 + 2(h _w /a) ² = 6 + 2(176.00/6000) ²	6.002 Fig 6.1(a)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 6.002 x 210000 x 8.00 ³ /176.00	3299.855 kN 6.4(1)
I _y = s _s + 2t _f (1 + √(m ₁ + m ₂)) ≤ a	m ₂ = 0.000 (first pass λ _F = 0.231 ≤ 0.5 with m ₂ = 9.680, so m ₂ = 0); 0.00 + 2 x 8.00 x (1 + √(6.250 + 0.000)) = 56.00	56.00 mm 6.5(3)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(56.00 x 8.00 x 275/3299.855e3)	0.193 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.193 = 2.588 → 1.000 ; 1.000 x 56.00	56.00 mm 6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 56.00 x 8.00/1.0 per web; load share to this web = 0.935 (lever rule, e = 40 mm); F _{Rd} for the load = 131.795 kN	123.200 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 45.000 kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.341 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 163.859/77.550 = 2.113) = 2.032 ≤ 1.4 (ULS: 1.35G + 1.5Q)	1.451 Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

M _{b,Rd} = M _{c,y,Rd}	closed hollow section, λ _{LT} = 0.291 ≤ 0.4: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M _{cr} (information, I _w = 0)	SN003a with C ₁ = 1.163, L _e = 6 m; z _g = 0 (load through the shear centre)	916.230 kN.m	SN003a
M _{y,Ed} /M _{b,Rd}	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m
W _t	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³

$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	275 x 251 / ($\sqrt{3} \times 1$)	39.852 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	4.260 / 39.852	0.107	OK
$V_{pl.T.Rd} = [1 - T_t / (f_y / \sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $T_t = 16.97 \text{ N/mm}^2$	423.507 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl.T.Rd}$	107.894 / 423.507	0.255	OK
$\theta \text{ (SLS)}$	$T_{Ed,SLS} = 1.760 \text{ kN.m}$; @ x = 2 m	0.11°	advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67 \text{ mm}$ @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	0.255	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L3t · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 3.210

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +100$ mm (D/2, destabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)

Member RHS-L3t RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 0$ mm $z_g = 100$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 0$ mm $z_g = 100$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = 100$ mm (kN)

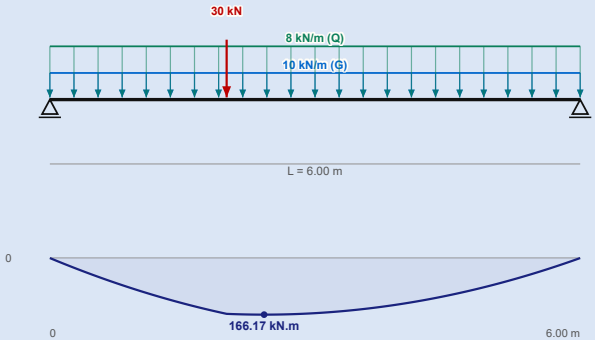
G SW 0.3443 kN/m 0–6 m (automatic)

$Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]

$Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m) 200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender) Class 1

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$	0.044 / 474.197 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	107.894 / 474.197 =	0.228	OK
$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} Y_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3}$ x 1)	474.197 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / Y_{M0}$	275 x 282.0/1	77.550 kN.m	
$M_{y.Ed}/M_{c.y.Rd}$	166.174 / 77.550 =	2.143	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017 \times 10^3)}$	0.298 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 528.000 =	0.204 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.146 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	876.701 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 876.701}$	0.297	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.297 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.297, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed} / M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +100$ mm (D/2, destabilising). $e = 0$

$M_{Y,Ed}/M_{c,Y,Rd}$	$166.174 / 77.550 =$	2.143	Warning
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NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3/176.00$	2974.017 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017 \text{e}3)}$	0.298	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00/1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 528.000 =	0.204	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.146	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	916.230 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L3b · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 3.210

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -100$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member RHS-L3b RHS 200 x 100 x 8.0 (HF)
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)

Q UDL 8.000 0-6 m $e = 0$ mm $z_g = -100$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -100$ mm (kN)

G SW 0.3443 kN/m 0-6 m (automatic)

$Y_{G,inf}$ companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]

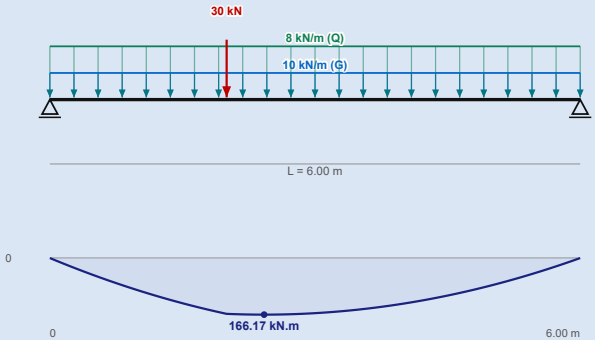
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]

$Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member

(single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m) 200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200
B=100 t=8

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 9.50, 22.00, 275, 0.00, 166.17, 0.00 (Axial: Non-Slender) Class 1

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$ 0.044 / 474.197 = 0.000 Low Shear

$V_{y.Ed,max}/V_{pl.y.Rd}$ 107.894 / 474.197 = 0.228 OK

$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} Y_{MO})$ 2986.7 mm² x 275 / ($\sqrt{3}$ x 1) 474.197 kN 6.2.6

$M_{c.y.Rd} = f_y \cdot W_{pl.y} / Y_{MO}$ 275 x 282.0/1 77.550 kN.m

$M_{y.Ed}/M_{c.y.Rd}$ 166.174 / 77.550 = 2.143 **Warning**

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3/176.00$	2974.017 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017 \text{e}3)}$	0.298 6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5/0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00/1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN 6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 528.000 =	0.204 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.146 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -100, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	921.883 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 921.883}$	0.290	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.290 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.290, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	53.49 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Web h_w, t_w, t_f, b_f 176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15\epsilon t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/176.00$	5.409 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.409 \times 210000 \times 8.00^3 / 176.00$	2974.017 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.409 \times 210000 \times 8.00^2 / (2 \times 275 \times 176.00) = 751.01 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.312 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $\min[201.00, 120.00]$	120.00 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(120.00 \times 8.00 \times 275 / 2974.017 \text{e}3)}$	0.298 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.298 = 1.678 \rightarrow 1.000$; 1.000×120.00	120.00 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 120.00 \times 8.00 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 528.000 kN	264.000 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.894$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 528.000 =	0.204 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.204 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 77.550 = 0.000) = 0.204 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.146 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	governs
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	246.400	45.000	ULS: 1.35G + 1.5Q	0.183	1.338	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = -100$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	916.230 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed} / h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. $N_{f,Ed} = M_{Ed} / h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 53.49 mm **Warning**

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.143	2.143	3.210	0.228	0.204	1.338	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member RHS-L4 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

G SW 0.3443 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

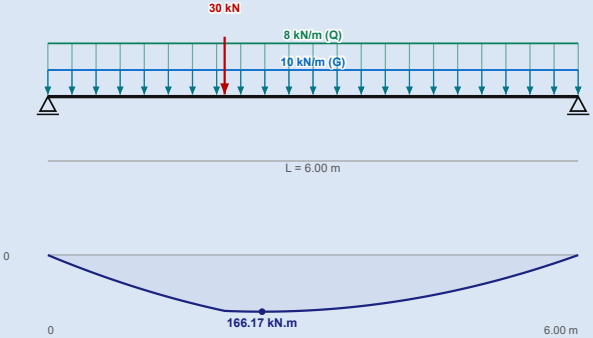
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member

(single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Upliftno end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9)

R_{min} ≥ 0OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200
B=100 t=8

Class = Fn(c/t, d/t, f_y, N, M_y, M_z)9.50, 22.00, 275, 0.00, 166.17, 0.00

(Axial: Non-Slender)

Class 1

Auto Design Load Cases1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd}0.044 / 474.197 =0.000Low Shear

V_{y,Ed,max}/V_{pl,y,Rd}107.894 / 474.197 =0.228OK

V_{pl,y,Rd} = A_vf_y/(√3V_{M0})2986.7 mm² x 275/(√3 x 1)474.197 kN6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/V_{M0}275 x 282.0/177.550 kN.m

M_{y,Ed}/M_{c,y,Rd}166.174 / 77.550 =2.143Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f176.00, 8.00, 8.00, 50.00 mm (B/2 = 50.00 ≤ t_w + 15εt_f = 118.93); f_{yw} = f_{yf} = 275; 2 webs (flat depth from the section table, corner geometry)Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)²275 x 50.00/(275 x 8.00) = 6.250 ; 0.02 x (176.00/8.00)² = 9.680 if λ_F > 0.5, else 06.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)	
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3/176.00$	3299.855 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275/3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5/\bar{\lambda}_F \leq 1; L_{eff} = \chi_F \cdot I_y$	$0.5/0.193 = 2.588 \rightarrow 1.000; 1.000 \times 56.00$	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00/1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 163.859/77.550 = 2.113) = 2.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.451	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{max}=166.2$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	876.701 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 876.701}$	0.297	
$\bar{\lambda}_{LT} \leq \bar{\lambda}_{LT,0}$	$0.297 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	1.000, 0.297, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t	SCI P385 Table A.7/A.8; I _t = 1800 cm ⁴	251 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 251/(√3 x 1)	39.852 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 39.852	0.107	OK
V _{pl.T.Rd} = [1 - T _t /(f _y √3)].V _{pl.Rd}	@ x = 0 m: T _t = 16.97 N/mm ²	423.507 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl.T.Rd}	107.894 / 423.507	0.255	OK
θ (SLS)	T _{Ed,SLS} = 1.760 kN.m; @ x = 2 m	0.11°	advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	53.49 mm	Warning
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MA/Mc	M _{-(y.Ed)/M_{-(b.Rd)}}	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Torsion	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	0.255	2.143

FAIL — governing Deflection = 3.210

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member RHS-L4 RHS 200 x 100 x 8.0 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

G SW 0.3443 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

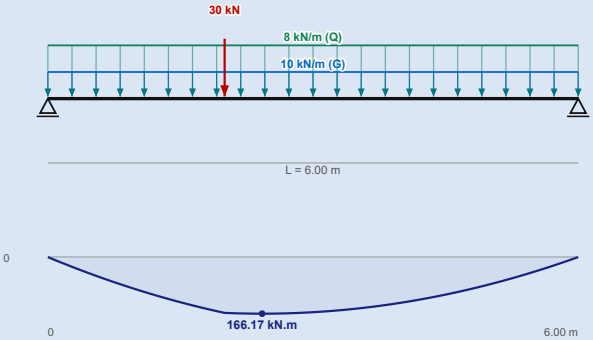
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	0.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m) 200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200
B=100 t=8

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 9.50, 22.00, 275, 0.00, 166.17, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.044 / 474.197 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 107.894 / 474.197 = 0.228 OK

V_{pl,y,Rd} = A_vf_y/(√3V_{M0}) 2986.7 mm² x 275/(√3 x 1) 474.197 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/V_{M0} 275 x 282.0/1 77.550 kN.m

M_{y,Ed}/M_{c,y,Rd} 166.174 / 77.550 = 2.143 Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
s_s , c; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN 6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm 6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855e3)}$	0.193 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.859 / 77.550 = 2.113) = 2.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.451 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.146	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.451	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.126	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	916.230 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w
W_{n0}, S_{w1}

1800 cm⁴, — dm⁶, — mm, —, —
—, —

P385 App A
P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)
W_t
T_{Rd} = f_yW_t/(√3Y_{M0})
T_{Ed}/T_{Rd}
V_{pl.T.Rd} = [1 - T_t/(f_y√3)].V_{pl.Rd}

V_{Ed}/V_{pl.T.Rd}
θ (SLS)

combination ULS: 1.35G + 1.5Q
SCI P385 Table A.7/A.8; I_t = 1800 cm⁴
275 x 251/(√3 x 1)
4.260 / 39.852
@ x = 0 m: T_t = 16.97 N/mm²

107.894 / 423.507
T_{Ed,SLS} = 1.760 kN.m; @ x = 2 m

4.260 kN.m
251 cm³
39.852 kN.m
0.107
423.507 kN

0.255
0.11°

6.2.7(7)

OK
6.2.7(9) Eq
6.28
OK
advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360

53.49 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m

53.49 mm

Warning

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	Max
2.143	2.143	3.210	0.228	0.341	1.451	0.107	0.255	2.143

FAIL — governing Deflection = 3.210

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +100 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
 RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN
 value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all
 loads at e = +40 mm, z_g = +100 mm

$M_{GZ,Rd} = f_v \cdot W_{pl,z} / \gamma_{M0}$	$275 \times 172.0 / 1$	47.300 kN.m
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$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	44.80 x 275/1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	282.0 x 275/1	77.550 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^{\alpha} + (M_{z.Ed} / M_{N.z.Rd})^{\beta}$	$(166.174 / 77.550)^{1.69} + (0)^1 =$	3.621	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 163.859 / 77.550 + N_{Ed} / N_{pl.Rd} = 150.000 / 1232.000 = 2.235) = 2.129 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.521	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.216	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.521	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.195	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b.y.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10/1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b.z.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10/1 =$	368.021 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	899.055 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
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$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.8, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.6$, $M(L/2)=161.8$, $M(3L/4)=110.1$, $M_{\max}=166.2$ kN·m	1.163	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.84$, $\psi = 0.897$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.84$, $\psi = 0.897$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	876.701 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{282.0 \times 275 / 876.701}$	0.297	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.297 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.297, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed}/M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	166.174 / 77.550 ($M_{y,Rk} = 77.550$)	2.143	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	0.000 / 47.300	0.000	OK
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(0.979-0.2)\times 0.179\} \leq 1.000(1+0.8\times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.702-0.2)\times 0.408\} \leq 1.000(1+0.8\times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny}+k_{yy} \cdot U_{M,y}+k_{yz} \cdot U_{M,z}$	$0.179+1.139\times 2.143+1.326\times 0.000$	2.620	Warning
$U_{Nz}+k_{zy} \cdot U_{M,y}+k_{zz} \cdot U_{M,z}$	$0.408+0.684\times 2.143+1.326\times 0.000$	1.873	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W_{n0} , S_{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	4.260 / 39.852	0.107	OK
$V_{pl,T,Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 16.97$ N/mm ²	423.507 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	107.894 / 423.507	0.255	OK
θ (SLS)	$T_{Ed,SLS} = 1.760$ kN.m; @ x = 2 m	0.11°	advisory

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$

$53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$

53.49 mm

Warning

N_Ed/N_(pl.Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.122	3.621	0.408	2.143	2.620	1.873	3.210	0.228	2.143	2.143	0.341	1.521	0.107	0.255	3.621

FAIL — governing Bending+axial cross-section (6.2.9) = 3.621

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS-L5 · RHS 200 x 100 x 8.0 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Bending+axial cross-section (6.2.9) = 3.621

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at $e = +40$ mm, $z_g = +100$ mm

Axial with Moments (Member) (FAIL)
Member RHS-L5 RHS 200 x 100 x 8.0 (HF)
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 40$ mm $z_g = 100$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 100$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 100$ mm (kN)

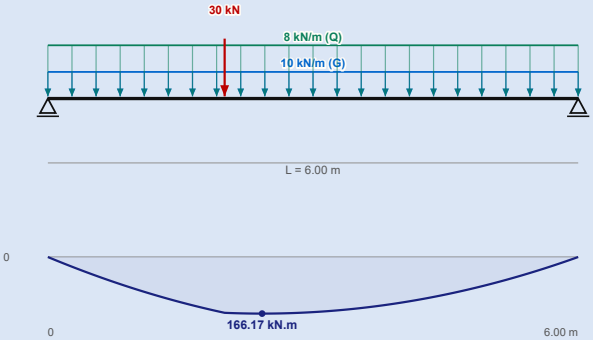
G SW 0.3443 kN/m 0–6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	-4.26	107.89	0.00	0.00	0.00	166.17	0.00	-53.49	107.89	—
	End 2, x = 6	150.00C	0.00	-92.89	0.00	0.00	0.00	@ 2.424	@ —	@ 2.850	92.89	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (35.1 kg/m)	200 x 100 x 8.0 RHS [Hot-finished] [S275] D=200 B=100 t=8		
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$	9.50, 22.00, 275, 150.00, 166.17, 0.00 (Axial: Non-Slender)		Class 1
Web classified for	bending + compression: $\alpha = 0.597$, $\psi = -0.756$; limits 58.6ε / 67.5ε / 99.9ε	Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1		

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$	0.044 / 474.197 =	0.000	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	107.894 / 474.197 =	0.228	OK
$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$	2986.7 mm ² x 275 / ($\sqrt{3}$ x 1)	474.197 kN	6.2.6
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 282.0 / 1	77.550 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	166.174 / 77.550 =	2.143	Warning
$V_{z,Ed} / V_{pl,z,Rd}$	0 / 237.099 = ($A_{v,z} = 1493.3$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 172.0 / 1	47.300 kN.m	

$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	44.80 x 275/1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	282.0 x 275/1	77.550 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^{\alpha} + (M_{z.Ed} / M_{N.z.Rd})^{\beta}$	$(166.174 / 77.550)^{1.69} + (0)^1 =$	3.621	Warning
NOT VERIFIED			
Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.			
NOT VERIFIED			

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)	
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)	
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.341	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.341 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 163.859 / 77.550 + N_{Ed} / N_{pl.Rd} = 150.000 / 1232.000 = 2.235) = 2.129 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.521	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	107.894	ULS: 1.35G + 1.5Q	0.204	0.216	OK
2	point load 3	(a)	0.00	6.002	56.00	0.193	1.000	131.795	45.000	ULS: 1.35G + 1.5Q	0.341	1.521	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	92.894	ULS: 1.35G + 1.5Q	0.176	0.195	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
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$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10 / 1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 = (z-z: \text{held in position at both ends, in direction at neither: } 1.0 L)$	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10 / 1 =$	368.021 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 161.84, \psi = 0.897, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M = 0.000, \psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 161.84, \psi = 0.897, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.291 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163, L_e = 6 \text{ m}; z_g = +100 \text{ mm}$ entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	916.230 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	166.174 / 77.550	2.143	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 838.121$ ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 368.021$	0.408	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$166.174 / 77.550$ ($M_{y,Rk} = 77.550$)	2.143	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$0.000 / 47.300$	0.000	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.979 - 0.2) \times 0.179\} \leq 1.000 (1 + 0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.702 - 0.2) \times 0.408\} \leq 1.000 (1 + 0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.179 + 1.139 \times 2.143 + 1.326 \times 0.000$	2.620	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.408 + 0.684 \times 2.143 + 1.326 \times 0.000$	1.873	Warning

Torsion Design

J, H, a, Q _f , Q _w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0}, S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1800 \text{ cm}^4$	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	$4.260 / 39.852$	0.107	OK
$V_{pl,T,Rd} = [1 - T_t / (f_y / \sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 16.97 \text{ N/mm}^2$	423.507 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl,T,Rd}$	$107.894 / 423.507$	0.255	OK
θ (SLS)	$T_{Ed,SLS} = 1.760 \text{ kN.m}; @ x = 2 \text{ m}$	0.11°	advisory

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $53.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 53.49 mm **Warning**

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.122	3.621	0.408	2.143	2.620	1.873	3.210	0.228	2.143	2.143	0.341	1.521	0.107	0.255	3.621

FAIL — governing Bending+axial cross-section (6.2.9) = 3.621

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +100, factored by the combination) plus an entered CONSTANT M_z,Ed = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_mz = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.100 = 1.03 kN.m characteristic) is not modelled

$M_{Y,Ed}/M_{c,Y,Rd}$	$220.241 / 77.550 =$	2.840	Warning
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$V_{z,Ed}/V_{pl,z,Rd}$	0 / 237.099 = ($A_{v,z} = 1493.3 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 x 172.0/1	47.300 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	44.80 x 275/1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	282.0 x 275/1	77.550 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	172.0, 14.933, 0.122	172.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	172.0 x 275/1	47.300 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^\alpha + (M_{z,Ed}/M_{N,z,Rd})^\beta$	$(220.241/77.550)^{1.69} + (20.521/47.300)^{1.69} =$	6.070	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3/176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275/3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w/\gamma_{M1}$	$275 \times 56.00 \times 8.00/1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 220.241/77.550 + N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 2.962) = 3.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.165	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	136.085	ULS: 1.35G + 1.5Q	0.258	0.254	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.165	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	106.990	ULS: 1.35G + 1.5Q	0.203	0.214	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y/N_{cr}}$	$\sqrt{44.80 \times 275/1283.87}$ ($N_{cr,y} = \pi^2 EI_y/L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y/\gamma_{M1}$	$44.80 \times 0.680 \times 275/10/1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y/N_{crz}}$	$\sqrt{44.80 \times 275/425.46}$ ($N_{cr,z} = \pi^2 EI_z/L_{ez}^2$)	1.702	

$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10 / 1 =$	368.021 kN	Curve a
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Equivalent Uniform Moment Factors C_1 , C_{mLT} , C_{mz} , and C_{my}

$C_1 = M_{cr} / M_{cr,uniform}$	934.976 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.187	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 204.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.9$, $M(L/2)=204.1$, $M(3L/4)=131.3$, $M_{max}=220.2$ kN.m	1.217	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 204.13$, $\psi = 0.822$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 204.13$, $\psi = 0.822$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check $M_{b,Rd}$

$L_e =$ portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
$M_{cr} =$ FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	910.839 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{282.0 \times 275 / 910.839}$	0.292	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.292 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.292, 0.918, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed} / M_{b,Rd}$	$220.241 / 77.550$	2.840	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 838.121$ ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 368.021$	0.408	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$220.241 / 77.550$ ($M_{y,Rk} = 77.550$)	2.840	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$20.521 / 47.300$	0.434	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.979 - 0.2) \times 0.179\} \leq 1.000 (1 + 0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.702 - 0.2) \times 0.408\} \leq 1.000 (1 + 0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{N_y + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}}$	$0.179 + 1.139 \times 2.840 + 1.326 \times 0.434$	3.990	Warning
$U_{N_z + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}}$	$0.408 + 0.684 \times 2.840 + 1.326 \times 0.434$	2.924	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm^4 , — dm^6 , — mm, —, —	P385 App A
W_{n0} , S_{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1800 \text{ cm}^4$	251 cm^3	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	$5.388 / 39.852$	0.135	OK
$V_{pl,T,Rd} = [1 - t_t / (f_y / \sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $t_t = 21.46 \text{ N/mm}^2$	410.089 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl,T,Rd}$	$136.085 / 410.089$	0.332	OK
θ (SLS)	$T_{Ed,SLS} = 2.512 \text{ kN.m}$; @ $x = 2$ m	0.17°	advisory

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $76.75 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 76.75 mm **Warning**

N_Ed/N_(pl.Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.122	6.070	0.408	2.840	3.990	2.924	4.605	0.287	2.840	2.840	0.662	2.165	0.135	0.332	6.070

FAIL — governing Biaxial bending + axial (6.2.9.1) = 6.070

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: ... and axial load on beam, load applied at angle on beam
 RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +100, factored by the combination) plus an entered CONSTANT M_z,Ed = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_mz = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.100 = 1.03 kN.m characteristic) is not modelled

$M_{V_{Ed}}/M_{C_{V_{Bd}}}$	$220.241 / 77.550 =$	2.840	Warning
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$V_{z,Ed}/V_{pl,z,Rd}$	0 / 237.099 = ($A_{v,z} = 1493.3 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 x 172.0/1	47.300 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	44.80 x 275/1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	282.0 x 275/1	77.550 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	172.0, 14.933, 0.122	172.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	172.0 x 275/1	47.300 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^\alpha + (M_{z,Ed}/M_{N,z,Rd})^\beta$	$(220.241/77.550)^{1.69} + (20.521/47.300)^{1.69} =$	6.070	Warning
NOT VERIFIED Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.			
NOT VERIFIED			

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior			
s_s, c ; k_F	$s_s = 0.00 \text{ mm}$ (default 0); $c = 2000.00 \text{ mm}$ (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3/176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275/3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w/\gamma_{M1}$	$275 \times 56.00 \times 8.00/1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 220.241/77.550 + N_{Ed}/N_{pl,Rd} = 150.000/1232.000 = 2.962) = 3.032 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.165	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	136.085	ULS: 1.35G + 1.5Q	0.258	0.254	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.165	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	106.990	ULS: 1.35G + 1.5Q	0.203	0.214	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an

end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10 / 1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10 / 1 =$	368.021 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 204.13$, $\psi = 0.822$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 204.13$, $\psi = 0.822$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.284 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.217$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	958.768 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	220.241 / 77.550	2.840	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	220.241 / 77.550 ($M_{y,Rk} = 77.550$)	2.840	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 47.300	0.434	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.979 - 0.2) \times 0.179\} \leq 1.000 (1 + 0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.702 - 0.2) \times 0.408\} \leq 1.000 (1 + 0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6 x 1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.179 + 1.139 \times 2.840 + 1.326 \times 0.434$	3.990	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.408 + 0.684 \times 2.840 + 1.326 \times 0.434$	2.924	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	5.388 / 39.852	0.135	OK

$V_{pl.T.Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $T_t = 21.46 \text{ N/mm}^2$	410.089 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl.T.Rd}$	136.085 / 410.089	0.332	OK
$\theta \text{ (SLS)}$	$T_{Ed,SLS} = 2.512 \text{ kN.m; @ x = 2 m}$	0.17°	advisory
NOT VERIFIED		Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	
		NOT VERIFIED	

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$76.75 \leq 6000 / 360 = 16.67 \text{ mm @ x = 2.85 m}$	76.75 mm	Warning
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N_Ed/N_(pl.Rd)_local	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.122	6.070	0.408	2.840	3.990	2.924	4.605	0.287	2.840	2.840	0.662	2.165	0.135	0.332	6.070

FAIL — governing Biaxial bending + axial (6.2.9.1) = 6.070

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; $L7 = L6 + \text{"EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM"}$: couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

$V_{z,Ed}/V_{pl,z,Rd}$	0 / 237.099 = ($A_{v,z}$ = 1493.3 mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
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$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 172.0/1	47.300 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	44.80 x 275/1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	282.0 x 275/1	77.550 kN.m	
$W_{pl.N.z} = F_n(W_{pl.z}, A_{vz}, n)$	172.0, 14.933, 0.122	172.0 cm ³	
$M_{N.z.Rd} = W_{pl.N.z} \cdot f_y / \gamma_{M0}$	172.0 x 275/1	47.300 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^\alpha + (M_{z.Ed} / M_{N.z.Rd})^\beta$	$(241.150 / 77.550)^{1.69} + (20.521 / 47.300)^{1.69} =$	7.033	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 240.241 / 77.550 + N_{Ed} / N_{pl.Rd} = 150.000 / 1232.000 = 3.220) = 3.238 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.313	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	146.085	ULS: 1.35G + 1.5Q	0.277	0.267	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.313	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	96.990	ULS: 1.35G + 1.5Q	0.184	0.201	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b.y.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.680 \times 275 / 10/1 =$	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b.z.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$44.80 \times 0.299 \times 275 / 10/1 =$	368.021 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr}/M_{cr,uniform}$	916.893 / 787.606 kN.m (shape only: $z_g = z_j = 0$)	1.164	k_c only
$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 234.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.9$, $M(L/2)=234.1$, $M(3L/4)=116.3$, $M_{max}=241.1$ kN.m	1.214	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 3 solve(s)	895.804 kN.m	converged
$\bar{\chi}_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{282.0 \times 275 / 895.804}$	0.294	
$\bar{\chi}_{LT} \leq \bar{\chi}_{LT,0}$	$0.294 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = Fn(\chi_{LT}, \bar{\chi}_{LT}, k_c, f)$	1.000, 0.294, 0.927, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 282.0 \times 275 \leq 77.550 =$	77.550 kN.m	
$M_{y,Ed}/M_{b,Rd}$	$241.150 / 77.550$	3.110	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	241.150 / 77.550 ($M_{y,Rk} = 77.550$)	3.110	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	20.521 / 47.300	0.434	OK
$k_{yy} = C_{my}\{1+(\bar{\chi}_y-0.2)U_{N,y}\}$	$1.000\{1+(0.979-0.2) \times 0.179\} \leq 1.000(1+0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz}\{1+(\bar{\chi}_z-0.2)U_{N,z}\}$	$1.000\{1+(1.702-0.2) \times 0.408\} \leq 1.000(1+0.8 \times 0.408)$	1.326	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.139	0.684	Table B.1
$U_{Ny}+k_{yy} \cdot U_{M,y}+k_{yz} \cdot U_{M,z}$	$0.179+1.139 \times 3.110+1.326 \times 0.434$	4.297	Warning
$U_{Nz}+k_{zy} \cdot U_{M,y}+k_{zz} \cdot U_{M,z}$	$0.408+0.684 \times 3.110+1.326 \times 0.434$	3.109	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 251 / (\sqrt{3} \times 1)$	39.852 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	5.388 / 39.852	0.135	OK
$V_{pl,T,Rd} = [1 - \tau_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $\tau_t = 21.46$ N/mm ²	410.089 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	146.085 / 410.089	0.356	OK
θ (SLS)	$T_{Ed,SLS} = 2.512$ kN.m; @ x = 2 m	0.17°	advisory

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ 87.49 $\leq$ 6000 / 360 = 16.67 mm @ x = 2.85 m 87.49 mm **Warning**

N_Ed/N_(pl,Rd)local	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.122	7.033	0.408	3.110	4.297	3.109	5.249	0.308	3.110	3.110	0.662	2.313	0.135	0.356	7.033

FAIL — governing Biaxial bending + axial (6.2.9.1) = 7.033

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

RHS 200 x 100 x 8.0 (HF), 6 m SS, unrestrained; $L7 = L6 + \text{"EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM"}$: couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

$V_{z,Ed}/V_{pl,z,Rd}$	$0 / 237.099 = (A_{v,z} = 1493.3 \text{ mm}^2; \text{no minor-axis shear in the single-plane model})$	0.000	Low Shear
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$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 172.0/1	47.300 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	44.80 x 275/1 = (No bearing / block tearing design)	1232.000 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 1232.000 =	0.122	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	282.0, 29.867, 0.122	282.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	282.0 x 275/1	77.550 kN.m	
$W_{pl.N.z} = F_n(W_{pl.z}, A_{vz}, n)$	172.0, 14.933, 0.122	172.0 cm ³	
$M_{N.z.Rd} = W_{pl.N.z} \cdot f_y / \gamma_{M0}$	172.0 x 275/1	47.300 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^\alpha + (M_{z.Ed} / M_{N.z.Rd})^\beta$	$(241.150/77.550)^{1.69} + (20.521/47.300)^{1.69} =$	7.033	Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	176.00, 8.00, 8.00, 50.00 mm ($B/2 = 50.00 \leq t_w + 15t_f = 118.93$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 50.00 / (275 \times 8.00) = 6.250$; $0.02 \times (176.00/8.00)^2 = 9.680$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

s_s, c ; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(176.00/6000)^2$	6.002	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.002 \times 210000 \times 8.00^3 / 176.00$	3299.855 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.231 \leq 0.5$ with $m_2 = 9.680$, so $m_2 = 0$); $0.00 + 2 \times 8.00 \times (1 + \sqrt{(6.250 + 0.000)}) = 56.00$	56.00 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.00 \times 8.00 \times 275 / 3299.855e3)}$	0.193	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.193 = 2.588 \rightarrow 1.000$; 1.000×56.00	56.00 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.00 \times 8.00 / 1.0$ per web; load share to this web = 0.935 (lever rule, e = 40 mm); F_{Rd} for the load = 131.795 kN	123.200 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 131.795 =	0.662	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.662 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 240.241/77.550 + N_{Ed} / N_{pl.Rd} = 150.000/1232.000 = 3.220) = 3.238 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.313	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	146.085	ULS: 1.35G + 1.5Q	0.277	0.267	OK
2	point load 3+4	(a)	0.00	6.002	56.00	0.193	1.000	131.795	87.286	ULS: 1.35G + 1.5Q	0.662	2.313	Warning
6	End 2 reaction	(c)	100.00	5.409	120.00	0.298	1.000	528.000	96.990	ULS: 1.35G + 1.5Q	0.184	0.201	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{44.80 \times 275 / 1283.87}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.979	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	44.80 x 0.680 x 275 / 10 / 1 =	838.121 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{44.80 \times 275 / 425.46}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.702	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	44.80 x 0.299 x 275 / 10 / 1 =	368.021 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 234.13$, $\psi = 0.701$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section, $\lambda_{LT} = 0.285 \leq 0.4$: LTB may be ignored	77.550 kN.m	6.3.2.2(4)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.214$, $L_e = 6$ m; $z_g = +100$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	955.991 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	241.150 / 77.550	3.110	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 838.121 ($N_{Rk} = 1232.000$)	0.179	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 368.021	0.408	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	241.150 / 77.550 ($M_{y,Rk} = 77.550$)	3.110	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 47.300	0.434	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (0.979 - 0.2) \times 0.179\} \leq 1.000 (1 + 0.8 \times 0.179)$	1.139	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.702 - 0.2) \times 0.408\} \leq 1.000 (1 + 0.8 \times 0.408)$	1.326	Table B.1
$k_{yz} = k_{zz}$	rectangular hollow section	1.326	Table B.1
$k_{zy} = 0.6 k_{yy}$	0.6 x 1.139	0.684	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	0.179 + 1.139 x 3.110 + 1.326 x 0.434	4.297	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	0.408 + 0.684 x 3.110 + 1.326 x 0.434	3.109	Warning

Torsion Design

J, H, a, Q_f , Q_w	1800 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1800$ cm ⁴	251 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	275 x 251 / ($\sqrt{3} \times 1$)	39.852 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	5.388 / 39.852	0.135	OK
$V_{pl,T,Rd} = [1 - T_t / (f_y / \sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 21.46$ N/mm ²	410.089 kN	6.2.7(9) Eq 6.28

$V_{Ed}/V_{pl.T.Rd}$	146.085 / 410.089	0.356	OK
θ (SLS)	$T_{Ed,SLS} = 2.512 \text{ kN.m; @ } x = 2 \text{ m}$	0.17°	advisory
NOT VERIFIED		Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	
		NOT VERIFIED	

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$87.49 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	87.49 mm	Warning
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N_Ed/N_(pl.Rd)local	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.122	7.033	0.408	3.110	4.297	3.109	5.249	0.308	3.110	3.110	0.662	2.313	0.135	0.356	7.033

FAIL — governing Biaxial bending + axial (6.2.9.1) = 7.033

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +100 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

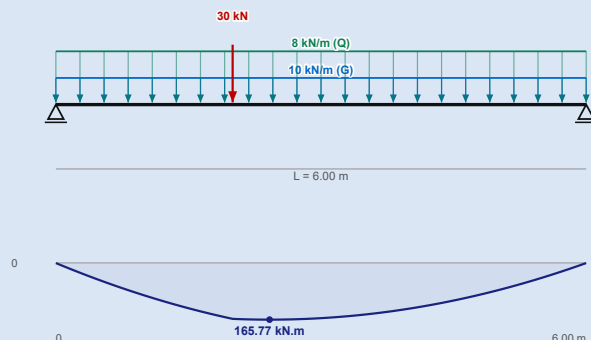
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS 150 x 150 x 6.3 (HF), 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre ($e = 0$, $z_g = 0$)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)

Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6.00C	0.00	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 0.00, 165.77, 0.00	Class 1
	(Axial: Non-Slender)	

Auto Design Load Cases 1; SLS 1

$V_{y,Ed}/V_{pl,y,Rd}$	$0.058 / 284.201 =$	0.000	Low Shear OK 6.2.6
$V_{y,Ed,max}/V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	
$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} Y_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / Y_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	

Web h_w, t_w, t_f, b_f 131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15\epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905 ; 0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
$s_s, c ; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000 ; 1.000 \times 56.07$	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	1190.076 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1190.076}$	0.211	
$\bar{\lambda}_{LT} \leq \bar{\lambda}_{LT,0}$	$0.211 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = Fn(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	1.000, 0.211, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 192.0 \times 275 \leq 52.800 =$	52.800 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	97.78 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Web h_w , t_w , t_f , b_f 131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15\epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
$s_s, c ; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\bar{\lambda}_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = 0$ (load through the shear centre)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed} / h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00$ mm. $N_{f,Ed} = M_{Ed} / h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span} / 360$	$97.78 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	97.78 mm	Warning
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MA/Mc	$M_{(y,Ed)} / M_{(b,Rd)}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

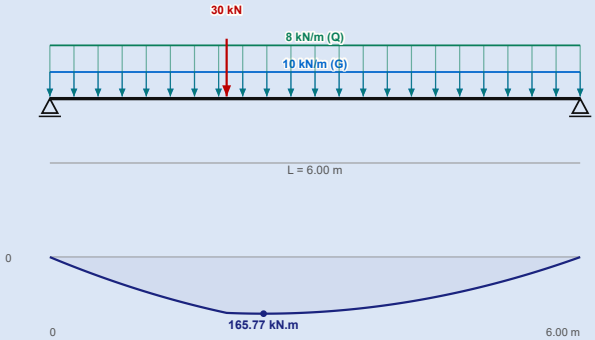
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)
Member SHS-L2 SHS 150 x 150 x 6.3 (HF)
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2757 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 20.81, 20.81, 275, 0.00, 165.77, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c.y.Rd}

V_{y.Ed}/V_{pl.y.Rd} 0.058 / 284.201 = 0.000 Low Shear
V_{y.Ed,max}/V_{pl.y.Rd} 107.617 / 284.201 = 0.379 OK
V_{pl.y.Rd} = A_vf_y/√3V_{M0} 1790 mm² x 275/(√3 x 1) 284.201 kN 6.2.6
M_{c.y.Rd} = f_yW_{pl.y}/V_{M0} 275 x 192.0/1 52.800 kN.m
M_{y.Ed}/M_{c.y.Rd} 165.773 / 52.800 = 3.140 Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 131.10, 6.30, 6.30, 75.00 mm (B/2 = 75.00 ≤ t_w + 15t_f = 93.66); f_{yw} = f_{yf} = 275; 2 webs (flat depth from the section table, corner geometry) Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.30) = 11.905 ; 0.02 x (131.10/6.30)² = 8.661 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)	
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s _s , c ; k _F	s _s = 0.00 mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; k _F = 6 + 2(h _w /a) ² = 6 + 2(131.10/6000) ²	6.001	Fig 6.1(a)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 6.001 x 210000 x 6.30 ³ /131.10	2163.222 kN	6.4(1)
I _y = s _s + 2t _f (1 + √(m ₁ + m ₂)) ≤ a	m ₂ = 0.000 (first pass λ _F = 0.236 ≤ 0.5 with m ₂ = 8.661, so m ₂ = 0); 0.00 + 2 x 6.30 x (1 + √(11.905 + 0.000)) = 56.07	56.07 mm	6.5(3)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(56.07 x 6.30 x 275/2163.222e3)	0.212	6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.212 = 2.359 → 1.000 ; 1.000 x 56.07	56.07 mm	6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 56.07 x 6.30/1.0 per web; load share to this web = 0.778 (lever rule, e = 40 mm); F _{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 45.000 kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
η ₂ + 0.8η ₁ ≤ 1.4	0.361 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 163.489/52.800 = 3.096) = 2.838 ≤ 1.4 (ULS: 1.35G + 1.5Q)	2.027	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	1190.076 / 1042.431 kN.m (shape only: z _g = z _j = 0)	1.142	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.3, M(L/2)=161.4, M(3L/4)=109.8, M _{max} =165.8 kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	1190.076 kN.m	converged
λ _{LT} = √W.f _y /M _{cr}	√ 192.0 x 275 / 1190.076	0.211	
λ _{LT} ≤ λ _{LT,0}	0.211 ≤ 0.4 : LTB may be ignored	χ _{LT} = 1.000	6.3.2.2(4)
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	1.000, 0.211, 0.936, 1.000	1.000	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	1.000 x 192.0 x 275 ≤ 52.800 =	52.800 kN.m	
M _{y,Ed} /M _{b,Rd}	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 38.105	0.112	OK
V _{pl.T.Rd} = [1 - T _t /(f _y √3)].V _{pl.Rd}	@ x = 0 m: T _t = 17.75 N/mm ²	252.428 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl.T.Rd}	107.617 / 252.428	0.426	OK
θ (SLS)	T _{Ed,SLS} = 1.760 kN.m; @ x = 2 m	0.11°	advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
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MA/Mc	M _{-(y.Ed)/M_{-(b.Rd)}}	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Torsion	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	0.426	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member SHS-L2 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

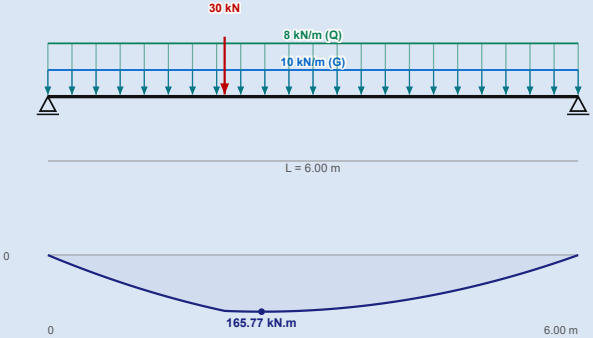
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 20.81, 20.81, 275, 0.00, 165.77, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.058 / 284.201 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 107.617 / 284.201 = 0.379 OK

V_{pl,y,Rd} = A_vf_y/((√3)V_{M0}) 1790 mm² x 275/(√3 x 1) 284.201 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/V_{M0} 275 x 192.0/1 52.800 kN.m

M_{y,Ed}/M_{c,y,Rd} 165.773 / 52.800 = 3.140 Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 131.10, 6.30, 6.30, 75.00 mm (B/2 = 75.00 ≤ t_w + 15t_f = 93.66); f_{yw} = f_{yf} = 275; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.30) = 11.905 ; 0.02 x (131.10/6.30)² = 8.661 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm ($d = 2000.00$ to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3/131.10$	2163.222 kN 6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275/2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1; L_{eff} = \chi_F \cdot I_y$	$0.5/0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30/1.0$ per web; load share to this web = 0.778 (lever rule, $e = 40$ mm); F_{Rd} for the load = 124.812 kN	97.148 kN 6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 163.489/52.800 = 3.096) = 2.838 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.027 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = 0$ (load through the shear centre)	1212.847 kN.m	SN003a
$M_{y,Ed}/M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W_{n0} , S_{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m
W_t	SCI P385 Table A.7/A.8; $I_t = 1910$ cm ⁴	240 cm ³

$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 240 / (\sqrt{3} \times 1)$	38.105 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	$4.260 / 38.105$	0.112	OK
$V_{pl.T.Rd} = [1 - \tau_t / (f_y / \sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $\tau_t = 17.75 \text{ N/mm}^2$	252.428 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl.T.Rd}$	$107.617 / 252.428$	0.426	OK
$\theta \text{ (SLS)}$	$T_{Ed,SLS} = 1.760 \text{ kN.m; @ x = 2 m}$	0.11°	advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67 \text{ mm @ x = 2.85 m}$	97.78 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	0.426	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

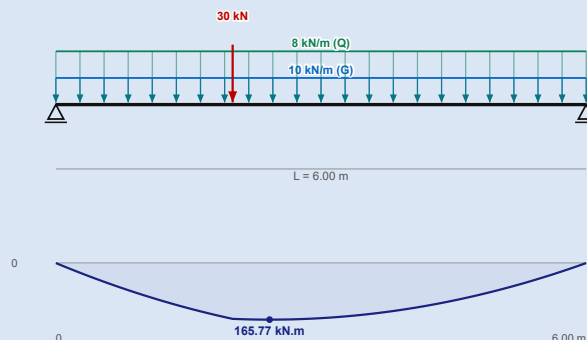
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +75$ mm (D/2, destabilising). $e = 0$

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

the moment and shear envelopes keep the entered $\gamma_{G'}$, so add a



Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6.00C	0.00	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 0.00, 165.77, 0.00	Class 1
	(Axial: Non-Slender)	

Auto Design Load Cases 1: SLS 1

$V_{y,Ed}/V_{pl,y,Rd}$	$0.058 / 284.201 =$	0.000	Low Shear OK 6.2.6
$V_{y,Ed,max}/V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	
$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	

Web h_w, t_w, t_f, b_f 131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15\epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905 ; 0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
$s_s, c ; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000 ; 1.000 \times 56.07$	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	1162.306 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1162.306}$	0.213	
$\bar{\lambda}_{LT} \leq \bar{\lambda}_{LT,0}$	$0.213 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	1.000, 0.213, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 192.0 \times 275 \leq 52.800 =$	52.800 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	97.78 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

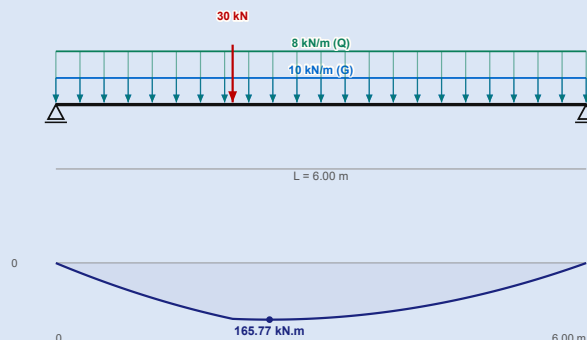
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange $z_g = +75$ mm (D/2, destabilising). $e = 0$

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

the moment and shear envelopes keep the entered $\gamma_{G'}$, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_{-} is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3

Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 0.00, 165.77, 0.00	Class 1
	(Axial: Non-Slender)	

Auto Design Load Cases 1: SLS 1

$V_{y,Ed}/V_{pl,y,Rd}$	$0.058 / 284.201 =$	0.000	Low Shear OK 6.2.6
$V_{y,Ed,max}/V_{pl,y,Rd}$	$107.617 / 284.201 =$	0.379	
$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} Y_{M0})$	$1790 \text{ mm}^2 \times 275 / (\sqrt{3} \times 1)$	284.201 kN	
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / Y_{M0}$	$275 \times 192.0 / 1$	52.800 kN.m	
$M_{y,Ed} / M_{c,y,Rd}$	$165.773 / 52.800 =$	3.140	

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
s_s , c; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +75$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 97.78 mm **Warning**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +75 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS-L3b · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: eigen · verdict **FAIL** · governing Deflection = 5.867

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -75$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member SHS-L3b SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 0$ mm $z_g = -75$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 0$ mm $z_g = -75$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -75$ mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

$Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]

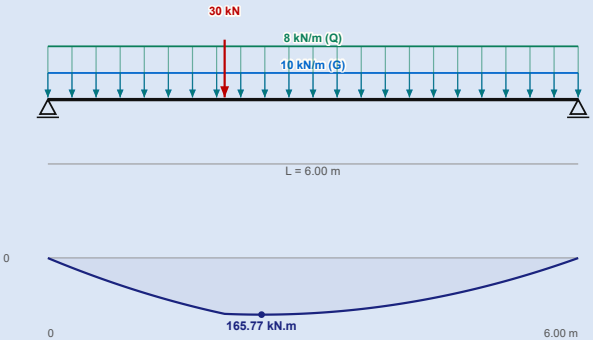
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]

$Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member

(single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 20.81, 20.81, 275, 0.00, 165.77, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$	0.058 / 284.201 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	107.617 / 284.201 =	0.379	OK
$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} Y_{MO})$	1790 mm ² x 275 / ($\sqrt{3}$ x 1)	284.201 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / Y_{MO}$	275 x 192.0/1	52.800 kN.m	
$M_{y.Ed}/M_{c.y.Rd}$	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905 ; 0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
$s_s, c ; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000 ; 1.000 \times 56.07$	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{max}=165.8$ kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -75, 0$ mm; 1 combination(s), 1 solve(s) + 2 cached	1218.400 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1218.400}$	0.208	
$\bar{\lambda}_{LT} \leq \bar{\lambda}_{LT,0}$	$0.208 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = Fn(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	1.000, 0.208, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 192.0 \times 275 \leq 52.800 =$	52.800 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	97.78 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS-L3b · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Deflection = 5.867

Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange $z_g = -75$ mm (-D/2, stabilising), $e = 0$

Beam & Beam-Portion (Member) (FAIL)
Member SHS-L3b SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 0$ mm $z_g = -75$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 0$ mm $z_g = -75$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 0$ mm $z_g = -75$ mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

$Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]

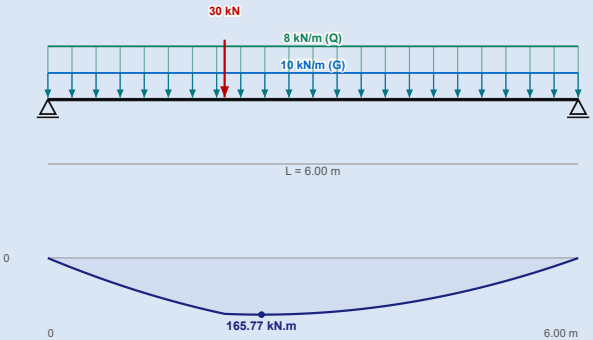
ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]

$Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member

(single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 20.81, 20.81, 275, 0.00, 165.77, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$ 0.058 / 284.201 = 0.000 Low Shear

$V_{y.Ed,max}/V_{pl.y.Rd}$ 107.617 / 284.201 = 0.379 OK

$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} Y_{MO})$ 1790 mm² x 275 / ($\sqrt{3}$ x 1) 284.201 kN 6.2.6

$M_{c.y.Rd} = f_y \cdot W_{pl.y} / Y_{MO}$ 275 x 192.0/1 52.800 kN.m

$M_{y.Ed}/M_{c.y.Rd}$ 165.773 / 52.800 = 3.140 **Warning**

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry) Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905 ; 0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
$s_s, c ; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\bar{\lambda}_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000 ; 1.000 \times 56.07$	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.500 (e = 0); F_{Rd} for the load = 194.297 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 194.297 =	0.232 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.232 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.709 \leq 1.4$ (ULS: 1.35G + 1.5Q)	1.935 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	194.297	45.000	ULS: 1.35G + 1.5Q	0.232	1.935	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\bar{\lambda}_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = -75$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed} / h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. $N_{f,Ed} = M_{Ed} / h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span} / 360$	$97.78 \leq 6000 / 360 = 16.67$ mm @ x = 2.85 m	97.78 mm	Warning
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MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
3.140	3.140	5.867	0.379	0.232	1.935	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member SHS-L4 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 75 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 75 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 75 mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

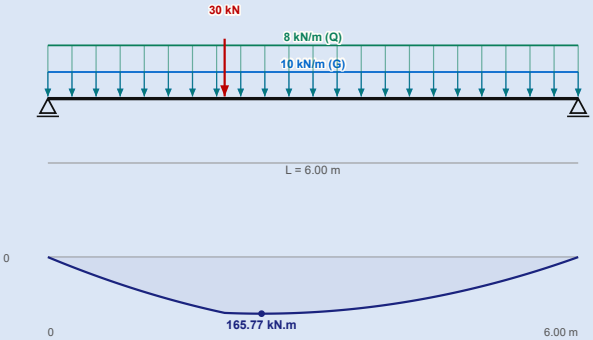
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 0.00, 165.77, 0.00 (Axial: Non-Slender)	Class 1
Auto Design Load Cases	1; SLS 1	

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.058 / 284.201 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.617 / 284.201 =	0.379	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	1790 mm² x 275/(√3 x 1)	284.201 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 192.0/1	52.800 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.773 / 52.800 =	3.140	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h _w , t _w , t _f , b _f	131.10, 6.30, 6.30, 75.00 mm (B/2 = 75.00 ≤ t _w + 15t _f = 93.66); f _{yw} = f _{yf} = 275; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
m ₁ = f _{yf} ·b _f /(f _{yw} ·t _w) ; m ₂ = 0.02(h _w /t _f)²	275 x 75.00/(275 x 6.30) = 11.905 ; 0.02 x (131.10/6.30)² = 8.661 if λ _F > 0.5, else 0	6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)	
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior			
s _s , c ; k _F	s _s = 0.00 mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end) ; k _F = 6 + 2(h _w /a) ² = 6 + 2(131.10/6000) ²	6.001	Fig 6.1(a)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 6.001 x 210000 x 6.30 ³ /131.10	2163.222 kN	6.4(1)
I _y = s _s + 2t _f (1 + √(m ₁ + m ₂)) ≤ a	m ₂ = 0.000 (first pass λ _F = 0.236 ≤ 0.5 with m ₂ = 8.661, so m ₂ = 0); 0.00 + 2 x 6.30 x (1 + √(11.905 + 0.000)) = 56.07	56.07 mm	6.5(3)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(56.07 x 6.30 x 275/2163.222e3)	0.212	6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.212 = 2.359 → 1.000 ; 1.000 x 56.07	56.07 mm	6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 56.07 x 6.30/1.0 per web; load share to this web = 0.778 (lever rule, e = 40 mm); F _{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 45.000 kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
η ₂ + 0.8η ₁ ≤ 1.4	0.361 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 163.489/52.800 = 3.096) = 2.838 ≤ 1.4 (ULS: 1.35G + 1.5Q)	2.027	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	1190.076 / 1042.431 kN.m (shape only: z _g = z _j = 0)	1.142	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.3, M(L/2)=161.4, M(3L/4)=109.8, M _{max} =165.8 kN.m	1.163	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 75, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	1162.306 kN.m	converged
λ _{LT} = √(W.f _y /M _{cr})	√(192.0 x 275 / 1162.306)	0.213	
λ _{LT} ≤ λ _{LT,0}	0.213 ≤ 0.4 : LTB may be ignored	χ _{LT} = 1.000	6.3.2.2(4)
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	1.000, 0.213, 0.936, 1.000	1.000	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	1.000 x 192.0 x 275 ≤ 52.800 =	52.800 kN.m	
M _{y,Ed} /M _{b,Rd}	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W _{n0} , S _{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W _t	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /(√3γ _{M0})	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	4.260 / 38.105	0.112	OK
V _{pl.T.Rd} = [1 - T _t /(f _y √3)].V _{pl.Rd}	@ x = 0 m: T _t = 17.75 N/mm ²	252.428 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl.T.Rd}	107.617 / 252.428	0.426	OK
θ (SLS)	T _{Ed,SLS} = 1.760 kN.m; @ x = 2 m	0.11°	advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	97.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	97.78 mm	Warning
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MA/Mc	M _{-(y.Ed)/M_{-(b.Rd)}}	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Torsion	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	0.426	3.140

FAIL — governing Deflection = 5.867

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member SHS-L4 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 75 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 75 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 75 mm (kN)

G SW 0.2757 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

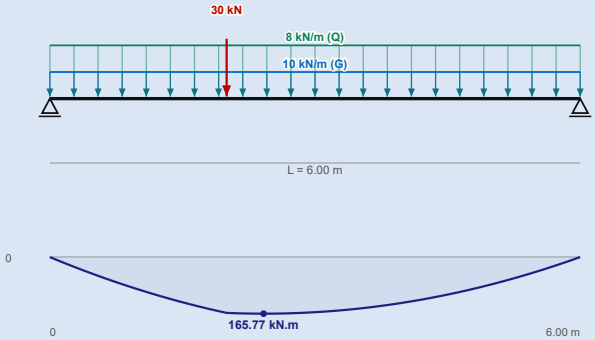
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	0.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 20.81, 20.81, 275, 0.00, 165.77, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.058 / 284.201 = 0.000 Low Shear

V_{y,Ed,max}/V_{pl,y,Rd} 107.617 / 284.201 = 0.379 OK

V_{pl,y,Rd} = A_vf_y/(√3V_{M0}) 1790 mm² x 275/(√3 x 1) 284.201 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/V_{M0} 275 x 192.0/1 52.800 kN.m

M_{y,Ed}/M_{c,y,Rd} 165.773 / 52.800 = 3.140 Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior		
s_s, c ; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001 Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN 6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm 6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489 / 52.800 = 3.096) = 2.838 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.027 Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.142	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.027	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.122	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +75$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w1910 cm⁴, — dm⁶, — mm, —, —P385 App A

W_{n0}, S_{w1}—, —P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)combination ULS: 1.35G + 1.5Q4.260 kN.m

W_tSCI P385 Table A.7/A.8; I_t = 1910 cm⁴240 cm³

T_{Rd} = f_yW_t/(√3Y_{M0})275 x 240/(√3 x 1)38.105 kN.m6.2.7(7)

T_{Ed}/T_{Rd}4.260 / 38.1050.112OK

V_{pl.T.Rd} = [1 - T_t/(f_y√3)].V_{pl.Rd}@ x = 0 m: T_t = 17.75 N/mm²252.428 kN6.2.7(9) Eq 6.28

V_{Ed}/V_{pl.T.Rd}107.617 / 252.4280.426OK

θ (SLS)T_{Ed,SLS} = 1.760 kN.m; @ x = 2 m0.11°advisory

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/36097.78 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m97.78 mmWarning

MA/Mc	M _{-(y,Ed)/M_{-(b,Rd)}}	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	Max
3.140	3.140	5.867	0.379	0.361	2.027	0.112	0.426	3.140

FAIL — governing Deflection = 5.867

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +75 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN
value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all
loads at e = +40 mm, z_g = +75 mm

$$M_{Cz,Rd} = f_{yk} \cdot W_{pl,z} / \gamma_{M0}$$

$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl.N.y} = F_n(W_{pl.y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N.y.Rd} = W_{pl.N.y} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y.Ed} / M_{N.y.Rd})^{\alpha} + (M_{z.Ed} / M_{N.z.Rd})^{\beta}$	$(165.773/52.800)^{1.7} + (0)^1 =$	7.031	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 163.489/52.800 + N_{Ed} / N_{pl.Rd} = 150.000/984.500 = 3.249) = 2.960 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.114	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.229	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.114	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.209	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b.y.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10/1 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b.z.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10/1 =$	533.439 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	1190.076 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.142	k_c only
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$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.3$, $M(L/2)=161.4$, $M(3L/4)=109.8$, $M_{\max}=165.8$ kN·m	1.163	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.42$, $\psi = 0.895$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.42$, $\psi = 0.895$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	1162.306 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$	$\sqrt{192.0 \times 275 / 1162.306}$	0.213	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.213 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.213, 0.936, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 192.0 \times 275 \leq 52.800 =$	52.800 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(X_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed}/(X_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed}/(X_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	165.773 / 52.800 ($M_{y,Rk} = 52.800$)	3.140	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	0.000 / 52.800	0.000	OK
$k_{yy} = C_{my}\{1+(\lambda_y-0.2)U_{N,y}\}$	$1.000\{1+(1.181-0.2)\times 0.281\} \leq 1.000(1+0.8\times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz}\{1+(\lambda_z-0.2)U_{N,z}\}$	$1.000\{1+(1.181-0.2)\times 0.281\} \leq 1.000(1+0.8\times 0.281)$	1.225	Table B.1
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny}+k_{yy} \cdot U_{M,y}+k_{yz} \cdot U_{M,z}$	$0.281+1.225\times 3.140+1.225\times 0.000$	4.127	Warning
$U_{Nz}+k_{zy} \cdot U_{M,y}+k_{zz} \cdot U_{M,z}$	$0.281+0.735\times 3.140+1.225\times 0.000$	2.589	Warning

Torsion Design

J, H, a, Q_f , Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —	P385 App A
W_{n0} , S_{w1}	—, —	P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1910$ cm ⁴	240 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 240 / (\sqrt{3} \times 1)$	38.105 kN.m	6.2.7(7)
T_{Ed}/T_{Rd}	$4.260 / 38.105$	0.112	OK
$V_{pl,T,Rd} = [1 - T_t/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 17.75$ N/mm ²	252.428 kN	6.2.7(9) Eq 6.28
$V_{Ed}/V_{pl,T,Rd}$	$107.617 / 252.428$	0.426	OK
θ (SLS)	$T_{Ed,SLS} = 1.760$ kN.m; @ x = 2 m	0.11°	advisory

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$

$97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$

97.78 mm

Warning

N_Ed/N _(pl.Rd) local	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.152	7.031	0.281	3.140	4.127	2.589	5.867	0.379	3.140	3.140	0.361	2.114	0.112	0.426	7.031

FAIL — governing Bending+axial cross-section (6.2.9) = 7.031

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS-L5 · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Bending+axial cross-section (6.2.9) = 7.031

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam

SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at $e = +40$ mm, $z_g = +75$ mm

Axial with Moments (Member) (FAIL)
Member SHS-L5 SHS 150 x 150 x 6.3 (HF)
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m $e = 40$ mm $z_g = 75$ mm (kN/m)

Q UDL 8.000 0-6 m $e = 40$ mm $z_g = 75$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 75$ mm (kN)

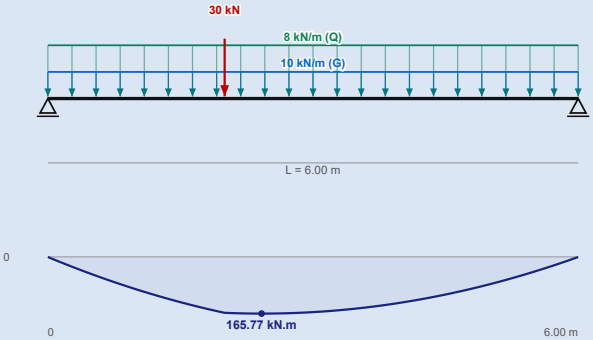
G SW 0.2757 kN/m 0-6 m (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	-4.26	107.62	0.00	0.00	0.00	165.77	0.00	-97.78	107.62	—
	End 2, x = 6	150.00C	0.00	-92.62	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.62	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m) 150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150
B=150 t=6.3

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 20.81, 20.81, 275, 150.00, 165.77, 0.00 Class 1
(Axial: Non-Slender)

Web classified for bending + compression: $\alpha = 0.665, \psi = -0.695$; Class 1 Table 5.2
limits 51.8ε / 59.6ε / 95.3ε

Auto Design Load Cases 1; SLS 1

Local Capacity Check

$V_{y,Ed}/V_{pl,y,Rd}$	0.058 / 284.201 =	0.000	Low Shear
$V_{y,Ed,max}/V_{pl,y,Rd}$	107.617 / 284.201 =	0.379	OK
$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$	1790 mm² x 275 / ($\sqrt{3}$ x 1)	284.201 kN	6.2.6
$M_{c,y,Rd} = f_y \cdot W_{pl,y} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$M_{y,Ed}/M_{c,y,Rd}$	165.773 / 52.800 =	3.140	Warning
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790$ mm²; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	

$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed} / N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y,Ed} / M_{N,y,Rd})^{\alpha} + (M_{z,Ed} / M_{N,z,Rd})^{\beta}$	$(165.773/52.800)^{1.7} + (0)^1 =$	7.031	Warning

NOT VERIFIED

Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75$ mm is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15\epsilon t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\bar{\lambda}_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3, load type (a) Fig 6.1(a) interior

$s_s, c; k_F$	$s_s = 0.00$ mm (default 0); $c = 2000.00$ mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$l_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\bar{\lambda}_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\bar{\lambda}_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \bar{\lambda}_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 45.000$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.361	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.361 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 163.489/52.800 + N_{Ed} / N_{pl,Rd} = 150.000/984.500 = 3.249) = 2.960 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.114	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	$\bar{\lambda}_F$	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	107.617	ULS: 1.35G + 1.5Q	0.199	0.229	OK
2	point load 3	(a)	0.00	6.001	56.07	0.212	1.000	124.812	45.000	ULS: 1.35G + 1.5Q	0.361	2.114	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	92.617	ULS: 1.35G + 1.5Q	0.171	0.209	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
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$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	$1 \times 6 =$ (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.42$, $\psi = 0.895$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6 + 0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.42$, $\psi = 0.895$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$	closed hollow section — not susceptible to LTB ($\lambda_{LT} = 0.209 \leq 0.4$)	52.800 kN.m	6.3.2.1(2)
M_{cr} (information, $I_w = 0$)	SN003a with $C_1 = 1.163$, $L_e = 6$ m; $z_g = +75$ mm entered — NOT applied: C_2 not published for this moment diagram; M_{cr} is at the shear centre (PASS blocked)	1212.847 kN.m	SN003a
$M_{y,Ed} / M_{b,Rd}$	165.773 / 52.800	3.140	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	165.773 / 52.800 ($M_{y,Rk} = 52.800$)	3.140	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	0.000 / 52.800	0.000	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.281 + 1.225 \times 3.140 + 1.225 \times 0.000$	4.127	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.281 + 0.735 \times 3.140 + 1.225 \times 0.000$	2.589	Warning

Torsion Design

J, H, a, Q_f , Q_w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1910$ cm ⁴	240 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	$275 \times 240 / (\sqrt{3} \times 1)$	38.105 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	$4.260 / 38.105$	0.112	OK
$V_{pl,T,Rd} = [1 - T_t / (f_y / \sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 17.75$ N/mm ²	252.428 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl,T,Rd}$	107.617 / 252.428	0.426	OK
θ (SLS)	$T_{Ed,SLS} = 1.760$ kN.m; @ x = 2 m	0.11°	advisory

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $97.78 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 97.78 mm **Warning**

N_Ed/N _{pl,Rd}	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.152	7.031	0.281	3.140	4.127	2.589	5.867	0.379	3.140	3.140	0.361	2.114	0.112	0.426	7.031

FAIL — governing Bending+axial cross-section (6.2.9) = 7.031

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$M_{YEd}/M_{c,YRd}$	$219.871 / 52.800 =$	4.164	Warning
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$(M_{y,Ed}/M_{N.V.y,Rd})^{\alpha} + (M_{z,Ed}/M_{N.V.z,Rd})^{\beta}$ @ x	@ x = 0.528 m (ULS: 1.35G + 1.5Q): V = 122.147, ρ = 0.000; $N_{V,Rd} = (A - \rho A_v)f_y = 984.499$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2)f_y/\gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t))f_y/\gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N.V.y,Rd} = 52.800$, $M_{N.V.z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 68.100$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho)f_y$ on A_v)	1.743	Warning 6.2.10
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z}/\gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y/\gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y/\gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y/\gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y,Ed}/M_{N.y,Rd})^{\alpha} + (M_{z,Ed}/M_{N.z,Rd})^{\beta}$	$(219.871/52.800)^{1.7} + (20.521/52.800)^{1.7} =$	11.579	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

s_s , c; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3/131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275/2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w/\gamma_{M1}$	$275 \times 56.07 \times 6.30/1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 219.871/52.800 + N_{Ed}/N_{pl,Rd} = 150.000/984.500 = 4.317) = 4.153 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.966	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	135.808	ULS: 1.35G + 1.5Q	0.251	0.266	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	2.966	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	106.712	ULS: 1.35G + 1.5Q	0.197	0.228	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	1237.844 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.187	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 203.7, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.6$, $M(L/2)=203.7$, $M(3L/4)=131.0$, $M_{max}=219.9$ kN.m	1.218	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.71$, $\psi = 0.821$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.71$, $\psi = 0.821$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e =$ portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
$M_{cr} =$ FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	1207.870 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1207.870}$	0.209	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.209 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.209, 0.918, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 192.0 \times 275 \leq 52.800 =$	52.800 kN.m	
$M_{y,Ed} / M_{b,Rd}$	219.871 / 52.800	4.164	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	219.871 / 52.800 ($M_{y,Rk} = 52.800$)	4.164	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 52.800	0.389	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.281 + 1.225 \times 4.164 + 1.225 \times 0.389$	5.858	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.281 + 0.735 \times 4.164 + 1.225 \times 0.389$	3.818	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1910\text{ cm}^4$	240 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	275 x 240 / ($\sqrt{3} \times 1$)	38.105 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	5.388 / 38.105	0.141	OK
$V_{pl.T.Rd} = [1 - T_t / (f_y / \sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $T_t = 22.45\text{ N/mm}^2$	244.018 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl.T.Rd}$	135.808 / 244.018	0.557	OK
θ (SLS)	$T_{Ed,SLS} = 2.512\text{ kN.m}$; @ x = 2 m	0.16°	advisory

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$140.29 \leq 6000 / 360 = 16.67\text{ mm}$ @ x = 2.85 m	140.29 mm	Warning
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N_Ed/N_(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.152	11.579	0.281	4.164	5.858	3.818	8.417	0.478	4.164	4.164	1.743	0.699	2.966	0.141	0.557	11.579

FAIL — governing Biaxial bending + axial (6.2.9.1) = 11.579

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

SHS-L6 · SHS 150 x 150 x 6.3 (HF) · M_{cr} method: standard · verdict **FAIL** · governing Biaxial bending + axial (6.2.9.1) = 11.579

Owner case: ... and axial load on beam, load applied at angle on beam
SHS 150 x 150 x 6.3 (HF), 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +75, factored by the combination) plus an entered CONSTANT M_{z,Ed} = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_{mz} = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.075 = 0.77 kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)
Member SHS-L6 SHS 150 x 150 x 6.3 (HF)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

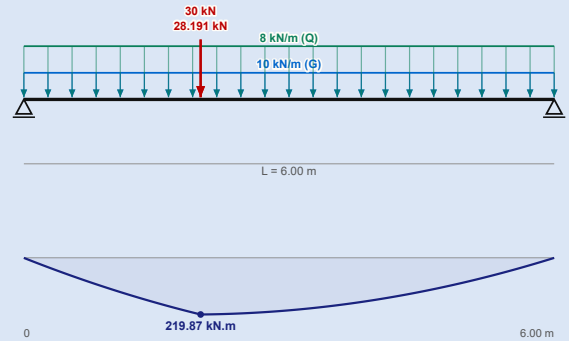
Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 75 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 75 mm (kN)
Q PY 28.191 @ 2 m e = 40 mm z_g = 75 mm (kN)
G SW 0.2757 kN/m 0-6 m (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions;



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	-5.39	135.81	0.00	0.00	20.52	219.87	20.52	-140.29	135.81	—
	End 2, x = 6	150.00C	0.00	-106.71	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	106.71	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (28.1 kg/m)	150 x 150 x 6.3 SHS [Hot-finished] [S275] D=150 B=150 t=6.3	
Class = Fn(c/t, d/t, f _y , N, M _y , M _z)	20.81, 20.81, 275, 150.00, 219.87, 20.52 (Axial: Non-Slender)	Class 1
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε	Class 1 Table 5.2
Auto Design Load Cases	1; SLS 1	

Local Capacity Check

V _{y,Ed} /V _{pl,y,Rd}	3.223 / 284.201 =	0.011	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	135.808 / 284.201 =	0.478	OK
V _{pl,y,Rd} = A _v f _y /((√3)Y _{M0})	1790 mm² x 275/(√3 x 1)	284.201 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /Y _{M0}	275 x 192.0/1	52.800 kN.m	
M _{y,Ed} /M _{c,y,Rd}	219.871 / 52.800 =	4.164	Warning

$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$ @ x	@ x = 0.528 m (ULS: 1.35G + 1.5Q): V = 122.147, $\rho = 0.000$; $N_{V,Rd} = (A - \rho A_v) f_y = 984.499$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,y,Rd} = 52.800$, $M_{N,v,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 68.100$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	1.743	Warning 6.2.10
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$	$(219.871/52.800)^{1.7} + (20.521/52.800)^{1.7} =$	11.579	Warning
NOT VERIFIED			
Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.			NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior			
s_s , c; k_F	$s_s = 0.00 \text{ mm}$ (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 219.871/52.800 + N_{Ed}/N_{pl,Rd} = 150.000/984.500 = 4.317) = 4.153 \leq 1.4$ (ULS: 1.35G + 1.5Q)	2.966	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	135.808	ULS: 1.35G + 1.5Q	0.251	0.266	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	2.966	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	106.712	ULS: 1.35G + 1.5Q	0.197	0.228	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √A.f _y /N _{cr}	√35.80x275/702.39 (N _{cr,y} = π²EI _y /L _{ey} ²)	1.181	
N _{b,y,Rd} = Area.χ.f _y /γ _{M1}	35.80x0.542x275/10/1 =	533.439 kN	Curve a
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A.f _y /N _{crz}	√35.80x275/702.39 (N _{cr,z} = π²EI _z /L _{ez} ²)	1.181	
N _{b,z,Rd} = Area.χ.f _y /γ _{M1}	35.80x0.542x275/10/1 =	533.439 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 203.71, ψ = 0.821, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 203.71, ψ = 0.821, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

M _{b,Rd} = M _{c,y,Rd}	closed hollow section — not susceptible to LTB (λ _{LT} = 0.204 ≤ 0.4)	52.800 kN.m	6.3.2.1(2)
M _{cr} (information, I _w = 0)	SN003a with C ₁ = 1.218, L _e = 6 m; z _g = +75 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	1269.389 kN.m	SN003a
M _{y,Ed} /M _{b,Rd}	219.871 / 52.800	4.164	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

U _{N,y} = N _{Ed} /(χ _y .N _{Rk} /γ _{M1})	150.000 / 533.439 (N _{Rk} = 984.500)	0.281	OK
U _{N,z} = N _{Ed} /(χ _z .N _{Rk} /γ _{M1})	150.000 / 533.439	0.281	OK
U _{M,y} = M _{y,Ed} /(χ _{LT} .M _{y,Rk} /γ _{M1})	219.871 / 52.800 (M _{y,Rk} = 52.800)	4.164	Warning
U _{M,z} = M _{z,Ed} /(M _{z,Rk} /γ _{M1})	20.521 / 52.800	0.389	OK
k _{yy} = C _{my} {1+(λ _y -0.2)U _{N,y} }	1.000{1+(1.181-0.2)x0.281} ≤ 1.000(1+0.8x0.281)	1.225	Table B.1
k _{zz} = C _{mz} {1+(λ _z -0.2)U _{N,z} }	1.000{1+(1.181-0.2)x0.281} ≤ 1.000(1+0.8x0.281)	1.225	Table B.1 (RHS)

$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.281 + 1.225 \times 4.164 + 1.225 \times 0.389$	5.858	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.281 + 0.735 \times 4.164 + 1.225 \times 0.389$	3.818	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W _t	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /(√3Y _{M0})	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	5.388 / 38.105	0.141	OK
V _{pl.T.Rd} = [1 - T _t /(f _y √3)].V _{pl.Rd}	@ x = 0 m: T _t = 22.45 N/mm ²	244.018 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl.T.Rd}	135.808 / 244.018	0.557	OK
θ (SLS)	T _{Ed,SLS} = 2.512 kN.m; @ x = 2 m	0.16°	advisory

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$140.29 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	140.29 mm	Warning
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N _{Ed} /N _{pl,Rd}	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	V+T	Max	
0.152	11.579	0.281	4.164	5.858	3.818	8.417	0.478	4.164	4.164	1.743	0.699	2.966	0.141	0.557	11.579

FAIL — governing Biaxial bending + axial (6.2.9.1) = 11.579

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +75 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$ @ x	@ x = 0.918 m (ULS: 1.35G + 1.5Q): V = 122.057, ρ = 0.000; $N_{V,Rd} = (A - \rho A_v) f_y = 984.500$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,y,Rd} = 52.800$, $M_{N,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 122.950$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	4.424	Warning 6.2.10
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790$ mm ² ; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$	$(240.758/52.800)^{1.7} + (20.521/52.800)^{1.7} =$	13.483	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior

s_s , c; k_F	$s_s = 0.00$ mm (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286$ kN (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 239.871/52.800 + N_{Ed}/N_{pl,Rd} = 150.000/984.500 = 4.695) = 4.456 \leq 1.4$ (ULS: 1.35G + 1.5Q)	3.183	Warning 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	145.808	ULS: 1.35G + 1.5Q	0.270	0.280	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	3.183	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	96.712	ULS: 1.35G + 1.5Q	0.179	0.215	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	1.181	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{35.80 \times 275 / 702.39}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	1.181	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$35.80 \times 0.542 \times 275 / 10 / 1 =$	533.439 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	1213.717 / 1042.431 kN.m (shape only: $z_g = z_j = 0$)	1.164	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 233.7, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.6$, $M(L/2)=233.7$, $M(3L/4)=116.0$, $M_{max}=240.8$ kN.m	1.214	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.71$, $\psi = 0.699$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.71$, $\psi = 0.699$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e =$ portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
$M_{cr} =$ FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 75$, 0 mm; 1 combination(s), 3 solve(s)	1187.543 kN.m	converged
$\lambda_{LT} = \sqrt{W_y \cdot f_y / M_{cr}}$	$\sqrt{192.0 \times 275 / 1187.543}$	0.211	
$\lambda_{LT} \leq \lambda_{LT,0}$	$0.211 \leq 0.4$: LTB may be ignored	$\chi_{LT} = 1.000$	6.3.2.2(4)
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	1.000, 0.211, 0.927, 1.000	1.000	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$1.000 \times 192.0 \times 275 \leq 52.800 =$	52.800 kN.m	
$M_{y,Ed} / M_{b,Rd}$	240.758 / 52.800	4.560	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 150.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 533.439 ($N_{Rk} = 984.500$)	0.281	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 533.439	0.281	OK
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	240.758 / 52.800 ($M_{y,Rk} = 52.800$)	4.560	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	20.521 / 52.800	0.389	OK
$k_{yy} = C_{my} \{1 + (\lambda_y - 0.2) U_{N,y}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1
$k_{zz} = C_{mz} \{1 + (\lambda_z - 0.2) U_{N,z}\}$	$1.000 \{1 + (1.181 - 0.2) \times 0.281\} \leq 1.000 (1 + 0.8 \times 0.281)$	1.225	Table B.1 (RHS)
$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6 k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{M,y} + k_{yz} \cdot U_{M,z}$	$0.281 + 1.225 \times 4.560 + 1.225 \times 0.389$	6.343	Warning
$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	$0.281 + 0.735 \times 4.560 + 1.225 \times 0.389$	4.109	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W_{n0} , S_{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W_t	SCI P385 Table A.7/A.8; $I_t = 1910\text{ cm}^4$	240 cm ³	
$T_{Rd} = f_y W_t / (\sqrt{3} \gamma_{M0})$	275 x 240 / ($\sqrt{3}$ x 1)	38.105 kN.m	6.2.7(7)
T_{Ed} / T_{Rd}	5.388 / 38.105	0.141	OK
$V_{pl.T.Rd} = [1 - T_t / (f_y / \sqrt{3})] \cdot V_{pl.Rd}$	@ x = 0 m: $T_t = 22.45\text{ N/mm}^2$	244.018 kN	6.2.7(9) Eq 6.28
$V_{Ed} / V_{pl.T.Rd}$	145.808 / 244.018	0.598	OK
θ (SLS)	$T_{Ed,SLS} = 2.512\text{ kN.m}$; @ x = 2 m	0.16°	advisory

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$159.91 \leq 6000 / 360 = 16.67\text{ mm}$ @ x = 2.85 m	159.91 mm	Warning
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N_Ed/N_(pl,Rd)	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F_Rd	Web 7.2	Torsion	V+T	Max	
0.152	13.483	0.281	4.560	6.343	4.109	9.595	0.513	4.560	4.560	4.424	0.699	3.183	0.141	0.598	13.483

FAIL — governing Biaxial bending + axial (6.2.9.1) = 13.483

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$ @ x	@ x = 0.918 m (ULS: 1.35G + 1.5Q): V = 122.057, ρ = 0.000; $N_{V,Rd} = (A - \rho A_v) f_y = 984.500$, $n_v = 0.152$; $M_{v,y,Rd} = (W_{pl,y} - \rho t(h - 2t)^2/2) f_y / \gamma_{M0}$ [two webs] = 52.800; $M_{v,z,Rd} = (W_{pl,z} - \rho t(h - 2t)(b - t)) f_y / \gamma_{M0}$ [two webs] = 52.800; $a_v = 0.472$, $a_{f,v} = 0.472$; $M_{N,y,Rd} = 52.800$, $M_{N,v,z,Rd} = 52.800$, $\alpha = 1.7$, $\beta = 1.7$; $M_y = 122.950$, $M_z = 20.521$, $N = 150.000$ (plastic, Eq 6.39/6.40 with $(1 - \rho) f_y$ on A_v)	4.424	Warning 6.2.10
$V_{z,Ed}/V_{pl,z,Rd}$	0 / 284.201 = ($A_{v,z} = 1790 \text{ mm}^2$; no minor-axis shear in the single-plane model)	0.000	Low Shear
$M_{c,z,Rd} = f_y \cdot W_{pl,z} / \gamma_{M0}$	275 x 192.0/1	52.800 kN.m	
$N_{pl,Rd} = A_g \cdot f_y / \gamma_{M0}$	35.80 x 275/1 = (No bearing / block tearing design)	984.500 kN	
$n = N_{Ed}/N_{pl,Rd}$	150.000 / 984.500 =	0.152	OK
$W_{pl,N,y} = F_n(W_{pl,y}, A_{vy}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,y,Rd} = W_{pl,N,y} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$W_{pl,N,z} = F_n(W_{pl,z}, A_{vz}, n)$	192.0, 17.900, 0.152	192.0 cm ³	
$M_{N,z,Rd} = W_{pl,N,z} \cdot f_y / \gamma_{M0}$	192.0 x 275/1	52.800 kN.m	
$(M_{y,Ed}/M_{N,y,Rd})^{\alpha} + (M_{z,Ed}/M_{N,z,Rd})^{\beta}$	$(240.758/52.800)^{1.7} + (20.521/52.800)^{1.7} =$	13.483	Warning
NOT VERIFIED			
Standard (closed-form) M_{cr} : a destabilising load height $z_g = +75 \text{ mm}$ is entered, but SN003a publishes C_2 only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr} , or the destabilising L_E switch (BS 5950 practice); PASS is blocked.			NOT VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	131.10, 6.30, 6.30, 75.00 mm ($B/2 = 75.00 \leq t_w + 15t_f = 93.66$); $f_{yw} = f_{yf} = 275$; 2 webs (flat depth from the section table, corner geometry)		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.30) = 11.905$; $0.02 \times (131.10/6.30)^2 = 8.661$ if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)
Governing station x = 2 m: point load 3+4, load type (a) Fig 6.1(a) interior			
s_s , c; k_F	$s_s = 0.00 \text{ mm}$ (default 0); c = 2000.00 mm (d = 2000.00 to the member end); $k_F = 6 + 2(h_w/a)^2 = 6 + 2(131.10/6000)^2$	6.001	Fig 6.1(a)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 6.001 \times 210000 \times 6.30^3 / 131.10$	2163.222 kN	6.4(1)
$I_y = s_s + 2t_f(1 + \sqrt{(m_1 + m_2)}) \leq a$	$m_2 = 0.000$ (first pass $\lambda_F = 0.236 \leq 0.5$ with $m_2 = 8.661$, so $m_2 = 0$); $0.00 + 2 \times 6.30 \times (1 + \sqrt{(11.905 + 0.000)}) = 56.07$	56.07 mm	6.5(3)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(56.07 \times 6.30 \times 275 / 2163.222e3)}$	0.212	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.212 = 2.359 \rightarrow 1.000$; 1.000×56.07	56.07 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 56.07 \times 6.30 / 1.0$ per web; load share to this web = 0.778 (lever rule, e = 40 mm); F_{Rd} for the load = 124.812 kN	97.148 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 87.286 \text{ kN}$ (ULS: 1.35G + 1.5Q; on the top flange, in compression) / 124.812 =	0.699	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.699 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 239.871/52.800 + N_{Ed}/N_{pl,Rd} = 150.000/984.500 = 4.695) = 4.456 \leq 1.4$ (ULS: 1.35G + 1.5Q)	3.183	Warning 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	X _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	145.808	ULS: 1.35G + 1.5Q	0.270	0.280	OK
2	point load 3+4	(a)	0.00	6.001	56.07	0.212	1.000	124.812	87.286	ULS: 1.35G + 1.5Q	0.699	3.183	Warning
6	End 2 reaction	(a)	100.00	6.001	156.07	0.354	1.000	540.797	96.712	ULS: 1.35G + 1.5Q	0.179	0.215	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

L _{ey} = K _y .L _y	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _y = √A.f _y /N _{cr}	√35.80x275/702.39 (N _{cr,y} = π²EI _y /L _{ey} ²)	1.181	
N _{b,y,Rd} = Area.χ.f _y /γ _{M1}	35.80x0.542x275/10/1 =	533.439 kN	Curve a
L _{ez} = K _z .L _z	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
λ _z = √A.f _y /N _{crz}	√35.80x275/702.39 (N _{cr,z} = π²EI _z /L _{ez} ²)	1.181	
N _{b,z,Rd} = Area.χ.f _y /γ _{M1}	35.80x0.542x275/10/1 =	533.439 kN	Curve a

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

C _{mLT} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 233.71, ψ = 0.699, α _h = 0.000	1.000	Table B.3
C _{mz} = Max(0.6+0.4ψ, 0.4)	M = 20.521, ψ = 1.000	1.000	Table B.3
C _{my} = 1.0 (diagram outside Table B.3)	M _h = 0.00, M _s = 233.71, ψ = 0.699, α _h = 0.000	1.000	Table B.3

Lateral Buckling Check M.b.Rd

M _{b,Rd} = M _{c,y,Rd}	closed hollow section — not susceptible to LTB (λ _{LT} = 0.204 ≤ 0.4)	52.800 kN.m	6.3.2.1(2)
M _{cr} (information, I _w = 0)	SN003a with C ₁ = 1.214, L _e = 6 m; z _g = +75 mm entered — NOT applied: C ₂ not published for this moment diagram; M _{cr} is at the shear centre (PASS blocked)	1265.547 kN.m	SN003a
M _{y,Ed} /M _{b,Rd}	240.758 / 52.800	4.560	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 150.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

U _{N,y} = N _{Ed} /(χ _y .N _{Rk} /γ _{M1})	150.000 / 533.439 (N _{Rk} = 984.500)	0.281	OK
U _{N,z} = N _{Ed} /(χ _z .N _{Rk} /γ _{M1})	150.000 / 533.439	0.281	OK
U _{M,y} = M _{y,Ed} /(χ _{LT} .M _{y,Rk} /γ _{M1})	240.758 / 52.800 (M _{y,Rk} = 52.800)	4.560	Warning
U _{M,z} = M _{z,Ed} /(M _{z,Rk} /γ _{M1})	20.521 / 52.800	0.389	OK
k _{yy} = C _{my} {1+(λ _y -0.2)U _{N,y} }	1.000{1+(1.181-0.2)x0.281} ≤ 1.000(1+0.8x0.281)	1.225	Table B.1
k _{zz} = C _{mz} {1+(λ _z -0.2)U _{N,z} }	1.000{1+(1.181-0.2)x0.281} ≤ 1.000(1+0.8x0.281)	1.225	Table B.1 (RHS)

$k_{yz} = k_{zz}$	rectangular hollow section	1.225	Table B.1 (RHS)
$k_{zy} = 0.6k_{yy}$	0.6×1.225	0.735	Table B.1
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.281 + 1.225 \times 4.560 + 1.225 \times 0.389$	6.343	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$0.281 + 0.735 \times 4.560 + 1.225 \times 0.389$	4.109	Warning

Torsion Design

J, H, a, Q _f , Q _w	1910 cm ⁴ , — dm ⁶ , — mm, —, —		P385 App A
W _{n0} , S _{w1}	—, —		P385 App A

Torsion Bending Design @ 0 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.388 kN.m	
W _t	SCI P385 Table A.7/A.8; I _t = 1910 cm ⁴	240 cm ³	
T _{Rd} = f _y W _t /(√3Y _{M0})	275 x 240/(√3 x 1)	38.105 kN.m	6.2.7(7)
T _{Ed} /T _{Rd}	5.388 / 38.105	0.141	OK
V _{pl.T.Rd} = [1 - T _t /(f _y √3)].V _{pl.Rd}	@ x = 0 m: T _t = 22.45 N/mm ²	244.018 kN	6.2.7(9) Eq 6.28
V _{Ed} /V _{pl.T.Rd}	145.808 / 244.018	0.598	OK
θ (SLS)	T _{Ed,SLS} = 2.512 kN.m; @ x = 2 m	0.16°	advisory

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$159.91 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$	159.91 mm	Warning
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N _{Ed} /N _{pl,Rd}	UNyz	UMyz	Ax+M_6.61	Ax+M_6.62	Deflection	V/Vpl	MA/Mc	LTB	M-V-N	F/F _{Rd}	Web 7.2	Torsion	V+T	Max	
0.152	13.483	0.281	4.560	6.343	4.109	9.595	0.513	4.560	4.560	4.424	0.699	3.183	0.141	0.598	13.483

FAIL — governing Biaxial bending + axial (6.2.9.1) = 13.483

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: Standard (closed-form) M_{cr}: a destabilising load height z_g = +75 mm is entered, but SN003a publishes C₂ only for the simply supported and fixed-ended full-span UDL and central point-load diagrams (Table 3.2). The closed form cannot include the load height for this moment diagram, so the printed M_{cr} is computed with the load at the shear centre (unconservative). Use the FE eigenvalue M_{cr} method, a verified M_{cr}, or the destabilising L_E switch (BS 5950 practice); PASS is blocked.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member PFC-L1 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m (kN/m)

Q UDL 8.000 0–6 m (kN/m)

Q PY 30.000 @ 2 m (kN)

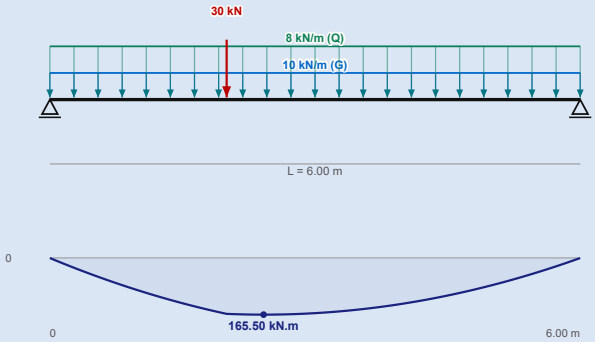
G SW 0.2296 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	0.00	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x (175.00/12.50)² = 3.920 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners

no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)

6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00

5.429

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 5.429 x 210000 x 6.00³/175.00

1266.377 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19]

144.19 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(144.19 x 6.00 x 275/1266.377e3)

0.433

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19

144.19 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 144.19 x 6.00/1.0

237.920 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =

0.452

OK

η₂ + 0.8η₁ ≤ 1.4

0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.323

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

35.819 / 31.304 kN.m (shape only: z_g = z_j = 0)

1.144

k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN·m

1.164

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

End 1 v, φ = 0; End 2 v, φ = 0

6 m

FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached

35.819 kN.m

converged

λ_{LT} = √W.f_y/M_{cr}

√ 227.0 x 275 / 35.819

1.320

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

1.320, 1.503, 0.76

0.403

Curve d

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

0.403, 1.320, 0.935, 0.985

0.409

6.3.2.3

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c,y,Rd}

0.409 x 227.0 x 275 ≤ 62.425 =

25.562 kN.m

M_{y,Ed}/M_{b,Rd}

165.504 / 25.562

6.475

Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{r,Ed} = M _{Ed} /h (kN)	2.5 % N _{r,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 60.86 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.651	6.475	3.652	0.505	0.452	1.666	6.475

FAIL — governing LTB $M_{Ed}/M_{b,Rd} = 6.475$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

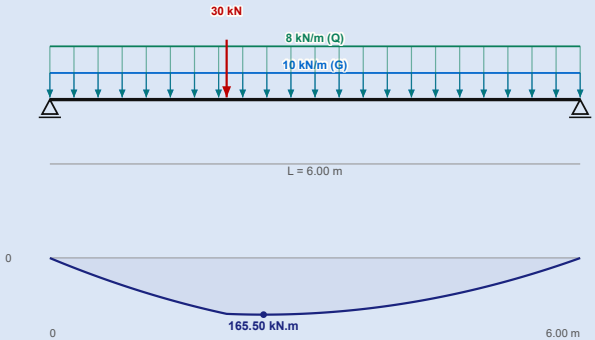
ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L1 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB M_{Ed}/M_{b,Rd} = 18.794
Owner case: vertical loads resulting in no torsion
PFC 200 x 75 x 23, 6 m SS fork ends, unrestrained; L1 "vertical loads resulting in no torsion": G 10 + Q 8 kN/m + Q 30 kN at 2.0 m through the shear centre (e = 0, z_g = 0)

Beam & Beam-Portion (Member) (FAIL)
Member PFC-L1 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)
End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m (kN/m)
Q UDL 8.000 0–6 m (kN/m)
Q PY 30.000 @ 2 m (kN)
G SW 0.2296 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	0.00	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)
Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x (175.00/12.50)² = 3.920 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
l _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(l _y .t _w .f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M _o , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _f .f _y /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{cr} route (load through the shear centre)	M _{cr} = 35.680, λ ⁻ = 1.323, χ = 0.402, f = 0.983, χ _{mod} = 0.409	25.538 kN.m	SN003a (z ₁ = 0)
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ $60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$ 60.86 mm Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Max
2.651	18.794	3.652	0.505	0.452	1.666	18.794

FAIL — governing LTB $M_{Ed}/M_{b,Rd} = 18.794$

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$$(175.00/12.50)^2 = 3.920 \text{ if } \lambda_E > 0.5, \text{ else } 0$$

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00 5.429 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.429 x 210000 x 6.00³/175.00 1266.377 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19] 144.19 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(144.19 x 6.00 x 275/1266.377e3) 0.433 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19 144.19 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 144.19 x 6.00/1.0 237.920 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 = 0.452 0.452 OK

η₂ + 0.8η₁ ≤ 1.4 0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 35.819 / 31.304 kN.m (shape only: z_g = z_j = 0) 1.144 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN.m 1.164 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached 35.819 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 227.0 x 275 / 35.819 1.320

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 1.320, 1.503, 0.76 0.403 Curve d

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.403, 1.320, 0.935, 0.985 0.409 6.3.2.3

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c,y,Rd} 0.409 x 227.0 x 275 ≤ 62.425 = 25.562 kN.m

M_{y,Ed}/M_{b,Rd} 165.504 / 25.562 6.475 Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{r,Ed} = M _{Ed} /h (kN)	2.5 % N _{r,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

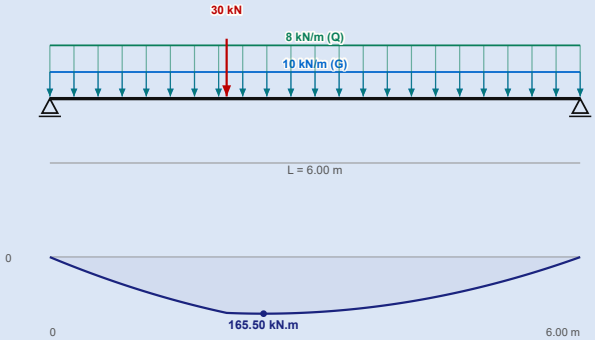
ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)
Member PFC-L2 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)
End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning
M _{y,Ed} /M _{v,y,Rd} @ x	@ x = 2 m: V _{Ed} = 55.810 > V _{pl,Rd} = 20.965: pure shear governs	2.662	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x 6.5(1)
(175.00/12.50)² = 3.920 if λ_F > 0.5, else 0

a = distance between transverse stiffeners

no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)

6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00

5.429

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 5.429 x 210000 x 6.00³/175.00

1266.377 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19]

144.19 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(144.19 x 6.00 x 275/1266.377e3)

0.433

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19

144.19 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 144.19 x 6.00/1.0

237.920 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =

0.452

OK

η₂ + 0.8η₁ ≤ 1.4

0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.323

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = fn(M₁, M₂, M_o, ψ, μ)

0.0, 0.0, 161.1, 1.000, 300.000

1.164

Serna

C₁ basis

channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN·m

SN003a

Lateral Buckling Check M.b.Rd

L_e = 1 L

1 x 6 =

6 m

λ_{LT} = (L_e/i_z)/κ

(6000/23.9)/96 (S275)

2.615

P362 channel

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

2.615, 3.906, 0.76

0.141

Curve d

M_{b,Rd} = χW_{pl,y}.f_y (κ chain, no f)

0.141 x 227.0 x 275 =

8.806 kN.m

M_{cr} (back-calculated) = W_f.f_y/λ_{LT}²

227.0 x 275 / 2.615²

9.128 kN.m

Annex A only

M_{y,Ed}/M_{b,Rd}

165.504 / 8.806

18.794

Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A

station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT
VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L2w · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L2 twin "load through the web line": e = +29.1 mm = P385 Table A.3 e₀ (shear centre to the web line; the centroid is at e_{sc} = 50.9 mm), z_g = 0

Beam & Beam-Portion (Member) (FAIL)

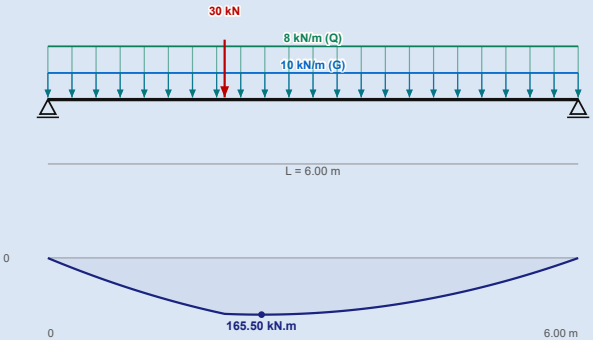
Member PFC-L2w PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 29 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 29 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 29 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	2.75	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-2.34	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.021 / 212.754 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 107.430 / 212.754 = 0.505 OK
V_{pl,y,Rd} = A_yf_y/(√3V_{M0}) 1340 mm² x 275/(√3 x 1) 212.754 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/V_{M0} 275 x 227.0/1 62.425 kN.m
M_{y,Ed}/M_{c,y,Rd} 165.504 / 62.425 = 2.651 Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x (175.00/12.50)² = 3.920 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners no transverse stiffeners declared: a = L = 6000 mm, 6.4(1)
the full member length (conservative)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00 5.429 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.429 x 210000 x 6.00³/175.00 1266.377 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19] 144.19 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(144.19 x 6.00 x 275/1266.377e3) 0.433 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19 144.19 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 144.19 x 6.00/1.0 237.920 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 = 0.452 0.452 OK

η₂ + 0.8η₁ ≤ 1.4 0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform} 35.819 / 31.304 kN.m (shape only: z_g = z_j = 0) 1.144 k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison) 0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN.m 1.164 Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints End 1 v, φ = 0; End 2 v, φ = 0 6 m FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error) 64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached 35.819 kN.m converged

λ_{LT} = √W.f_y/M_{cr} √ 227.0 x 275 / 35.819 1.320

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 1.320, 1.503, 0.76 0.403 Curve d

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f) 0.403, 1.320, 0.935, 0.985 0.409 6.3.2.3

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c,y,Rd} 0.409 x 227.0 x 275 ≤ 62.425 = 25.562 kN.m

M_{y,Ed}/M_{b,Rd} 165.504 / 25.562 6.475 Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{r,Ed} = M _{Ed} /h (kN)	2.5 % N _{r,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	3.146 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.4854 rad = 27.81°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	0.498 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	2.657 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	80.134 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 1.690, 79.767 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	12.165	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)].V _{pl,Rd}	@ x = 0 m: τ _t = 142.45 (τ _w = 3.82) N/mm ² ; V _{pl,Rd} = 212.754	107.911 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 107.911	0.996	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 35.819 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 3.146 / 2.710 kN.m	2.754 / 2.339 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	11.62°	review

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _α is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _{y,Ed} /M _{b,Rd}	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	0.452	1.666	12.165	0.996	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 11.6° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L2w · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at horizontal eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L2 twin "load through the web line": e = +29.1 mm = P385 Table A.3 e₀ (shear centre to the web line; the centroid is at e_{sc} = 50.9 mm), z_g = 0

Beam & Beam-Portion (Member) (FAIL)

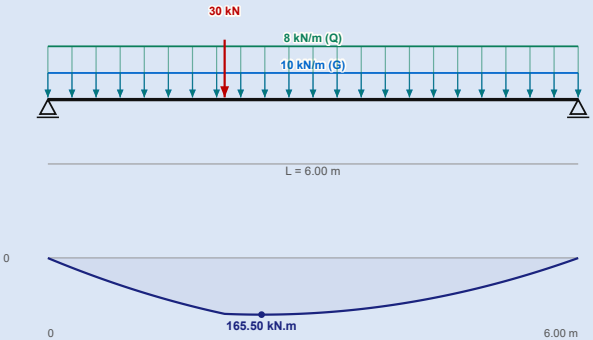
Member PFC-L2w PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 29 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 29 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 29 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	2.75	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-2.34	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd}

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _y f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x 6.5(1)
(175.00/12.50)² = 3.920 if λ_F > 0.5, else 0

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
I _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
I _y = min[I _e + t _f ³ /(m ₁ /2 + (I _e /t _f) ² + m ₂), I _e + t _f ³ /(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
λ _{LT} = (L _e /i _z)/κ	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} .f _y (κ chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _{f,y} /λ _{LT} ²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A

station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	3.146 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.4854 rad = 27.81°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.498 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	2.657 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	80.134 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 1.690, 79.767 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	12.165	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 142.45$ ($\tau_w = 3.82$) N/mm ² ; $V_{pl,Rd} = 212.754$	107.911 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 107.911	0.996	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 3.146 / 2.710$ kN.m	2.754 / 2.339 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	11.62°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT
VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	12.165	0.996	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 11.6^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L3t · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L3 "vertical + eccentricity from shear center": all loads on the top flange z_g = +100 mm (D/2, destabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

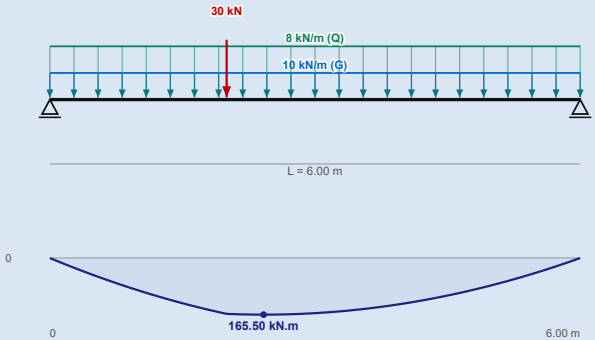
Member PFC-L3t PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 0 mm z_g = 100 mm (kN/m)
Q UDL 8.000 0-6 m e = 0 mm z_g = 100 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = 100 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.04	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-0.04	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _f f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x (175.00/12.50)² = 3.920 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners

no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)

6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429	Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN	6.4(1)
l _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm	6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm	6.5(4)
λ _F = √(l _y .t _w .f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433	6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm	6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN	6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	35.819 / 31.304 kN.m (shape only: z _g = z _j = 0)	1.144	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m	1.164	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 100, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	30.940 kN.m	converged
λ _{LT} = √W.f _y /M _{cr}	√ 227.0 x 275 / 30.940	1.420	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.420, 1.644, 0.76	0.366	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.366, 1.420, 0.935, 0.993	0.368	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.368 x 227.0 x 275 ≤ 62.425 =	22.992 kN.m	
M _{y,Ed} /M _{b,Rd}	165.504 / 22.992	7.198	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{r,Ed} = M _{Ed} /h (kN)	2.5 % N _{r,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
$B_{Ed} = EI_w\varphi''$ (max)	@ $x = 2.988$ m	0.004 kN.m ²	
$M_{w,Ed} = EI_w\varphi'''/(h-t_f)$	max over span	0.020 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	1.166 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 0.020, 1.137 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	7.101	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 2.05$ ($\tau_w = 0.08$) N/mm ² ; $V_{pl,Rd} = 212.754$	211.552 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 211.552	0.508	OK
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 30.940$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T\varphi'$ at $x = 0 / x = L$; total $T = GI_T\varphi' - EI_w\varphi''' = 0.047 / 0.047$ kN.m	0.040 / 0.040 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 0$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	0.00°	OK

NOT VERIFIED	$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	7.198	3.652	0.505	0.452	1.666	7.101	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 75.00 / (275 \times 6.00) = 12.500 ; 0.02 \times (175.00/12.50)^2 = 3.920 \text{ if } \lambda_{F_e} > 0.5, \text{ else } 0 \quad 6.5(1)$$

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00 5.429 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.429 x 210000 x 6.00³/175.00 1266.377 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19] 144.19 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(144.19 x 6.00 x 275/1266.377e3) 0.433 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19 144.19 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 144.19 x 6.00/1.0 237.920 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 = 0.452 OK

η₂ + 0.8η₁ ≤ 1.4 0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = fn(M₁, M₂, M_o, ψ, μ) 0.0, 0.0, 161.1, 1.000, 300.000 1.164 Serna

C₁ basis channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN·m SN003a

Lateral Buckling Check M.b.Rd

L_e = 1 L 1 x 6 = 6 m

z_g z_g = +100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered load height

λ_{LT} = (L_e/i_z)/κ (6000/23.9)/96 (S275) 2.615 P362 channel

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 2.615, 3.906, 0.76 0.141 Curve d

M_{b,Rd} = χW_{pl,y}.f_y (κ chain, no f) 0.141 x 227.0 x 275 = 8.806 kN.m

M_{cr} (back-calculated) = W.f_y/λ_{LT}² 227.0 x 275 / 2.615² 9.128 kN.m Annex A only

M_{y,Ed}/M_{b,Rd} 165.504 / 8.806 18.794 Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
B _{Ed} = EI _w φ'' (max)	@ x = 2.988 m	0.004 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	0.020 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	1.166 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 0.020, 1.137 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	7.101	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)]. V _{pl,Rd}	@ x = 0 m: τ _t = 2.05 (τ _w = 0.08) N/mm ² ; V _{pl,Rd} = 212.754	211.552 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 211.552	0.508	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 0.047 / 0.047 kN.m	0.040 / 0.040 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 0 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	0.00°	OK

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _{alpha} is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _y (y,Ed)/M _y (b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	7.101	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_{alpha} is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$$m_1 = f_{yf} \cdot b_f (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 75.00 / (275 \times 6.00) = 12.500 ; 0.02 \times (175.00/12.50)^2 = 3.920 \text{ if } \lambda_{\text{E}} > 0.5, \text{ else } 0 \quad 6.5(1)$$

a = distance between transverse stiffeners

no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)

6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429	Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN	6.4(1)
l _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm	6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm	6.5(4)
λ _F = √(l _y .t _w .f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433	6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm	6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN	6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	35.819 / 31.304 kN.m (shape only: z _g = z _j = 0)	1.144	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m	1.164	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = -100, 0 mm; 1 combination(s), 1 solve(s) + 2 cached	41.391 kN.m	converged
λ _{LT} = √W.f _y /M _{cr}	√ 227.0 x 275 / 41.391	1.228	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.228, 1.380, 0.76	0.442	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.442, 1.228, 0.935, 0.979	0.452	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.452 x 227.0 x 275 ≤ 62.425 =	28.204 kN.m	
M _{y,Ed} /M _{b,Rd}	165.504 / 28.204	5.868	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{r,Ed} = M _{Ed} /h (kN)	2.5 % N _{r,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2.988$ m	0.004 kN.m ²	
$M_{w,Ed} = EI_w \varphi''' / (h - t_f)$	max over span	0.020 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	1.166 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 0.020, 1.137 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	7.101	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 2.05$ ($\tau_w = 0.08$) N/mm ² ; $V_{pl,Rd} = 212.754$	211.552 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 211.552	0.508	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 41.391$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 0.047 / 0.047$ kN.m	0.040 / 0.040 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 0$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	0.00°	OK

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	5.868	3.652	0.505	0.452	1.666	7.101	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L3b · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical + - eccentricity from shear center
PFC 200 x 75 x 23, 6 m SS, unrestrained; L3 twin "vertical - eccentricity from shear center": all loads hung from the bottom flange z_g = -100 mm (-D/2, stabilising), e = 0

Beam & Beam-Portion (Member) (FAIL)

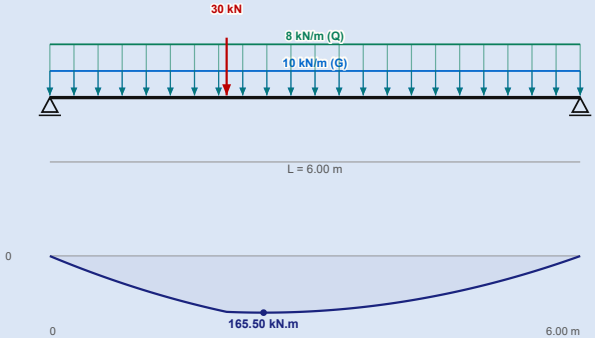
Member PFC-L3b PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 0 mm z_g = -100 mm (kN/m)
Q UDL 8.000 0–6 m e = 0 mm z_g = -100 mm (kN/m)
Q PY 30.000 @ 2 m e = 0 mm z_g = -100 mm (kN)
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.04	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-0.04	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x 6.5(1)
(175.00/12.50)² = 3.920 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00 5.429 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.429 x 210000 x 6.00³/175.00 1266.377 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19] 144.19 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(144.19 x 6.00 x 275/1266.377e3) 0.433 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19 144.19 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 144.19 x 6.00/1.0 237.920 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 = 0.452 OK

η₂ + 0.8η₁ ≤ 1.4 0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = fn(M₁, M₂, M_o, ψ, μ) 0.0, 0.0, 161.1, 1.000, 300.000 1.164 Serna

C₁ basis channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN·m SN003a

Lateral Buckling Check M.b.Rd

L_e = 1 L 1 x 6 = 6 m

z_g z_g = -100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered load height

λ_{LT} = (L_e/i_z)/κ (6000/23.9)/96 (S275) 2.615 P362 channel

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 2.615, 3.906, 0.76 0.141 Curve d

M_{b,Rd} = χW_{pl,y}.f_y (κ chain, no f) 0.141 x 227.0 x 275 = 8.806 kN.m

M_{cr} (back-calculated) = W.f_y/λ_{LT}² 227.0 x 275 / 2.615² 9.128 kN.m Annex A only

M_{y,Ed}/M_{b,Rd} 165.504 / 8.806 18.794 Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	0.047 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.0072 rad = 0.41°	
B _{Ed} = EI _w φ'' (max)	@ x = 2.988 m	0.004 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	0.020 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	1.166 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 0.020, 1.137 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	7.101	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)]. V _{pl,Rd}	@ x = 0 m: τ _t = 2.05 (τ _w = 0.08) N/mm ² ; V _{pl,Rd} = 212.754	211.552 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 211.552	0.508	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 0.047 / 0.047 kN.m	0.040 / 0.040 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 0 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	0.00°	OK

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _α is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _y (y,Ed)/M _y (b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	7.101	0.508	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)

Member PFC-L4 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

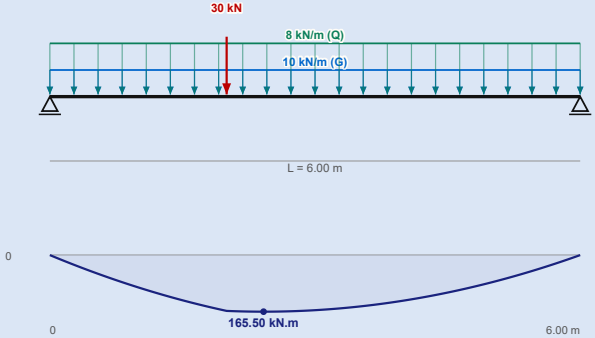
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only

- uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _v f _y /(√3V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning
M _{y,Ed} /M _{v,y,Rd} @ x	@ x = 2 m: V _{Ed} = 55.810 > V _{pl,Rd} = 20.965: pure shear governs	2.662	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x (175.00/12.50)² = 3.920 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
I _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
I _y = min[I _e + t _f ³ /(m ₁ /2 + (I _e /t _f) ² + m ₂), I _e + t _f ³ /(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	35.819 / 31.304 kN.m (shape only: z _g = z _j = 0)	1.144	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ = 0; End 2 v, φ = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; z _g = 100, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	30.940 kN.m	converged
λ _{LT} = √(W _y .f _y /M _{cr})	√(227.0 x 275 / 30.940)	1.420	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.420, 1.644, 0.76	0.366	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.366, 1.420, 0.935, 0.993	0.368	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.368 x 227.0 x 275 ≤ 62.425 =	22.992 kN.m	
M _{y,Ed} /M _{b,Rd}	165.504 / 22.992	7.198	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{r,Ed} = M _{Ed} /h (kN)	2.5 % N _{r,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 30.940$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	7.198	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC 200 x 75 x 23, 6 m SS, unrestrained; L4 "vertical horizontal eccentricity": all loads at $e = +40$ mm and $z_g = +100$ mm (top flange)

$$(175.00/12.50)^2 = 3.920 \text{ if } \lambda_F^- > 0.5, \text{ else } 0$$

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00 5.429 Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w 0.9 x 5.429 x 210000 x 6.00³/175.00 1266.377 kN 6.4(1)

I_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c 5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00 100.00 mm 6.5(4)

I_y = min[I_e + t_f³/(m₁/2 + (I_e/t_f)² + m₂), I_e + t_f³/(m₁ + m₂)] m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19] 144.19 mm 6.5(4)

λ_F = √(I_y.t_w.f_{yw}/F_{cr}) √(144.19 x 6.00 x 275/1266.377e3) 0.433 6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.I_y 0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19 144.19 mm 6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1} 275 x 144.19 x 6.00/1.0 237.920 kN 6.2(1)

F_{Ed}/F_{Rd} F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 = 0.452 OK

η₂ + 0.8η₁ ≤ 1.4 0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q) 0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = fn(M₁, M₂, M_o, ψ, μ) 0.0, 0.0, 161.1, 1.000, 300.000 1.164 Serna

C₁ basis channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 kN·m SN003a

Lateral Buckling Check M.b.Rd

L_e = 1 L 1 x 6 = 6 m

z_g z_g = +100 mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered load height

λ_{LT} = (L_e/i_z)/κ (6000/23.9)/96 (S275) 2.615 P362 channel

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT}) 2.615, 3.906, 0.76 0.141 Curve d

M_{b,Rd} = χW_{pl,y}.f_y (κ chain, no f) 0.141 x 227.0 x 275 = 8.806 kN.m

M_{cr} (back-calculated) = W.f_y/λ_{LT}² 227.0 x 275 / 2.615² 9.128 kN.m Annex A only

M_{y,Ed}/M_{b,Rd} 165.504 / 8.806 18.794 Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
W_{n0}, S_{w1} 40.3 cm², 115 cm⁴; W_{n2} = 27.2 cm², S_{w2} = 62.5, S_{w3} = 46.3 cm⁴, e₀ = 29.1, e_{sc} = 50.9 mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 12.27 P385 App C
φ_{max} (ULS) L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 0.683 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 3.645 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 109.717 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M_{pl,y} = 62.425, M_{pl,f} = 4.834, M_{pl,z} = 16.665 14.062 **Warning** P385 3.1.2
V_{pl,T.Rd} = [√(1 - τ_t/(1.25f_y/√3)) - τ_w/(f_y/√3)] · V_{pl,Rd} @ x = 0 m: τ_t = 195.04 (τ_w = 5.22) N/mm²; V_{pl,Rd} = 212.754 20.965 kN 6.2.7(9)
V_{Ed}/V_{pl,T.Rd} 107.430 / 20.965 5.124 **Warning**
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_αk_{zw}k_αM_w/M_{f,Rk} k_α unbounded: M_{y,Ed} reaches M_{cr} = 9.128 kN.m — **BLOCKED**
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.307 / 3.707 kN.m 3.771 / 3.200 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 15.97° review

NOT VERIFIED M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged. **NOT VERIFIED**

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 60.86 mm **Warning**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
PFC 200 x 75 x 23, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_Ed = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e_y = +40 mm, z_g = +100 mm

Low Shear

$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 60.6/1	16.665 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	29.90 x 275/1 = (No bearing / block tearing design)	822.250 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 822.250 =	0.182	OK
$M_{c.y.Rd}, M_{c.z.Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y.Ed} / M_{c.y.Rd} + M_{z.Ed} / M_{c.z.Rd}$	0.182 + 165.504/62.425 + 0 =	2.834	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	275 x 75.00/(275 x 6.00) = 12.500; 0.02 x (175.00/12.50) ² = 3.920 if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377 \text{e}3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 0.000/62.425 + N_{Ed} / N_{pl.Rd} = 150.000/822.250 = 0.182) = 0.597 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.427	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.427	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.770	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.382	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.852	
$N_{b.y.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10/1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	2.892	
$N_{b.z.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10/1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4

$N_{cr,T} = (GI_T + \pi^2 EI_w / L_T^2) / i_0^2$	(81000 x 11.6e4 + π^2 x 210000 x 0.0107e12/6000 ²)/9739.2 (I_T , I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T}^2)}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\chi_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	29.90 x 0.497 x 275 / 10 / 1 = (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
$N_{Ed} / N_{b,T,Rd}$	150.000 / 408.410 = (the lower of χ_T and the flexural χ feeds $U_{N,y}$, $U_{N,z}$)	0.367	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.14$, $\psi = 0.892$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 161.14$, $\psi = 0.892$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 0 solve(s) + 3 cached	30.940 kN.m	converged
$\chi_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{227.0 \times 275 / 30.940}$	1.420	
$\chi_{LT} = \text{Fn}(\chi_{LT}, \Phi_{LT}, \alpha_{LT})$	1.420, 1.644, 0.76	0.366	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \chi_{LT}, k_c, f)$	0.366, 1.420, 0.935, 0.993	0.368	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	0.368 x 227.0 x 275 $\leq$ 62.425 =	22.992 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 22.992	7.198	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	165.504 / 22.992 ($M_{y,Rk} = 62.425$)	7.198	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	0.000 / 9.295	0.000	OK
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.852 \times 0.367\} \leq 1.000 (1 + 0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 2.892 \times 1.793\} \leq 1.000 (1 + 0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\chi_z, 1) \cdot n_z / (C_{mLT} - 0.25)$; $\chi_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.367 + 1.188 \times 7.198 + 2.076 \times 0.000$	8.917	Warning
$U_{Nz} + k_{zy} \cdot U_{My} + k_{zz} \cdot U_{Mz}$	$1.793 + 0.880 \times 7.198 + 2.076 \times 0.000$	8.131	Warning

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
V _{pl,T,Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)].V _{pl,Rd}	@ x = 0 m: τ _t = 195.04 (τ _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 20.965	5.124	Warning
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 30.940 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT VERIFIED

NOT VERIFIED

M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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N_Ed/N_(pLRd)	UNyz	UMyz	Ax+M_6.4x+M_6.4z	Deflection	V/Vpl	MA/Mc	LTB	M-V	N_b.T	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max	
0.182	2.834	1.793	7.198	8.917	8.131	3.652	0.505	2.651	7.198	2.662	0.367	0.452	1.770	0.367	5.124	99.00099.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L5 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: udl point vertical loads acting at vertical horizontal eccentricity from shear center, and axial load on beam
PFC 200 x 75 x 23, 6 m SS, unrestrained; L5 = L4 + "axial load on beam": N_{Ed} = 150 kN compression (U_x at End 1; the ULS DESIGN value entered directly, not factored by the combination - in MasterSeries an axial load whose factored value is 150 kN, e.g. 100 kN imposed), all loads at e = +40 mm, z_g = +100 mm

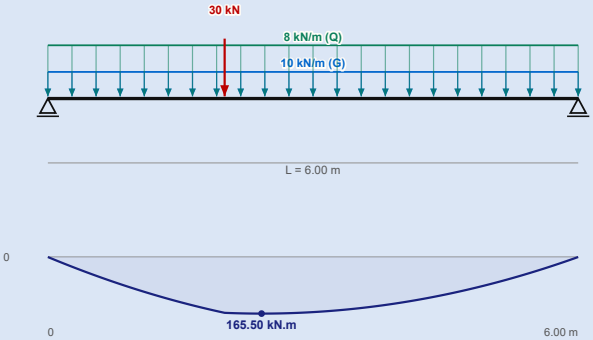
Axial with Moments (Member) (FAIL)
Member PFC-L5 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	150.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 150.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Web classified for bending + compression: α = 0.801, ψ = -0.635; Class 1 Table 5.2
limits 42.1ε / 48.5ε / 91.2ε
Auto Design Load Cases 1; SLS 1

Local Capacity Check

V_{y,Ed}/V_{pl,y,Rd} 0.021 / 212.754 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 107.430 / 212.754 = 0.505 OK
V_{pl,y,Rd} = A_vf_y/((√3)Y_{M0}) 1340 mm² x 275/((√3) x 1) 212.754 kN 6.2.6
M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 227.0/1 62.425 kN.m
M_{y,Ed}/M_{c,y,Rd} 165.504 / 62.425 = 2.651 Warning
M_{y,Ed}/M_{v,y,Rd} @ x 2 m: V_{Ed} = 55.810 > V_{pl,Rd} = 20.965: pure shear governs 2.662 Warning
V_{z,Ed}/V_{pl,z,Rd} 0 / 308.016 = (A_{v,z} = 1940 mm²; no minor-axis shear in the single-plane model) 0.000 Low Shear

$M_{c.z.Rd} = f_y \cdot W_{pl.z} / \gamma_{M0}$	275 x 60.6/1	16.665 kN.m	
$N_{pl.Rd} = A_g \cdot f_y / \gamma_{M0}$	29.90 x 275/1 = (No bearing / block tearing design)	822.250 kN	
$n = N_{Ed} / N_{pl.Rd}$	150.000 / 822.250 =	0.182	OK
$M_{c.y.Rd}, M_{c.z.Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y.Ed} / M_{c.y.Rd} + M_{z.Ed} / M_{c.z.Rd}$	0.182 + 165.504/62.425 + 0 =	2.834	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web		Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	275 x 75.00/(275 x 6.00) = 12.500; 0.02 x (175.00/12.50) ² = 3.920 if $\lambda_F > 0.5$, else 0		6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)		6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377 \text{e}3)}$	0.433	6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c.y.Rd} = 0.000/62.425 + N_{Ed} / N_{pl.Rd} = 150.000/822.250 = 0.182) = 0.597 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.427	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.427	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.770	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.382	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c.y.Rd}$ ($W_{pl.y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.852	
$N_{b.y.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10/1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	2.892	
$N_{b.z.Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10/1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4

$N_{cr,T} = (GI_T + \pi^2 EI_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6 \text{ m} = L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T}^2)}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0 / i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\lambda_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 10 / 1 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c
$N_{Ed} / N_{b,T,Rd}$	$150.000 / 408.410 =$ (the lower of χ_T and the flexural χ feeds $U_{N,y}, U_{N,z}$)	0.367	OK

Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M_{max}=165.5 \text{ kN}\cdot\text{m}$		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 161.14, \psi = 0.892, \alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 0.000, \psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00, M_s = 161.14, \psi = 0.892, \alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
z_g	$z_g = +100 \text{ mm}$ entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered		load height
$\lambda_{LT} = (L_e / i_z) / \kappa$	$(6000 / 23.9) / 96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) $= W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed} / M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed} / h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed} / h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed} / (\chi_y \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 408.410$ ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed} / (\chi_z \cdot N_{Rk} / \gamma_{M1})$	$150.000 / 83.643$ (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed} / (\chi_{LT} \cdot M_{y,Rk} / \gamma_{M1})$	$165.504 / 8.806$ ($M_{y,Rk} = 62.425$)	18.794	Warning
$U_{M,z} = M_{z,Ed} / (M_{z,Rk} / \gamma_{M1})$	$0.000 / 9.295$	0.000	OK
$k_{yy} = C_{my} \{1 + 0.6 \chi_y U_{N,y}\}$	$1.000 \{1 + 0.6 \times 0.852 \times 0.367\} \leq 1.000 (1 + 0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz} \{1 + 0.6 \chi_z U_{N,z}\}$	$1.000 \{1 + 0.6 \times 2.892 \times 1.793\} \leq 1.000 (1 + 0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1 \chi_z U_{N,z} / (C_{mLT} - 0.25)$	Table B.2 (susceptible): $1 - 0.05 \cdot \min(\chi_z, 1) \cdot \eta_z / (C_{mLT} - 0.25)$; $\chi_z = 2.892, U_{N,z} = 1.793, C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny} + k_{yy} \cdot U_{My} + k_{yz} \cdot U_{Mz}$	$0.367 + 1.188 \times 18.794 + 2.076 \times 0.000$	22.691	Warning

$U_{Nz} + k_{zy} \cdot U_{M,y} + k_{zz} \cdot U_{M,z}$	1.793+0.880x18.794+2.076x0.000	18.341	Warning
Torsion Design			
J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm		P385 App A
Torsion Bending Design @ 2.415 m			
T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ" (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ"/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ · M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
V _{pl,T,Rd} = [√(1 - T _t /(1.25f _y /√3)) - T _w /(f _y /√3)] · V _{pl,Rd}	@ x = 0 m: T _t = 195.04 (T _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	107.430 / 20.965	5.124	Warning
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review
NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.		NOT VERIFIED
NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _α is unbounded; the member is inadequate as arranged.		NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$				$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$							60.86 mm		Warning		
N_Ed/N_(pL_Rd)	UNyz	UMyz	Ax+M_6.6.1x+M_6.6.2	Deflection	V/Vpl	MA/Mc	LTB	M-V	N_b.T	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.182	2.834	1.793	18.794	22.691	18.341	0.505	2.651	18.794	2.662	0.367	0.452	1.770	0.367	5.124	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L6 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing Pure shear failure at M-V check point (6.2.6) = >= 99 (capacity exhausted)

Owner case: ... and axial load on beam, load applied at angle on beam

PFC 200 x 75 x 23, 6 m SS, unrestrained; L6 = L5 + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +100, factored by the combination) plus an entered CONSTANT M_{z,Ed} = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_{mz} = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.100 = 1.03 kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)

Member PFC-L6 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

Q PY 28.191 @ 2 m e = 40 mm z_g = 100 mm (kN)

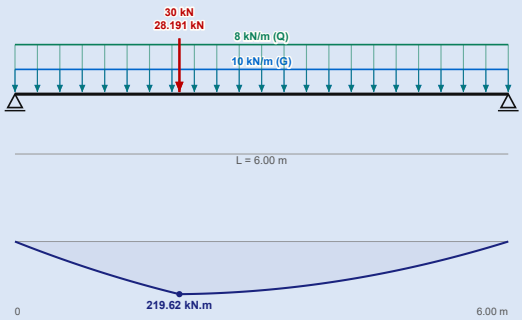
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	4.87	135.62	0.00	0.00	20.52	219.62	20.52	-87.32	135.62	—
	End 2, x = 6	150.00C	-3.76	-106.53	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	106.53	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = F _n (c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 150.00, 219.62, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε	Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1		

Local Capacity Check

V _{y,Ed} /V _{pl,y,Rd}	3.286 / 212.754 =	0.015	High Shear
V _{y,Ed,max} /V _{pl,y,Rd}	135.621 / 212.754 =	0.637	OK
V _{pl,y,Rd} = A _v f _y / (√3 Y _{M0})	1340 mm² x 275/ (√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y · W _{pl,y} /Y _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	219.622 / 62.425 =	3.518	Warning
M _{y,Ed} /M _{Vy,Rd} @ x	@ x = 2 m: V _{Ed} = 3.286 > V _{pl,Rd} = 0.000: pure shear governs	3285542509.905	Warning
V _{z,Ed} /V _{pl,z,Rd}	0 / 308.016 = (A _{V,z} = 1940 mm²; no minor-axis shear in the single-plane model)	0.000	Low Shear
M _{c,z,Rd} = f _y · W _{pl,z} /Y _{M0}	275 x 60.6/1	16.665 kN.m	
N _{pl,Rd} = A _g · f _y /Y _{M0}	29.90 x 275/1 = (No bearing / block tearing design)	822.250 kN	
n = N _{Ed} /N _{pl,Rd}	150.000 / 822.250 =	0.182	OK

$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed}/M_{c,y,Rd} + M_{z,Ed}/M_{c,z,Rd}$	$0.182 + 219.622/62.425 + 20.521/16.665 =$	4.932	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3/175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 135.621$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.570	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.570 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.716 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.511	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	135.621	ULS: 1.35G + 1.5Q	0.570	0.511	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.448	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	106.525	ULS: 1.35G + 1.5Q	0.448	0.424	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.852	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10 / 1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	2.892	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10 / 1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4
$N_{cr,T} = (G I_T + \pi^2 E I_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T})^2}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\lambda_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 10 / 1 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c

$N_{Ed}/N_{b.T.Rd}$	150.000 / 408.410 = (the lower of χ_T and the flexural χ feeds $U_{N.y}$, $U_{N.z}$)	0.367	OK
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Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr}/M_{cr,uniform}$	37.280 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.191	k_c only
$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 203.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.4$, $M(L/2)=203.4$, $M(3L/4)=130.8$, $M_{max}=219.6$ kN·m	1.218	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.43$, $\psi = 0.817$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.43$, $\psi = 0.817$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 2 solve(s) + 1 cached	32.041 kN.m	converged
$\bar{\lambda}_{LT} = \sqrt{W_{f,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 32.041}$	1.396	
$\chi_{LT} = Fn(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	1.396, 1.609, 0.76	0.374	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \bar{\lambda}_{LT}, k_c, f)$	0.374, 1.396, 0.916, 0.988	0.379	6.3.2.3
$M_{b.Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y.Rd}$	$0.379 \times 227.0 \times 275 \leq 62.425$	23.659 kN.m	
$M_{y.Ed}/M_{b.Rd}$	219.622 / 23.659	9.283	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f.Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f.Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f.Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f.Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N.y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b.T.Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N.z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 83.643 (min with $N_{b.T.Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M.y} = M_{y.Ed}/(\chi_{LT} \cdot M_{y.Rk}/\gamma_{M1})$	219.622 / 23.659 ($M_{y.Rk} = 62.425$)	9.283	Warning
$U_{M.z} = M_{z.Ed}/(M_{z.Rk}/\gamma_{M1})$	20.521 / 9.295	2.208	Warning
$k_{yy} = C_{my}\{1+0.6\bar{\lambda}_y U_{N.y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\bar{\lambda}_z U_{N.z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\bar{\lambda}_z U_{N.z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\bar{\lambda}_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\bar{\lambda}_z = 2.892$, $U_{N.z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny}+k_{yy} \cdot U_{M.y}+k_{yz} \cdot U_{M.z}$	$0.367+1.188 \times 9.283+2.076 \times 2.208$	15.977	Warning
$U_{Nz}+k_{zy} \cdot U_{M.y}+k_{zz} \cdot U_{M.z}$	$1.793+0.880 \times 9.283+2.076 \times 2.208$	14.550	Warning

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.097 kN.m ²	
$M_{w.Ed} = EI_w \phi''/(h-t_f)$	max over span	5.850 kN.m	
$M_{z.Ed} = \phi \cdot M_{y.Ed}$	max coincident	185.324 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2$ m: 219.622, 5.850, 180.013 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	24.390	Warning P385 3.1.2
$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl.Rd}$	@ $x = 0$ m: $\tau_t = 251.90$ ($\tau_w = 5.50$) N/mm ² ; $V_{pl.Rd} = 212.754$	0.000 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	135.621 / 0.000	∞	Warning
$M_y/M_{b.Rd} + C_{mz}M_z/M_{z.Rk} + k_{wz}k_{wz}M_w/M_{f.Rk}$	k_a unbounded: $M_{y.Ed}$ reaches $M_{cr} = 32.041$ kN.m	—	BLOCKED

End torques T_t	St Venant part $GI_T\phi'$ at $x = 0 / x = L$; total $T = GI_T\phi' -$ $El_w\phi'' = 5.435 / 4.271$ kN.m	4.870 / 3.764 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.34$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	23.48°	review
NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.		NOT VERIFIED
NOT VERIFIED	$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_{α} is unbounded; the member is inadequate as arranged.		NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$87.32 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	87.32 mm	Warning
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N_Ed/N_(pl.Ed)	N/N_d	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	M-V	N_b.T	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.182	4.932	1.793	9.283	15.977	14.550	5.239	0.637	3.518	9.283	3285542509.905	42357905	570	2.448	0.367	99.000	99.000	3285542509.905

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 3285542509.905

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_{α} is unbounded; the member is inadequate as arranged.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 23.5^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L6 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing Pure shear failure at M-V check point (6.2.6) = >= 99 (capacity exhausted)

Owner case: ... and axial load on beam, load applied at angle on beam

PFC 200 x 75 x 23, 6 m SS, unrestrained; L₆ = L₅ + "load applied at angle on beam": a further 30 kN (Q) point load at 2.0 m inclined 20 deg from the vertical - APPROXIMATION: modelled as its vertical component 28.191 kN (Q, e = +40, z_g = +100, factored by the combination) plus an entered CONSTANT M_{z,Ed} = 20.521 kN.m (= 1.5 x 13.681, the ULS design value of the Q-case moment H a b / L = 10.261 x 2 x 4 / 6 = 13.681 kN.m, the simply supported minor-axis moment at 2.0 m; the tool's M_z input is a design value not run through the combinations); the tool has no lateral load, so the minor-axis moment is applied as a constant along the member (C_{mz} = 1.0), not as the true triangular diagram, and the torque of the horizontal component acting at z_g (H x z_g = 10.261 x 0.100 = 1.03 kN.m characteristic) is not modelled

Axial with Moments (Member) (FAIL)

Member PFC-L6 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

Q PY 28.191 @ 2 m e = 40 mm z_g = 100 mm (kN)

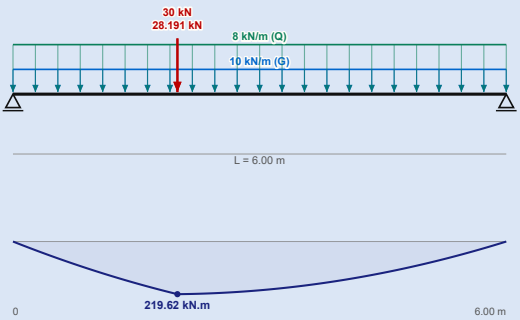
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	4.87	135.62	0.00	0.00	20.52	219.62	20.52	-87.32	135.62	—
	End 2, x = 6	150.00C	-3.76	-106.53	0.00	0.00	20.52	@ 2.000	@ —	@ 2.850	106.53	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m)	200 x 75 x 23 PFC [S275]		
Class = F _n (c/t, d/t, f _y , N, M _y , M _z)	4.56, 25.20, 275, 150.00, 219.62, 20.52		Class 1
	(Axial: Non-Slender)		
Web classified for	biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε	Class 1	Table 5.2
Auto Design Load Cases	1; SLS 1		

Local Capacity Check

V _{y,Ed} /V _{pl,y,Rd}	3.286 / 212.754 =	0.015	High Shear
V _{y,Ed,max} /V _{pl,y,Rd}	135.621 / 212.754 =	0.637	OK
V _{pl,y,Rd} = A _v f _y / (√3 Y _{M0})	1340 mm² x 275/ (√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y · W _{pl,y} /Y _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	219.622 / 62.425 =	3.518	Warning
M _{y,Ed} /M _{Vy,Rd} @ x	@ x = 2 m: V _{Ed} = 3.286 > V _{pl,Rd} = 0.000: pure shear governs	3285542509.905	Warning
V _{z,Ed} /V _{pl,z,Rd}	0 / 308.016 = (A _{V,z} = 1940 mm²; no minor-axis shear in the single-plane model)	0.000	Low Shear
M _{c,z,Rd} = f _y · W _{pl,z} /Y _{M0}	275 x 60.6/1	16.665 kN.m	
N _{pl,Rd} = A _g · f _y /Y _{M0}	29.90 x 275/1 = (No bearing / block tearing design)	822.250 kN	
n = N _{Ed} /N _{pl,Rd}	150.000 / 822.250 =	0.182	OK

$M_{c,y,Rd}$; $M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed}/M_{c,y,Rd} + M_{z,Ed}/M_{c,z,Rd}$	$0.182 + 219.622/62.425 + 20.521/16.665 =$	4.932	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: $a = L = 6000$ mm, the full member length (conservative)	6.4(1)

Governing station $x = 0$ m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3/175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 135.621$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.570	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.570 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.716 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.511	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	135.621	ULS: 1.35G + 1.5Q	0.570	0.511	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.448	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	106.525	ULS: 1.35G + 1.5Q	0.448	0.424	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 E I_y / L_{ey}^2$)	0.852	
$N_{b,y,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10 / 1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 E I_z / L_{ez}^2$)	2.892	
$N_{b,z,Rd} = \text{Area} \cdot \chi \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10 / 1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4
$N_{cr,T} = (G I_T + \pi^2 E I_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T})^2}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\lambda_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 10 / 1 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c

$N_{Ed}/N_{b.T.Rd}$	150.000 / 408.410 = (the lower of χ_T and the flexural χ feeds $U_{N.y}$, $U_{N.z}$)	0.367	OK
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Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 203.4, 1.000, 300.000	1.218	Serna
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=174.4$, $M(L/2)=203.4$, $M(3L/4)=130.8$, $M_{max}=219.6$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.43$, $\psi = 0.817$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 203.43$, $\psi = 0.817$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
z_g	$z_g = +100$ mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered		load height
$\lambda_{LT} = (L_e/i_z)/\kappa$	(6000/23.9)/96 (S275)	2.615	P362 channel
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b.Rd} = \chi W_{pl.y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y.Ed}/M_{b.Rd}$	219.622 / 8.806	24.940	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f.Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f.Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f.Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f.Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N.y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b.T.Rd} =$ 408.410 (6.3.1.4))	0.367	OK
$U_{N.z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 83.643 (min with $N_{b.T.Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M.y} = M_{y.Ed}/(\chi_{LT} \cdot M_{y.Rk}/\gamma_{M1})$	219.622 / 8.806 ($M_{y.Rk} = 62.425$)	24.940	Warning
$U_{M.z} = M_{z.Ed}/(M_{z.Rk}/\gamma_{M1})$	20.521 / 9.295	2.208	Warning
$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N.y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N.z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\lambda_z U_{N.z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 2.892$, $U_{N.z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny}+k_{yy} \cdot U_{M.y}+k_{yz} \cdot U_{M.z}$	$0.367+1.188 \times 24.940+2.076 \times 2.208$	34.574	Warning
$U_{Nz}+k_{zy} \cdot U_{M.y}+k_{zz} \cdot U_{M.z}$	$1.793+0.880 \times 24.940+2.076 \times 2.208$	28.335	Warning

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	1.097 kN.m ²	
$M_{w.Ed} = EI_w \varphi''/(h-t_f)$	max over span	5.850 kN.m	
$M_{z.Ed} = \varphi \cdot M_{y.Ed}$	max coincident	185.324 kN.m	
$(M_y/M_{pl.y})^2 + M_w/M_{pl.f} + M_z/M_{pl.z}$	@ $x = 2$ m: 219.622, 5.850, 180.013 kN.m; $M_{pl.y} = 62.425$, $M_{pl.f} = 4.834$, $M_{pl.z} = 16.665$	24.390	Warning P385 3.1.2
$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl.Rd}$	@ $x = 0$ m: $\tau_t = 251.90$ ($\tau_w = 5.50$) N/mm ² ; $V_{pl.Rd} = 212.754$	0.000 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	135.621 / 0.000	∞	Warning

$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_\alpha M_w/M_{t,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T\phi'$ at $x = 0 / x = L$; total $T = GI_T\phi' - EI_w\phi'' = 5.435 / 4.271$ kN.m	4.870 / 3.764 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.34$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	23.48°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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NOT VERIFIED	$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$87.32 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	87.32 mm	Warning
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N_Ed/N_(pl.Ed)	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	M-V	N_b.T	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.182	4.932	1.793	24.940	34.574	28.335	5.239	0.637	3.518	24.940	3285542509.905	0.570	2.448	0.367	99.000	99.000	3285542509.905

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 3285542509.905

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 23.5^\circ$ exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L7 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing Pure shear failure at M-V check point (6.2.6) = >= 99 (capacity exhausted)
Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER
PFC 200 x 75 x 23, 6 m SS, unrestrained; L7 = L6 + "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

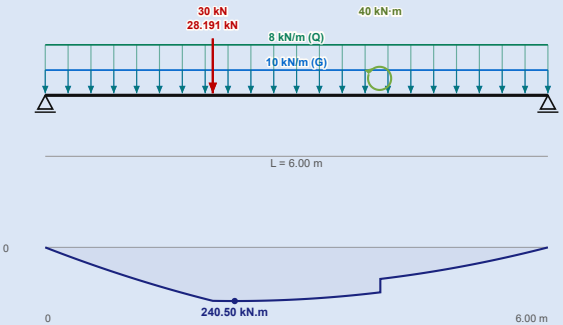
Axial with Moments (Member) (FAIL)
Member PFC-L7 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)
Q PY 28.191 @ 2 m e = 40 mm z_g = 100 mm (kN)
Q M 40.000 @ 4 m (kN.m)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	4.87	145.62	0.00	0.00	20.52	240.50	20.52	-99.54	145.62	—
	End 2, x = 6	150.00C	-3.76	-96.53	0.00	0.00	20.52	@ 2.262	@ —	@ 2.850	96.53	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 150.00, 240.50, 20.52 (Axial: Non-Slender) Class 1
Web classified for biaxial: uniform-compression web bound; limits 33ε / 38ε / 42ε Class 1 Table 5.2
Auto Design Load Cases 1; SLS 1

Local Capacity Check

V _{y,Ed} /V _{pl,y,Rd}	0.048 / 212.754 =	0.000	High Shear
V _{y,Ed,max} /V _{pl,y,Rd}	145.621 / 212.754 =	0.684	OK
V _{pl,y,Rd} = A _v f _y /(√3Y _{M0})	1340 mm² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /Y _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	240.495 / 62.425 =	3.853	Warning
M _{y,Ed} /M _{v,y,Rd} @ x	@ x = 2.262 m: V _{Ed} = 0.048 > V _{pl,Rd} = 0.000: pure shear governs	47749441.757	Warning
V _{z,Ed} /V _{pl,z,Rd}	0 / 308.016 = (A _{v,z} = 1940 mm²; no minor-axis shear in the single-plane model)	0.000	Low Shear
M _{c,z,Rd} = f _y ·W _{pl,z} /Y _{M0}	275 x 60.6/1	16.665 kN.m	
N _{pl,Rd} = A _g ·f _y /Y _{M0}	29.90 x 275/1 = (No bearing / block tearing design)	822.250 kN	
n = N _{Ed} /N _{pl,Rd}	150.000 / 822.250 =	0.182	OK

$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed}/M_{c,y,Rd} + M_{z,Ed}/M_{c,z,Rd}$	$0.182 + 240.495/62.425 + 20.521/16.665 =$	5.266	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3/175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 145.621$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.612	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.612 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.758 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.541	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	145.621	ULS: 1.35G + 1.5Q	0.612	0.541	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.631	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	96.525	ULS: 1.35G + 1.5Q	0.406	0.394	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.852	
$N_{b,y,Rd} = Area \cdot \chi_y \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 1.0 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	2.892	
$N_{b,z,Rd} = Area \cdot \chi_z \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 1.0 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4
$N_{cr,T} = (GI_T + \pi^2 EI_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T})^2}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\lambda_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = Area \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 1.0 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c

$N_{Ed}/N_{b,T,Rd}$	150.000 / 408.410 = (the lower of χ_T and the flexural χ feeds $U_{N,y}$, $U_{N,z}$)	0.367	OK
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Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = M_{cr}/M_{cr,uniform}$	36.575 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.168	k_c only
$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 233.4, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.4$, $M(L/2)=233.4$, $M(3L/4)=115.8$, $M_{max}=240.5$ kN·m	1.214	Serna
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.43$, $\psi = 0.695$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.43$, $\psi = 0.695$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.000 %; $z_g = 100$, 0 mm; 1 combination(s), 3 solve(s)	31.947 kN.m	converged
$\lambda_{LT} = \sqrt{W_{f,y}/M_{cr}}$	$\sqrt{227.0 \times 275 / 31.947}$	1.398	
$\chi_{LT} = Fn(\lambda_{LT}, \phi_{LT}, \alpha_{LT})$	1.398, 1.612, 0.76	0.374	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.374, 1.398, 0.925, 0.989	0.378	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.378 \times 227.0 \times 275 \leq 62.425 =$	23.577 kN.m	
$M_{y,Ed}/M_{b,Rd}$	240.495 / 23.577	10.201	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	150.000 / 408.410 ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	150.000 / 83.643 (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	240.495 / 23.577 ($M_{y,Rk} = 62.425$)	10.201	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	20.521 / 9.295	2.208	Warning
$k_{yy} = C_{my}\{1+0.6\lambda_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\lambda_z U_{N,z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\lambda_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\lambda_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\lambda_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny}+k_{yy} \cdot U_{M,y}+k_{yz} \cdot U_{M,z}$	$0.367+1.188 \times 10.201+2.076 \times 2.208$	17.067	Warning
$U_{Nz}+k_{zy} \cdot U_{M,y}+k_{zz} \cdot U_{M,z}$	$1.793+0.880 \times 10.201+2.076 \times 2.208$	15.358	Warning

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —		P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm		P385 App A

Torsion Bending Design @ 2.241 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.8570 rad = 49.10°	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2$ m	1.097 kN.m ²	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	5.850 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	205.657 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.241$ m: 240.490, 4.090, 204.012 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	27.929	Warning P385 3.1.2

$V_{pl.T.Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})].V_{pl.Rd}$	@ x = 0 m: $\tau_t = 251.90$ ($\tau_w = 5.50$) N/mm ² ; $V_{pl.Rd} = 212.754$	0.000 kN	6.2.7(9)
$V_{Ed}/V_{pl.T.Rd}$	145.621 / 0.000	∞	Warning
$M_y/M_{b.Rd} + C_{mz}M_z/M_{z.Rk} + k_wk_{zw}k_\alpha M_w/M_{f.Rk}$	k_α unbounded: $M_{y.Ed}$ reaches $M_{cr} = 31.947$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T\phi'$ at x = 0 / x = L; total T = $GI_T\phi' - EI_w\phi'' = 5.435 / 4.271$ kN.m	4.870 / 3.764 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ x = 2.34 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	23.48°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
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NOT VERIFIED	$M_{y.Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	99.54 $\leq$ 6000 / 360 = 16.67 mm @ x = 2.85 m	99.54 mm	Warning
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N _{Ed} /N _(pl.Ed)	UNyz	UMyz	Ax+M _{6.6}	Ax+M _{6.6}	Deflection	V/Vpl	MA/Mc	LTB	M-V	N _{b.T}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.182	5.266	1.793	10.201	17.067	15.358	5.972	0.684	3.853	10.201	4774941.757	0.612	2.631	0.367	99.000	99.000	477494

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 47749441.757

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: $M_{y.Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 23.5^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-L7 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing Pure shear failure at M-V check point (6.2.6) = >= 99 (capacity exhausted)

Owner case: ... load applied at angle on beam AND EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM, ON SHEAR CENTER AND ABOVE OR BELOW SHEAR CENTER

PFC 200 x 75 x 23, 6 m SS, unrestrained; L7 = L6 + "EXTERNAL BENDING MOMENT AT ANY POINT ON BEAM": couple 40 kN.m (Q, factored by the combination) at 4 m, applied as a pure M_y in the diagram - the "on / above / below the shear centre" variant is NOT a distinct case for a member tool: a couple has no line of action, so it produces no torque (P.e) and no load-height term (C2 z_g); only forces with a lever arm do

Axial with Moments (Member) (FAIL)
Member PFC-L7 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q UDL 8.000 0-6 m e = 40 mm z_g = 100 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 100 mm (kN)

Q PY 28.191 @ 2 m e = 40 mm z_g = 100 mm (kN)

Q M 40.000 @ 4 m (kN.m)

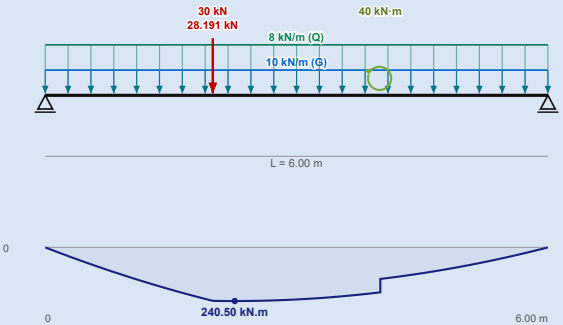
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)

Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	150.00C	4.87	145.62	0.00	0.00	20.52	240.50	20.52	-99.54	145.62	—
	End 2, x = 6	150.00C	-3.76	-96.53	0.00	0.00	20.52	@ 2.262	@ —	@ 2.850	96.53	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 150.00, 240.50, 20.52 Class 1

(Axial: Non-Slender)

Web classified for biaxial: uniform-compression web bound; limits 33ε / 38ε Class 1 Table 5.2

Auto Design Load Cases 1; SLS 1

Local Capacity Check

V_{y,Ed}/V_{pl,y,Rd} 0.048 / 212.754 = 0.000 High Shear

V_{y,Ed,max}/V_{pl,y,Rd} 145.621 / 212.754 = 0.684 OK

V_{pl,y,Rd} = A_vf_y/(√3 Y_{M0}) 1340 mm² x 275/(√3 x 1) 212.754 kN 6.2.6

M_{c,y,Rd} = f_y·W_{pl,y}/Y_{M0} 275 x 227.0/1 62.425 kN.m

M_{y,Ed}/M_{c,y,Rd} 240.495 / 62.425 = 3.853 Warning

M_{y,Ed}/M_{v,y,Rd} @ x @ x = 2.262 m: V_{Ed} = 0.048 > V_{pl,y,Rd} = 0.000: pure shear governs 47749441.757 Warning

V_{z,Ed}/V_{pl,z,Rd} 0 / 308.016 = (A_{v,z} = 1940 mm²; no minor-axis shear in the single-plane model) 0.000 Low Shear

M_{c,z,Rd} = f_y·W_{pl,z}/Y_{M0} 275 x 60.6/1 16.665 kN.m

N_{pl,Rd} = A_g·f_y/Y_{M0} 29.90 x 275/1 = (No bearing / block tearing design) 822.250 kN

n = N_{Ed}/N_{pl,Rd} 150.000 / 822.250 = 0.182 OK

$M_{c,y,Rd}, M_{c,z,Rd}$ (channel, no M_N reduction)	cl 6.2.1(7) linear interaction: channel: linear interaction (cl 6.2.1(7); $\alpha = \beta = 1$, conservative — verified commercial-software basis)	62.425, 16.665 kN.m	6.2.1(7)
$n + M_{y,Ed}/M_{c,y,Rd} + M_{z,Ed}/M_{c,z,Rd}$	$0.182 + 240.495/62.425 + 20.521/16.665 =$	5.266	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3/175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 145.621$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.612	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.612 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 + N_{Ed}/N_{pl,Rd} = 150.000/822.250 = 0.182) = 0.758 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.541	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	145.621	ULS: 1.35G + 1.5Q	0.612	0.541	governs
2	point load 3+4	(a)	0.00	6.002	113.39	0.366	1.000	187.091	87.286	ULS: 1.35G + 1.5Q	0.467	2.631	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	96.525	ULS: 1.35G + 1.5Q	0.406	0.394	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Compression Resistance N.b.Rd

$L_{ey} = K_y \cdot L_y$	1 x 6 = (y-y: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_y = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 1128.42}$ ($N_{cr,y} = \pi^2 EI_y / L_{ey}^2$)	0.852	
$N_{b,y,Rd} = \text{Area} \cdot \chi_y \cdot f_y / \gamma_{M1}$	$29.90 \times 0.629 \times 275 / 10/1 =$	517.570 kN	Curve c
$L_{ez} = K_z \cdot L_z$	1 x 6 = (z-z: held in position at both ends, in direction at neither: 1.0 L)	6 m	end fixities [verify]
$\lambda_z = \sqrt{A \cdot f_y / N_{crz}}$	$\sqrt{29.90 \times 275 / 97.87}$ ($N_{cr,z} = \pi^2 EI_z / L_{ez}^2$)	2.892	
$N_{b,z,Rd} = \text{Area} \cdot \chi_z \cdot f_y / \gamma_{M1}$	$29.90 \times 0.102 \times 275 / 10/1 =$	83.643 kN	Curve c
$i_0^2 = i_y^2 + i_z^2 + y_0^2$	$81.10^2 + 23.90^2 + 50.90^2$ ($y_0 = e_{sc}$ = shear centre to centroid, SCI P385 Table A.3 (Blue Book $e_0 + c_y$))	9739.2 mm ²	6.3.1.4
$N_{cr,T} = (GI_T + \pi^2 EI_w / L_T^2) / i_0^2$	$(81000 \times 11.6e4 + \pi^2 \times 210000 \times 0.0107e12 / 6000^2) / 9739.2$ (I_T, I_w : SCI P385 Table A.3; $L_T = 6$ m = $L_{cr,y}$ (no intermediate twist restraint))	1028.010 kN	6.3.1.4
$N_{cr,TF} = (N_{cr,y} + N_{cr,T}) / 2\beta [1 - \sqrt{1 - 4\beta N_{cr,y} N_{cr,T} / (N_{cr,y} + N_{cr,T})^2}]$	$N_{cr,y} = 1128.425$ (flexure about the axis of symmetry y-y, L_{ey}), $\beta = 1 - (y_0/i_0)^2 = 0.734$; $N_{cr} = \min(N_{cr,T}, N_{cr,TF}) = 709.070$ (TF mode)	709.070 kN	6.3.1.4
$\lambda_T = \sqrt{A \cdot f_y / N_{cr}}$	$\sqrt{29.90 \times 275 / 709.070}$	1.077	
$N_{b,T,Rd} = \text{Area} \cdot \chi_T \cdot f_y / \gamma_{M1}$	$29.90 \times 0.497 \times 275 / 10/1 =$ (curve related to z-z: Table 6.2 U-sections, any axis [verify])	408.410 kN	Curve c

$N_{Ed}/N_{b,T,Rd}$	150.000 / 408.410 = (the lower of χ_T and the flexural χ feeds $U_{N,y}$, $U_{N,z}$)	0.367	OK
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Equivalent Uniform Moment Factors C1, C.mLT, C.mz, and C.my

$C_1 = fn(M_1, M_2, M_0, \psi, \mu)$	0.0, 0.0, 233.4, 1.000, 300.000	1.214	Serna
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=189.4$, $M(L/2)=233.4$, $M(3L/4)=115.8$, $M_{max}=240.5$ kN·m		SN003a
$C_{mLT} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.43$, $\psi = 0.695$, $\alpha_h = 0.000$	1.000	Table B.3
$C_{mz} = \text{Max}(0.6+0.4\psi, 0.4)$	$M = 20.521$, $\psi = 1.000$	1.000	Table B.3
$C_{my} = 1.0$ (diagram outside Table B.3)	$M_h = 0.00$, $M_s = 233.43$, $\psi = 0.695$, $\alpha_h = 0.000$	1.000	Table B.3

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
z_g	$z_g = +100$ mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered		load height
$\bar{\lambda}_{LT} = (L_e/i_z)/\kappa$	$(6000/23.9)/96$ (S275)	2.615	P362 channel
$\chi_{LT} = Fn(\bar{\lambda}_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W_{f,y}/\bar{\lambda}_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	$240.495 / 8.806$	27.310	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Buckling Resistance

$U_{N,y} = N_{Ed}/(\chi_y \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 408.410$ ($N_{Rk} = 822.250$; min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	0.367	OK
$U_{N,z} = N_{Ed}/(\chi_z \cdot N_{Rk}/\gamma_{M1})$	$150.000 / 83.643$ (min with $N_{b,T,Rd} = 408.410$ (6.3.1.4))	1.793	Warning
$U_{M,y} = M_{y,Ed}/(\chi_{LT} \cdot M_{y,Rk}/\gamma_{M1})$	$240.495 / 8.806$ ($M_{y,Rk} = 62.425$)	27.310	Warning
$U_{M,z} = M_{z,Ed}/(M_{z,Rk}/\gamma_{M1})$	$20.521 / 9.295$	2.208	Warning
$k_{yy} = C_{my}\{1+0.6\bar{\lambda}_y U_{N,y}\}$	$1.000\{1+0.6 \times 0.852 \times 0.367\} \leq 1.000(1+0.6 \times 0.367)$	1.188	Table B.1 (Class 3)
$k_{zz} = C_{mz}\{1+0.6\bar{\lambda}_z U_{N,z}\}$	$1.000\{1+0.6 \times 2.892 \times 1.793\} \leq 1.000(1+0.6 \times 1.793)$	2.076	
$k_{yz} = k_{zz}$		2.076	
$k_{zy} = 1 - 0.1\bar{\lambda}_z U_{N,z}/(C_{mLT}-0.25)$	Table B.2 (susceptible): $1-0.05 \cdot \min(\bar{\lambda}_z, 1) \cdot n_z/(C_{mLT}-0.25)$; $\bar{\lambda}_z = 2.892$, $U_{N,z} = 1.793$, $C_{mLT} = 1.000$	0.880	Table B.2
$U_{Ny}+k_{yy} \cdot U_{M,y}+k_{yz} \cdot U_{M,z}$	$0.367+1.188 \times 27.310+2.076 \times 2.208$	37.389	Warning
$U_{Nz}+k_{zy} \cdot U_{M,y}+k_{zz} \cdot U_{M,z}$	$1.793+0.880 \times 27.310+2.076 \times 2.208$	30.422	Warning

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm^4 , 0.0107 dm^6 , 489 mm , —, —		P385 App A
W_{n0}, S_{w1}	40.3 cm^2 , 115 cm^4 ; $W_{n2} = 27.2 \text{ cm}^2$, $S_{w2} = 62.5$, $S_{w3} = 46.3 \text{ cm}^4$, $e_0 = 29.1$, $e_{sc} = 50.9 \text{ mm}$		P385 App A

Torsion Bending Design @ 2.241 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	5.435 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\phi = 0$, warping free)	$L/a = 12.27$	P385 App C
ϕ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	$0.8570 \text{ rad} = 49.10^\circ$	
$B_{Ed} = EI_w \phi''$ (max)	@ $x = 2 \text{ m}$	1.097 kN.m^2	
$M_{w,Ed} = EI_w \phi''/(h-t_f)$	max over span	5.850 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	205.657 kN.m	

$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.241 m: 240.490, 4.090, 204.012 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	27.929	Warning P385
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ x = 0 m: $\tau_t = 251.90$ ($\tau_w = 5.50$) N/mm ² ; $V_{pl,Rd} = 212.754$	0.000 kN	3.1.2
$V_{Ed}/V_{pl,T,Rd}$	145.621 / 0.000	∞	Warning
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \phi'$ at x = 0 / x = L; total T = $GI_T \phi' - EI_w \phi'' = 5.435 / 4.271$ kN.m	4.870 / 3.764 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ x = 2.34 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	23.48°	review

NOT VERIFIED	Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.	NOT VERIFIED
NOT VERIFIED	$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.	NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$					99.54 $\leq$ 6000 / 360 = 16.67 mm @ x = 2.85 m							99.54 mm		Warning		
N_Ed/N_{pl,Rd}	UNyz	UMyz	Ax+M_6.6	Ax+M_6.6	Deflection	V/Vpl	MA/Mc	LTB	M-V	N_b,T	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.182	5.266	1.793	27.310	37.389	30.422	5.972	0.684	3.853	27.310	4774941.757	0.612	2.631	0.367	99.000	99.000	477494

FAIL — governing Pure shear failure at M-V check point (6.2.6) = 47749441.757

NOT COVERED: Combined torsion with direct axial force or imposed minor-axis bending is not implemented as one interaction. Separate checks do not establish adequacy; PASS is blocked.

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: For biaxial bending of this section family each wall parallel to the web is conservatively classified using the uniform-compression limits in Table 5.2; no favourable biaxial stress distribution is assumed.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 23.5^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member)

Member UB-R1 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm (kN)

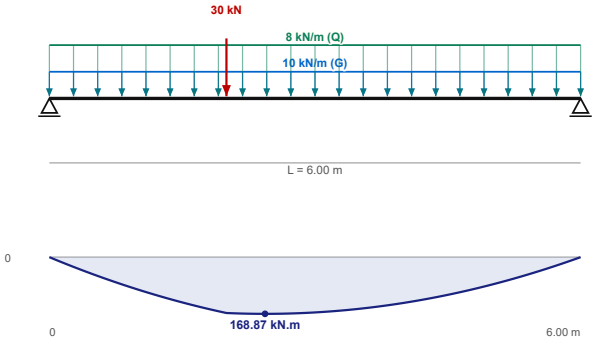
G SW 0.8044 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_c.y.Rd - Fully Restrained Beam

V_y.Ed/V_{pl}.y.Rd 0.006 / 756.260 = 0.000 Low Shear

V_y.Ed,max/V_{pl}.y.Rd 109.758 / 756.260 = 0.145 OK

V_{pl}.y.Rd = A_v.f_y/(√3Y_{M0}) 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6

M_c.y.Rd = f_y.W_{pl}.y/Y_{M0} 275 x 1830.0/1 503.250 kN.m

M_y.Ed/M_c.y.Rd 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

m₁ = f_{yf}.b_f/(f_{yw}.t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x (428.00/16.00)² = 14.311 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

3.402

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 3.402 x 210000 x 9.90³/428.00

1457.610 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

192.79 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(192.79 x 9.90 x 275/1457.610e3)

0.600

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.600 = 0.833 ; 0.833 x 192.79

160.64 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 160.64 x 9.90/1.0

437.340 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251

OK

η₂ + 0.8η₁ ≤ 1.4

0.251 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.179

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

M_{b.Rd} = M_{c.y.Rd}

Fully Restrained

503.250 kN.m

Torsion Design

J, H, a, Q_f, Q_w

69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, —

P385 App A

W_{n0}, S_{w1}

212 cm², 1620 cm⁴

P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)

combination ULS: 1.35G + 1.5Q

4.260 kN.m

Torsion analysis

SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)

L/a = 3.23

P385 App C

φ_{max} (ULS)

L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)

0.0605 rad = 3.47°

B_{Ed} = EI_wφ" (max)

@ x = 2 m

3.435 kN.m²

M_{w.Ed} = EI_wφ"/(h-t_f)

max over span

7.736 kN.m

M_{z.Ed} = φ.M_{y.Ed}

max coincident

10.129 kN.m

(M_y/M_{pl.y})² + M_w/M_{pl.f} + M_z/M_{pl.z}

@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl.y} = 503.250, M_{pl.f} = 41.800, M_{pl.z} = 83.600

0.408

OK P385 3.1.2

V_{pl.T.Rd} = √(1 - τ_t/(1.25f_y/√3)).V_{pl.Rd}

@ x = 0 m: τ_t = 27.22 N/mm²; V_{pl.Rd} = 756.260

702.492 kN

6.2.7(9)

V_{Ed}/V_{pl.T.Rd}

109.758 / 702.492

0.156

OK

End torques T_t

St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m

1.902 / 1.724 kN.m

θ_{ser} ≤ θ_{limit}

@ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)

1.44°

OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360

3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m

3.22 mm

OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	Max
0.336	0.336	0.193	0.145	0.251	0.257	0.408	0.156	0.408

PASS — governing Bending+torsion cross-section (P385 3.1.2) = 0.408

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R2 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.897
Owner case: (reference) unrestrained between fork ends - the base of the restraint variants
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R2 unrestrained between fork ends (U_y + R_x held at both ends, warping free) - the reference for R3..R11

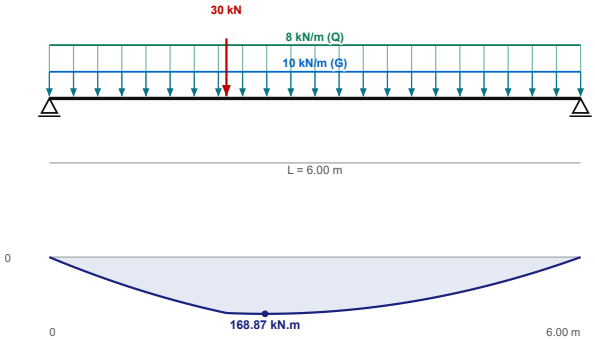
Beam & Beam-Portion (Member)
Member UB-R2 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.8044 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the **END REACTIONS** only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V_{y,Ed}/V_{pl,y}.R_d 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y}.R_d 109.758 / 756.260 = 0.145 OK
V_{pl,y}.R_d = A_v.f_y/(√3Y_{M0}) 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y}.R_d = f_y.W_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y}.R_d 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}.b_f/(f_{yw}.t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

F_{cr} = 0.9k_F.E.t_w³/h_w

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

F_{Ed}/F_{Rd}

η₂ + 0.8η₁ ≤ 1.4

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

0.9 x 3.402 x 210000 x 9.90³/428.00

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

√(192.79 x 9.90 x 275/1457.610e3)

0.5/0.600 = 0.833 ; 0.833 x 192.79

275 x 160.64 x 9.90/1.0

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

3.402

1457.610 kN

100.00 mm

192.79 mm

0.600

160.64 mm

437.340 kN

0.251

0.179

Fig 6.1(c)

6.4(1)

6.5(4)

6.5(4)

6.4(1)

6.4(1)

6.2(1)

OK

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.147

1.162

k_c only

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

λ_{LT} = √W.f_y/M_{cr}

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

M_{b.Rd} = χW_{pl.y}.f_y ≤ M_{c.y.Rd}

M_{y.Ed}/M_{b.Rd}

End 1 v, φ = 0; End 2 v, φ = 0

64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached

√ 1830.0 x 275 / 393.107

1.131, 1.159, 0.49

0.562, 1.131, 0.934, 0.974

0.577 x 1830.0 x 275 ≤ 503.250 =

168.868 / 290.421

6 m

393.107 kN.m

1.131

0.562

0.577

290.421 kN.m

0.581

FE

converged

Curve c

6.3.2.3

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
φ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2.365$ m: 168.802/282.919 + 1.000x9.873/83.600 + 0.665x0.882x1.753x7.403/41.800	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.788$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm	OK
---------------------------------------	--	---------	----

MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.408	0.156	0.897	0.897

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Owner case: (reference) unrestrained between fork ends - the base of the restraint variants
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R2 unrestrained between fork ends ($U_y + R_x$ held at both ends, warping free) - the reference for R3..R11

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$$

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ" (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.365 m: 168.802/285.039 + 0.890 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.408	0.156	0.890	0.890

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$ $s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$ 3.402 Fig 6.1(c)

$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$ $0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$ 1457.610 kN 6.4(1)

$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$ $3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$ 100.00 mm 6.5(4)

$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$ $m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$ 192.79 mm 6.5(4)

$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$ $\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$ 0.600 6.4(1)

$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$ $0.5/0.600 = 0.833$; 0.833×192.79 160.64 mm 6.4(1)

$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w/Y_{M1}$ $275 \times 160.64 \times 9.90/1.0$ 437.340 kN 6.2(1)

F_{Ed}/F_{Rd} $F_{Ed} = 109.758$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 = 0.251 OK

$\eta_2 + 0.8\eta_1 \leq 1.4$ $0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q) 0.179 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

$C_1 = M_{cr}/M_{cr,uniform}$ (critical bay) bay between restraints containing the eigenmode peak (x = 2.952 m); whole-member ratio 1.376 1.036 NA 2.18

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison) 136.3, 166.3, 13.3, 0.820, 0.080 [segment 2–4 m] — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=168.8$, $M(L/2)=164.6$, $M(3L/4)=153.8$, $M_{max}=168.9$ kN·m 1.035 Serna

Lateral Buckling Check M.b.Rd

Governing bay 2–4 m (isolated, fork ends) $M_{y,Ed} = 168.868$, $M_{cr} = 2355.701$, $\chi = 0.462$, $\chi = 0.965$ 485.737 kN.m bay by bay

L_e = portion between restraints End 1 v, $\phi = 0$; End 2 v, $\phi = 0$; lateral restraint points at x = 0, 2, 4, 6 m 2 m FE

M_{cr} = FE eigenvalue (n_{Elem} , mesh error) 63 elements, 0.001 %; 1 combination(s), 2 solve(s) + 1 cached 3128.975 kN.m (whole member merged)

$\lambda_{LT} = \sqrt{W_{pl,y}/M_{cr}}$ $\sqrt{1830.0 \times 275 / 2355.701}$ 0.462

$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$ 0.462, 0.595, 0.49 0.965 Curve c

$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$ 0.965, 0.462, —, 1.000 0.965 isolated bay: f = 1

$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$ $0.965 \times 1830.0 \times 275 \leq 503.250 =$ 485.737 kN.m

$M_{y,Ed}/M_{b,Rd}$ $168.868 / 485.737$ 0.348 OK

Lateral Restraint Portions (bay by bay, fork ends)

Portion	From – To (m)	L _e (m)	M _{y,Ed} (kN.m)	M _{cr} (kN.m)	λ _{LT}	χ _{LT}	M _{b,Rd} (kN.m)	M _{y,Ed} /M _{b,Rd}	
1	0 – 2	2	166.344	3763.377	0.366	1.000	503.250	0.331	
2	2 – 4	2	168.868	2355.701	0.462	0.965	485.737	0.348	governs
3	4 – 6	2	136.344	3675.482	0.370	1.000	503.250	0.271	

χ_{LT} per portion without the f-factor. Whole-member M_{cr} = 3128.975 kN.m. intermediately restrained member: lateral-torsional buckling checked BAY BY BAY between the lateral restraints. Each bay is solved in isolation with fork ends, its own M_{cr} and its own peak moment; the worst bay governs. Governing bay 2–4 m: M_{Ed} = 168.9, M_{cr} = 2355.7, M_{b,Rd} = 485.7 kN·m, utilisation 0.35. (Whole-member eigen M_{cr} = 3129.0 kN·m retained for reference.)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	166.344	ULS: 1.35G + 1.5Q	361.617	9.040	restraint design force, advisory
4	lateral restraint 2	136.344	ULS: 1.35G + 1.5Q	296.399	7.410	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w
W_{n0}, S_{w1}

69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, —
212 cm², 1620 cm⁴

P385 App A
P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)
Torsion analysis
φ_{max} (ULS)
B_{Ed} = EI_wφ'' (max)
M_{w,Ed} = EI_wφ''/(h-t_f)
M_{z,Ed} = φ.M_{y,Ed}
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z}
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)).V_{pl,Rd}
V_{Ed}/V_{pl,T,Rd}
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} +
k_wk_{zw}k_αM_w/M_{f,Rk}
End torques T_t
θ_{ser} ≤ θ_{limit}

combination ULS: 1.35G + 1.5Q
SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)
L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)
@ x = 2 m
max over span
max coincident
@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600
@ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260
109.758 / 702.492
@ x = 2.44 m: 168.868/502.958 + 1.000x9.973/83.600 + 0.665x0.881x1.057x7.332/41.800
St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m
@ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)

4.260 kN.m
L/a = 3.23
0.0605 rad = 3.47°
3.435 kN.m²
7.736 kN.m
10.129 kN.m
0.408 OK P385 3.1.2
702.492 kN 6.2.7(9)
0.156 OK
0.564 OK EN 1993-6
A
1.902 / 1.724 kN.m
1.44° OK

P385 App C

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360

3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m

3.22 mm

OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.348	0.193	0.145	0.251	0.257	0.408	0.156	0.564	0.564

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.564

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R3 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.890
Owner case: RESTRAINED AT SOME POINTS
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R3 "RESTRAINED AT SOME POINTS": lateral restraints (v and phi held) at 2.0 and 4.0 m; the eigen route solves bay by bay, the standard route keeps one segment L_E = L (advisory)

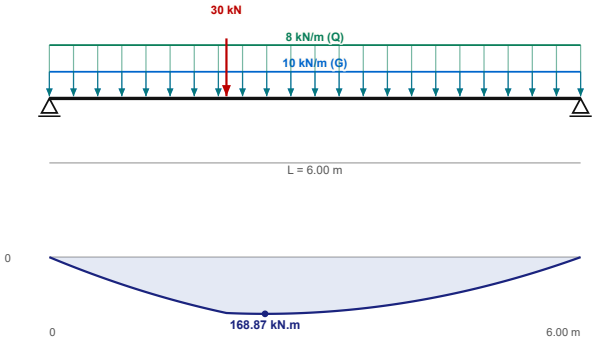
Beam & Beam-Portion (Member)
Member UB-R3 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	1.90	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	-1.72	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}R_d

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	166.344	ULS: 1.35G + 1.5Q	361.617	9.040	restraint design force, advisory
4	lateral restraint 2	136.344	ULS: 1.35G + 1.5Q	296.399	7.410	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ.M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - τ_t/(1.25f_y/√3)).V_{pl,Rd} @ x = 0 m: τ_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.365 m: 168.802/285.039 + 0.890 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.408	0.156	0.890	0.890

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: Standard (closed-form) M_{cr}: the member is treated as ONE segment of length L_E = 1.00×L with C₁ from the whole-member moment diagram; intermediate lateral restraints are not applied on this route (conservative on L_E). Use the FE eigenvalue method for the bay-by-bay M_{cr}, or verify each bay separately.

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr}/M_{cr,uniform}$	669.516 / 606.993 kN.m (shape only: $z_g = z_j = 0$)	1.103	k_c only
$C_1 = \text{fn}(M_1, M_2, M_0, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m [$k = k_w = 1$: clamped / warping-fixed end(s) taken as forks]	1.162	Serna

Lateral Buckling Check $M_{b,Rd}$

L_e = portion between restraints	End 1 $v, \phi, \phi' = 0$; End 2 $v, \phi, \phi' = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	669.516 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 669.516}$	0.867	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.867, 0.896, 0.49	0.722	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.722, 0.867, 0.952, 0.976	0.739	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.739 \times 1830.0 \times 275 \leq 503.250 =$	371.977 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 371.977	0.454	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.45 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	El _w φ''' - Gl _T φ' = m _t (x): warping-torsion FE (240 elements); twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed); closed forms not applicable: warping-fixed end; 2 FE solve(s)	mesh error 0.000 %	FE (≤ 0.5 %)
φ _{max} (ULS)	warping-torsion FE, twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed)	0.0204 rad = 1.17°	
B _{Ed} = El _w φ'' (max)	@ x = 0 m	4.007 kN.m ²	
M _{w,Ed} = El _w φ''/(h-t _f)	max over span	9.024 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	3.406 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.45 m: 168.866, 3.937, 3.326 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 41.800, M _{pl,z} = 83.600	0.247	OK P385 3.1.2
V _{pl,T,Rd} = √(1 - τ _t /(1.25f _y /√3)).V _{pl,Rd}	@ x = 0 m: τ _t = 0.00 N/mm ² ; V _{pl,Rd} = 756.260	756.260 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	109.758 / 756.260	0.145	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.45 m: 168.866/363.160 + 1.000x3.326/83.600 + 0.681x0.960x1.337x3.937/41.800	0.587	OK EN 1993-6 A
End torques T _t	St Venant part Gl _T φ' at x = 0 / x = L; total T = Gl _T φ' - El _w φ''' = 4.380 / 3.540 kN.m	0.000 / 0.000 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.75 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	0.49°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	3.22 mm	OK
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.454	0.193	0.145	0.251	0.257	0.247	0.145	0.587	0.587

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.587

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R4 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.739

Owner case: WARPING RESTRAINED

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R4 "WARPING RESTRAINED": warping ($\phi' = 0$) held at both ends (ends warp flags; LTB eigen $k_w = 0.5$ type and warping-torsion FE)

Beam & Beam-Portion (Member)
Member UB-R4 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x W$ restrained (pinned in plane; LTB fork, warping fixed); End 2: $U_y U_z R_x W$ restrained (pinned in plane; LTB fork, warping fixed) — custom end conditions

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)

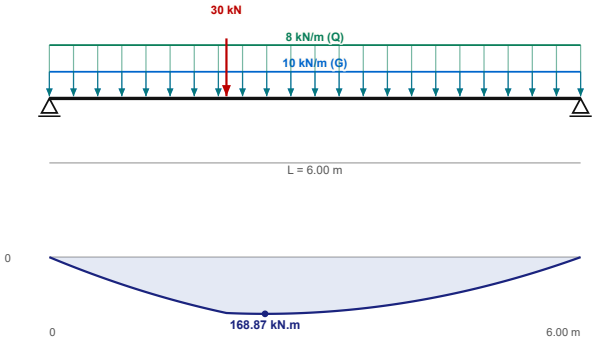
G SW 0.8044 kN/m 0–6 m (automatic)

$Y_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with $G > 1.0$, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 1.0$, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [$Y_{G,inf} = 0.9$, EQU set A]

$Y_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	0.00	0.00	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6.00C	0.00	0.00	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $Y_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c,y,Rd}$

$V_{y,Ed}/V_{pl,y,Rd}$ 0.006 / 756.260 = 0.000 Low Shear

$V_{y,Ed,max}/V_{pl,y,Rd}$ 109.758 / 756.260 = 0.145 OK

$V_{pl,y,Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$ 4763.2 mm² x 275 / ($\sqrt{3}$ x 1) ; $A_y \geq \eta h_w t_w = 4237.2$ mm² 756.260 kN 6.2.6

$M_{c,y,Rd} = f_y W_{pl,y} / \gamma_{M0}$ 275 x 1830.0 / 1 503.250 kN.m

$M_{y,Ed} / M_{c,y,Rd}$ 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30 \epsilon t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web Fig 5.1

$m_1 = f_{yf} b_f / (f_{yw} t_w)$; $m_2 = 0.02(h_w/t_f)^2$ 275 x 191.30 / (275 x 9.90) = 19.323 ; 0.02 x 6.5(1)

(428.00/16.00)² = 14.311 if $\lambda_F > 0.5$, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/428.00	3.402	Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 3.402 x 210000 x 9.90 ³ /428.00	1457.610 kN	6.4(1)
I _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	3.402 x 210000 x 9.90 ² /(2 x 275 x 428.00) = 297.44 ≤ 100.00	100.00 mm	6.5(4)
I _y = min[I _e + t _f ³ /(m ₁ /2 + (I _e /t _f) ² + m ₂), I _e + t _f ³ /(m ₁ + m ₂)]	m ₂ = 14.311 (λ _F > 0.5); min[227.03, 192.79]	192.79 mm	6.5(4)
λ _F = √(I _y .t _w .f _{yw} /F _{cr})	√(192.79 x 9.90 x 275/1457.610e3)	0.600	6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .I _y	0.5/0.600 = 0.833 ; 0.833 x 192.79	160.64 mm	6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 160.64 x 9.90/1.0	437.340 kN	6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
η ₂ + 0.8η ₁ ≤ 1.4	0.251 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	I _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C ₁ basis	general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M _{max} =168.9 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
M _{cr} = Fn(C ₁ , L _e , I _z , I _t , I _w , E)	1.162, 6.000, 1870, 69.2, 0.922, 210000; z _g = 0 (load through the shear centre)	398.376 kN.m	SN003a
λ _{LT} = √W.f _y /M _{cr}	√ 1830.0 x 275 / 398.376	1.124	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.124, 1.151, 0.49	0.566	Curve c
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.583 x 1830.0 x 275 ≤ 503.250 =	293.432 kN.m	
λ _{LT} (P362 6.55 simplified) ; M _{b,Rd} (comparison only)	(1/√C ₁)0.9λ _z √β _w = 1.364; χ _{LT,mod} = 0.451; M _{y,Ed} /M _{b,Rd,simp} = 0.744	226.988 kN.m	P362 6.55 comparison
M _{y,Ed} /M _{b,Rd}	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W _{n0} , S _{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.45 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	El _w φ''' - Gl _T φ' = m _t (x): warping-torsion FE (240 elements); twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed); closed forms not applicable: warping-fixed end; 0 FE solve(s) + 2 cached	mesh error 0.000 %	FE (≤ 0.5 %)
φ _{max} (ULS)	warping-torsion FE, twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed)	0.0204 rad = 1.17°	
B _{Ed} = El _w φ'' (max)	@ x = 0 m	4.007 kN.m ²	
M _{w,Ed} = El _w φ''/(h-t _f)	max over span	9.024 kN.m	
M _{z,Ed} = φ.M _{y,Ed}	max coincident	3.406 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.45 m: 168.866, 3.937, 3.326 kN.m; M _{pl,y} = 503.250, M _{pl,f} = 41.800, M _{pl,z} = 83.600	0.247	OK P385 3.1.2
V _{pl,T,Rd} = √(1 - τ _t /(1.25f _y /√3)).V _{pl,Rd}	@ x = 0 m: τ _t = 0.00 N/mm ² ; V _{pl,Rd} = 756.260	756.260 kN	6.2.7(9)
V _{Ed} /V _{pl,T,Rd}	109.758 / 756.260	0.145	OK
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	@ x = 2.425 m: 168.867/285.039 + 1.000x3.311/83.600 + 0.681x0.960x1.736x3.943/41.800	0.739	OK EN 1993-6 A
End torques T _t	St Venant part Gl _T φ' at x = 0 / x = L; total T = Gl _T φ' - El _w φ''' = 4.380 / 3.540 kN.m	0.000 / 0.000 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.75 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	0.49°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	3.22 mm	OK
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.247	0.145	0.739	0.739

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.739

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: Standard (closed-form) M_{cr}: warping-fixed ends (End 1 and End 2) taken as fork ends, k = k_w = 1 (conservative: the SN003a k = 0.5 / k_w = 0.5 rows are not applied; the FE eigen route uses v' = 0 / φ' = 0 and gives the higher M_{cr}).

6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

3.402

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 3.402 x 210000 x 9.90³/428.00

1457.610 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

192.79 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(192.79 x 9.90 x 275/1457.610e3)

0.600

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.600 = 0.833 ; 0.833 x 192.79

160.64 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 160.64 x 9.90/1.0

437.340 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251

OK

η₂ + 0.8η₁ ≤ 1.4

0.251 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.179

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

1.147

k_c only

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.162

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

End 1 v, φ = 0; End 2 v, φ = 0

6 m

FE

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached

393.107 kN.m

converged

λ_{LT} = √W.f_y/M_{cr}

√ 1830.0 x 275 / 393.107

1.131

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

1.131, 1.159, 0.49

0.562

Curve c

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

0.562, 1.131, 0.934, 0.974

0.577

6.3.2.3

M_{b.Rd} = χW_{pl.y}.f_y ≤ M_{c.y.Rd}

0.577 x 1830.0 x 275 ≤ 503.250 =

290.421 kN.m

M_{y.Ed}/M_{b.Rd}

168.868 / 290.421

0.581

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0}, S_{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
φ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2.365$ m: 168.802/282.919 + 1.000x9.873/83.600 + 0.665x0.882x1.753x7.403/41.800	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.788$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm	OK
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.408	0.156	0.897	0.897

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R5 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.890
Owner case: WARPING FREE
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R5 "WARPING FREE": warping free at both ends (the default fork end; identical input to R2, kept under the owner's heading)

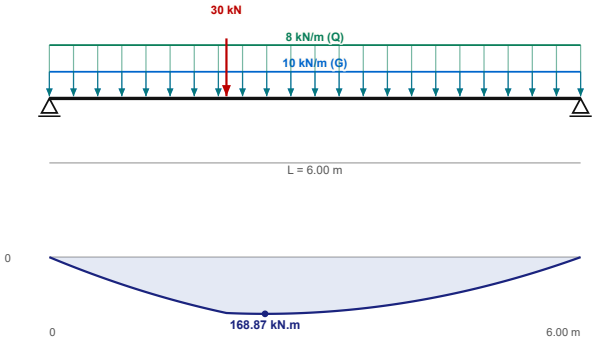
Beam & Beam-Portion (Member)
Member UB-R5 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.8044 kN/m 0–6 m (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the **END REACTIONS** only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V_{y,Ed}/V_{pl,y}.R_d 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y}.R_d 109.758 / 756.260 = 0.145 OK
V_{pl,y}.R_d = A_vf_y/(√3Y_{M0}) 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y}.R_d = f_y·W_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y}.R_d 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833$; 0.833×192.79	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 =$	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2+0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = Fn(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = 0$ (load through the shear centre)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ" (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ·M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)).V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.365 m: 168.802/285.039 + 0.890 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.408	0.156	0.890	0.890

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R6 "DESTABILIZING LOADS" as the switch alone with every $z_g = 0$: the eigen route BLOCKS this contradictory input (load height is carried by z_g), the standard route applies $L_E = 1.2 L$ (BS 5950 practice) - kept to record the refusal; the modelled destabilising case is L_{3t} / L_4 ($z_g = +D/2$)

OK

NOT VERIFIED

Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f	428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30t_f = 453.62$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $437.340 = 0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.251 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.179 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = M_{cr} / M_{cr,uniform}$	393.107 / 342.775 kN.m (shape only: $z_g = z_j = 0$)	1.147	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN.m	1.162	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	393.107 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 393.107}$	1.131	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.131, 1.159, 0.49	0.562	Curve c

$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.562, 1.131, 0.934, 0.974	0.577	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.577 \times 1830.0 \times 275 \leq 503.250 =$	290.421 kN.m	
$M_{y,Ed}/M_{b,Rd}$	168.868 / 290.421	0.581	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 460.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0} , S_{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 3.23$	P385 App C
φ_{max} (ULS)	$L/a = 3.23$; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi'''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.34$ m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ $x = 0$ m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_a M_w/M_{f,Rk}$	@ $x = 2.365$ m: 168.802/282.919 + 1.000x9.873/83.600 + 0.665x0.882x1.753x7.403/41.800	0.897	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.788$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$3.22 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	3.22 mm	OK
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.408	0.156	0.897	0.897

NOT VERIFIED — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

NOT COVERED: Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R6 "DESTABILIZING LOADS" as the switch alone with every $z_g = 0$: the eigen route BLOCKS this contradictory input (load height is carried by z_g), the standard route applies $L_E = 1.2 L$ (BS 5950 practice) - kept to record the refusal; the modelled destabilising case is L_{3t} / L_4 ($z_g = +D/2$)

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 191.30 / (275 \times 9.90) = 19.323$; $0.02 \times (428.00/16.00)^2 = 14.311$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3 / 428.00$	1457.610 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275 / 1457.610e3)}$	0.600 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.600 = 0.833$; 0.833×192.79	160.64 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90 / 1.0$	437.340 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check $M_{b,Rd}$

$L_e = 1.2 L$	$1.2 \times 6 =$	7.2 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	$1.162, 7.200, 1870, 69.2, 0.922, 210000$; $z_g = 0$ (load through the shear centre)	306.318 kN.m	SN003a
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{1830.0 \times 275 / 306.318}$	1.282	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.282, 1.332, 0.49	0.483	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.483, 1.282, 0.928, 0.981	0.493	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.493 \times 1830.0 \times 275 \leq 503.250 =$	248.097 kN.m	
λ_{LT} (P362 6.55 simplified) ; $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.637$; $\chi_{LT,mod} = 0.341$; $M_{y,Ed} / M_{b,Rd,simp} = 0.983$	171.787 kN.m	P362 6.55 comparison
$M_{y,Ed} / M_{b,Rd}$	168.868 / 248.097	0.681	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ" (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.315 m: 168.675/243.284 + 1.000x9.793/83.600 + 0.664x0.883x2.229x7.449/41.800 1.043 **Warning** EN 1993-6 A
k_wk_{zw}k_αM_w/M_{f,Rk}
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.681	0.193	0.145	0.251	0.257	0.408	0.156	1.043	1.043

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 1.043

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R7 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.776

Owner case: STABILIZING LOADS
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R7 "STABILIZING LOADS": all loads hung from the bottom flange z_g = -230 mm (-D/2) with e = +40 mm

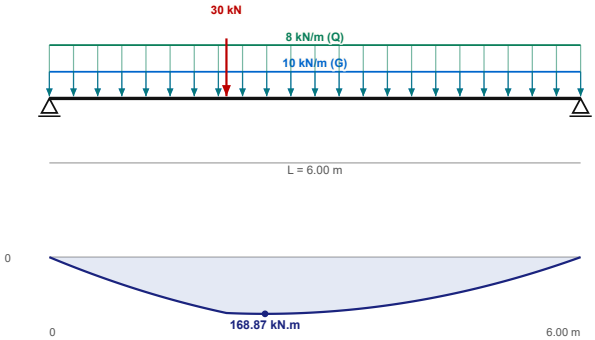
Beam & Beam-Portion (Member)
Member UB-R7 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = -230 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = -230 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = -230 mm (kN)
G SW 0.8044 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}R_d

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}b_f/(f_{yw}t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

F_{cr} = 0.9k_F.E.t_w³/h_w

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

F_{Ed}/F_{Rd}

η₂ + 0.8η₁ ≤ 1.4

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

0.9 x 3.402 x 210000 x 9.90³/428.00

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

√(192.79 x 9.90 x 275/1457.610e3)

0.5/0.600 = 0.833 ; 0.833 x 192.79

275 x 160.64 x 9.90/1.0

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251 + 0.8 x (M_{Ed}/M_{c.y,Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

3.402

1457.610 kN

100.00 mm

192.79 mm

0.600

160.64 mm

437.340 kN

0.251

0.179

Fig 6.1(c)

6.4(1)

6.5(4)

6.5(4)

6.4(1)

6.4(1)

6.2(1)

OK

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = M_{cr}/M_{cr,uniform}

C₁ = fn(M₁, M₂, M₀, ψ, μ) (standard, comparison)

393.107 / 342.775 kN.m (shape only: z_g = z_j = 0)

0.0, 0.0, 164.6, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 kN.m

1.147

1.162

k_c only

Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints

M_{cr} = FE eigenvalue (n_{Elem}, mesh error)

λ_{LT} = √W.f_y/M_{cr}

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

χ_{LT,mod} = Fn(χ_{LT}, λ_{LT}, k_c, f)

M_{b,Rd} = χW_{pl,y}.f_y ≤ M_{c.y,Rd}

M_{y,Ed}/M_{b,Rd}

End 1 v, φ = 0; End 2 v, φ = 0

64 elements, 0.000 %; z_g = -230, 0 mm; 1 combination(s), 0 solve(s) + 3 cached

√ 1830.0 x 275 / 542.681

0.963, 0.986, 0.49

0.662, 0.963, 0.934, 0.969

0.683 x 1830.0 x 275 ≤ 503.250 =

168.868 / 343.812

6 m

542.681 kN.m

0.963

0.662

0.683

343.812 kN.m

0.491

FE

converged

Curve c

6.3.2.3

OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	69.2 cm ⁴ , 0.922 dm ⁶ , 1859 mm, —, —	P385 App A
W_{n0} , S_{w1}	212 cm ² , 1620 cm ⁴	P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.260 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free)	L/a = 3.23	P385 App C
φ_{max} (ULS)	L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)	0.0605 rad = 3.47°	
$B_{Ed} = EI_w \varphi''$ (max)	@ x = 2 m	3.435 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	7.736 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	10.129 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; $M_{pl,y} = 503.250$, $M_{pl,f} = 41.800$, $M_{pl,z} = 83.600$	0.408	OK P385 3.1.2
$V_{pl,T,Rd} = \sqrt{(1 - T_t/(1.25f_y/\sqrt{3}))} \cdot V_{pl,Rd}$	@ x = 0 m: $T_t = 27.22$ N/mm ² ; $V_{pl,Rd} = 756.260$	702.492 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	109.758 / 702.492	0.156	OK
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_{\alpha} M_w/M_{f,Rk}$	@ x = 2.39 m: 168.840/333.034 + 1.000x9.909/83.600 + 0.665x0.881x1.452x7.379/41.800	0.776	OK EN 1993-6 A
End torques T_t	St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.260 / 3.660$ kN.m	1.902 / 1.724 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	1.44°	OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	3.22 mm	OK
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.491	0.193	0.145	0.251	0.257	0.408	0.156	0.776	0.776

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.776

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R7 · UB 457 x 191 x 82 · M_{cr} method: standard · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.890
Owner case: STABILIZING LOADS
UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R7 "STABILIZING LOADS": all loads hung from the bottom flange z_g = -230 mm (-D/2) with e = +40 mm

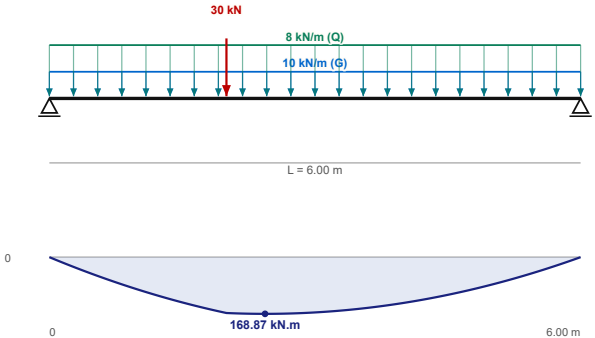
Beam & Beam-Portion (Member)
Member UB-R7 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = -230 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = -230 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = -230 mm (kN)
G SW 0.8044 kN/m 0-6 m (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y}.R_d

V_{y,Ed}/V_{pl,y}.R_d 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y}.R_d 109.758 / 756.260 = 0.145 OK
V_{pl,y}.R_d = A_vf_y/(√3γ_{M0}) 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 mm² 756.260 kN 6.2.6
M_{c,y}.R_d = f_y·W_{pl,y}/γ_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y}.R_d 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 191.30/(275 x 9.90) = 19.323 ; 0.02 x 6.5(1)
(428.00/16.00)² = 14.311 if λ_F > 0.5, else 0

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$s_s, c; k_F$	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00$	3.402	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 3.402 \times 210000 \times 9.90^3/428.00$	1457.610 kN	6.4(1)
$I_e = k_F \cdot E \cdot t_w^2/(2f_{yw} \cdot h_w) \leq s_s + c$	$3.402 \times 210000 \times 9.90^2/(2 \times 275 \times 428.00) = 297.44 \leq 100.00$	100.00 mm	6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 14.311$ ($\lambda_F > 0.5$); $\min[227.03, 192.79]$	192.79 mm	6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw}/F_{cr})}$	$\sqrt{(192.79 \times 9.90 \times 275/1457.610e3)}$	0.600	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1; L_{eff} = \chi_F \cdot I_y$	$0.5/0.600 = 0.833; 0.833 \times 192.79$	160.64 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 160.64 \times 9.90/1.0$	437.340 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 109.758$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =	0.251	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.179	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 164.6, 1.000, 300.000	1.162	Serna
C_1 basis	general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=134.7$, $M(L/2)=164.6$, $M(3L/4)=112.2$, $M_{max}=168.9$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E)$	1.162, 6.000, 1870, 69.2, 0.922, 210000; $z_g = -230$ mm (below the shear centre, stabilising) — not applied: C_2 not published for this moment diagram; M_{cr} is taken with the load at the shear centre (conservative)	398.376 kN.m	SN003a
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{1830.0 \times 275 / 398.376}$	1.124	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.124, 1.151, 0.49	0.566	Curve c
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.566, 1.124, 0.928, 0.971	0.583	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.583 \times 1830.0 \times 275 \leq 503.250 =$	293.432 kN.m	
λ_{LT} (P362 6.55 simplified); $M_{b,Rd}$ (comparison only)	$(1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.364$; $\chi_{LT,mod} = 0.451$; $M_{y,Ed}/M_{b,Rd,simp} = 0.744$	226.988 kN.m	P362 6.55 comparison
$M_{y,Ed}/M_{b,Rd}$	168.868 / 293.432	0.575	OK

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ" (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.365 m: 168.802/285.039 + 0.890 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.873/83.600 + 0.665x0.882x1.736x7.403/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.575	0.193	0.145	0.251	0.257	0.408	0.156	0.890	0.890

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.890

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R8 · UB 457 x 191 x 82 · M_{cr} method: eigen · verdict **PASS** · governing LTB+torsion (EN 1993-6 Annex A) = 0.897

Owner case: EFFECTIVE LENGTHS OVERRIDE

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R8 "EFFECTIVE LENGTHS OVERRIDE": L_E = 0.85 L entered (S.leFactor 0.85) - applied by the standard (closed-form) route; the eigen route derives its own buckling length from the end flags and does not use the factor for M_{cr} (shown for comparison)

Beam & Beam-Portion (Member)
Member UB-R8 UB 457 x 191 x 82

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)

Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)

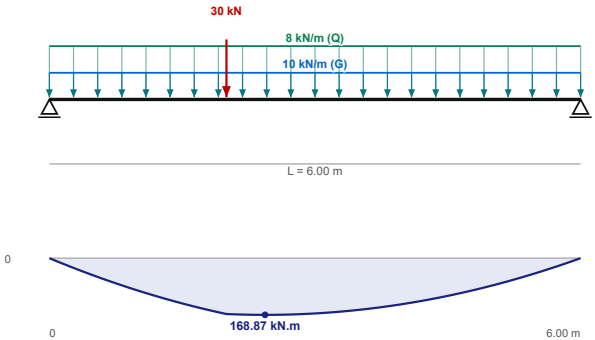
G SW 0.8044 kN/m 0–6 m (automatic)

Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]

ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]

Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only – uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	1.90	109.76	0.00	0.00	0.00	168.87	0.00	-3.22	109.76	—
	End 2, x = 6	0.00C	-1.72	-94.76	0.00	0.00	0.00	@ 2.436	@ —	@ 2.850	94.76	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (82.0 kg/m) 457 x 191 x 82 UB [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 5.03, 41.20, 275, 0.00, 168.87, 0.00 Class 1
(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.006 / 756.260 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 109.758 / 756.260 = 0.145 OK
V_{pl,y,Rd} = A_vf_y/√3Y_{M0} 4763.2 mm² x 275/(√3 x 1) ; A_v ≥ ηh_wt_w = 4237.2 756.260 kN 6.2.6
mm²
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 1830.0/1 503.250 kN.m
M_{y,Ed}/M_{c,y,Rd} 168.868 / 503.250 = 0.336 OK

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm (B = 191.30 ≤ t_w + 30εt_f = 453.62); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 = 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.179 \quad \text{OK 7.2} \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = M_{cr}/M_{cr,uniform} & \quad 393.107 / 342.775 \text{ kN.m (shape only: } z_g = z_j = 0) \quad 1.147 \quad k_c \text{ only} \\ C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) \text{ (standard, comparison)} & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \text{ — general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{max}=168.9 \text{ kN.m} \quad 1.162 \quad \text{Serna} \end{aligned}$$

Lateral Buckling Check $M_{b,Rd}$

$$\begin{aligned} L_e = \text{portion between restraints} & \quad \text{End 1 v, } \varphi = 0; \text{ End 2 v, } \varphi = 0 \quad 6 \text{ m} \quad \text{FE} \\ M_{cr} = \text{FE eigenvalue (n}_{Elem}, \text{ mesh error)} & \quad 64 \text{ elements, } 0.000 \% ; 1 \text{ combination(s), } 0 \text{ solve(s) + 3 cached} \quad 393.107 \text{ kN.m} \quad \text{converged} \\ \lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}} & \quad \sqrt{1830.0 \times 275 / 393.107} \quad 1.131 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 1.131, 1.159, 0.49 \quad 0.562 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.562, 1.131, 0.934, 0.974 \quad 0.577 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.577 \times 1830.0 \times 275 \leq 503.250 = 290.421 \text{ kN.m} \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 290.421 \quad 0.581 \quad \text{OK} \end{aligned}$$

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ" (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.365 m: 168.802/282.919 + 0.897 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.873/83.600 + 0.665x0.882x1.753x7.403/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.581	0.193	0.145	0.251	0.257	0.408	0.156	0.897	0.897

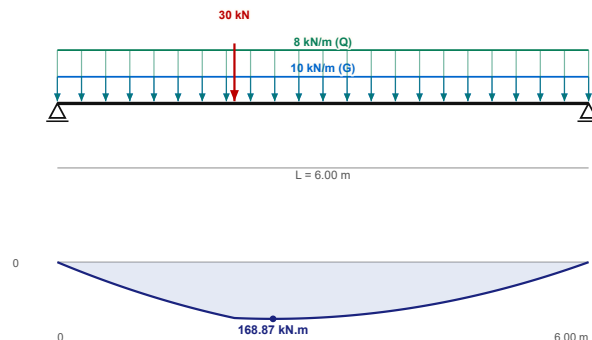
PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.897

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R8 "EFFECTIVE LENGTHS OVERRIDE": $L_E = 0.85 L$ entered (S.leFactor 0.85) - applied by the standard (closed-form) route; the eigen route derives its own buckling length from the end flags and does not use the factor for M_{cr} (shown for comparison)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported



Web h_w, t_w, t_f, b_f 428.00, 9.90, 16.00, 191.30 mm ($B = 191.30 \leq t_w + 30\epsilon t_f = 453.62$); $f_{vw} = f_{vf} = 275$; 1 web Fig. 5.1

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 191.30 / (275 \times 9.90) = 19.323 ; 0.02 \times (428.00/16.00)^2 = 14.311 \text{ if } \lambda_F > 0.5, \text{ else } 0 \quad 6.5(1)$$

$$a = \text{distance between transverse stiffeners} \quad \text{no transverse stiffeners declared: } a = L = 6000 \text{ mm, the full member length (conservative)} \quad 6.4(1)$$

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

$$\begin{aligned} s_s, c ; k_F & \quad s_s = 100.00 \text{ mm (entered); } c = 0.00 \text{ mm (d = 0.00 to the member end) ; } k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/428.00 \quad 3.402 \quad \text{Fig 6.1(c)} \\ F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w & \quad 0.9 \times 3.402 \times 210000 \times 9.90^3/428.00 \quad 1457.610 \text{ kN} \quad 6.4(1) \\ I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c & \quad 3.402 \times 210000 \times 9.90^2 / (2 \times 275 \times 428.00) = 297.44 \leq 100.00 \quad 100.00 \text{ mm} \quad 6.5(4) \\ I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}] & \quad m_2 = 14.311 (\lambda_F > 0.5); \min[227.03, 192.79] \quad 192.79 \text{ mm} \quad 6.5(4) \\ \lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})} & \quad \sqrt{(192.79 \times 9.90 \times 275 / 1457.610 \times 10^3)} \quad 0.600 \quad 6.4(1) \\ \chi_F = 0.5/\lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y & \quad 0.5/0.600 = 0.833 ; 0.833 \times 192.79 \quad 160.64 \text{ mm} \quad 6.4(1) \\ F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1} & \quad 275 \times 160.64 \times 9.90 / 1.0 \quad 437.340 \text{ kN} \quad 6.2(1) \\ F_{Ed}/F_{Rd} & \quad F_{Ed} = 109.758 \text{ kN (ULS: } 1.35G + 1.5Q; \text{ on the bottom flange, no coincident moment) / } 437.340 = 0.251 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/503.250 = 0.000) = 0.251 \leq 1.4 \text{ (ULS: } 1.35G + 1.5Q) \quad 0.251 \quad \text{OK} \\ \eta_2 + 0.8\eta_1 \leq 1.4 & \quad 0.179 \quad \text{OK } 7.2 \end{aligned}$$

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$$\begin{aligned} C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu) & \quad 0.0, 0.0, 164.6, 1.000, 300.000 \quad 1.162 \quad \text{Serna} \\ C_1 \text{ basis} & \quad \text{general moment diagram — Serna et al. quarter-point expression (SCI): } M(L/4)=134.7, M(L/2)=164.6, M(3L/4)=112.2, M_{\max}=168.9 \text{ kN}\cdot\text{m} \quad \text{SN003a} \end{aligned}$$

Lateral Buckling Check $M_{b,Rd}$

$$\begin{aligned} L_e = 0.85 L & \quad 0.85 \times 6 = 5.1 \text{ m} \\ M_{cr} = \text{Fn}(C_1, L_e, I_z, I_t, I_w, E) & \quad 1.162, 5.100, 1870, 69.2, 0.922, 210000; z_g = 0 \quad 510.582 \text{ kN}\cdot\text{m} \quad \text{SN003a} \\ & \quad \text{(load through the shear centre)} \\ \lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}} & \quad \sqrt{1830.0 \times 275 / 510.582} \quad 0.993 \\ \chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT}) & \quad 0.993, 1.015, 0.49 \quad 0.643 \quad \text{Curve c} \\ \chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f) & \quad 0.643, 0.993, 0.928, 0.966 \quad 0.666 \quad 6.3.2.3 \\ M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd} & \quad 0.666 \times 1830.0 \times 275 \leq 503.250 = 335.069 \text{ kN}\cdot\text{m} \\ \lambda_{LT} \text{ (P362 6.55 simplified) ; } M_{b,Rd} & \quad (1/\sqrt{C_1})0.9\lambda_z \sqrt{\beta_w} = 1.159; \chi_{LT,mod} = 0.562; \quad 282.726 \text{ kN}\cdot\text{m} \quad \text{P362 6.55} \\ & \quad M_{y,Ed}/M_{b,Rd,simp} = 0.597 \quad \text{comparison} \\ M_{y,Ed}/M_{b,Rd} & \quad 168.868 / 335.069 \quad 0.504 \quad \text{OK} \end{aligned}$$

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 460.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, — P385 App A
W_{n0}, S_{w1} 212 cm², 1620 cm⁴ P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.260 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 3.23 P385 App C
φ_{max} (ULS) L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10) 0.0605 rad = 3.47°
B_{Ed} = EI_wφ'' (max) @ x = 2 m 3.435 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 7.736 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 10.129 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl,y} = 503.250, M_{pl,f} = 41.800, M_{pl,z} = 83.600 0.408 OK P385 3.1.2
V_{pl,T,Rd} = √(1 - T_t/(1.25f_y/√3)) · V_{pl,Rd} @ x = 0 m: T_t = 27.22 N/mm²; V_{pl,Rd} = 756.260 702.492 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 109.758 / 702.492 0.156 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + @ x = 2.39 m: 168.840/323.840 + 0.794 OK EN 1993-6
k_wk_{zw}k_αM_w/M_{f,Rk} 1.000x9.909/83.600 + 0.665x0.881x1.494x7.379/41.800 A
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m 1.902 / 1.724 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 1.44° OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 3.22 mm OK

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
0.336	0.504	0.193	0.145	0.251	0.257	0.408	0.156	0.794	0.794

PASS — governing LTB+torsion (EN 1993-6 Annex A) = 0.794

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$$

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/428.00

3.402

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 3.402 x 210000 x 9.90³/428.00

1457.610 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

3.402 x 210000 x 9.90²/(2 x 275 x 428.00) = 297.44 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 14.311 (λ_F > 0.5); min[227.03, 192.79]

192.79 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(192.79 x 9.90 x 275/1457.610e3)

0.600

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.600 = 0.833 ; 0.833 x 192.79

160.64 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 160.64 x 9.90/1.0

437.340 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 109.758 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 437.340 =

0.251

OK

η₂ + 0.8η₁ ≤ 1.4

0.251 + 0.8 x (M_{Ed}/M_{c.y.Rd} = 0.000/503.250 = 0.000) = 0.251 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.179

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	109.758	ULS: 1.35G + 1.5Q	0.251	0.179	governs
2	point load 3	(a)	0.00	6.010	172.67	0.427	1.000	470.084	45.000	ULS: 1.35G + 1.5Q	0.096	0.257	OK
6	End 2 reaction	(c)	100.00	3.402	192.79	0.600	0.833	437.340	94.758	ULS: 1.35G + 1.5Q	0.217	0.155	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c.y.Rd} (W_{pl.y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

M_{b.Rd} = M_{c.y.Rd}

Fully Restrained

503.250 kN.m

Torsion Design

J, H, a, Q_f, Q_w

69.2 cm⁴, 0.922 dm⁶, 1859 mm, —, —

P385 App A

W_{n0}, S_{w1}

212 cm², 1620 cm⁴

P385 App A

Torsion Bending Design @ 2.34 m

T_{Ed} (max)

combination ULS: 1.35G + 1.5Q

4.260 kN.m

Torsion analysis

SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)

L/a = 3.23

P385 App C

φ_{max} (ULS)

L/a = 3.23; fork ends, warping free (P385 Cases 3/4/10)

0.0605 rad = 3.47°

B_{Ed} = EI_wφ" (max)

@ x = 2 m

3.435 kN.m²

M_{w.Ed} = EI_wφ"/(h-t_f)

max over span

7.736 kN.m

M_{z.Ed} = φ.M_{y.Ed}

max coincident

10.129 kN.m

(M_y/M_{pl.y})² + M_w/M_{pl.f} + M_z/M_{pl.z}

@ x = 2.34 m: 168.747, 7.426, 9.835 kN.m; M_{pl.y} = 503.250, M_{pl.f} = 41.800, M_{pl.z} = 83.600

0.408

OK P385 3.1.2

V_{pl.T.Rd} = √(1 - τ_t/(1.25f_y/√3)).V_{pl.Rd}

@ x = 0 m: τ_t = 27.22 N/mm²; V_{pl.Rd} = 756.260

702.492 kN

6.2.7(9)

V_{Ed}/V_{pl.T.Rd}

109.758 / 702.492

0.156

OK

End torques T_t

St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.260 / 3.660 kN.m

1.902 / 1.724 kN.m

θ_{ser} ≤ θ_{limit}

@ x = 2.788 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)

1.44°

OK

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360

3.22 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m

3.22 mm

OK

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	Max
0.336	0.336	0.193	0.145	0.251	0.257	0.408	0.156	0.408

PASS — governing Bending+torsion cross-section (P385 3.1.2) = 0.408

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

UB-R10 · UB 457 x 191 x 82 · M_{cr} method: refused · verdict **REFUSED**

Owner case: BOTTOM FLANGE RESTREAINED

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R10 "BOTTOM FLANGE RESTREAINED" = tension-flange restraint only: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Tension-flange (bottom flange of a sagging beam) restraint is not modelled: the tool's restraints (end flags $U_y / R_x / R_z /$ warping and the intermediate lateral restraints v, v', ϕ, ϕ') act at the shear centre and have no height in the cross-section, so a restraint that holds the tension flange laterally while the compression flange is free cannot be described; it needs a restraint-height term in the LTB eigen model (lateral constraint at $z = -h/2$, which couples v and ϕ) or the tension-flange-restraint method of BS 5950-1 Annex G / SCI P093. Treating it as unrestrained (R2) is the conservative bound; treating it as fully restrained (R9) is unsafe.

Owner case: BOTTOM FLANGE RESTREAINED

UB-R11 · UB 457 x 191 x 82 · M_{cr} method: refused · verdict **REFUSED**

Owner case: PARTIALLY RESTRAINED

UB 457 x 191 x 82, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R11 "PARTIALLY RESTRAINED" = elastic (spring) restraint: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Elastic (partial) restraint is not modelled: every restraint in the tool is rigid or absent (each of v , v' , ϕ , ϕ' is held or free), so a lateral or torsional spring of finite stiffness (a purlin, a sheeting rail, a slab with a stiffness) cannot be entered; it needs lateral / rotational spring stiffness terms added to the LTB eigen element matrices (and to the warping-torsion FE) with a stiffness input per restraint and per end, plus the EN 1993-1-1 BB.2 minimum-stiffness test to decide whether a spring counts as a full restraint. The bounds are R2 (no restraint) and R3 (rigid restraints at the same positions).

Owner case: PARTIALLY RESTRAINED

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500 ; 0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000 ; 1.000 \times 144.19$	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$ Fully Restrained 62.425 kN.m

Torsion Design

J, H, a, Q_f , Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
 W_{n0} , S_{w1} 40.3 cm², 115 cm⁴; $W_{n2} = 27.2$ cm², $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm⁴, $e_0 = 29.1$, $e_{sc} = 50.9$ mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free) L/a = 12.27 P385 App C
 φ_{max} (ULS) L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
 $B_{Ed} = EI_w \varphi''$ (max) @ x = 2 m 0.683 kN.m²
 $M_{w,Ed} = EI_w \varphi'' / (h - t_f)$ max over span 3.645 kN.m
 $M_{z,Ed} = \varphi \cdot M_{y,Ed}$ max coincident 109.717 kN.m
 $(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$ @ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$ 14.062 **Warning** P385 3.1.2
 $V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$ @ x = 0 m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm²; $V_{pl,Rd} = 212.754$ 20.965 kN 6.2.7(9)
 $V_{Ed}/V_{pl,T,Rd}$ 107.430 / 20.965 5.124 **Warning**
End torques T_t St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m 3.771 / 3.200 kN.m

$\theta_{ser} \leq \theta_{limit}$

@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)

15.97°

review

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$

$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$

60.86 mm

Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	Max
2.651	2.651	3.652	0.505	2.662	0.452	1.666	14.062	5.124	14.062

FAIL — governing Bending+torsion cross-section (P385 3.1.2) = 14.062

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-R2 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: (reference) unrestrained between fork ends - the base of the restraint variants
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R2 unrestrained between fork ends ($U_y + R_x$ held at both ends, warping free) - the reference for R3..R11

Beam & Beam-Portion (Member) (FAIL)

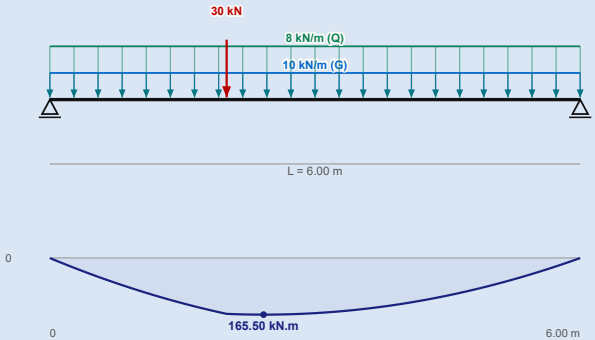
Member PFC-R2 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)
Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)
G SW 0.2296 kN/m 0–6 m $e = 50.9$ mm (automatic)
 $\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]
 $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$	0.021 / 212.754 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	107.430 / 212.754 =	0.505	OK
$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3}$ x 1)	212.754 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 227.0/1	62.425 kN.m	
$M_{y.Ed}/M_{c.y.Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y.Ed}/M_{v.y.Rd} @ x$	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl.Rd} = 20.965$: pure shear governs	2.662	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s_s, c ; k_F	$s_s = 100.00$ mm (entered); $c = 0.00$ mm ($d = 0.00$ to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $237.920 = 0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)] . V _{pl,Rd}	@ x = 0 m: τ _t = 195.04 (τ _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 20.965	5.124	Warning
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 35.819 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _{alpha} is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_{alpha} is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R2 unrestrained between fork ends ($U_y + R_x$ held at both ends, warping free) - the reference for R3..R11

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F · E · t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
l _e = k _F · E · t _w ² /(2f _{yw} · h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(l _y · t _w · f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F · l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} · L _{eff} · t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1 L	1 x 6 =	6 m	
λ _{LT} = (L _e /i _z)/k	(6000/23.9)/96 (S275)	2.615	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.615, 3.906, 0.76	0.141	Curve d
M _{b,Rd} = χW _{pl,y} · f _y (k chain, no f)	0.141 x 227.0 x 275 =	8.806 kN.m	
M _{cr} (back-calculated) = W _f · λ _{LT} ⁻²	227.0 x 275 / 2.615 ²	9.128 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 8.806	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h = \text{overall depth}$, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2 \text{ cm}^2$, $S_{w2} = 62.5$, $S_{w3} = 46.3 \text{ cm}^4$, $e_0 = 29.1$, $e_{sc} = 50.9 \text{ mm}$	P385 App A

Torsion Bending Design @ 2.415 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415 \text{ m}$: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128 \text{ kN.m}$	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707 \text{ kN.m}$	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465 \text{ m}$, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm}$ @ $x = 2.85 \text{ m}$	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Fig 5.1

$m_1 = f_{yf} \cdot b_F / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$ (critical bay)	bay between restraints containing the eigenmode peak (x = 2.952 m); whole-member ratio 1.345	1.037	NA 2.18
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	133.2, 163.2, 12.9, 0.816, 0.079 [segment 2–4 m] — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=165.4$, $M(L/2)=161.1$, $M(3L/4)=150.4$, $M_{max}=165.5$ kN·m	1.036	Serna

Lateral Buckling Check M.b.Rd

Governing bay 2–4 m (isolated, fork ends)	$M_{y,Ed} = 165.503$, $M_{cr} = 118.953$, $\chi = 0.724$, $\chi = 0.742$	46.302 kN.m	bay by bay
L_e = portion between restraints	End 1 v, $\phi = 0$; End 2 v, $\phi = 0$; lateral restraint points at x = 0, 2, 4, 6 m	2 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	63 elements, 0.001 %; 1 combination(s), 2 solve(s) + 1 cached	154.263 kN.m (whole member)	converged
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{227.0 \times 275 / 118.953}$	0.724	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	0.724, 0.820, 0.76	0.742	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.742, 0.724, —, 1.000	0.742	isolated bay: f = 1
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.742 \times 227.0 \times 275 \leq 62.425 =$	46.302 kN.m	
$M_{y,Ed} / M_{b,Rd}$	$165.503 / 46.302$	3.574	Warning

Lateral Restraint Portions (bay by bay, fork ends)

Portion	From – To (m)	L _e (m)	M _{y,Ed} (kN.m)	M _{cr} (kN.m)	λ _{LT}	χ _{LT}	M _{b,Rd} (kN.m)	M _{y,Ed} /M _{b,Rd}	
1	0 – 2	2	163.240	188.358	0.576	0.855	53.375	3.058	
2	2 – 4	2	165.503	118.953	0.724	0.742	46.302	3.574	governs
3	4 – 6	2	133.240	183.987	0.582	0.850	53.040	2.512	

χ_{LT} per portion without the f-factor. Whole-member M_{cr} = 154.263 kN.m. intermediately restrained member: lateral-torsional buckling checked BAY BY BAY between the lateral restraints. Each bay is solved in isolation with fork ends, its own M_{cr} and its own peak moment; the worst bay governs. Governing bay 2–4 m: M_{Ed} = 165.5, M_{cr} = 119.0, M_{b,Rd} = 46.3 kN·m, utilisation 3.57. (Whole-member eigen M_{cr} = 154.3 kN·m retained for reference.)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	163.240	ULS: 1.35G + 1.5Q	816.199	20.405	restraint design force, advisory
4	lateral restraint 2	133.240	ULS: 1.35G + 1.5Q	666.199	16.655	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w
W_{n0}, S_{w1}

11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, —
40.3 cm², 115 cm⁴; W_{n2} = 27.2 cm², S_{w2} = 62.5, S_{w3} = 46.3 cm⁴, e₀ = 29.1, e_{sc} = 50.9 mm

P385 App A
P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)
Torsion analysis
φ_{max} (ULS)
B_{Ed} = EI_wφ'' (max)
M_{w,Ed} = EI_wφ''/(h-t_f)
M_{z,Ed} = φ.M_{y,Ed}
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z}
V_{pl,T.Rd} = [√(1 - τ_t/(1.25f_y/√3)) - τ_w/(f_y/√3)].V_{pl,Rd}
V_{Ed}/V_{pl,T.Rd}
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_αM_w/M_{f,Rk}
End torques T_t
θ_{ser} ≤ θ_{limit}

combination ULS: 1.35G + 1.5Q
SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)
L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)
@ x = 2 m
max over span
max coincident
@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M_{pl,y} = 62.425, M_{pl,f} = 4.834, M_{pl,z} = 16.665
@ x = 0 m: τ_t = 195.04 (τ_w = 5.22) N/mm²; V_{pl,Rd} = 212.754
107.430 / 20.965
k_α unbounded: M_{y,Ed} reaches M_{cr} = 154.263 kN.m
St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.307 / 3.707 kN.m
@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)

4.307 kN.m
L/a = 12.27
0.6646 rad = 38.08°
0.683 kN.m²
3.645 kN.m
109.717 kN.m
14.062
20.965 kN
5.124
—
3.771 / 3.200 kN.m
15.97°

P385 App C
P385 App C
Warning P385 3.1.2
6.2.7(9)
Warning
BLOCKED
review

NOT VERIFIED

M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360

60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m

60.86 mm

Warning

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	3.574	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_{α} is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^{\circ}$ exceeds the $\sim 2^{\circ}$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-R3 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)

Owner case: RESTRAINED AT SOME POINTS

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R3 "RESTRAINED AT SOME POINTS": lateral restraints (v and ϕ held) at 2.0 and 4.0 m; the eigen route solves bay by bay, the standard route keeps one segment $L_E = L$ (advisory)

Beam & Beam-Portion (Member) (FAIL)

Member PFC-R3 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork) — simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q

G UDL 10.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)

Q UDL 8.000 0–6 m $e = 40$ mm $z_g = 0$ mm (kN/m)

Q PY 30.000 @ 2 m $e = 40$ mm $z_g = 0$ mm (kN)

G SW 0.2296 kN/m 0–6 m $e = 50.9$ mm (automatic)

$\gamma_{G,inf}$ companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G

> 1.0, variable factors as entered

ULS C1: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 1.0$, STR set B]

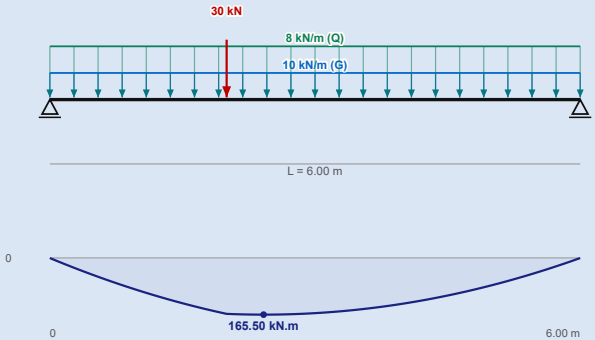
ULS C2: ULS: 1.35G + 1.5Q [$\gamma_{G,inf} = 0.9$, EQU set A]

$\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member

(single-source permanent action) for the END REACTIONS only

– uplift / hold-down forces and the web-bearing reactions;

the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9) $R_{min} \geq 0$ OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]

Class = $F_n(c/t, d/t, f_y, N, M_y, M_z)$ 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1

(Axial: Non-Slender)

Auto Design Load Cases 1; SLS 1

Moment Capacity Check $M_{c.y.Rd}$

$V_{y.Ed}/V_{pl.y.Rd}$	0.021 / 212.754 =	0.000	Low Shear
$V_{y.Ed,max}/V_{pl.y.Rd}$	107.430 / 212.754 =	0.505	OK
$V_{pl.y.Rd} = A_y f_y / (\sqrt{3} \gamma_{M0})$	1340 mm ² x 275 / ($\sqrt{3}$ x 1)	212.754 kN	6.2.6
$M_{c.y.Rd} = f_y \cdot W_{pl.y} / \gamma_{M0}$	275 x 227.0/1	62.425 kN.m	
$M_{y.Ed}/M_{c.y.Rd}$	165.504 / 62.425 =	2.651	Warning
$M_{y.Ed}/M_{v.y.Rd} @ x$	@ x = 2 m: $V_{Ed} = 55.810 > V_{pl.Rd} = 20.965$: pure shear governs	2.662	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\epsilon t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s_s, c ; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$\lambda_{LT} = (L_e / i_z) / k$	$(6000 / 23.9) / 96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (k chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed} / M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
2	lateral restraint 1	163.240	ULS: 1.35G + 1.5Q	816.199	20.405	restraint design force, advisory
4	lateral restraint 2	133.240	ULS: 1.35G + 1.5Q	666.199	16.655	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
W_{n0}, S_{w1} 40.3 cm², 115 cm⁴; W_{n2} = 27.2 cm², S_{w2} = 62.5, S_{w3} = 46.3 cm⁴, e₀ = 29.1, e_{sc} = 50.9 mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free) L/a = 12.27 P385 App C
φ_{max} (ULS) L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
B_{Ed} = EI_wφ" (max) @ x = 2 m 0.683 kN.m²
M_{w,Ed} = EI_wφ"/(h-t_f) max over span 3.645 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 109.717 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M_{pl,y} = 62.425, M_{pl,f} = 4.834, M_{pl,z} = 16.665 14.062 **Warning** P385 3.1.2
V_{pl,T,Rd} = [√(1 - τ_t/(1.25f_y/√3)) - τ_w/(f_y/√3)] · V_{pl,Rd} @ x = 0 m: τ_t = 195.04 (τ_w = 5.22) N/mm²; V_{pl,Rd} = 212.754 20.965 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 107.430 / 20.965 5.124 **Warning**
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_αM_w/M_{f,Rk} k_α unbounded: M_{y,Ed} reaches M_{cr} = 9.128 kN.m — **BLOCKED**
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.307 / 3.707 kN.m 3.771 / 3.200 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 15.97° review

NOT VERIFIED

M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 60.86 mm **Warning**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down

forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: Standard (closed-form) M_{cr} : the member is treated as ONE segment of length $L_E = 1.00 \times L$ with C_1 from the whole-member moment diagram; intermediate lateral restraints are not applied on this route (conservative on L_E). Use the FE eigenvalue method for the bay-by-bay M_{cr} , or verify each bay separately.

$$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w) ; m_2 = 0.02(h_w/t_f)^2 \quad 275 \times 75.00 / (275 \times 6.00) = 12.500 ; 0.02 \times (175.00/12.50)^2 = 3.920 \text{ if } \lambda_{\bar{c}} > 0.5, \text{ else } 0 \quad 6.5(1)$$

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429	Fig 6.1(c)
F _{cr} = 0.9k _F .E.t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN	6.4(1)
l _e = k _F .E.t _w ² /(2f _{yw} .h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm	6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm	6.5(4)
λ _F = √(l _y .t _w .f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433	6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F .l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm	6.4(1)
F _{Rd} = f _{yw} .L _{eff} .t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN	6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452	OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323	OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = M _{cr} /M _{cr,uniform}	41.619 / 37.351 kN.m (shape only: z _g = z _j = 0)	1.114	k _c only
C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ) (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN.m [k = k _w = 1: clamped / warping-fixed end(s) taken as forks]	1.164	Serna

Lateral Buckling Check M.b.Rd

L _e = portion between restraints	End 1 v, φ, φ' = 0; End 2 v, φ, φ' = 0	6 m	FE
M _{cr} = FE eigenvalue (n _{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 2 solve(s) + 1 cached	41.619 kN.m	converged
λ _{LT} = √(W.f _y /M _{cr})	√ 227.0 x 275 / 41.619	1.225	
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	1.225, 1.376, 0.76	0.444	Curve d
χ _{LT,mod} = Fn(χ _{LT} , λ _{LT} , k _c , f)	0.444, 1.225, 0.947, 0.983	0.452	6.3.2.3
M _{b,Rd} = χW _{pl,y} .f _y ≤ M _{c,y,Rd}	0.452 x 227.0 x 275 ≤ 62.425 =	28.191 kN.m	
M _{y,Ed} /M _{b,Rd}	165.504 / 28.191	5.871	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
W_{n0}, S_{w1} 40.3 cm², 115 cm⁴; W_{n2} = 27.2 cm², S_{w2} = 62.5, S_{w3} = 46.3 cm⁴, e₀ = 29.1, e_{sc} = 50.9 mm P385 App A

Torsion Bending Design @ 2.425 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis EI_wφ''' - GI_Tφ' = m_t(x): warping-torsion FE (240 mesh error 0.008 % FE (≤ 0.5 %)
elements); twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed); closed forms not applicable: warping-fixed end; 2 FE solve(s)
φ_{max} (ULS) warping-torsion FE, twist restrained at x = 0 m (φ = 0, φ' = 0: warping fixed), x = 6 m (φ = 0, φ' = 0: warping fixed) 0.4823 rad = 27.63°
B_{Ed} = EI_wφ'' (max) @ x = 0 m 1.871 kN.m²
M_{w,Ed} = EI_wφ''/(h-t_f) max over span 9.979 kN.m
M_{z,Ed} = φ · M_{y,Ed} max coincident 79.532 kN.m
(M_y/M_{pl,y})² + M_w/M_{pl,f} + M_z/M_{pl,z} @ x = 2.425 m: 165.503, 2.219, 78.950 kN.m; M_{pl,y} = 62.425, M_{pl,f} = 4.834, M_{pl,z} = 16.665 12.226 **Warning** P385 3.1.2
V_{pl,T,Rd} = [√(1 - τ_t/(1.25f_y/√3)) - τ_w/(f_y/√3)] · V_{pl,Rd} @ x = 0.625 m: τ_t = 126.74 (τ_w = 12.32) N/mm²; 111.396 kN 6.2.7(9)
V_{Ed}/V_{pl,T,Rd} 91.299 / 111.396 0.820 OK
M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_αM_w/M_{f,Rk} k_α unbounded: M_{y,Ed} reaches M_{cr} = 41.619 kN.m — **BLOCKED**
End torques T_t St Venant part GI_Tφ' at x = 0 / x = L; total T = GI_Tφ' - EI_wφ''' = 4.363 / 3.652 kN.m 0.000 / 0.000 kN.m
θ_{ser} ≤ θ_{limit} @ x = 2.5 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 11.73° review

NOT VERIFIED

M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360 60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m 60.86 mm **Warning**

MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	5.871	3.652	0.505	0.452	1.666	12.226	0.820	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 11.7° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down

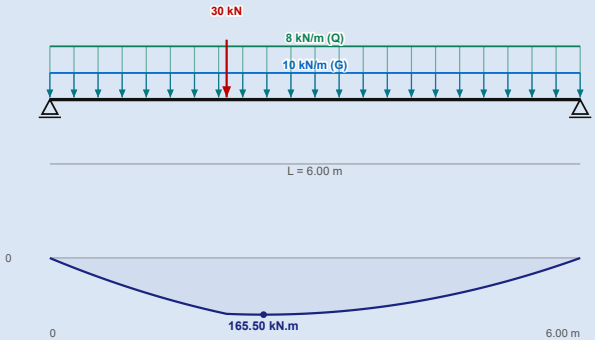
forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-R4 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: WARPING RESTRAINED
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R4 "WARPING RESTRAINED": warping (phi' = 0) held at both ends (ends warp flags; LTB eigen k_w = 0.5 type and warping-torsion FE)

Beam & Beam-Portion (Member) (FAIL)
Member PFC-R4 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)
End conditions: End 1: U_x U_y U_z R_x W restrained (pinned in plane; LTB fork, warping fixed); End 2: U_y U_z R_x W restrained (pinned in plane; LTB fork, warping fixed) — custom end conditions

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	0.00	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	0.00	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)
Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd
V_{y,Ed}/V_{pl,y,Rd} 0.021 / 212.754 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 107.430 / 212.754 = 0.505 OK
V_{pl,y,Rd} = A_ff_y/(√3V_{M0}) 1340 mm² x 275/(√3 x 1) 212.754 kN 6.2.6
M_{c,y,Rd} = f_y·W_{pl,y}/V_{M0} 275 x 227.0/1 62.425 kN.m
M_{y,Ed}/M_{c,y,Rd} 165.504 / 62.425 = 2.651 **Warning**

Web Transverse Forces (EN 1993-1-5 cl 6)
Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15t_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1
m₁ = f_{yf}·b_f/(f_{yw}·t_w) ; m₂ = 0.02(h_w/t_f)² 275 x 75.00/(275 x 6.00) = 12.500 ; 0.02 x (175.00/12.50)² = 3.920 if λ_F > 0.5, else 0 6.5(1)

a = distance between transverse stiffeners
no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative) 6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s, c ; k_F

s_s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k_F = 2 + 6(s_s + c)/h_w ≤ 6 = 2 + 6 x 100.00/175.00

5.429

Fig 6.1(c)

F_{cr} = 0.9k_F.E.t_w³/h_w

0.9 x 5.429 x 210000 x 6.00³/175.00

1266.377 kN

6.4(1)

l_e = k_F.E.t_w²/(2f_{yw}.h_w) ≤ s_s + c

5.429 x 210000 x 6.00²/(2 x 275 x 175.00) = 426.39 ≤ 100.00

100.00 mm

6.5(4)

l_y = min[l_e + t_f√(m₁/2 + (l_e/t_f)² + m₂), l_e + t_f√(m₁ + m₂)]

m₂ = 0.000 (first pass λ_F = 0.443 ≤ 0.5 with m₂ = 3.920, so m₂ = 0); min[204.77, 144.19]

144.19 mm

6.5(4)

λ_F = √(l_y.t_w.f_{yw}/F_{cr})

√(144.19 x 6.00 x 275/1266.377e3)

0.433

6.4(1)

χ_F = 0.5/λ_F ≤ 1 ; L_{eff} = χ_F.l_y

0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19

144.19 mm

6.4(1)

F_{Rd} = f_{yw}.L_{eff}.t_w/γ_{M1}

275 x 144.19 x 6.00/1.0

237.920 kN

6.2(1)

F_{Ed}/F_{Rd}

F_{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =

0.452

OK

η₂ + 0.8η₁ ≤ 1.4

0.452 + 0.8 x (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)

0.323

OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C₁ = fn(M₁, M₂, M_o, ψ, μ)

0.0, 0.0, 161.1, 1.000, 300.000

1.164

Serna

C₁ basis

channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI):
M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6,
M_{max}=165.5 kN·m

SN003a

Lateral Buckling Check M.b.Rd

L_e = 1 L

1 x 6 =

6 m

λ_{LT} = (L_e/i_z)/κ

(6000/23.9)/96 (S275)

2.615

P362 channel

χ_{LT} = Fn(λ_{LT}, Φ_{LT}, α_{LT})

2.615, 3.906, 0.76

0.141

Curve d

M_{b,Rd} = χW_{pl,y}.f_y (κ chain, no f)

0.141 x 227.0 x 275 =

8.806 kN.m

M_{cr} (back-calculated) = W.f_y/λ_{LT}²

227.0 x 275 / 2.615²

9.128 kN.m

Annex A only

M_{y,Ed}/M_{b,Rd}

165.504 / 8.806

18.794

Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A

station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0} , S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.425 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	$EI_w\phi''' - GI_T\phi' = m_t(x)$: warping-torsion FE (240 elements); twist restrained at $x = 0$ m ($\phi = 0$, $\phi' = 0$: warping fixed), $x = 6$ m ($\phi = 0$, $\phi' = 0$: warping fixed); closed forms not applicable: warping-fixed end; 0 FE solve(s) + 2 cached	mesh error 0.008 %	FE (≤ 0.5 %)
ϕ_{max} (ULS)	warping-torsion FE, twist restrained at $x = 0$ m ($\phi = 0$, $\phi' = 0$: warping fixed), $x = 6$ m ($\phi = 0$, $\phi' = 0$: warping fixed)	0.4823 rad = 27.63°	
$B_{Ed} = EI_w\phi''$ (max)	@ $x = 0$ m	1.871 kN.m ²	
$M_{w,Ed} = EI_w\phi''/(h-t_f)$	max over span	9.979 kN.m	
$M_{z,Ed} = \phi \cdot M_{y,Ed}$	max coincident	79.532 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.425$ m: 165.503, 2.219, 78.950 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	12.226	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0.625$ m: $\tau_t = 126.74$ ($\tau_w = 12.32$) N/mm ² ; $V_{pl,Rd} = 212.754$	111.396 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	91.299 / 111.396	0.820	OK
$M_y/M_{b,Rd} + C_{mz}M_z/M_{z,Rk} + k_wk_{zw}k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T\phi'$ at $x = 0$ / $x = L$; total $T = GI_T\phi' - EI_w\phi''' = 4.363$ / 3.652 kN.m	0.000 / 0.000 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.5$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	11.73°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	0.452	1.666	12.226	0.820	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 11.7^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

ADVISORY: Standard (closed-form) M_{cr} : warping-fixed ends (End 1 and End 2) taken as fork ends, $k = k_w = 1$ (conservative: the SN003a $k = 0.5$ / $k_w = 0.5$ rows are not applied; the FE eigen route uses $v' = 0$ / $\phi' = 0$ and gives the higher M_{cr}).

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)] . V _{pl,Rd}	@ x = 0 m: τ _t = 195.04 (τ _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 20.965	5.124	Warning
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 35.819 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _{alpha} is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_{alpha} is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

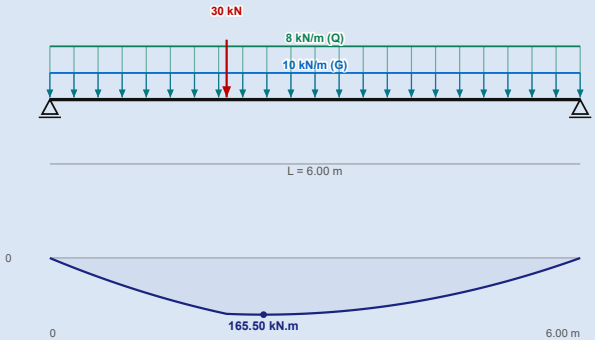
ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Beam & Beam-Portion (Member) (FAIL)
Member PFC-R5 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)
End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0–6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0–6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 – G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member
(single-source permanent action) for the END REACTIONS only
– uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V _{y,Ed} /V _{pl,y,Rd}	0.021 / 212.754 =	0.000	Low Shear
V _{y,Ed,max} /V _{pl,y,Rd}	107.430 / 212.754 =	0.505	OK
V _{pl,y,Rd} = A _v f _y /(√3 V _{M0})	1340 mm ² x 275/(√3 x 1)	212.754 kN	6.2.6
M _{c,y,Rd} = f _y ·W _{pl,y} /V _{M0}	275 x 227.0/1	62.425 kN.m	
M _{y,Ed} /M _{c,y,Rd}	165.504 / 62.425 =	2.651	Warning
M _{y,Ed} /M _{v,y,Rd} @ x	@ x = 2 m: V _{Ed} = 55.810 > V _{pl,Rd} = 20.965: pure shear governs	2.662	Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15εt_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s_s, c ; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $237.920 = 0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C_1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
$\lambda_{LT} = (L_e/i_z)/k$	$(6000/23.9)/96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (k chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed} / M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 9.128$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED	$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

NOT VERIFIED

Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

NOT
VERIFIED

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w , t_w , t_f , b_f	175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15t_f = 179.33$); $f_{yw} = f_{yf} = 275$; 1 web	Fig 5.1
$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)

Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]

s_s , c; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end); $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429	Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN	6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39$ ≤ 100.00	100.00 mm	6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm	6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377)} = 0.433$	0.433	6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot l_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm	6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN	6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / $237.920 = 0.452 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.452	OK
$\eta_2 + 0.8\eta_1 \leq 1.4$		0.323	OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr}/M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d

$\chi_{LT,mod} = F_n(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed}/M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 35.819$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{y,Ed}/M_{b,Rd}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: Destabilising loading is ticked but every load height z_g is 0 (load at the shear centre): the eigenvalue M_{cr} carries the load height exactly through z_g and does not apply the $\times 1.2 L_E$ device of the closed-form route, so this solve would treat the load as non-destabilising. Enter the load height (e.g. $z_g = +h/2$ for a top-flange load, with eccentricity/height inputs on) or untick the switch; PASS is blocked.

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{\max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,\inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

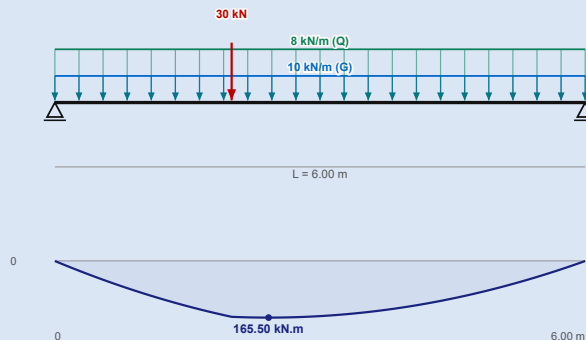
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R6 "DESTABLIZING LOADS" as the switch alone with every $z_g = 0$: the eigen route BLOCKS this contradictory input (load height is carried by z_g), the standard route applies $L_E = 1.2 L$ (BS 5950 practice) - kept to record the refusal; the modelled destabilising case is L_{3t} / L_4 ($z_g = +D/2$)

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: $U_x U_y U_z R_x$ restrained (pinned in plane; LTB fork); End 2: $U_y U_z R_x$ restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

the moment and shear envelopes keep the entered γ_G , so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)

Member No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0.00C	3.77	107.43	0.00	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6.00C	-3.20	-92.43	0.00	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: $1.35G + 1.5Q$ at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. $V_z = 0$: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.

Uplift	no end lifts in any of the 4 combinations (all reactions ≥ 0 ; incl. the 2 $\gamma_{G,inf}$ companions with G at 1.0 and 0.9)	$R_{min} \geq 0$	OK
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Classification and Effective Area (EN 1993: 2006)

(Axial: Non-Slender)

Class 1

Auto Design Load Cases 1: SLS 1

Moment Capacity Check M.c.y.Rd

shear governs

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm ($B = 75.00 \leq t_w + 15\epsilon t_f = 179.33$); $f_{vw} = f_{vf} = 275$; 1 web

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F · E · t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
l _e = k _F · E · t _w ² /(2f _{yw} · h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(l _y · t _w · f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F · l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} · L _{eff} · t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C₁

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 1.2 L	1.2 x 6 =	7.2 m	
λ _{LT} = (L _e /i _z)/k	(7200/23.9)/96 (S275)	3.138	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	3.138, 5.233, 0.76	0.102	Curve d
M _{b,Rd} = χW _{pl,y} · f _y (k chain, no f)	0.102 x 227.0 x 275 =	6.339 kN.m	
M _{cr} (back-calculated) = W _f · λ _{LT} ⁻²	227.0 x 275 / 3.138 ²	6.339 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 6.339	26.108	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00 \text{ mm}$. $N_{f,Ed} = M_{Ed}/h$ at the restraint station ($h =$ overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2 \text{ cm}^2$, $S_{w2} = 62.5$, $S_{w3} = 46.3 \text{ cm}^4$, $e_0 = 29.1$, $e_{sc} = 50.9 \text{ mm}$	P385 App A

Torsion Bending Design @ 2.415 m

$T_{Ed} \text{ (max)}$	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
$\varphi_{max} \text{ (ULS)}$	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi'' \text{ (max)}$	@ $x = 2 \text{ m}$	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415 \text{ m}$: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0 \text{ m}$: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 6.339 \text{ kN.m}$	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707 \text{ kN.m}$	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465 \text{ m}$, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67 \text{ mm}$ @ $x = 2.85 \text{ m}$	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	26.108	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; $z_g = -100, 0$ mm; 1 combination(s), 0 solve(s) + 3 cached	41.391 kN.m	converged
$\lambda_{LT} = \sqrt{W \cdot f_y / M_{cr}}$	$\sqrt{227.0 \times 275 / 41.391}$	1.228	
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.228, 1.380, 0.76	0.442	Curve d
$\chi_{LT,mod} = \text{Fn}(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.442, 1.228, 0.935, 0.979	0.452	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.452 \times 227.0 \times 275 \leq 62.425 =$	28.204 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 28.204	5.868	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M_{Ed} (kN.m)	Load case	$N_{f,Ed} = M_{Ed}/h$ (kN)	2.5 % $N_{f,Ed}$ (kN)	
0	End 1 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R_x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f , Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
 W_{n0} , S_{w1} 40.3 cm², 115 cm⁴; $W_{n2} = 27.2$ cm², $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm⁴, $e_0 = 29.1$, $e_{sc} = 50.9$ mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free) $L/a = 12.27$ P385 App C
 φ_{max} (ULS) $L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
 $B_{Ed} = EI_w \varphi''$ (max) @ $x = 2$ m 0.683 kN.m²
 $M_{w,Ed} = EI_w \varphi''/(h-t_f)$ max over span 3.645 kN.m
 $M_{z,Ed} = \varphi \cdot M_{y,Ed}$ max coincident 109.717 kN.m
 $(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$ @ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$ 14.062 **Warning** P385 3.1.2
 $V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3})) - \tau_w/(f_y/\sqrt{3})}] \cdot V_{pl,Rd}$ @ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm²; $V_{pl,Rd} = 212.754$ 20.965 kN 6.2.7(9)
 $V_{Ed}/V_{pl,T,Rd}$ 107.430 / 20.965 5.124 **Warning**
 $M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_\alpha k_{zw} k_\alpha M_w/M_{f,Rk}$ k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 41.391$ kN.m — **BLOCKED**
End torques T_t St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m 3.771 / 3.200 kN.m
 $\theta_{ser} \leq \theta_{limit}$ @ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory) 15.97° review

NOT VERIFIED $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged. **NOT VERIFIED**

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$ 60.86 $\leq$ 6000 / 360 = 16.67 mm @ $x = 2.85$ m 60.86 mm **Warning**

MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	5.868	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s_s, c ; k_F	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3/h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3/175.00$	1266.377 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5/\lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5/0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed}/F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed}/M_{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1)/1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = \text{fn}(M_1, M_2, M_o, \psi, \mu)$	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C_1 basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN·m		SN003a

Lateral Buckling Check M.b.Rd

$L_e = 1 L$	$1 \times 6 =$	6 m	
Z_g	$Z_g = -100$ mm entered — not used by the P385/P362 κ chain (load height enters the channel route only through the destabilising L_E switch); the shear-centre M_{cr} route is not offered		load height
$\lambda_{LT} = (L_e/i_z)/\kappa$	$(6000/23.9)/96$ (S275)	2.615	P362 channel
$\chi_{LT} = \text{Fn}(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	2.615, 3.906, 0.76	0.141	Curve d
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y$ (κ chain, no f)	$0.141 \times 227.0 \times 275 =$	8.806 kN.m	
M_{cr} (back-calculated) = $W \cdot f_y / \lambda_{LT}^2$	$227.0 \times 275 / 2.615^2$	9.128 kN.m	Annex A only
$M_{y,Ed}/M_{b,Rd}$	$165.504 / 8.806$	18.794	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)]. V _{pl,Rd}	@ x = 0 m: τ _t = 195.04 (τ _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 20.965	5.124	Warning
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 9.128 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _{alpha} is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	18.794	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_{alpha} is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

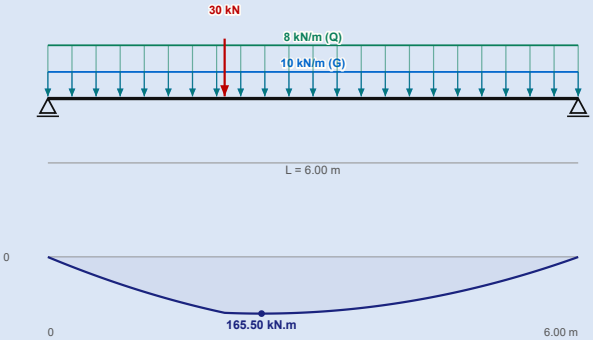
PFC-R8 · PFC 200 x 75 x 23 · M_{cr} method: eigen · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: EFFECTIVE LENGTHS OVERRIDE
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R8 "EFFECTIVE LENGTHS OVERRIDE": L_E = 0.85 L entered (S.leFactor 0.85) - applied by the standard (closed-form) route; the eigen route derives its own buckling length from the end flags and does not use the factor for M_{cr} (shown for comparison)

Beam & Beam-Portion (Member) (FAIL)
Member PFC-R8 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.021 / 212.754 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 107.430 / 212.754 = 0.505 OK
V_{pl,y,Rd} = A_vf_y/(√3 Y_{M0}) 1340 mm² x 275/(√3 x 1) 212.754 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 227.0/1 62.425 kN.m
M_{y,Ed}/M_{c,y,Rd} 165.504 / 62.425 = 2.651 Warning
M_{y,Ed}/M_{v,y,Rd} @ x @ x = 2 m: V_{Ed} = 55.810 > V_{pl,Rd} = 20.965: pure shear governs 2.662 Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15εt_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$l_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$l_y = \min[l_e + t_f \sqrt{(m_1/2 + (l_e/t_f)^2 + m_2)}, l_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(l_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1 ; L_{eff} = \chi_F \cdot l_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: $1.35G + 1.5Q$; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: $1.35G + 1.5Q$)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	l_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed}/F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

$C_1 = M_{cr} / M_{cr,uniform}$	35.819 / 31.304 kN.m (shape only: $z_g = z_j = 0$)	1.144	k_c only
$C_1 = fn(M_1, M_2, M_o, \psi, \mu)$ (standard, comparison)	0.0, 0.0, 161.1, 1.000, 300.000 — general moment diagram — Serna et al. quarter-point expression (SCI): $M(L/4)=132.1$, $M(L/2)=161.1$, $M(3L/4)=109.6$, $M_{max}=165.5$ kN.m	1.164	Serna

Lateral Buckling Check M.b.Rd

L_e = portion between restraints	End 1 v, $\varphi = 0$; End 2 v, $\varphi = 0$	6 m	FE
M_{cr} = FE eigenvalue (n_{Elem} , mesh error)	64 elements, 0.000 %; 1 combination(s), 0 solve(s) + 3 cached	35.819 kN.m	converged
$\lambda_{LT} = \sqrt{W_{pl,y} / M_{cr}}$	$\sqrt{227.0 \times 275 / 35.819}$	1.320	
$\chi_{LT} = Fn(\lambda_{LT}, \Phi_{LT}, \alpha_{LT})$	1.320, 1.503, 0.76	0.403	Curve d
$\chi_{LT,mod} = Fn(\chi_{LT}, \lambda_{LT}, k_c, f)$	0.403, 1.320, 0.935, 0.985	0.409	6.3.2.3
$M_{b,Rd} = \chi W_{pl,y} \cdot f_y \leq M_{c,y,Rd}$	$0.409 \times 227.0 \times 275 \leq 62.425 =$	25.562 kN.m	
$M_{y,Ed} / M_{b,Rd}$	165.504 / 25.562	6.475	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

h = 200.00 mm. N_{f,Ed} = M_{Ed}/h at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % N_{f,Ed} (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with M_{Ed} = 0 (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q _f , Q _w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W _{n0} , S _{w1}	40.3 cm ² , 115 cm ⁴ ; W _{n2} = 27.2 cm ² , S _{w2} = 62.5, S _{w3} = 46.3 cm ⁴ , e ₀ = 29.1, e _{sc} = 50.9 mm	P385 App A

Torsion Bending Design @ 2.415 m

T _{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L (φ = 0, warping free)	L/a = 12.27	P385 App C
φ _{max} (ULS)	L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
B _{Ed} = EI _w φ'' (max)	@ x = 2 m	0.683 kN.m ²	
M _{w,Ed} = EI _w φ''/(h-t _f)	max over span	3.645 kN.m	
M _{z,Ed} = φ . M _{y,Ed}	max coincident	109.717 kN.m	
(M _y /M _{pl,y}) ² + M _w /M _{pl,f} + M _z /M _{pl,z}	@ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; M _{pl,y} = 62.425, M _{pl,f} = 4.834, M _{pl,z} = 16.665	14.062	Warning P385 3.1.2
V _{pl,T.Rd} = [√(1 - τ _t /(1.25f _y /√3)) - τ _w /(f _y /√3)] . V _{pl,Rd}	@ x = 0 m: τ _t = 195.04 (τ _w = 5.22) N/mm ² ; V _{pl,Rd} = 212.754	20.965 kN	6.2.7(9)
V _{Ed} /V _{pl,T.Rd}	107.430 / 20.965	5.124	Warning
M _y /M _{b,Rd} + C _{mz} M _z /M _{z,Rk} + k _w k _{zw} k _α M _w /M _{f,Rk}	k _α unbounded: M _{y,Ed} reaches M _{cr} = 35.819 kN.m	—	BLOCKED
End torques T _t	St Venant part GI _T φ' at x = 0 / x = L; total T = GI _T φ' - EI _w φ''' = 4.307 / 3.707 kN.m	3.771 / 3.200 kN.m	
θ _{ser} ≤ θ _{limit}	@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED	M _{y,Ed} reaches the elastic critical moment M _{cr} : the Annex A amplifier k _{alpha} is unbounded; the member is inadequate as arranged.	NOT VERIFIED
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Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span δ ≤ Span/360	60.86 ≤ 6000 / 360 = 16.67 mm @ x = 2.85 m	60.86 mm	Warning
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MA/Mc	M _(y,Ed) /M _(b,Rd)	Deflection	V/V _{pl}	M-V	F/F _{Rd}	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	6.475	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: M_{y,Ed} reaches the elastic critical moment M_{cr}: the Annex A amplifier k_{alpha} is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist θ_{max} = 16.0° exceeds the ~2° guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: γ_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G, so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

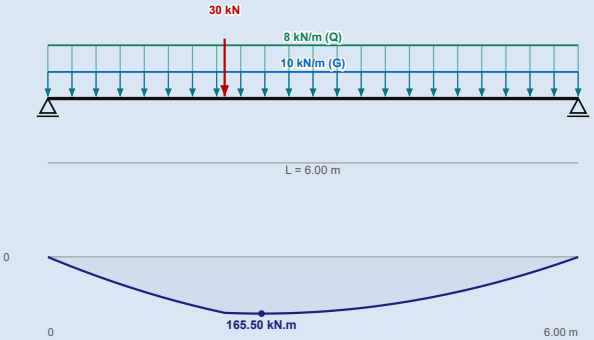
PFC-R8 · PFC 200 x 75 x 23 · M_{cr} method: standard · verdict **FAIL** · governing LTB+torsion (EN 1993-6 Annex A) = >= 99 (capacity exhausted)
Owner case: EFFECTIVE LENGTHS OVERRIDE
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R8 "EFFECTIVE LENGTHS OVERRIDE": L_E = 0.85 L entered (S.leFactor 0.85) - applied by the standard (closed-form) route; the eigen route derives its own buckling length from the end flags and does not use the factor for M_{cr} (shown for comparison)

Beam & Beam-Portion (Member) (FAIL)
Member PFC-R8 PFC 200 x 75 x 23
Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm z_g = 0 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm z_g = 0 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G > 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M.c.y.Rd

V_{y,Ed}/V_{pl,y,Rd} 0.021 / 212.754 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 107.430 / 212.754 = 0.505 OK
V_{pl,y,Rd} = A_vf_y/(√3 Y_{M0}) 1340 mm² x 275/(√3 x 1) 212.754 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/Y_{M0} 275 x 227.0/1 62.425 kN.m
M_{y,Ed}/M_{c,y,Rd} 165.504 / 62.425 = 2.651 Warning
M_{y,Ed}/M_{v,y,Rd} @ x @ x = 2 m: V_{Ed} = 55.810 > V_{pl,Rd} = 20.965: pure shear governs 2.662 Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15εt_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
s _s , c ; k _F	s _s = 100.00 mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; k _F = 2 + 6(s _s + c)/h _w ≤ 6 = 2 + 6 x 100.00/175.00	5.429 Fig 6.1(c)
F _{cr} = 0.9k _F · E · t _w ³ /h _w	0.9 x 5.429 x 210000 x 6.00 ³ /175.00	1266.377 kN 6.4(1)
l _e = k _F · E · t _w ² /(2f _{yw} · h _w) ≤ s _s + c	5.429 x 210000 x 6.00 ² /(2 x 275 x 175.00) = 426.39 ≤ 100.00	100.00 mm 6.5(4)
l _y = min[l _e + t _f √(m ₁ /2 + (l _e /t _f) ² + m ₂), l _e + t _f √(m ₁ + m ₂)]	m ₂ = 0.000 (first pass λ _F = 0.443 ≤ 0.5 with m ₂ = 3.920, so m ₂ = 0); min[204.77, 144.19]	144.19 mm 6.5(4)
λ _F = √(l _y · t _w · f _{yw} /F _{cr})	√(144.19 x 6.00 x 275/1266.377e3)	0.433 6.4(1)
χ _F = 0.5/λ _F ≤ 1 ; L _{eff} = χ _F · l _y	0.5/0.433 = 1.154 → 1.000 ; 1.000 x 144.19	144.19 mm 6.4(1)
F _{Rd} = f _{yw} · L _{eff} · t _w /γ _{M1}	275 x 144.19 x 6.00/1.0	237.920 kN 6.2(1)
F _{Ed} /F _{Rd}	F _{Ed} = 107.430 kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
η ₂ + 0.8η ₁ ≤ 1.4	0.452 + 0.8 x (M _{Ed} /M _{c,y,Rd} = 0.000/62.425 = 0.000) = 0.452 ≤ 1.4 (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s _s (mm)	k _F	l _y (mm)	λ _F	χ _F	F _{Rd} (kN)	F _{Ed} (kN)	Load case	F _{Ed} /F _{Rd}	(η ₂ +0.8η ₁)/1.4	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; s_s ≤ h_w (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever s_s + c < 2h_w/3 (k_F(c) < 6) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with F_{Ed} = max(P, R). η₁ uses the unreduced M_{c,y,Rd} (W_{pl,y}) [verify: EN 1993-1-5 4.6 writes η₁ with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Equivalent Uniform Moment Factor C1

C ₁ = fn(M ₁ , M ₂ , M ₀ , ψ, μ)	0.0, 0.0, 161.1, 1.000, 300.000	1.164	Serna
C ₁ basis	channel — P385/P362 kappa chain; general moment diagram — Serna et al. quarter-point expression (SCI): M(L/4)=132.1, M(L/2)=161.1, M(3L/4)=109.6, M _{max} =165.5 kN·m		SN003a

Lateral Buckling Check M.b.Rd

L _e = 0.85 L	0.85 x 6 =	5.1 m	
λ _{LT} = (L _e /i _z)/k	(5100/23.9)/96 (S275)	2.223	P362 channel
χ _{LT} = Fn(λ _{LT} , Φ _{LT} , α _{LT})	2.223, 3.045, 0.76	0.185	Curve d
M _{b,Rd} = χW _{pl,y} · f _y (k chain, no f)	0.185 x 227.0 x 275 =	11.549 kN.m	
M _{cr} (back-calculated) = W · f _y /λ _{LT} ²	227.0 x 275 / 2.223 ²	12.634 kN.m	Annex A only
M _{y,Ed} /M _{b,Rd}	165.504 / 11.549	14.331	Warning

Lateral Restraint Portions (restraint design forces)

x (m)	Restraint	M _{Ed} (kN.m)	Load case	N _{f,Ed} = M _{Ed} /h (kN)	2.5 % N _{f,Ed} (kN)	
0	End 1 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory
6	End 2 (torsional restraint, R _x held)	0.000	ULS: 1.35G + 1.5Q	0.000	0.000	restraint design force, advisory

$h = 200.00$ mm. $N_{f,Ed} = M_{Ed}/h$ at the restraint station (h = overall depth, EN 1993-1-1 5.3.3(3)); restraint design force 2.5 % $N_{f,Ed}$ (6.3.5.2(5)(b), SCI practice) — advisory, not part of the member verdict; the bracing system must also satisfy the 5.3.3 stiffness/imperfection requirements. A station with $M_{Ed} = 0$ (simply supported end) gets no flange force from this rule; its torsional restraint must still prevent twist.

Torsion Design

J, H, a, Q_f, Q_w	11.6 cm ⁴ , 0.0107 dm ⁶ , 489 mm, —, —	P385 App A
W_{n0}, S_{w1}	40.3 cm ² , 115 cm ⁴ ; $W_{n2} = 27.2$ cm ² , $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm ⁴ , $e_0 = 29.1$, $e_{sc} = 50.9$ mm	P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max)	combination ULS: 1.35G + 1.5Q	4.307 kN.m	
Torsion analysis	SCI P385 App C closed forms (Cases 3/4/10); fork ends at $x = 0$ and $x = L$ ($\varphi = 0$, warping free)	$L/a = 12.27$	P385 App C
φ_{max} (ULS)	$L/a = 12.27$; fork ends, warping free (P385 Cases 3/4/10)	0.6646 rad = 38.08°	
$B_{Ed} = EI_w \varphi''$ (max)	@ $x = 2$ m	0.683 kN.m ²	
$M_{w,Ed} = EI_w \varphi''/(h-t_f)$	max over span	3.645 kN.m	
$M_{z,Ed} = \varphi \cdot M_{y,Ed}$	max coincident	109.717 kN.m	
$(M_y/M_{pl,y})^2 + M_w/M_{pl,f} + M_z/M_{pl,z}$	@ $x = 2.415$ m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$	14.062	Warning P385 3.1.2
$V_{pl,T,Rd} = [\sqrt{(1 - \tau_t/(1.25f_y/\sqrt{3}))} - \tau_w/(f_y/\sqrt{3})] \cdot V_{pl,Rd}$	@ $x = 0$ m: $\tau_t = 195.04$ ($\tau_w = 5.22$) N/mm ² ; $V_{pl,Rd} = 212.754$	20.965 kN	6.2.7(9)
$V_{Ed}/V_{pl,T,Rd}$	107.430 / 20.965	5.124	Warning
$M_y/M_{b,Rd} + C_{mz} M_z/M_{z,Rk} + k_w k_{zw} k_\alpha M_w/M_{f,Rk}$	k_α unbounded: $M_{y,Ed}$ reaches $M_{cr} = 12.634$ kN.m	—	BLOCKED
End torques T_t	St Venant part $GI_T \varphi'$ at $x = 0 / x = L$; total $T = GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m	3.771 / 3.200 kN.m	
$\theta_{ser} \leq \theta_{limit}$	@ $x = 2.465$ m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)	15.97°	review

NOT VERIFIED

$M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

NOT VERIFIED

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$	$60.86 \leq 6000 / 360 = 16.67$ mm @ $x = 2.85$ m	60.86 mm	Warning
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MA/Mc	$M_{(y,Ed)}/M_{(b,Rd)}$	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	LTB+T	Max
2.651	14.331	3.652	0.505	2.662	0.452	1.666	14.062	5.124	99.000	99.000

FAIL — governing LTB+torsion (EN 1993-6 Annex A) = 99.000

NOT COVERED: $M_{y,Ed}$ reaches the elastic critical moment M_{cr} : the Annex A amplifier k_α is unbounded; the member is inadequate as arranged.

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-R9 · PFC 200 x 75 x 23 · M_{cr} method: full · verdict **FAIL** · governing Bending+torsion cross-section (P385 3.1.2) = 14.062

Owner case: TOP FLANGE RESTRAINED
PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at e = +40 mm, z_g = 0); R9 "TOP FLANGE RESTRAINED": for a sagging simply supported beam the top flange IS the compression flange, so this is the fully restrained case (restraint = full); the tool has no flange-level restraint input

Beam & Beam-Portion (Member) (FAIL)

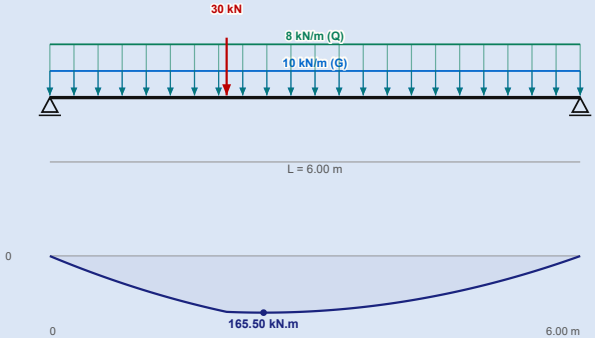
Member PFC-R9 PFC 200 x 75 x 23

Between 0 and 6 m, in Load Case 1 (ULS: 1.35G + 1.5Q)

End conditions: End 1: U_x U_y U_z R_x restrained (pinned in plane; LTB fork); End 2: U_y U_z R_x restrained (pinned in plane; LTB fork)
— simply supported

Member Loading and Member Forces

Loading Combination : ULS: 1.35G + 1.5Q
G UDL 10.000 0-6 m e = 40 mm (kN/m)
Q UDL 8.000 0-6 m e = 40 mm (kN/m)
Q PY 30.000 @ 2 m e = 40 mm (kN)
G SW 0.2296 kN/m 0-6 m e = 50.9 mm (automatic)
Y_{G,inf} companions (reactions only): 2 - G at 1.0 (STR set B) and 0.9 (EQU set A) of every ULS combination with G
> 1.0, variable factors as entered
ULS C1: ULS: 1.35G + 1.5Q [Y_{G,inf} = 1.0, STR set B]
ULS C2: ULS: 1.35G + 1.5Q [Y_{G,inf} = 0.9, EQU set A]
Y_{G,inf} companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only
- uplift / hold-down forces and the web-bearing reactions;
the moment and shear envelopes keep the entered Y_G, so add a



Member Forces in Load Case 1 (ULS: 1.35G + 1.5Q) and Maximum Deflection from Load Case 1 (SLS: 1.0Q)												
Mem ber No.	Node End1 End2	Axial Force (kN)	Torque Moment (kN.m)	Shear Force (kN)		Bending Moment (kN.m)		Maximum Moment (kN.m @ m)		Maximum Deflection (mm @ m)	Reaction R (kN), M (kN.m)	
				y-y	z-z	y-y	z-z	y-y	z-z		R	M
1	End 1, x = 0	0.00C	3.77	107.43	0.00	0.00	0.00	165.50	0.00	-60.86	107.43	—
	End 2, x = 6	0.00C	-3.20	-92.43	0.00	0.00	0.00	@ 2.418	@ —	@ 2.850	92.43	—

Reactions of ULS: 1.35G + 1.5Q at the member ends: R positive upward (a negative R is uplift), M = the end bending moment in the diagram convention (sagging positive, hogging negative); a pinned end carries no M, a guided end no R, a free end neither. V_z = 0: no minor-axis shear in the single-plane model; M_z is the entered constant design moment.
Uplift no end lifts in any of the 4 combinations (all reactions ≥ 0; incl. the 2 Y_{G,inf} companions with G at 1.0 and 0.9) R_{min} ≥ 0 OK

Classification and Effective Area (EN 1993: 2006)

Section (23.4 kg/m) 200 x 75 x 23 PFC [S275]
Class = Fn(c/t, d/t, f_y, N, M_y, M_z) 4.56, 25.20, 275, 0.00, 165.50, 0.00 Class 1
(Axial: Non-Slender)
Auto Design Load Cases 1; SLS 1

Moment Capacity Check M_{c,y,Rd} - Fully Restrained Beam

V_{y,Ed}/V_{pl,y,Rd} 0.021 / 212.754 = 0.000 Low Shear
V_{y,Ed,max}/V_{pl,y,Rd} 107.430 / 212.754 = 0.505 OK
V_{pl,y,Rd} = A_ff_y/(√3V_{M0}) 1340 mm² x 275/(√3 x 1) 212.754 kN 6.2.6
M_{c,y,Rd} = f_yW_{pl,y}/V_{M0} 275 x 227.0/1 62.425 kN.m
M_{y,Ed}/M_{c,y,Rd} 165.504 / 62.425 = 2.651 Warning
M_{y,Ed}/M_{v,y,Rd} @ x 2 m: V_{Ed} = 55.810 > V_{pl,Rd} = 20.965: pure shear governs 2.662 Warning

Web Transverse Forces (EN 1993-1-5 cl 6)

Web h_w, t_w, t_f, b_f 175.00, 6.00, 12.50, 75.00 mm (B = 75.00 ≤ t_w + 15εt_f = 179.33); f_{yw} = f_{yf} = 275; 1 web Fig 5.1

$m_1 = f_{yf} \cdot b_f / (f_{yw} \cdot t_w)$; $m_2 = 0.02(h_w/t_f)^2$	$275 \times 75.00 / (275 \times 6.00) = 12.500$; $0.02 \times (175.00/12.50)^2 = 3.920$ if $\lambda_F > 0.5$, else 0	6.5(1)
a = distance between transverse stiffeners	no transverse stiffeners declared: a = L = 6000 mm, the full member length (conservative)	6.4(1)
Governing station x = 0 m: End 1 reaction, load type (c) Fig 6.1(c) end [types (a), (c) evaluated, lower F_{Rd} governs]		
$s_s, c ; k_F$	$s_s = 100.00$ mm (entered); c = 0.00 mm (d = 0.00 to the member end) ; $k_F = 2 + 6(s_s + c)/h_w \leq 6 = 2 + 6 \times 100.00/175.00$	5.429 Fig 6.1(c)
$F_{cr} = 0.9k_F \cdot E \cdot t_w^3 / h_w$	$0.9 \times 5.429 \times 210000 \times 6.00^3 / 175.00$	1266.377 kN 6.4(1)
$I_e = k_F \cdot E \cdot t_w^2 / (2f_{yw} \cdot h_w) \leq s_s + c$	$5.429 \times 210000 \times 6.00^2 / (2 \times 275 \times 175.00) = 426.39 \leq 100.00$	100.00 mm 6.5(4)
$I_y = \min[I_e + t_f \sqrt{(m_1/2 + (I_e/t_f)^2 + m_2)}, I_e + t_f \sqrt{(m_1 + m_2)}]$	$m_2 = 0.000$ (first pass $\lambda_F = 0.443 \leq 0.5$ with $m_2 = 3.920$, so $m_2 = 0$); $\min[204.77, 144.19]$	144.19 mm 6.5(4)
$\lambda_F = \sqrt{(I_y \cdot t_w \cdot f_{yw} / F_{cr})}$	$\sqrt{(144.19 \times 6.00 \times 275 / 1266.377e3)}$	0.433 6.4(1)
$\chi_F = 0.5 / \lambda_F \leq 1$; $L_{eff} = \chi_F \cdot I_y$	$0.5 / 0.433 = 1.154 \rightarrow 1.000$; 1.000×144.19	144.19 mm 6.4(1)
$F_{Rd} = f_{yw} \cdot L_{eff} \cdot t_w / \gamma_{M1}$	$275 \times 144.19 \times 6.00 / 1.0$	237.920 kN 6.2(1)
F_{Ed} / F_{Rd}	$F_{Ed} = 107.430$ kN (ULS: 1.35G + 1.5Q; on the bottom flange, no coincident moment) / 237.920 =	0.452 OK
$\eta_2 + 0.8\eta_1 \leq 1.4$	$0.452 + 0.8 \times (M_{Ed} / M_{c,y,Rd} = 0.000 / 62.425 = 0.000) = 0.452 \leq 1.4$ (ULS: 1.35G + 1.5Q)	0.323 OK 7.2

x (m)	Station	Type	s_s (mm)	k_F	I_y (mm)	λ_F	χ_F	F_{Rd} (kN)	F_{Ed} (kN)	Load case	F_{Ed} / F_{Rd}	$(\eta_2 + 0.8\eta_1) / 1.4$	
0	End 1 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	107.430	ULS: 1.35G + 1.5Q	0.452	0.323	governs
2	point load 3	(a)	0.00	6.002	113.39	0.366	1.000	187.091	45.000	ULS: 1.35G + 1.5Q	0.241	1.666	Warning
6	End 2 reaction	(c)	100.00	5.429	144.19	0.433	1.000	237.920	92.430	ULS: 1.35G + 1.5Q	0.388	0.277	OK

Point loads act on the top flange (bottom flange for an upward load), reactions on the bottom flange; $s_s \leq h_w$ (6.3(1)); c = distance from the bearing edge to the member end; type (c) is evaluated whenever $s_s + c < 2h_w/3$ ($k_F(c) < 6$) and the lower F_{Rd} of types (a) and (c) governs; a load over an end holding U_z is type (b) with $F_{Ed} = \max(P, R)$. η_1 uses the unreduced $M_{c,y,Rd}$ ($W_{pl,y}$) [verify: EN 1993-1-5 4.6 writes η_1 with W_{eff}]. Distributed loads, hanger loads, the closely-spaced total-load check (6.3(3)) and flange-induced buckling (section 8) are not evaluated.

Lateral Buckling Check M.b.Rd

$M_{b,Rd} = M_{c,y,Rd}$ Fully Restrained 62.425 kN.m

Torsion Design

J, H, a, Q_f , Q_w 11.6 cm⁴, 0.0107 dm⁶, 489 mm, —, — P385 App A
 W_{n0} , S_{w1} 40.3 cm², 115 cm⁴; $W_{n2} = 27.2$ cm², $S_{w2} = 62.5$, $S_{w3} = 46.3$ cm⁴, $e_0 = 29.1$, $e_{sc} = 50.9$ mm P385 App A

Torsion Bending Design @ 2.415 m

T_{Ed} (max) combination ULS: 1.35G + 1.5Q 4.307 kN.m
Torsion analysis SCI P385 App C closed forms (Cases 3/4/10); fork ends at x = 0 and x = L ($\varphi = 0$, warping free) L/a = 12.27 P385 App C
 φ_{max} (ULS) L/a = 12.27; fork ends, warping free (P385 Cases 3/4/10) 0.6646 rad = 38.08°
 $B_{Ed} = EI_w \varphi''$ (max) @ x = 2 m 0.683 kN.m²
 $M_{w,Ed} = EI_w \varphi'' / (h - t_f)$ max over span 3.645 kN.m
 $M_{z,Ed} = \varphi \cdot M_{y,Ed}$ max coincident 109.717 kN.m
 $(M_y / M_{pl,y})^2 + M_w / M_{pl,f} + M_z / M_{pl,z}$ @ x = 2.415 m: 165.503, 2.315, 109.220 kN.m; $M_{pl,y} = 62.425$, $M_{pl,f} = 4.834$, $M_{pl,z} = 16.665$ 14.062 **Warning** P385 3.1.2
 $V_{pl,T,Rd} = [\sqrt{(1 - \tau_t / (1.25f_y / \sqrt{3}))} - \tau_t / (f_y / \sqrt{3})] \cdot V_{pl,Rd}$ @ x = 0 m: $\tau_t = 195.04$ ($T_w = 5.22$) N/mm²; $V_{pl,Rd} = 212.754$ 20.965 kN 6.2.7(9)
 $V_{Ed} / V_{pl,T,Rd}$ 107.430 / 20.965 5.124 **Warning**
End torques T_t St Venant part $GI_T \varphi'$ at x = 0 / x = L; total T = $GI_T \varphi' - EI_w \varphi''' = 4.307 / 3.707$ kN.m 3.771 / 3.200 kN.m

$\theta_{ser} \leq \theta_{limit}$

@ x = 2.465 m, SLS: 1.0Q; limit 2° (P385 guidance, advisory)

15.97°

review

Deflection Check - Load Case 1 (SLS: 1.0Q)

In-span $\delta \leq \text{Span}/360$

$60.86 \leq 6000 / 360 = 16.67 \text{ mm @ } x = 2.85 \text{ m}$

60.86 mm

Warning

MA/Mc	M_(y.Ed)/M_(b.Rd)	Deflection	V/Vpl	M-V	F/F_Rd	Web 7.2	Torsion	V+T	Max
2.651	2.651	3.652	0.505	2.662	0.452	1.666	14.062	5.124	14.062

FAIL — governing Bending+torsion cross-section (P385 3.1.2) = 14.062

ADVISORY: Web transverse forces (EN 1993-1-5 clause 6) are checked at every point load and support as patch loads on the loaded flange: distributed loads, loads hung from the bottom flange (hangers), the total-load check of closely spaced loads (6.3(3)) and flange-induced buckling (section 8) are not evaluated; a declared bearing stiffener must be designed to 9.4.

ADVISORY: Serviceability twist $\theta_{max} = 16.0^\circ$ exceeds the $\sim 2^\circ$ guideline (SCI P385): check that the rotation is acceptable for whatever the member supports.

ADVISORY: $\gamma_{G,inf}$ companions: 2 ULS combination(s) re-solved with G at 1.0 (STR set B, EN 1990 6.4.3.1(4) / Table A1.2(B)) and at 0.9 (EQU set A, Table A1.2(A)) over the whole member (single-source permanent action) for the END REACTIONS only - uplift / hold-down forces and the web-bearing reactions; the moment and shear envelopes keep the entered γ_G , so add a reduced-G combination by hand where a relieving permanent action could increase a moment.

PFC-R10 · PFC 200 x 75 x 23 · M_{cr} method: refused · verdict **REFUSED**

Owner case: BOTTOM FLANGE RESTREAINED

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R10 "BOTTOM FLANGE RESTREAINED" = tension-flange restraint only: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Tension-flange (bottom flange of a sagging beam) restraint is not modelled: the tool's restraints (end flags $U_y / R_x / R_z /$ warping and the intermediate lateral restraints v, v', ϕ, ϕ') act at the shear centre and have no height in the cross-section, so a restraint that holds the tension flange laterally while the compression flange is free cannot be described; it needs a restraint-height term in the LTB eigen model (lateral constraint at $z = -h/2$, which couples v and ϕ) or the tension-flange-restraint method of BS 5950-1 Annex G / SCI P093. Treating it as unrestrained (R2) is the conservative bound; treating it as fully restrained (R9) is unsafe.

Owner case: BOTTOM FLANGE RESTREAINED

PFC-R11 · PFC 200 x 75 x 23 · M_{cr} method: refused · verdict **REFUSED**

Owner case: PARTIALLY RESTRAINED

PFC 200 x 75 x 23, 6 m SS, L2 loading (all loads at $e = +40$ mm, $z_g = 0$); R11 "PARTIALLY RESTRAINED" = elastic (spring) restraint: NOT MODELLED by the tool (documented refusal)

REFUSED — not run

Elastic (partial) restraint is not modelled: every restraint in the tool is rigid or absent (each of v , v' , ϕ , ϕ' is held or free), so a lateral or torsional spring of finite stiffness (a purlin, a sheeting rail, a slab with a stiffness) cannot be entered; it needs lateral / rotational spring stiffness terms added to the LTB eigen element matrices (and to the warping-torsion FE) with a stiffness input per restraint and per end, plus the EN 1993-1-1 BB.2 minimum-stiffness test to decide whether a spring counts as a full restraint. The bounds are R2 (no restraint) and R3 (rigid restraints at the same positions).

Owner case: PARTIALLY RESTRAINED